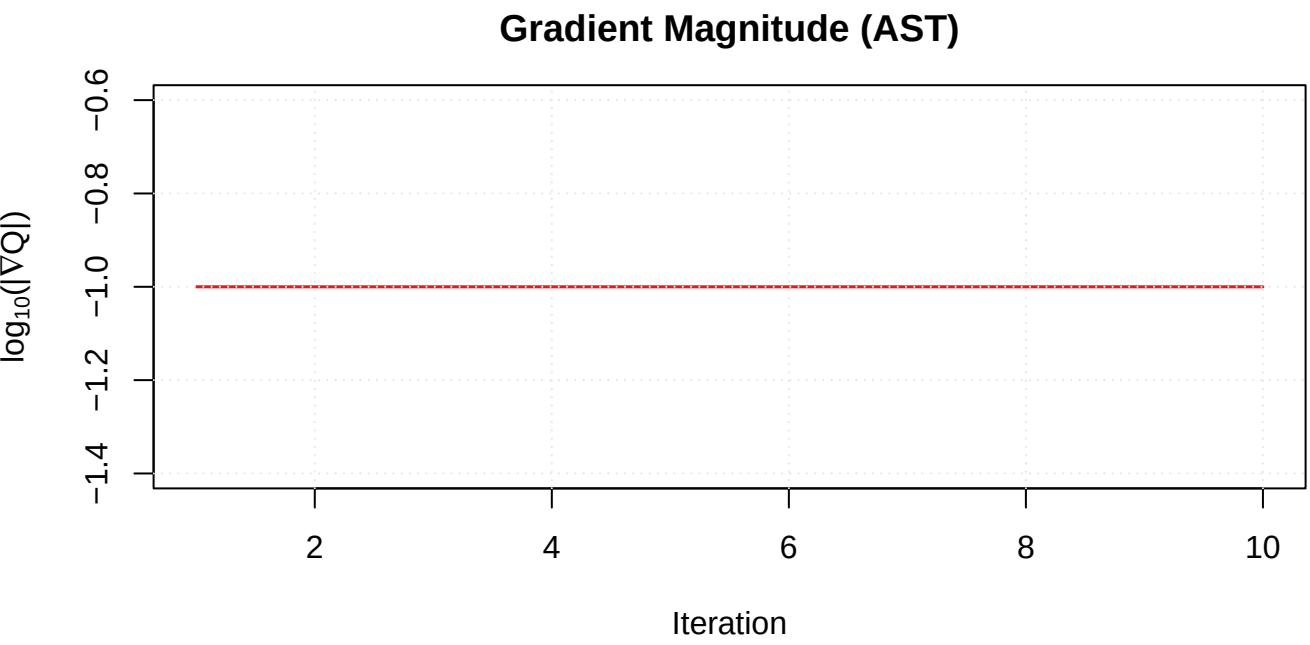
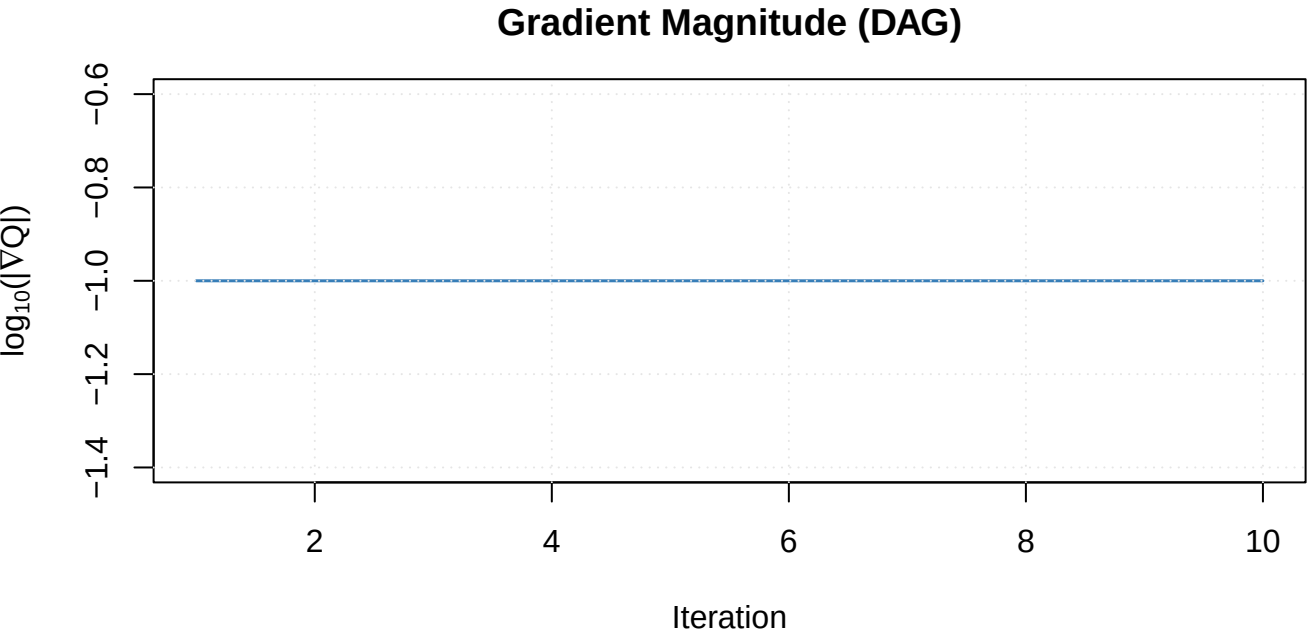
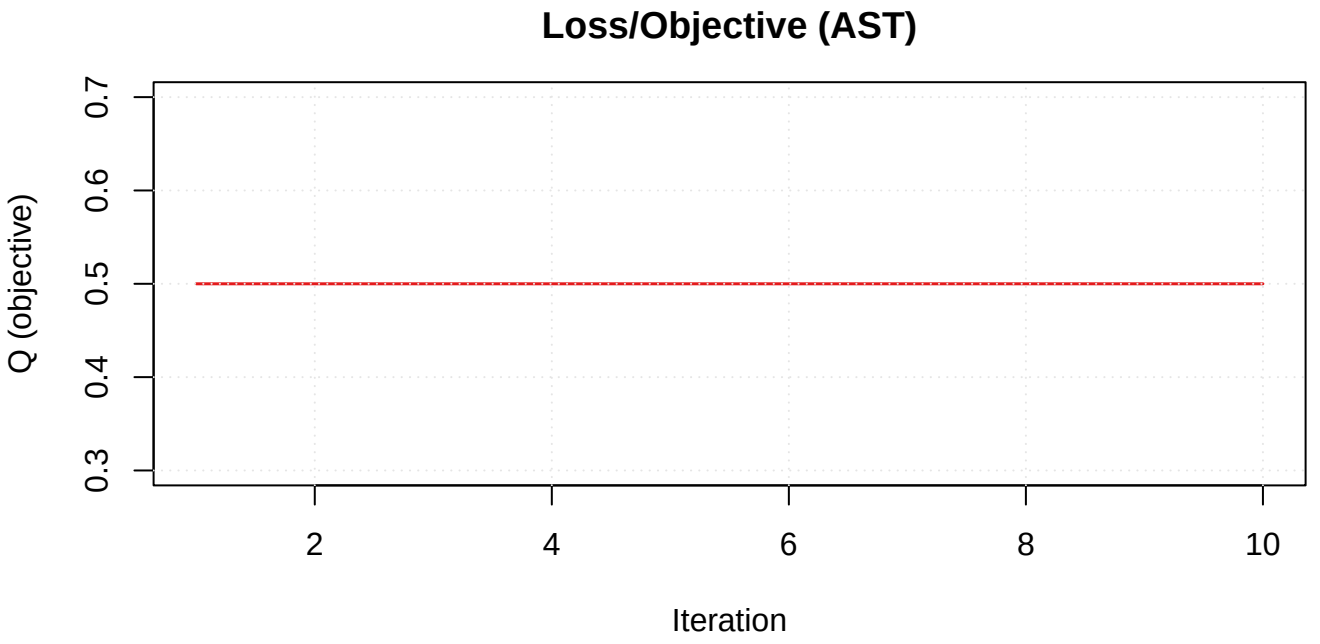
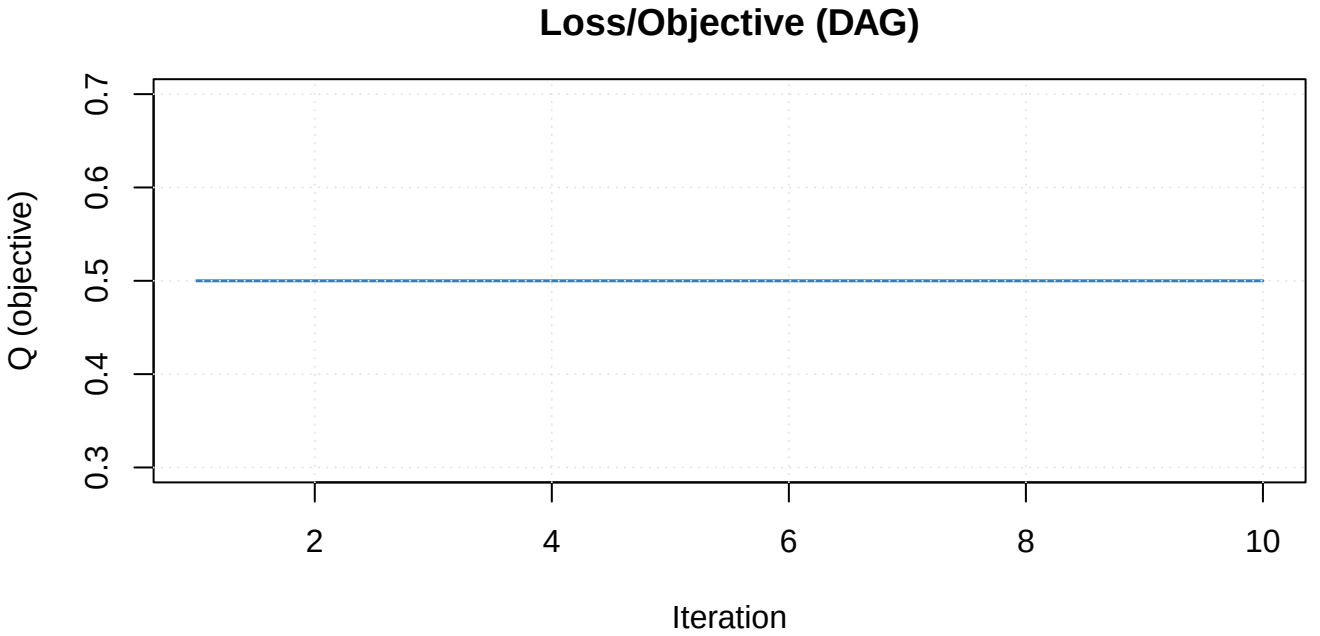
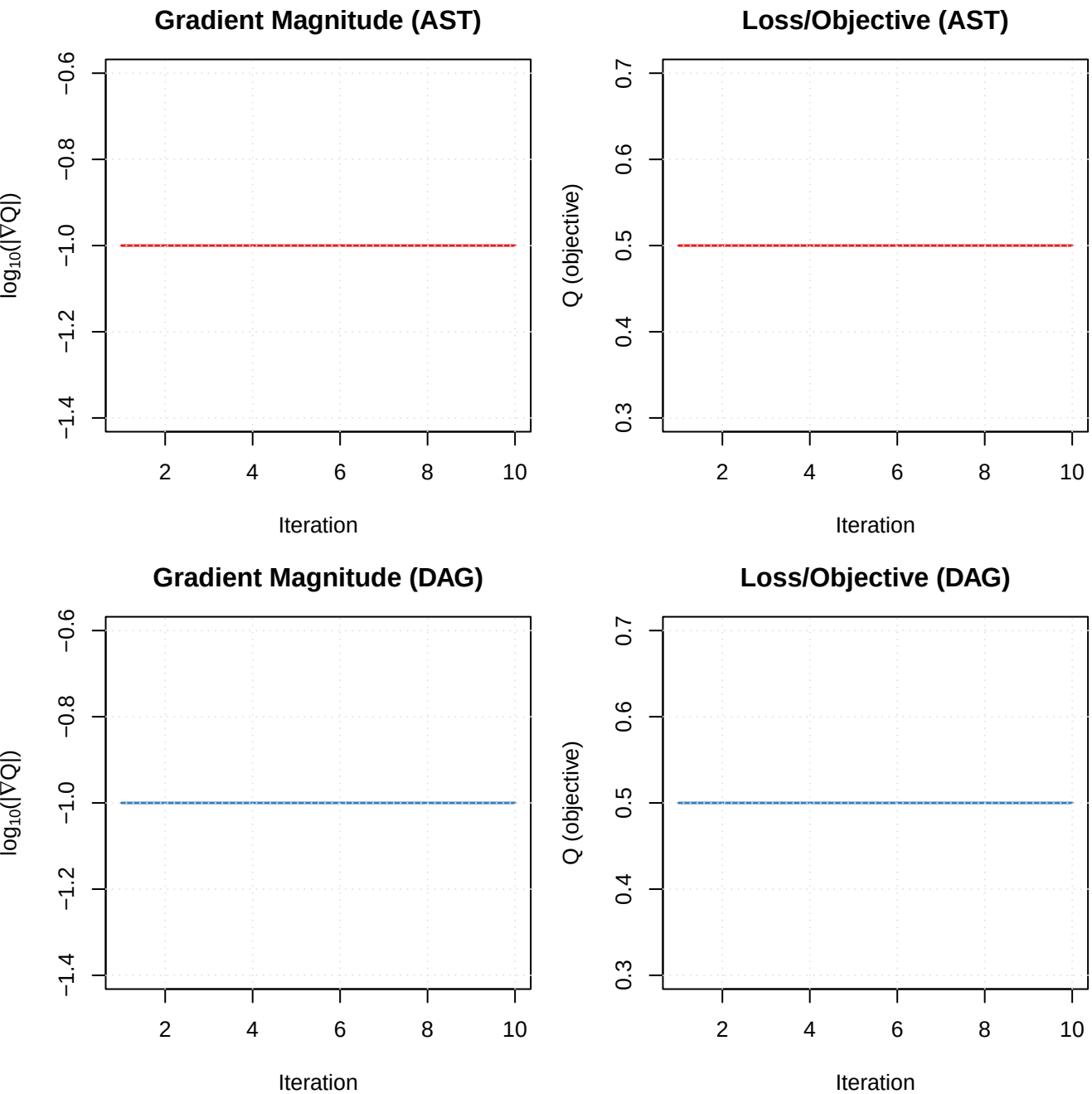
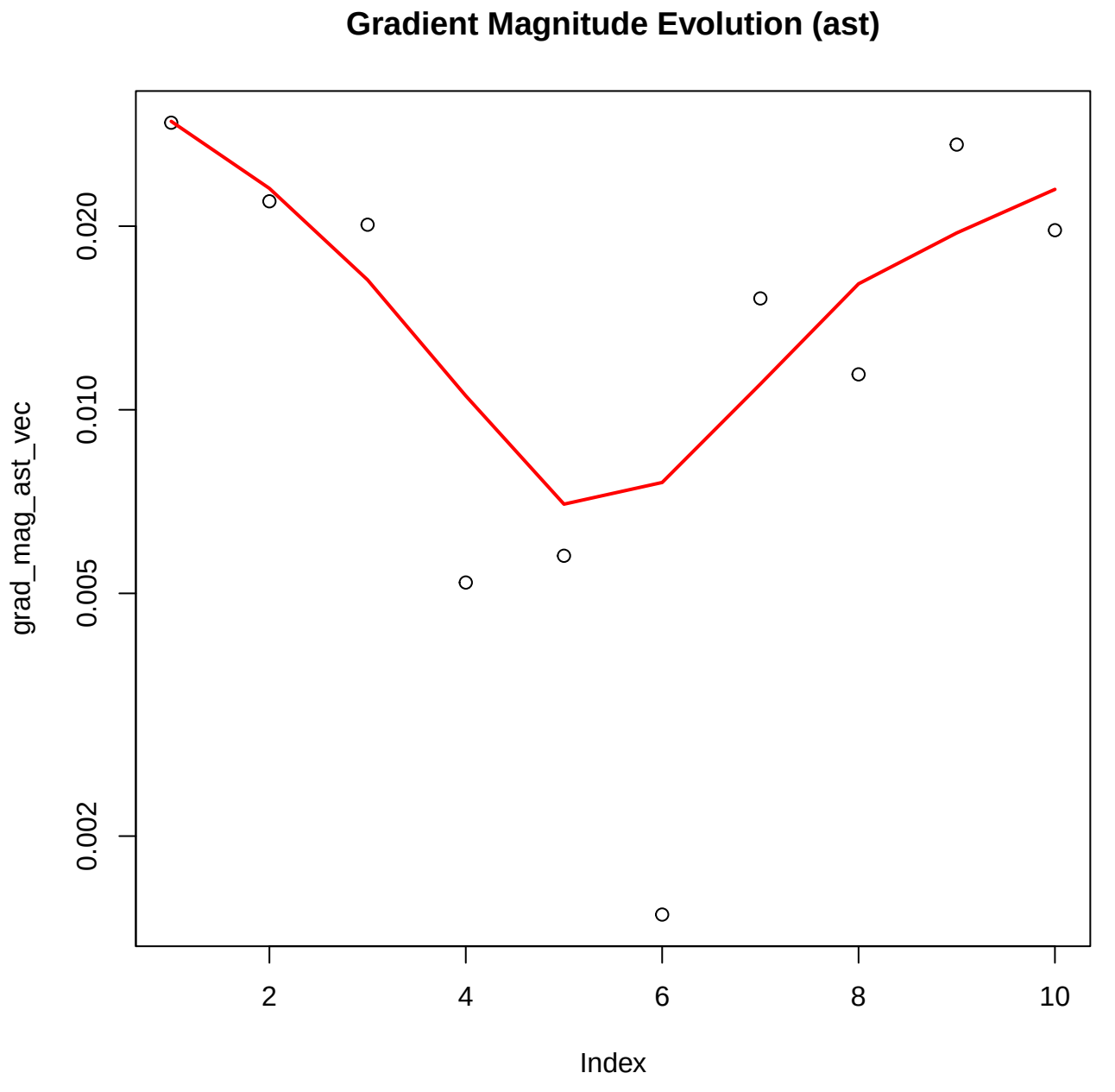

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

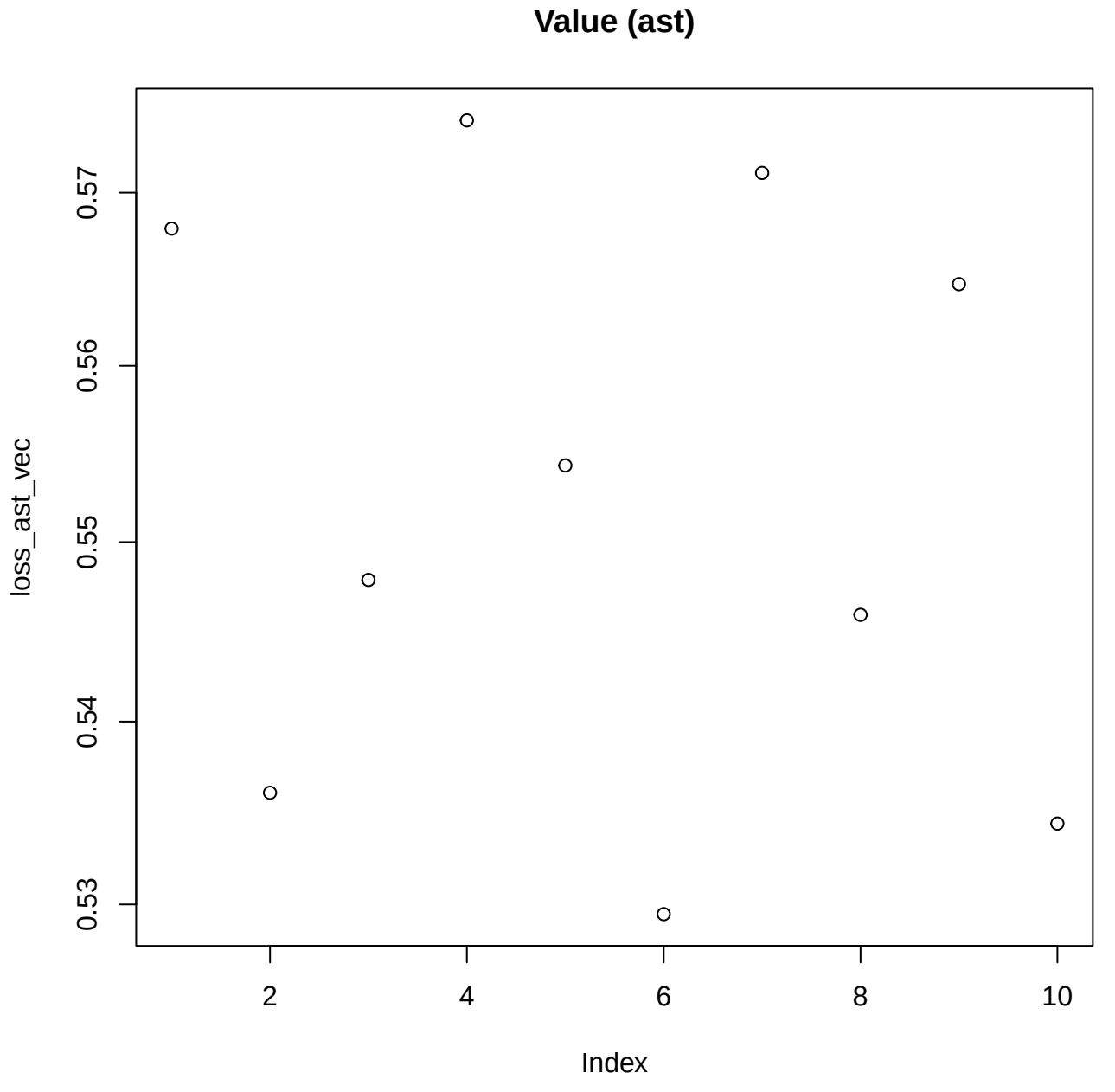

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

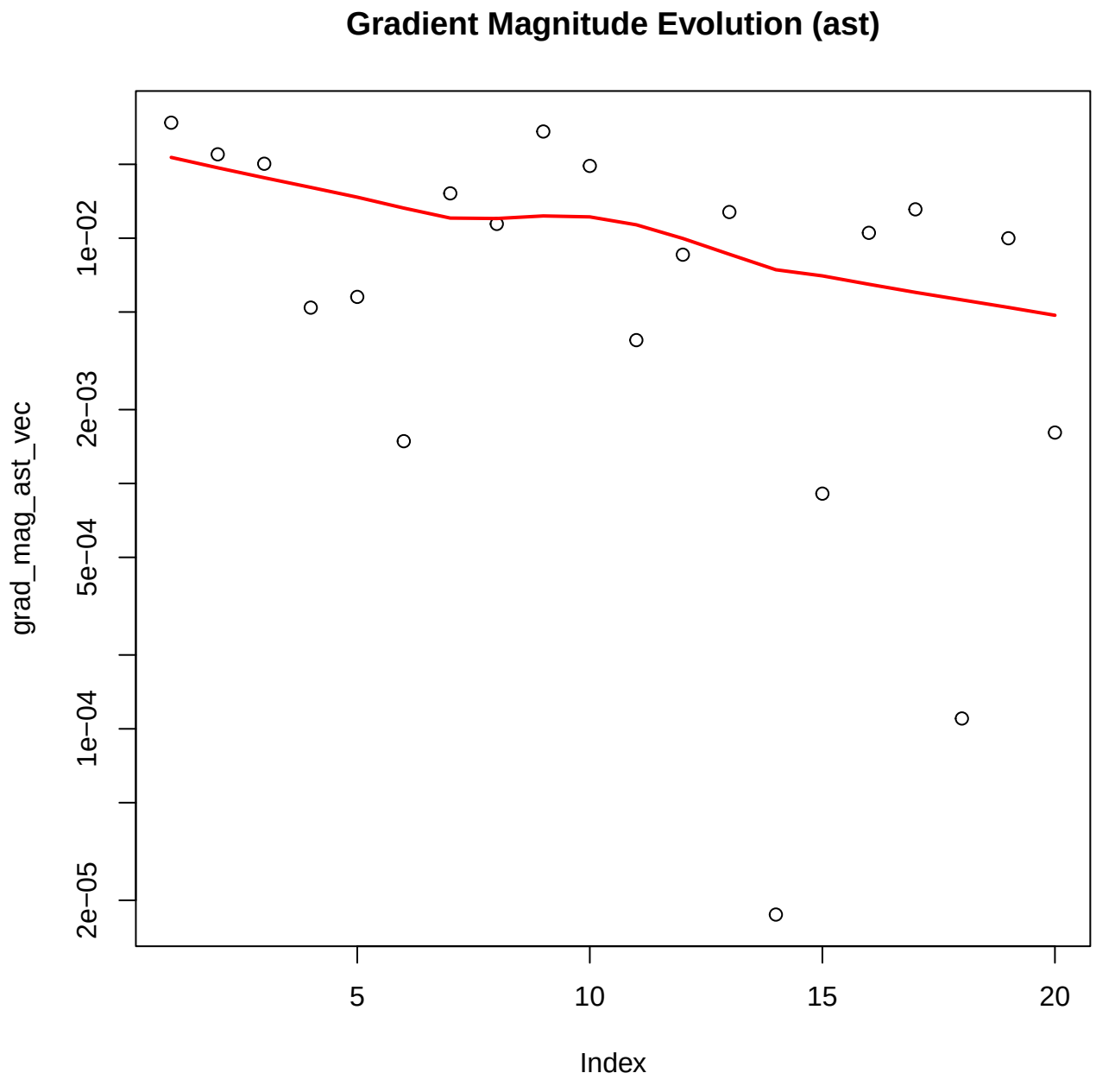

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

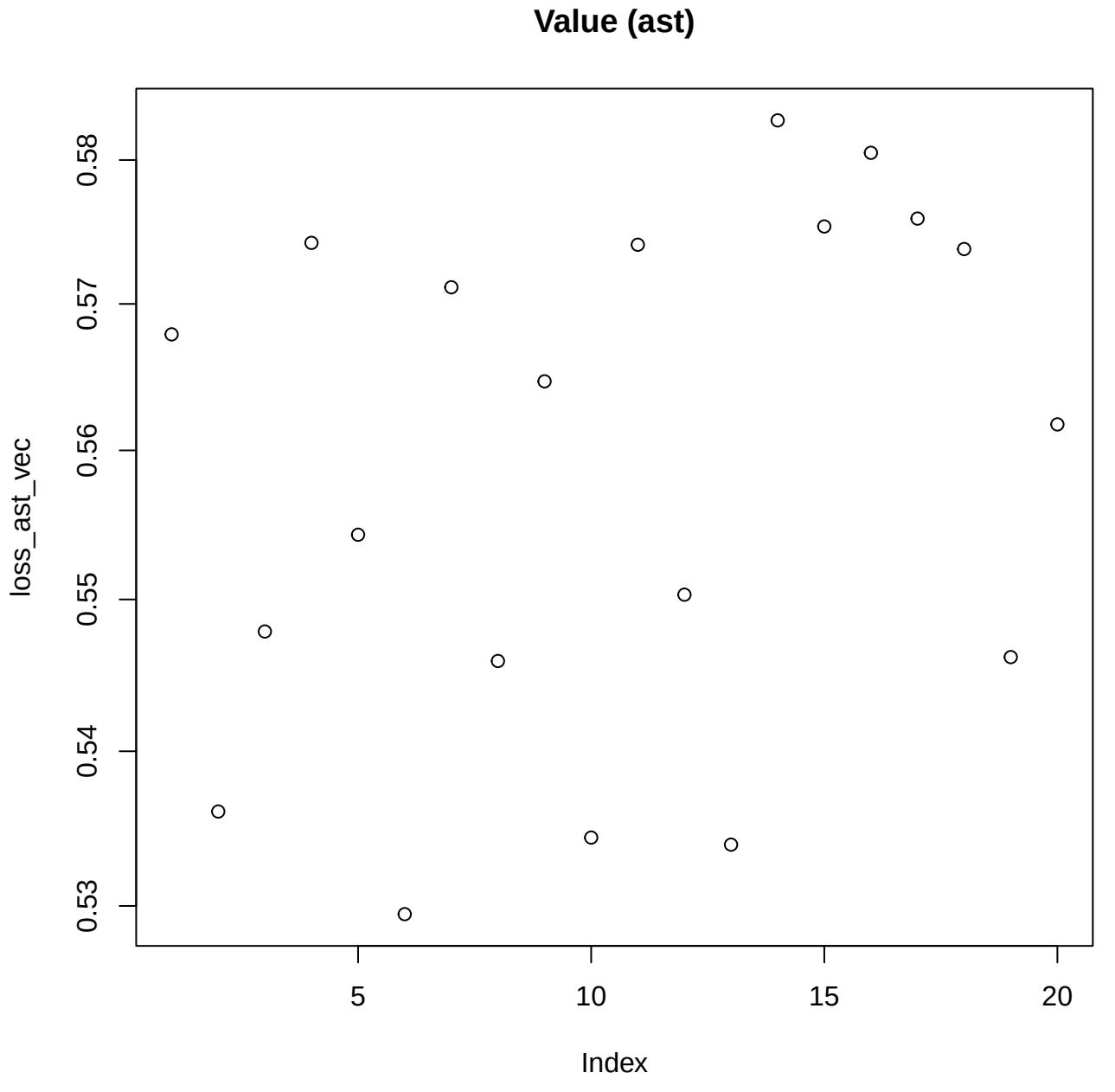

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

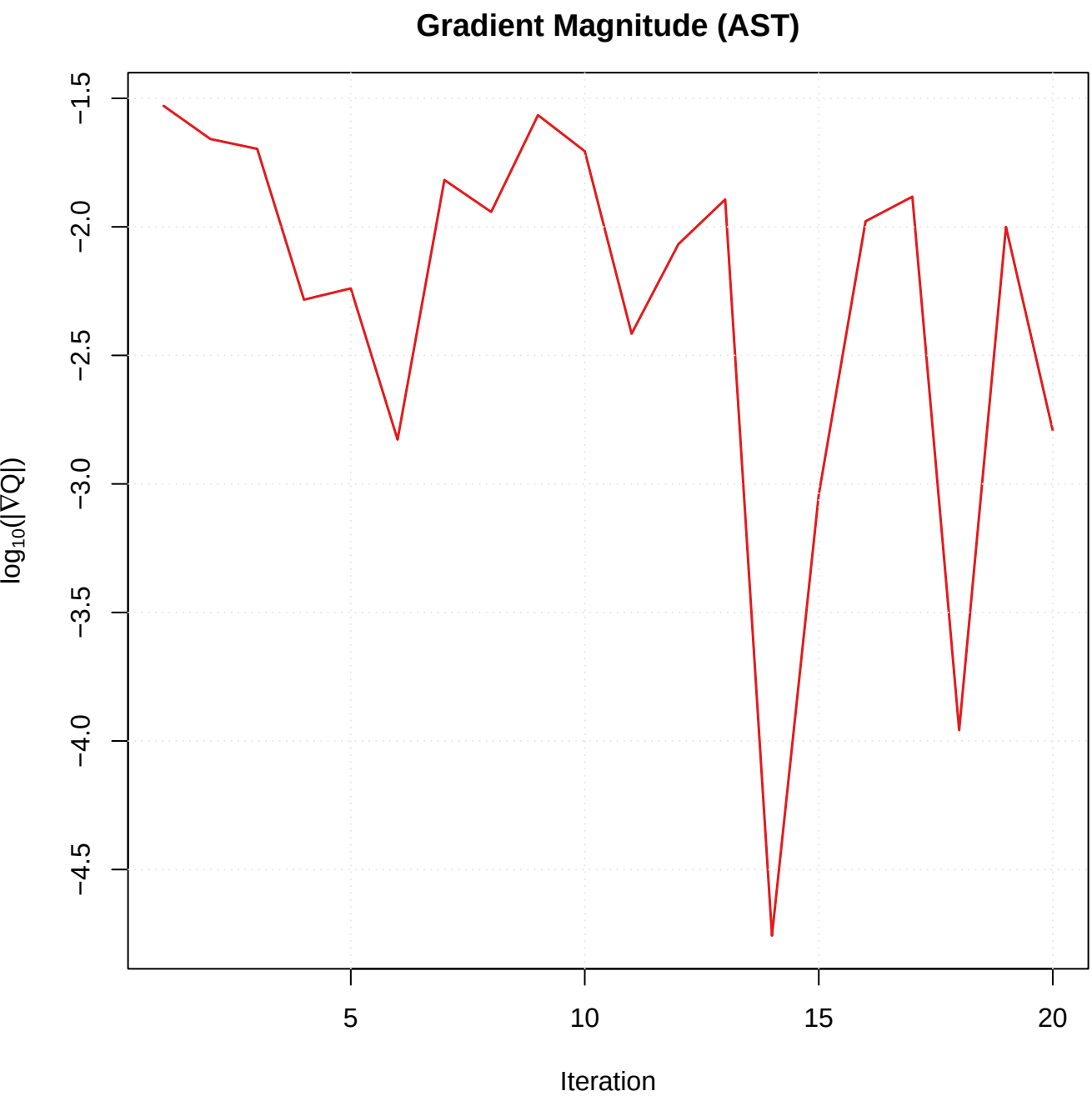

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

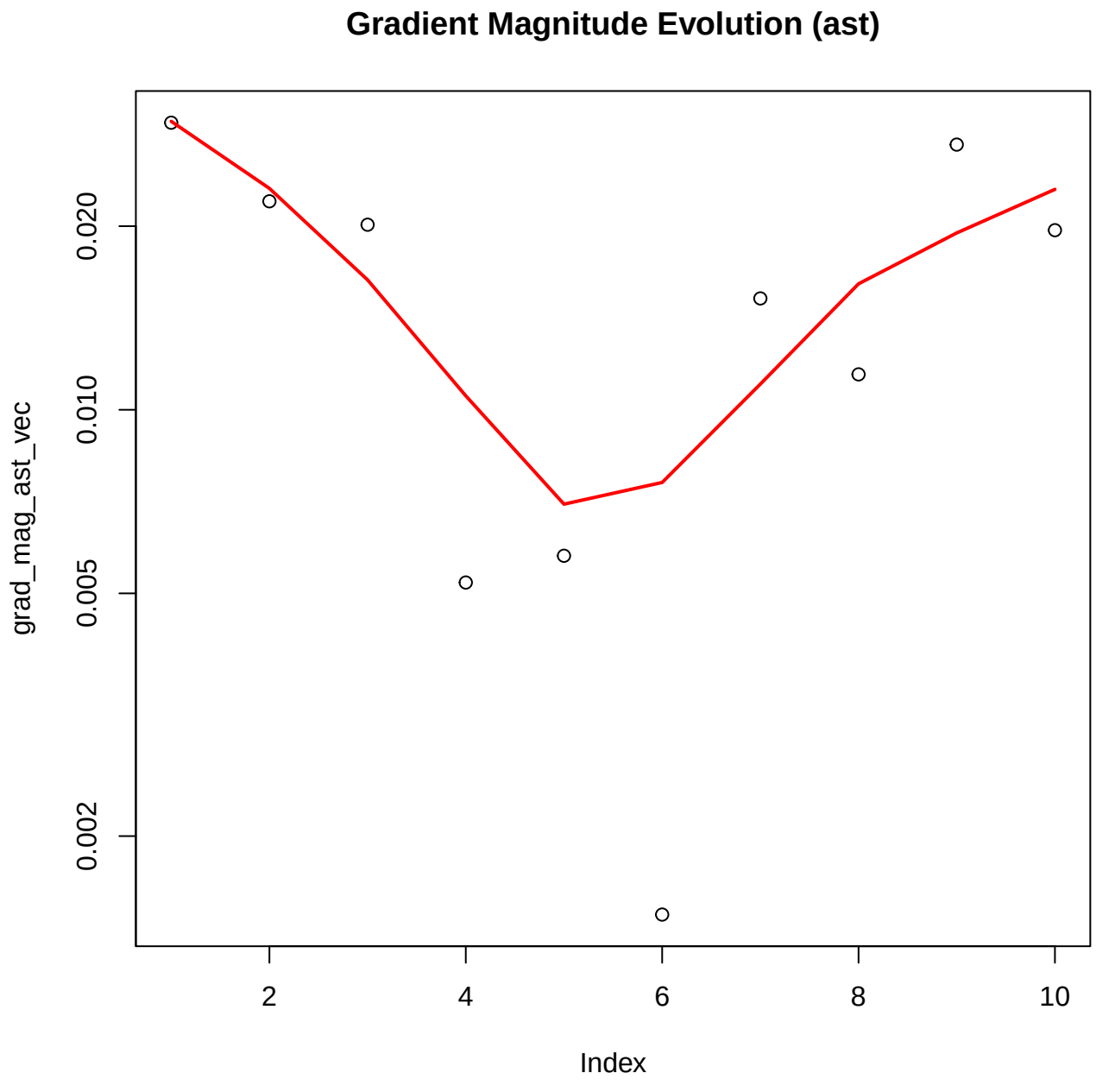

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

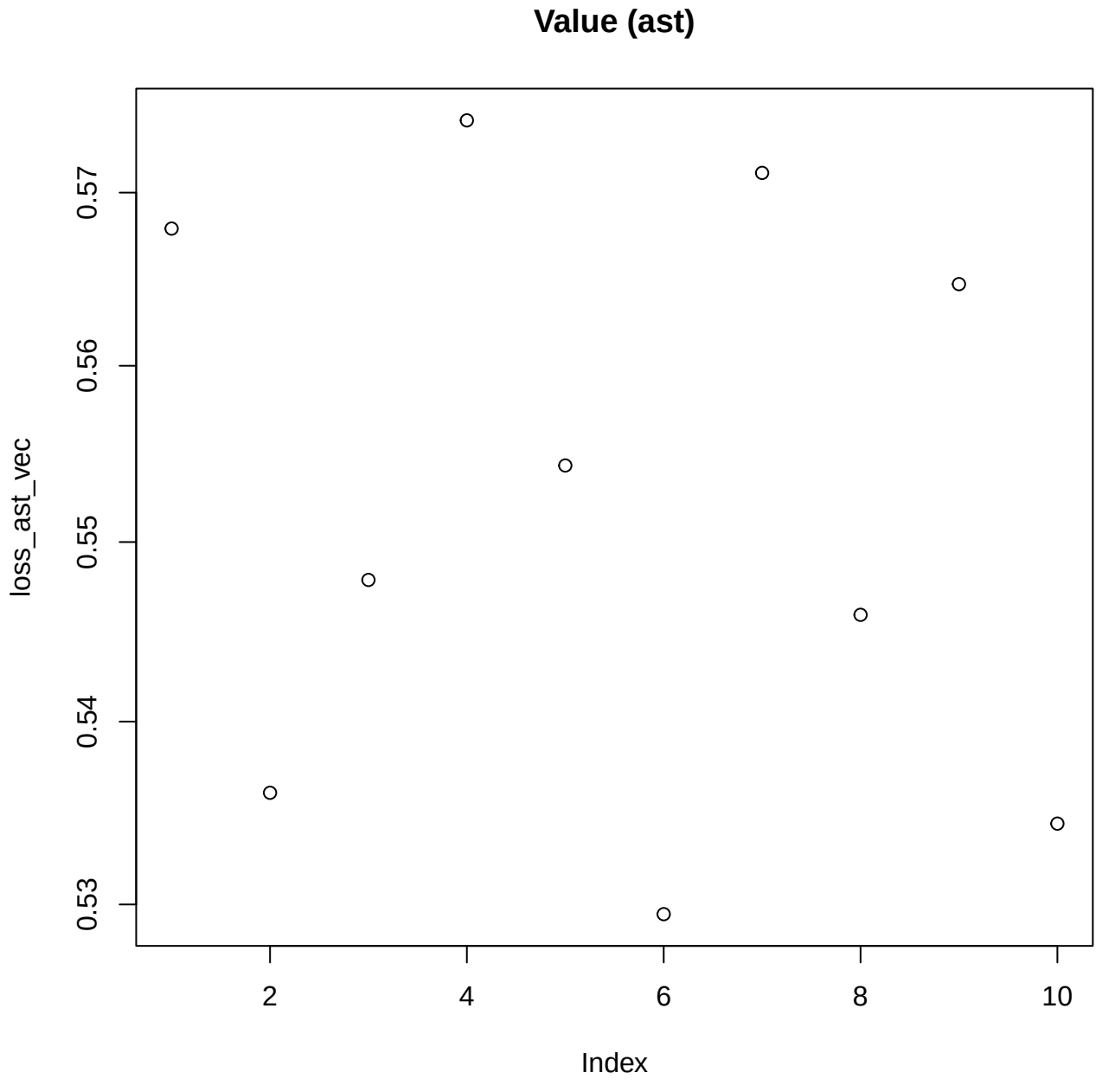

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

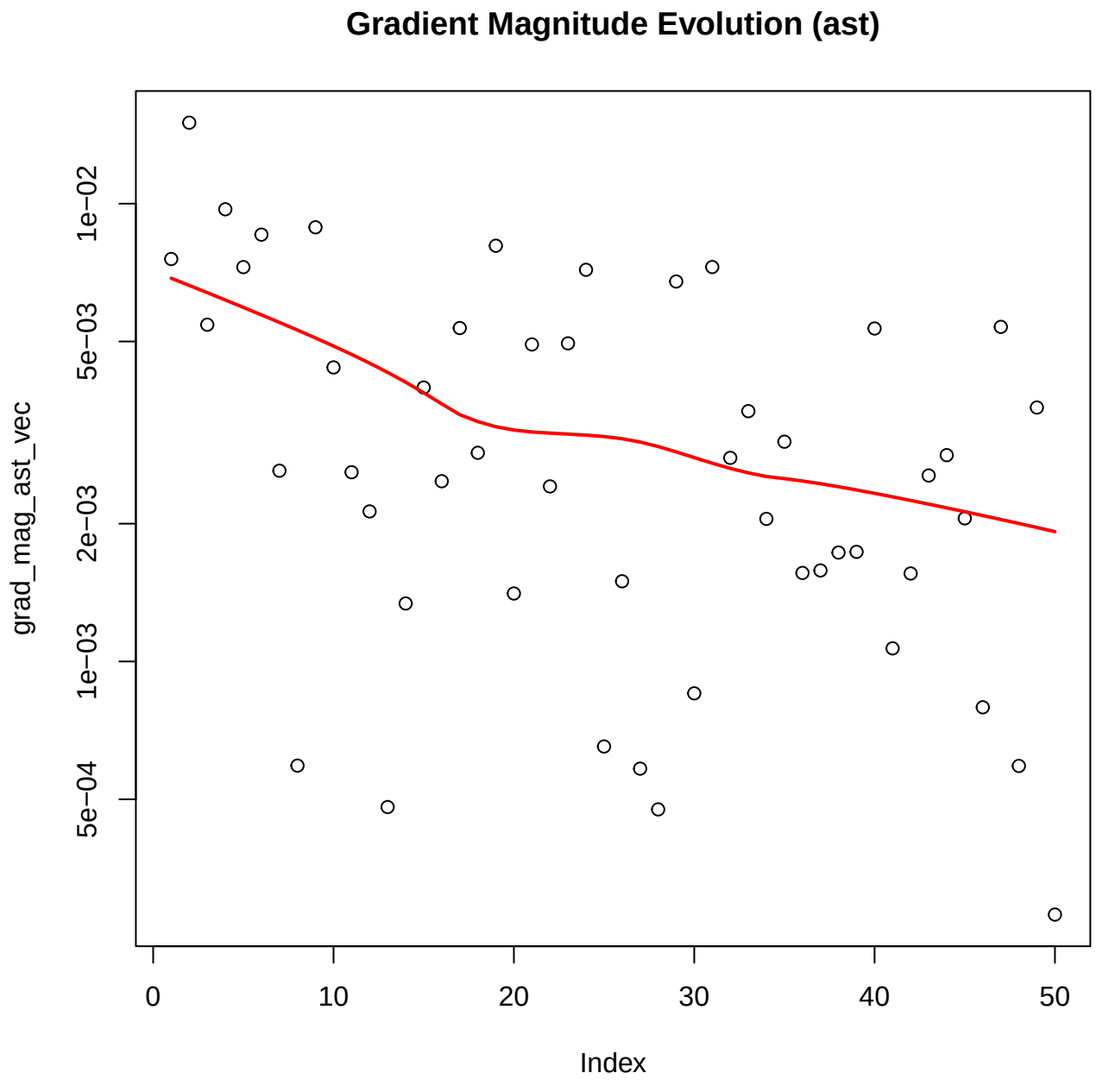

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
2e−03
5e−04
1e−04
2e−05
5
10
15
20
Index

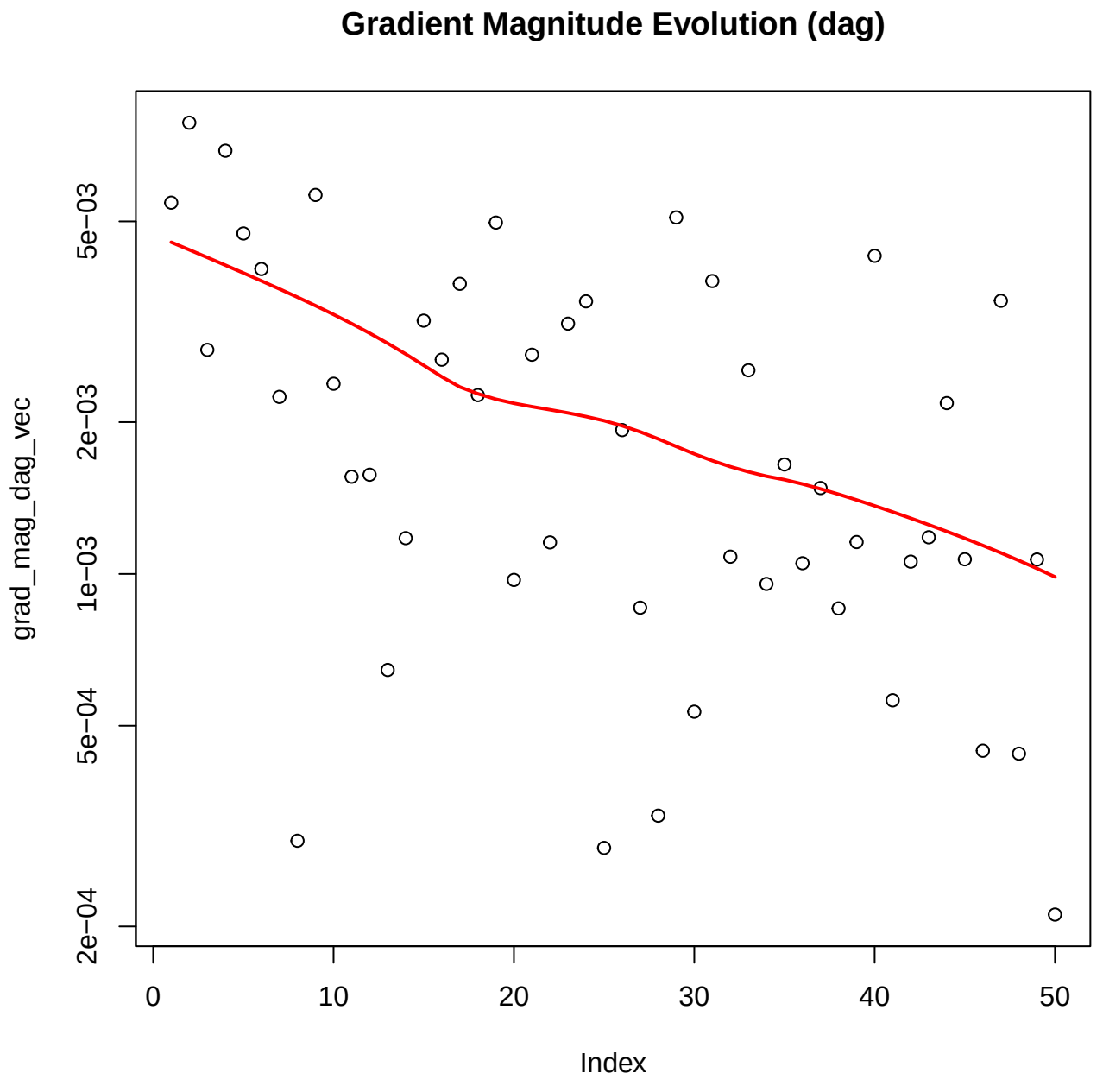

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
0.58
5
10
15
20
Index

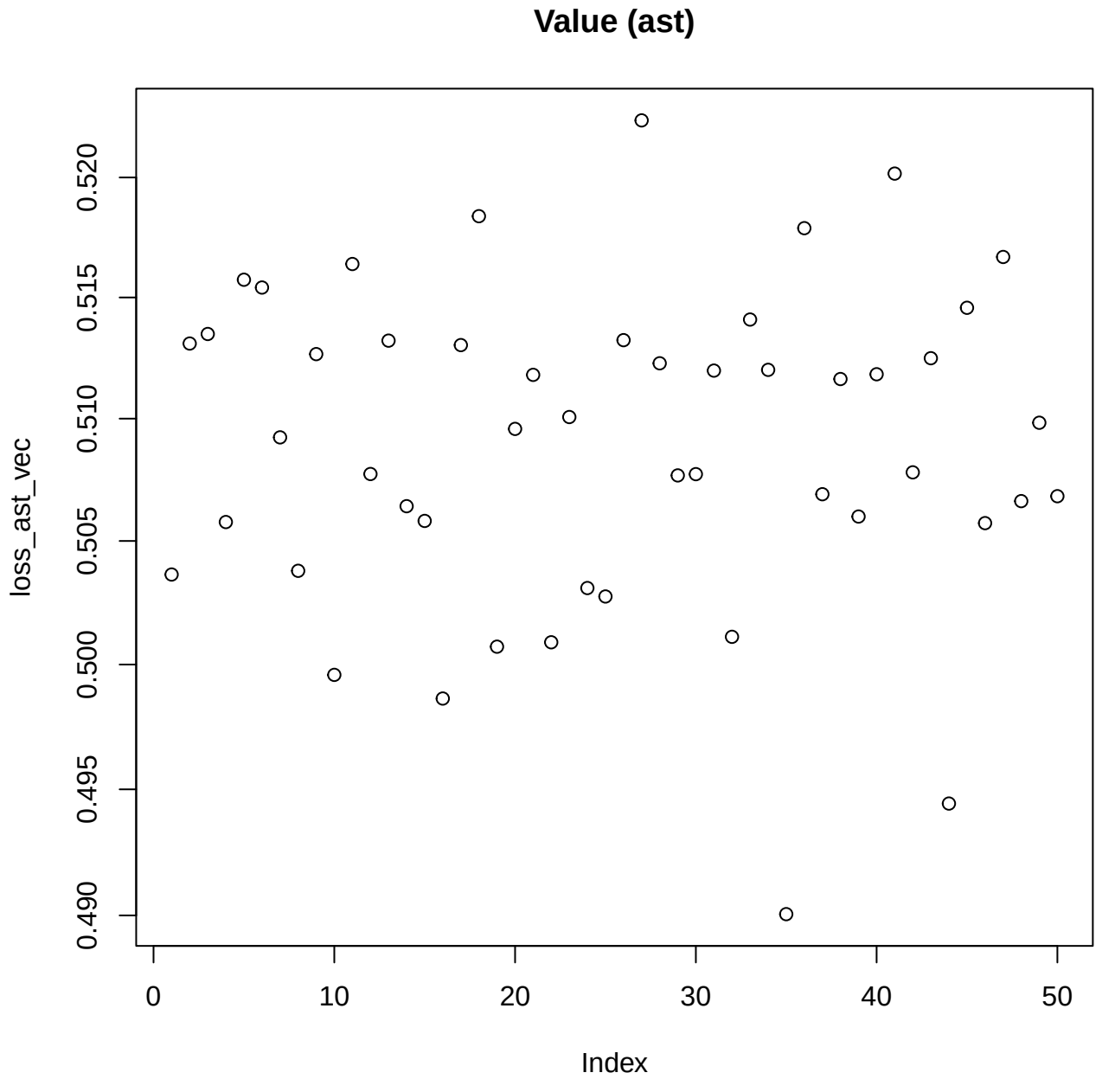

Gradient Magnitude (AST)
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
−4.5
log10(|∇Q|)
5
10
15
20
Iteration

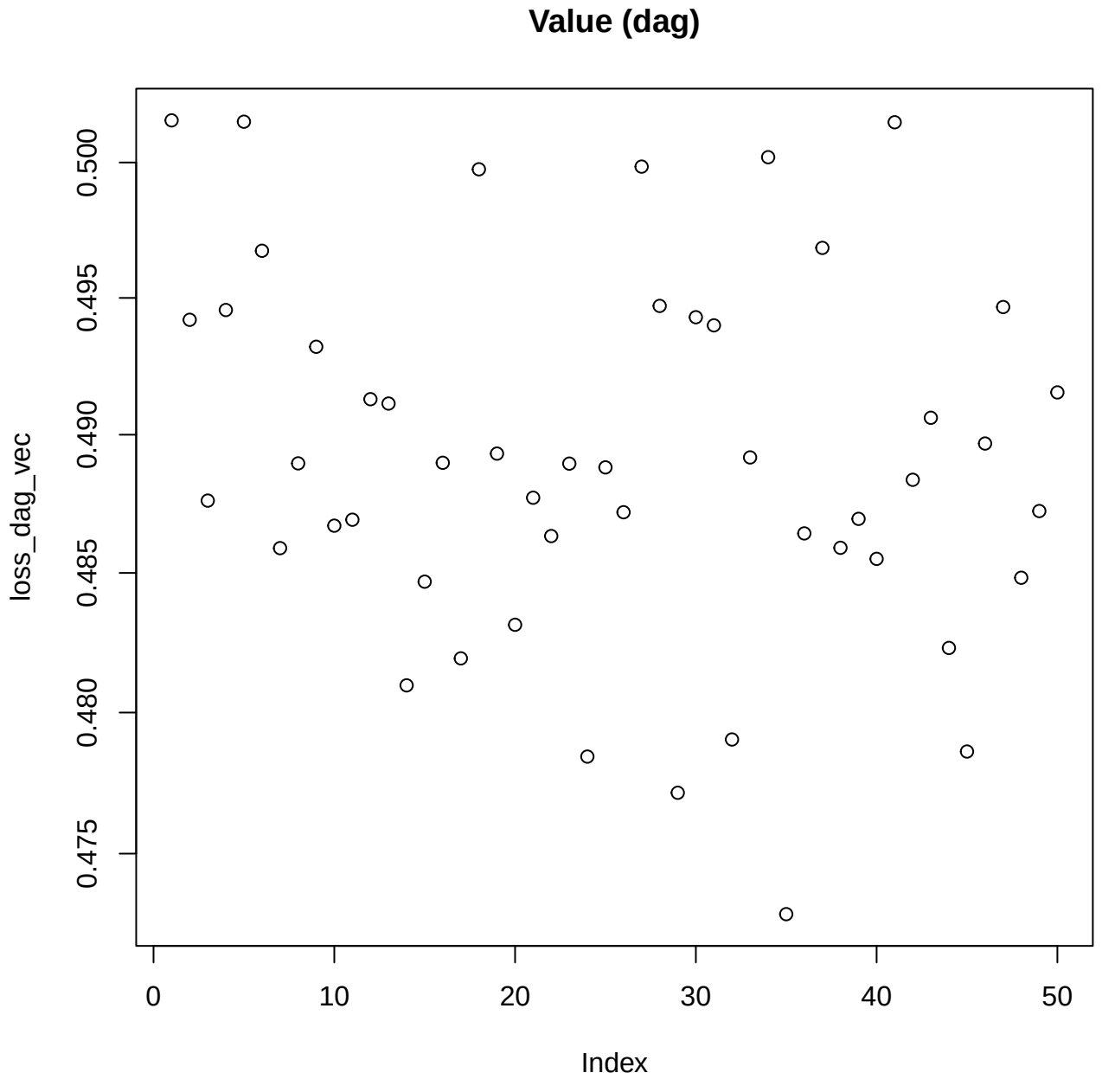

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.002
0.005
0.010
0.020
2
4
6
8
10
Index

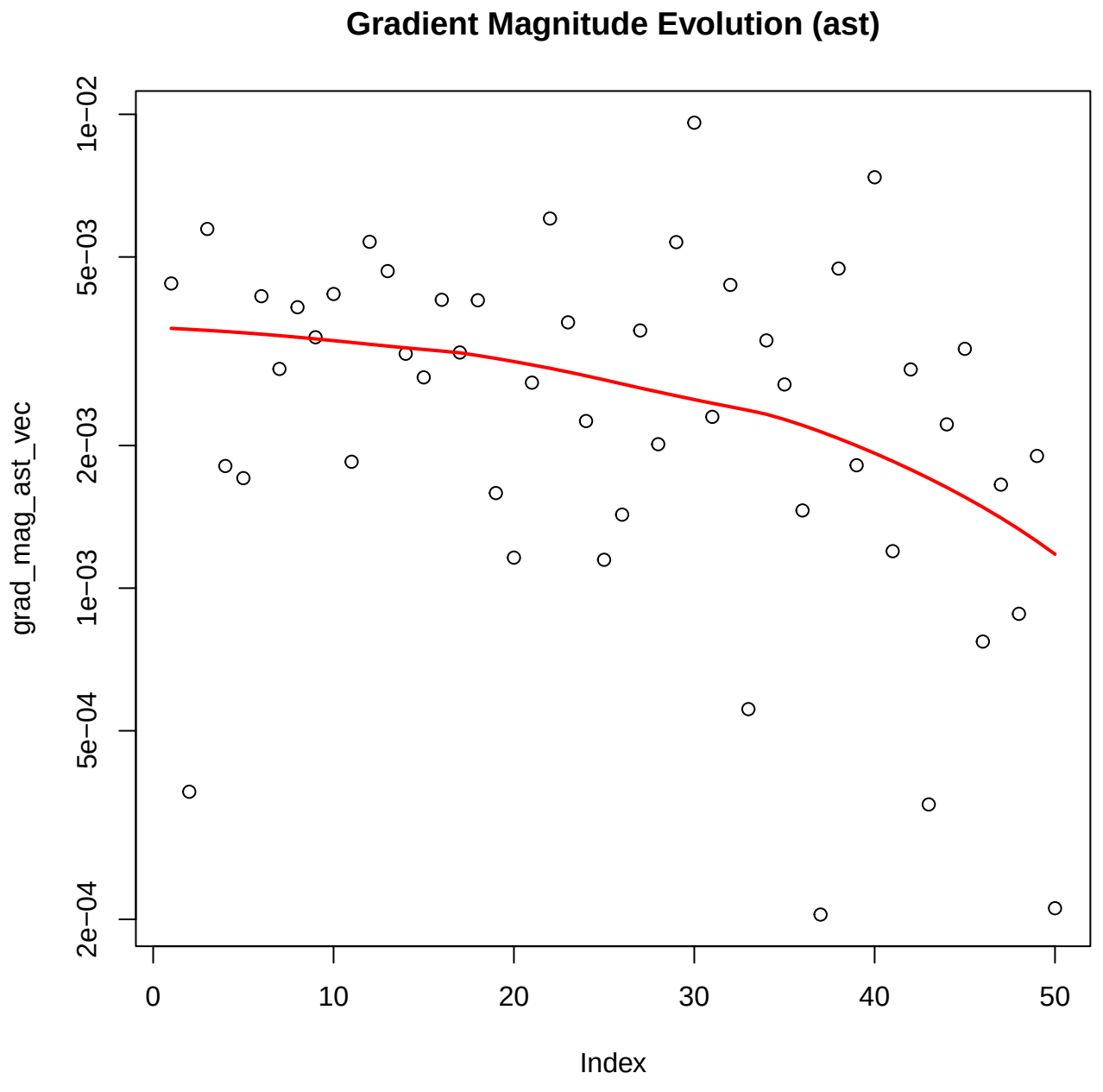

Value (ast)
loss_ast_vec
0.53
0.54
0.55
0.56
0.57
2
4
6
8
10
Index

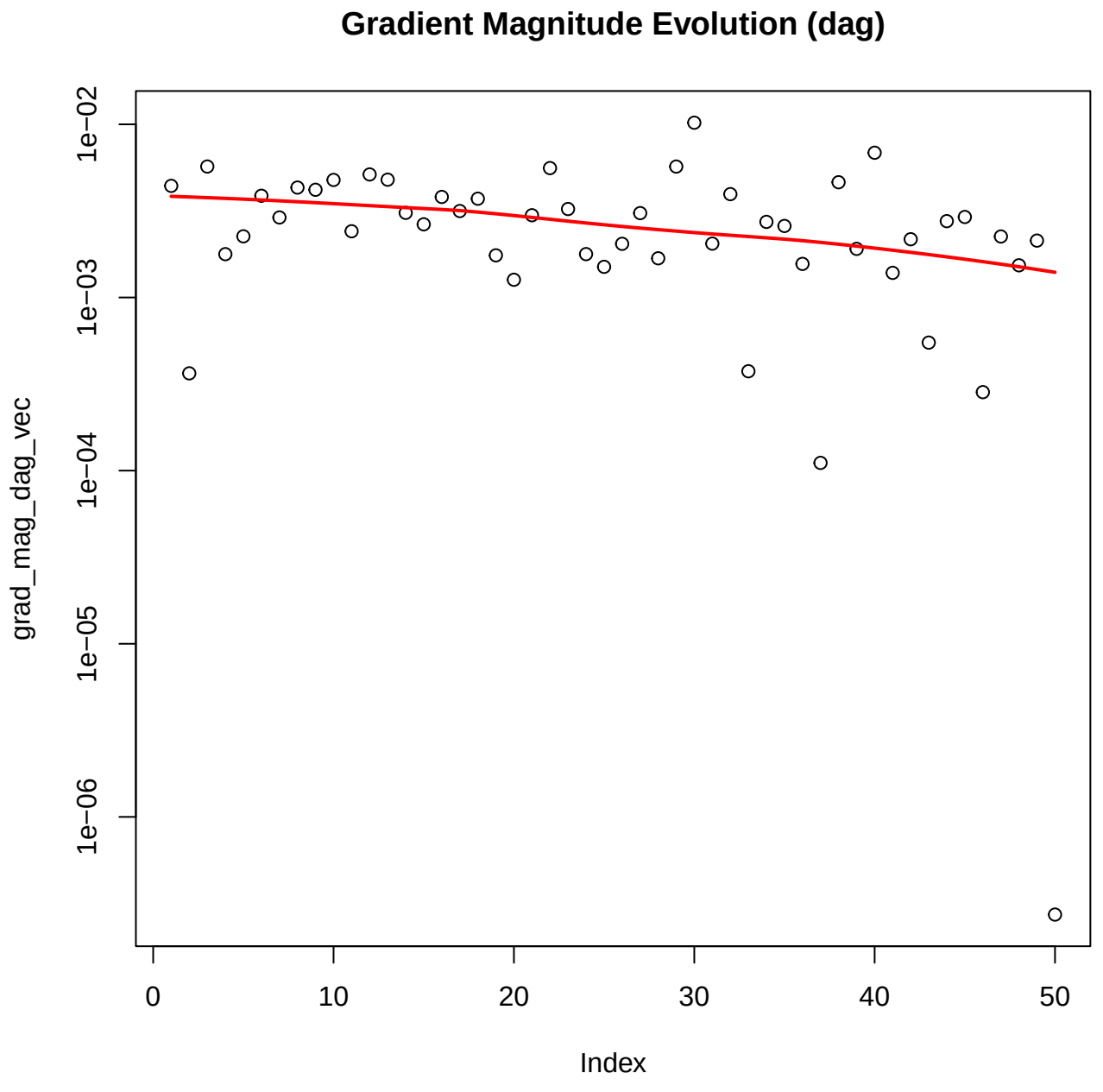

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
0
10
20
30
40
50
Index

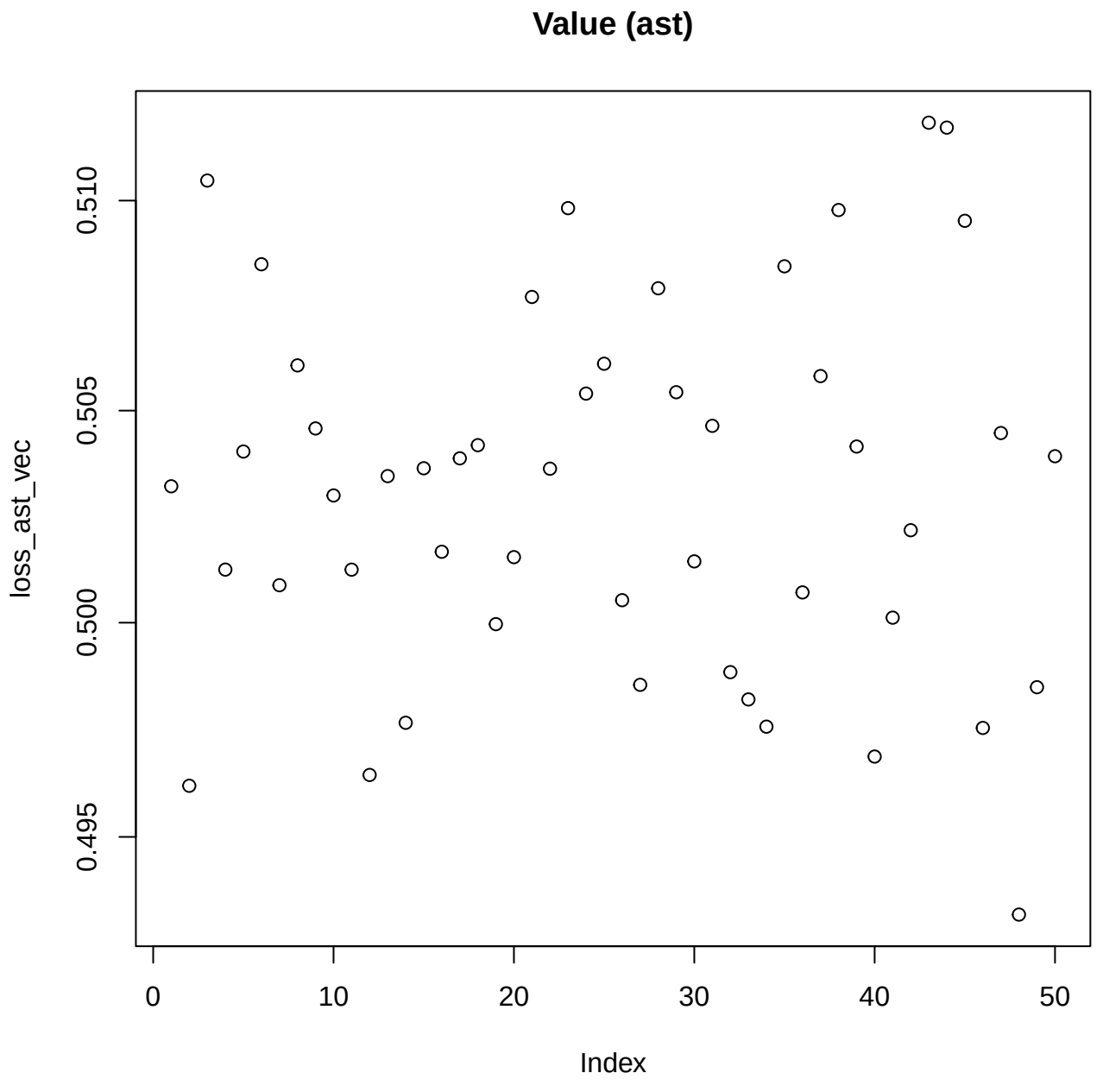

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

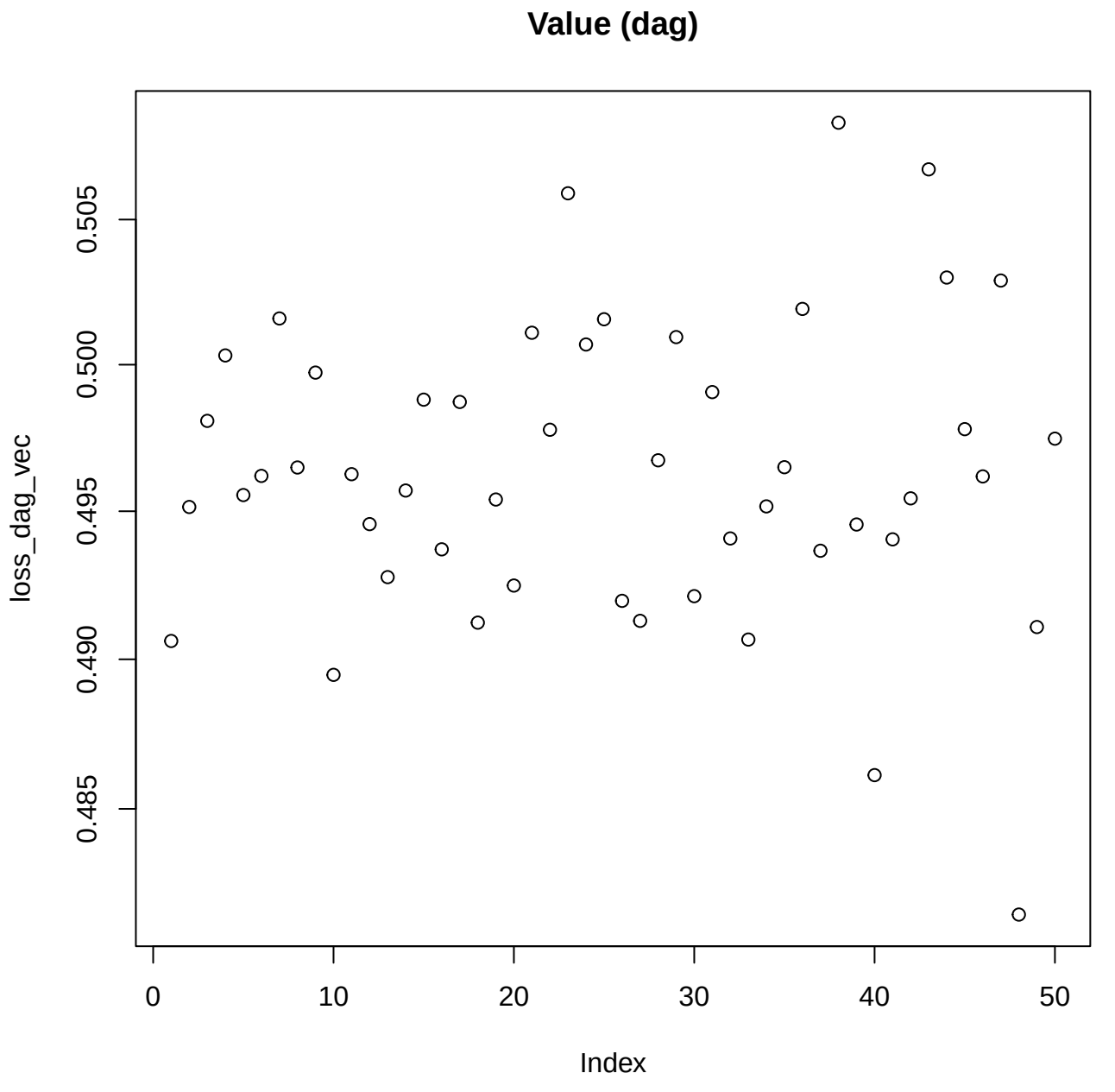

Value (ast)
loss_ast_vec
0.490
0.495
0.500
0.505
0.510
0.515
0.520
0
10
20
30
40
50
Index

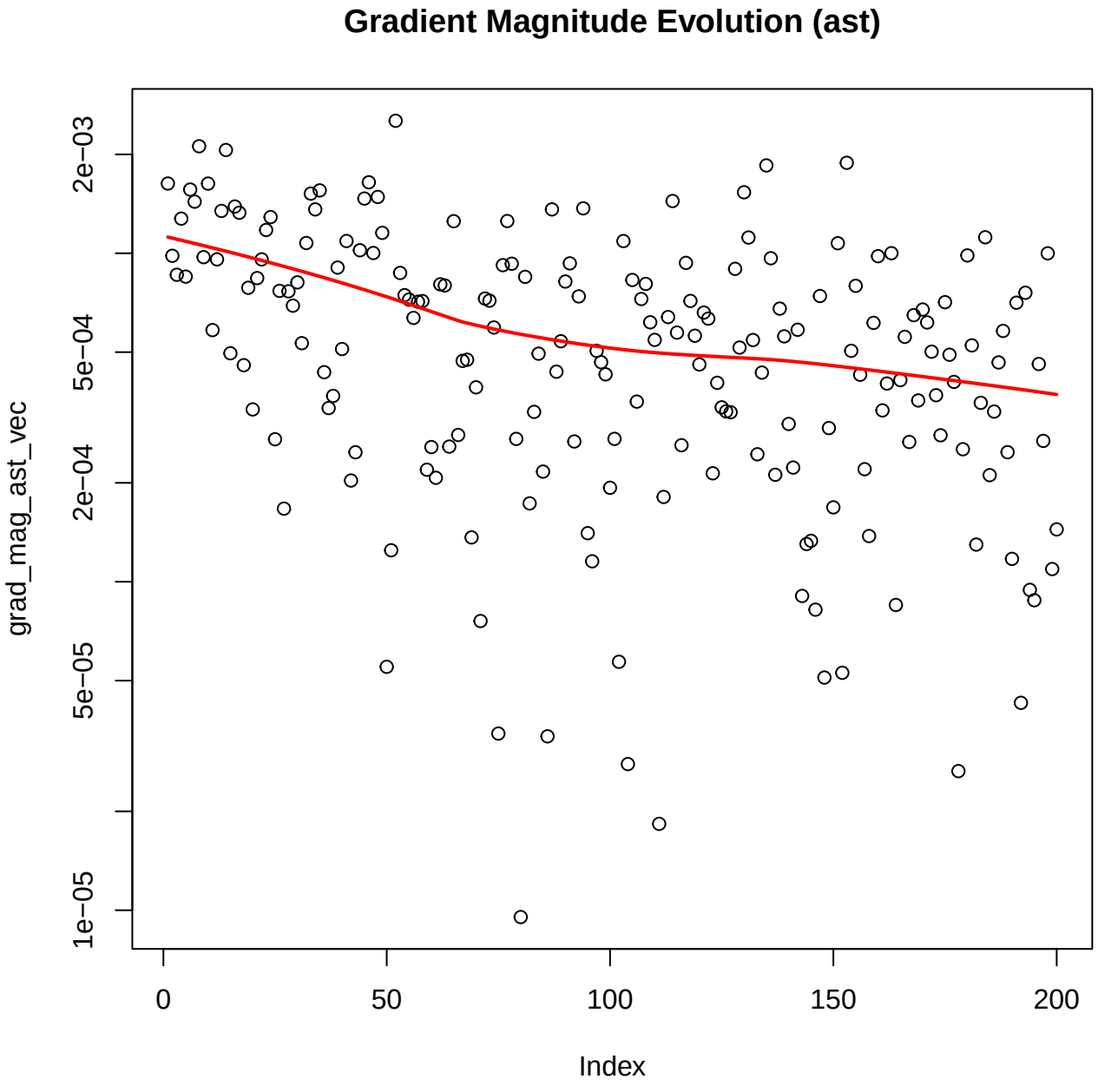

Value (dag)
loss_dag_vec
0.475
0.480
0.485
0.490
0.495
0.500
0
10
20
30
40
50
Index

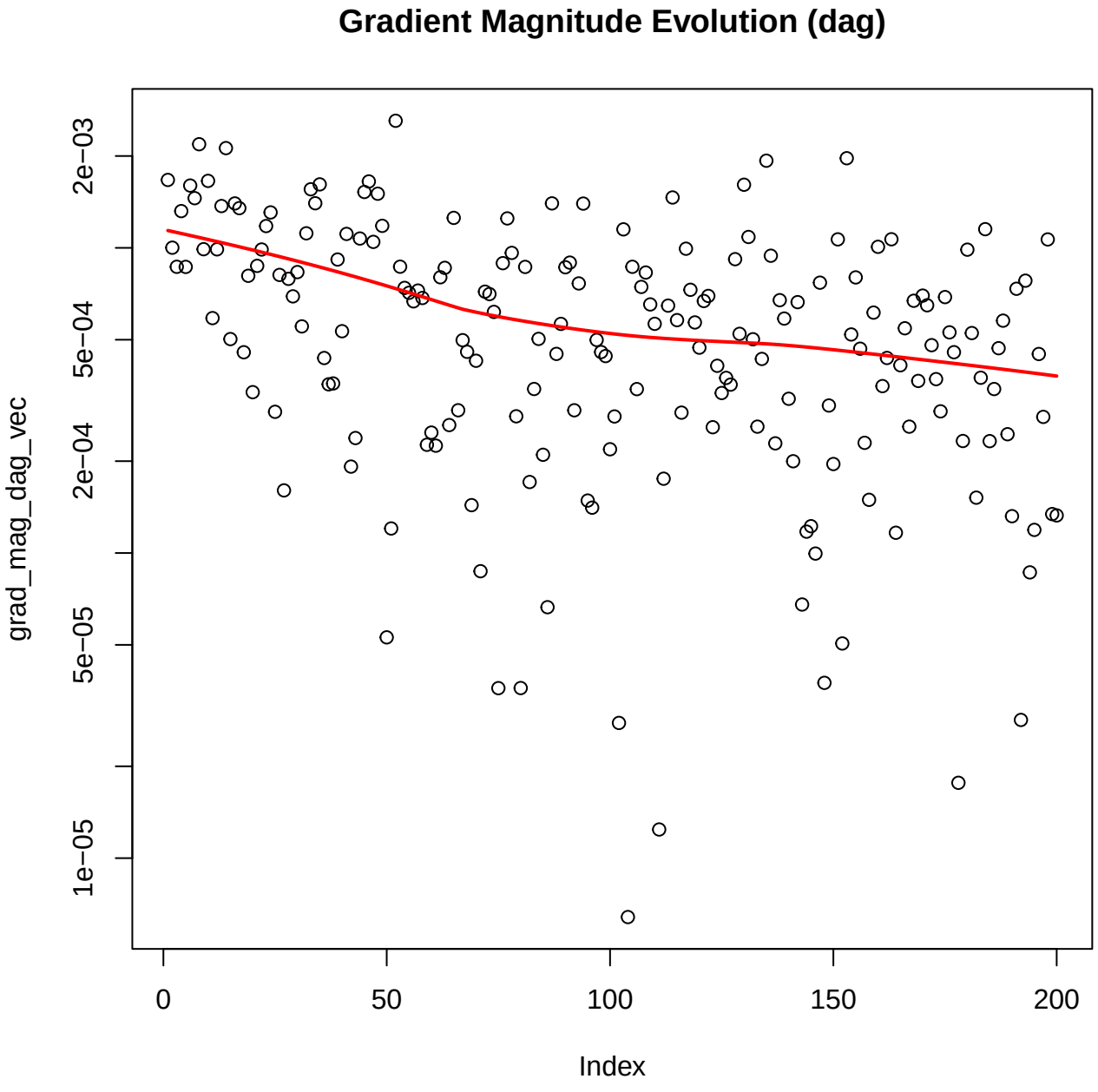

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e−02
5e−03
2e−03
1e−03
5e−04
2e−04
0
10
20
30
40
50
Index

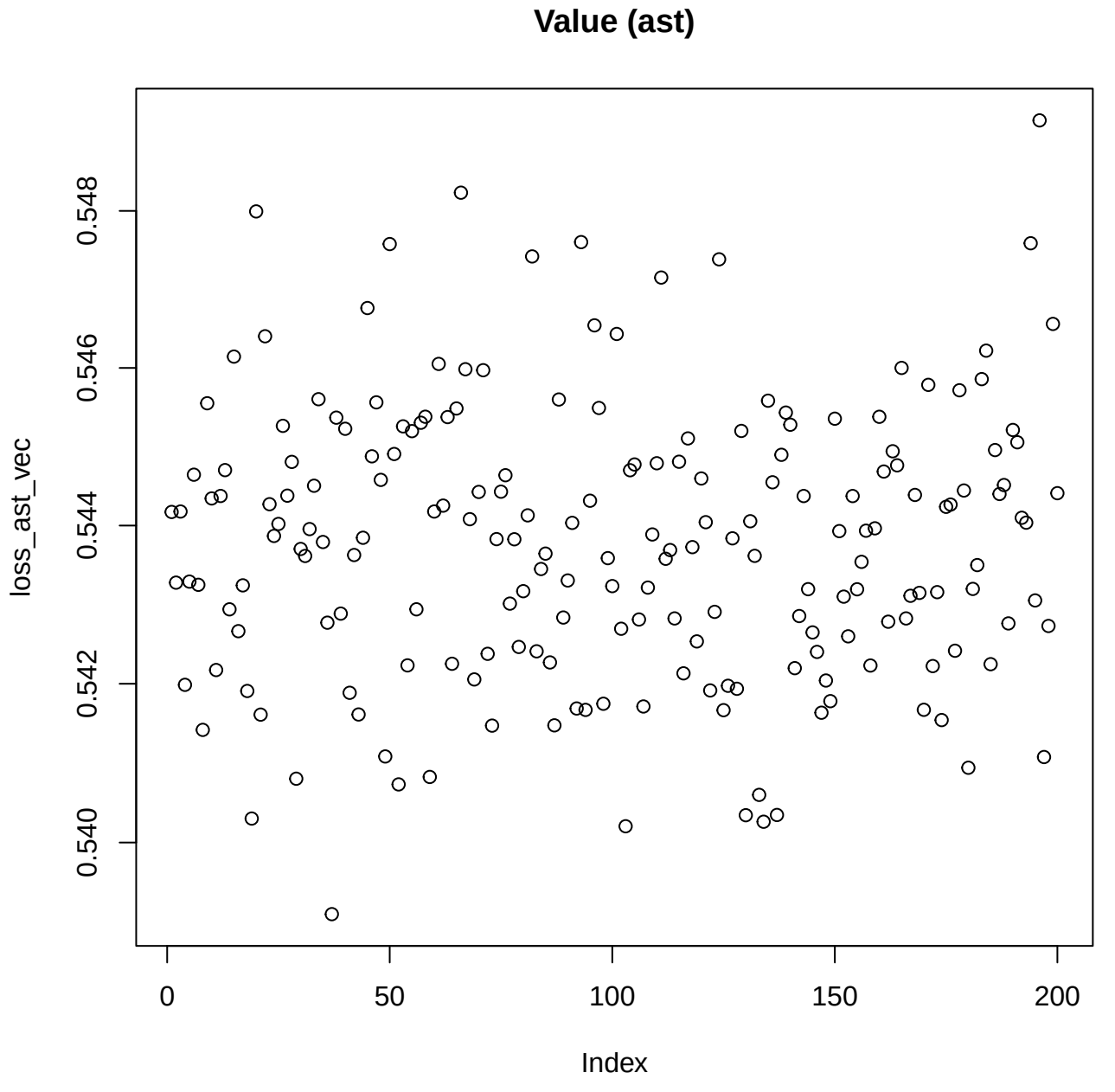

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
1e−02
1e−03
1e−04
1e−05
1e−06
0
10
20
30
40
50
Index

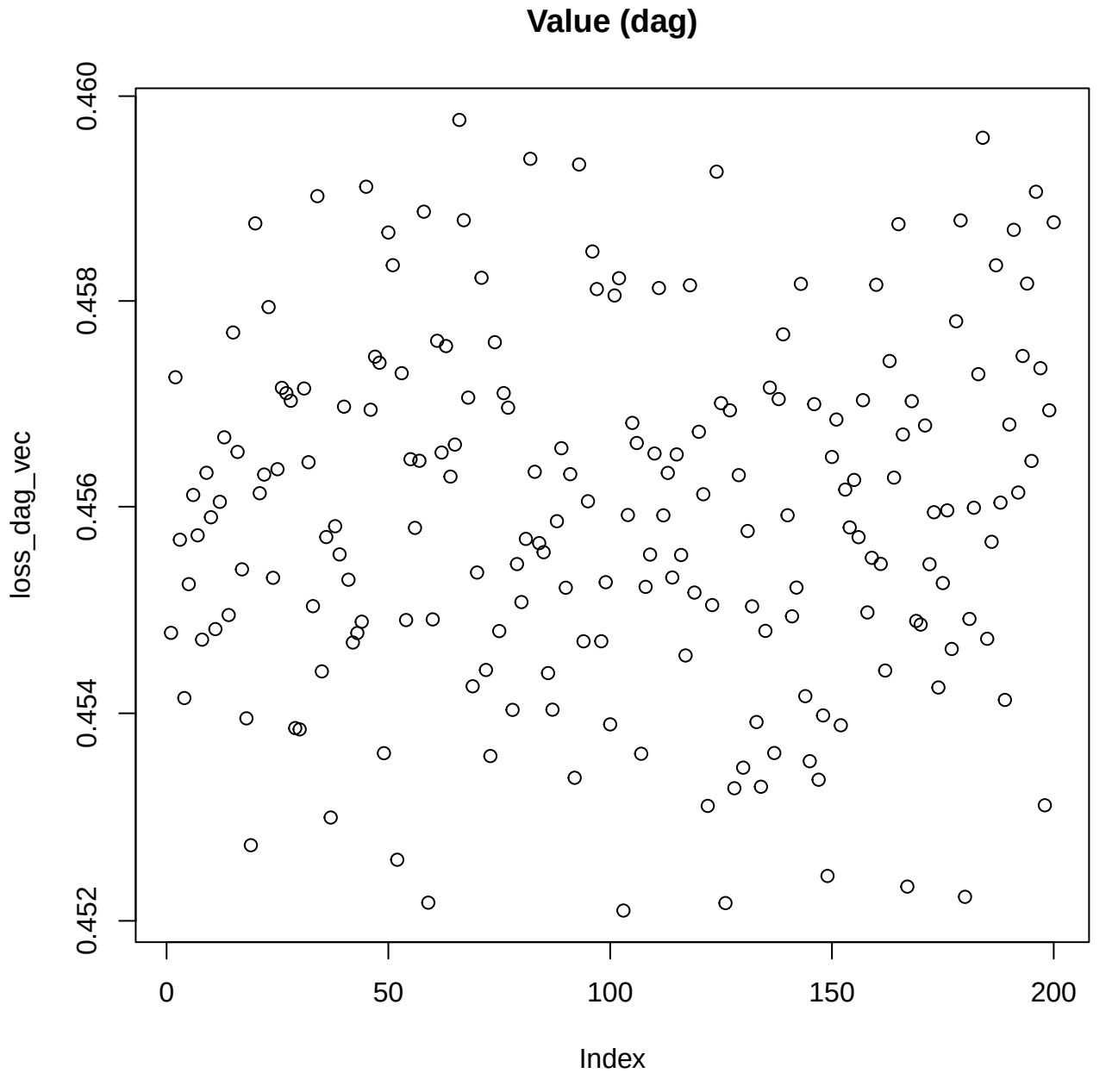

Value (ast)
loss_ast_vec
0.495
0.500
0.505
0.510
0
10
20
30
40
50
Index

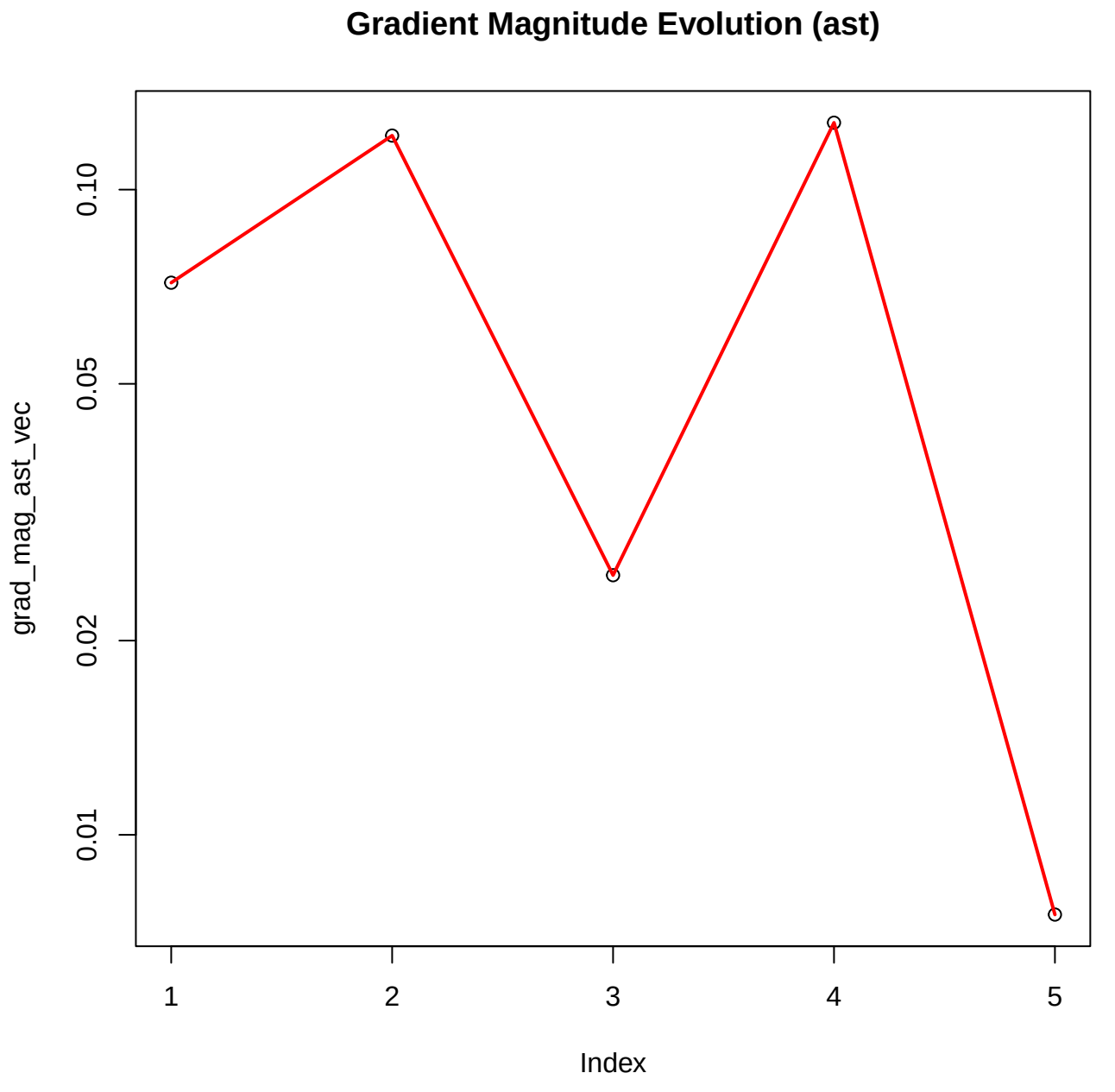

Value (dag)
loss_dag_vec
0.485
0.490
0.495
0.500
0.505
0
10
20
30
40
50
Index

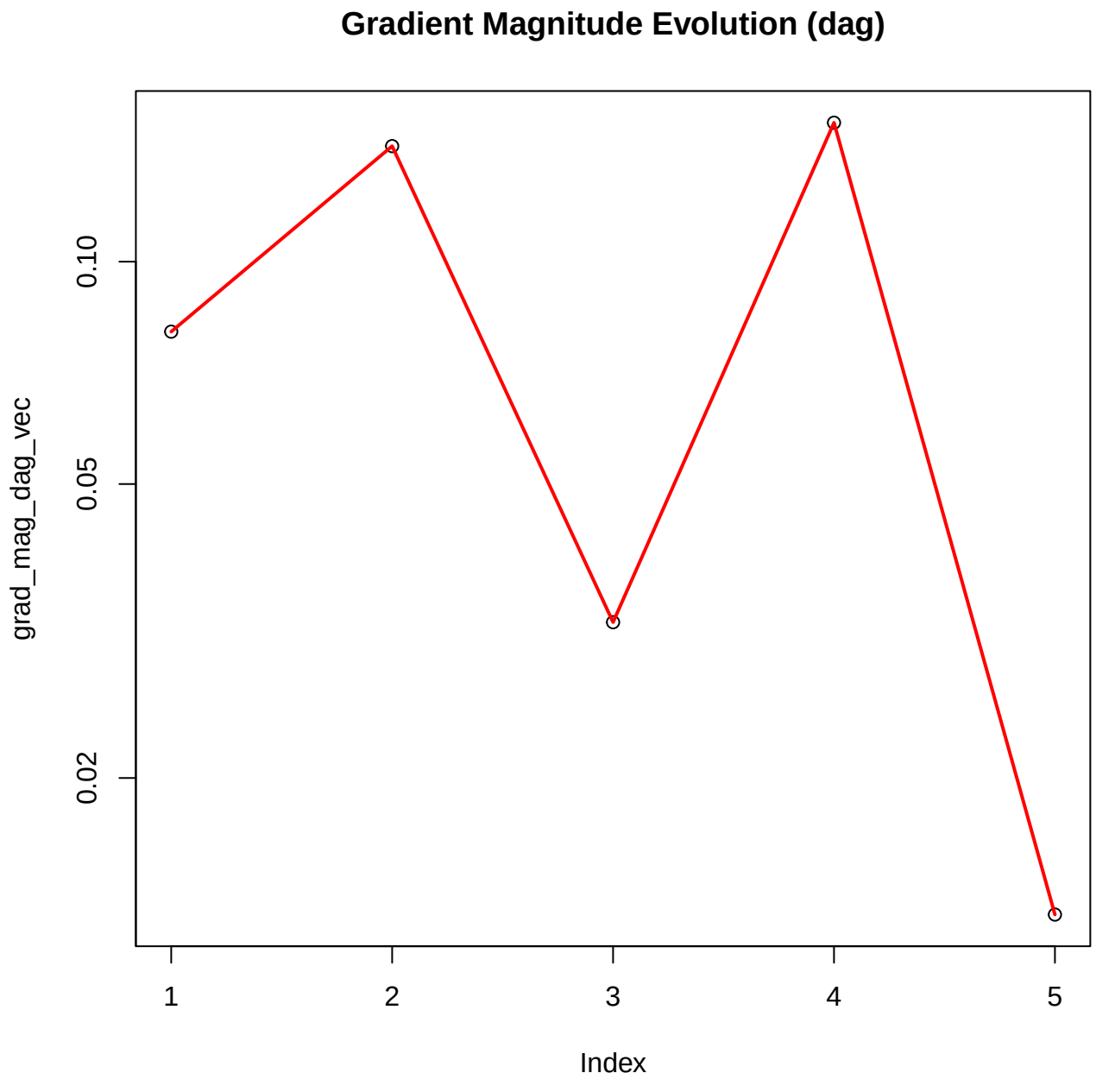

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

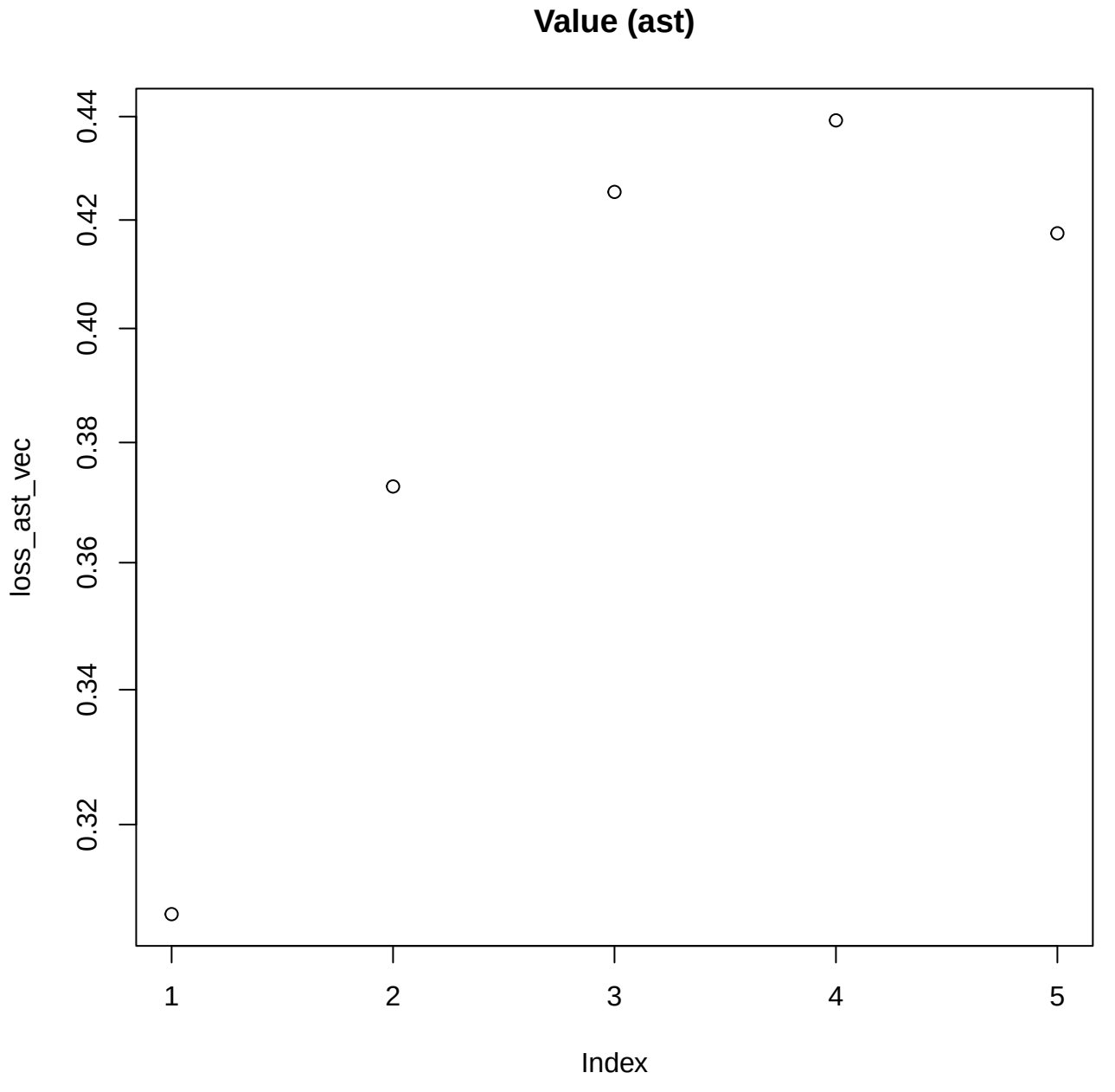

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

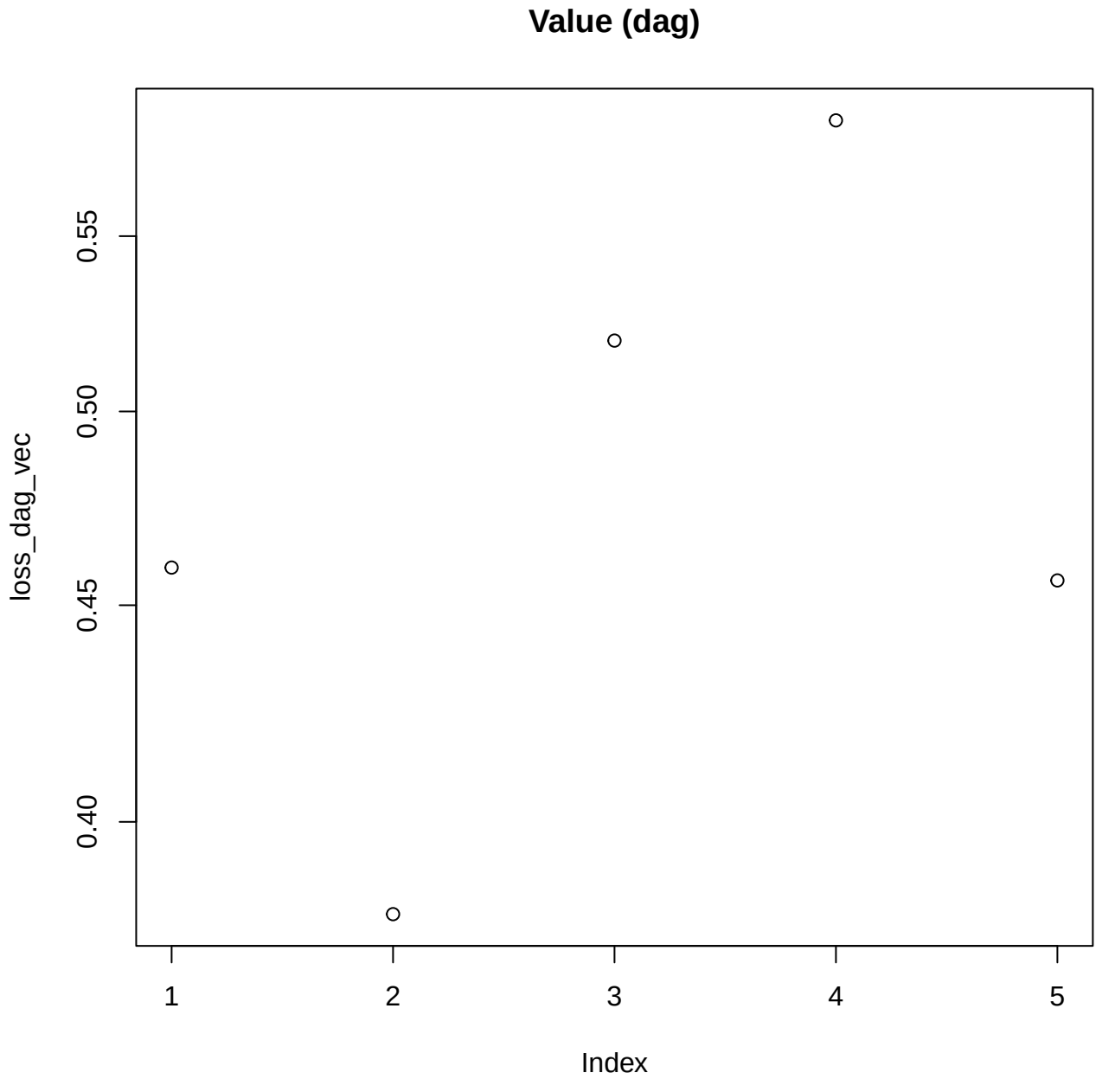

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

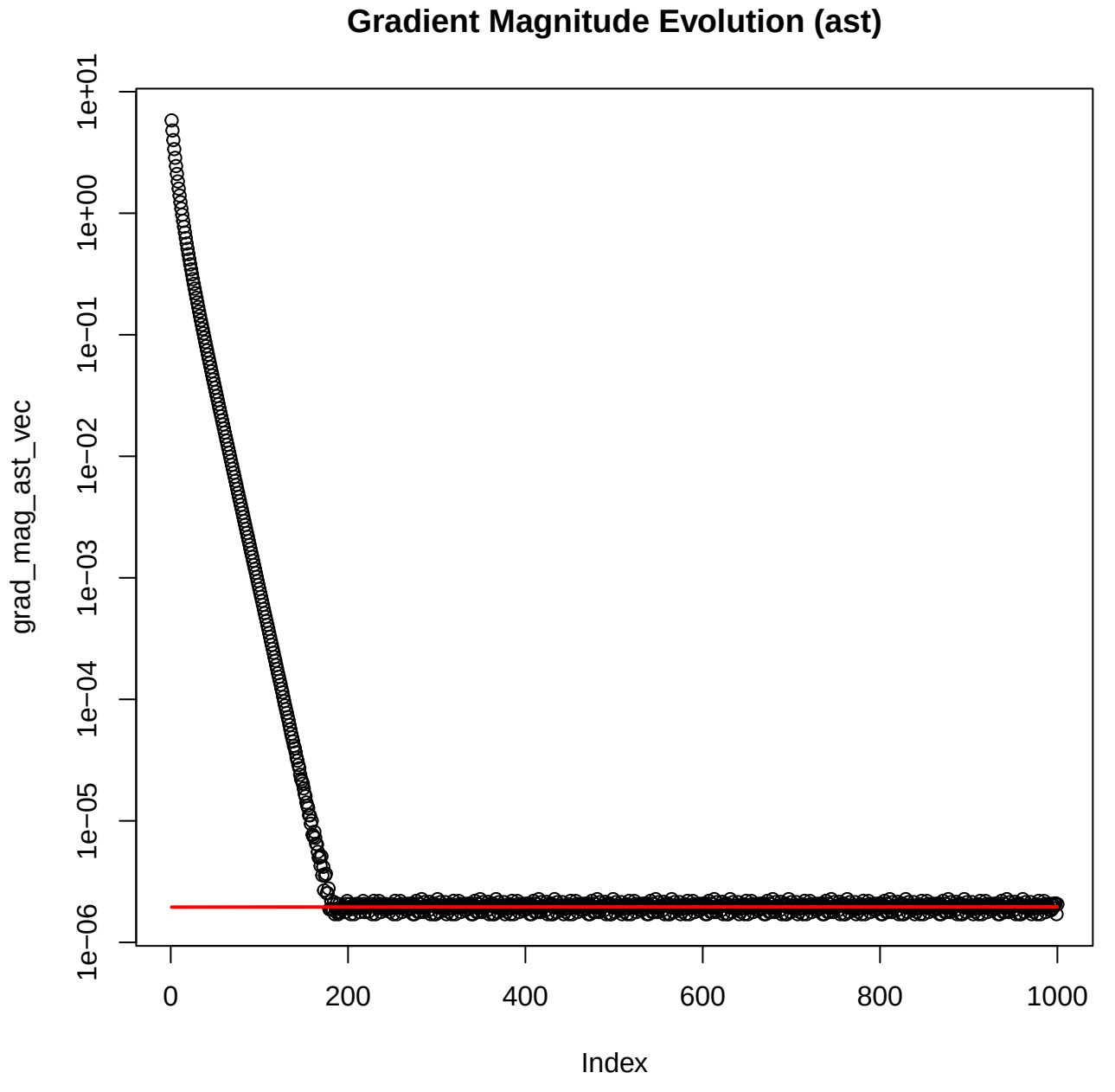

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

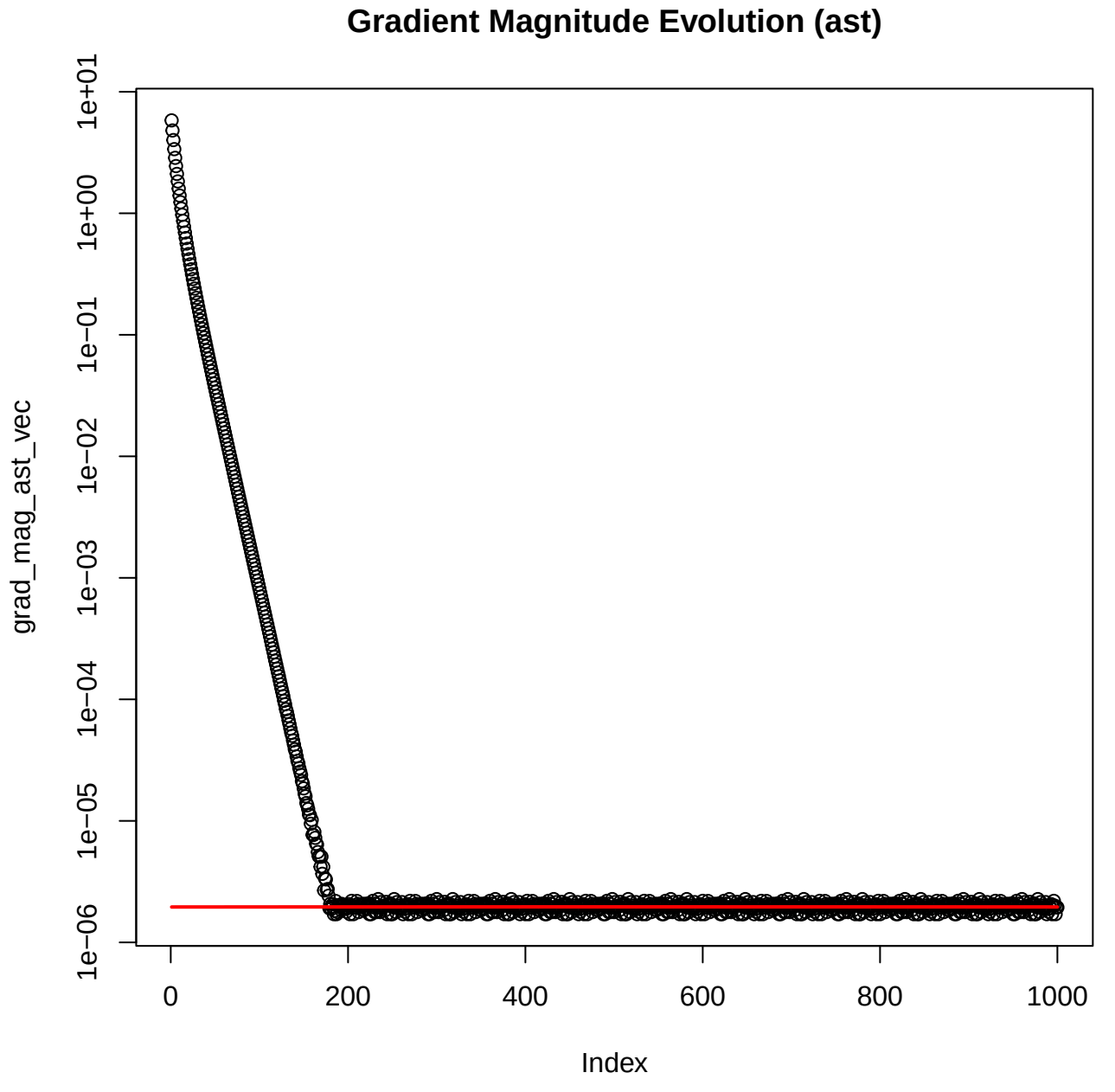

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

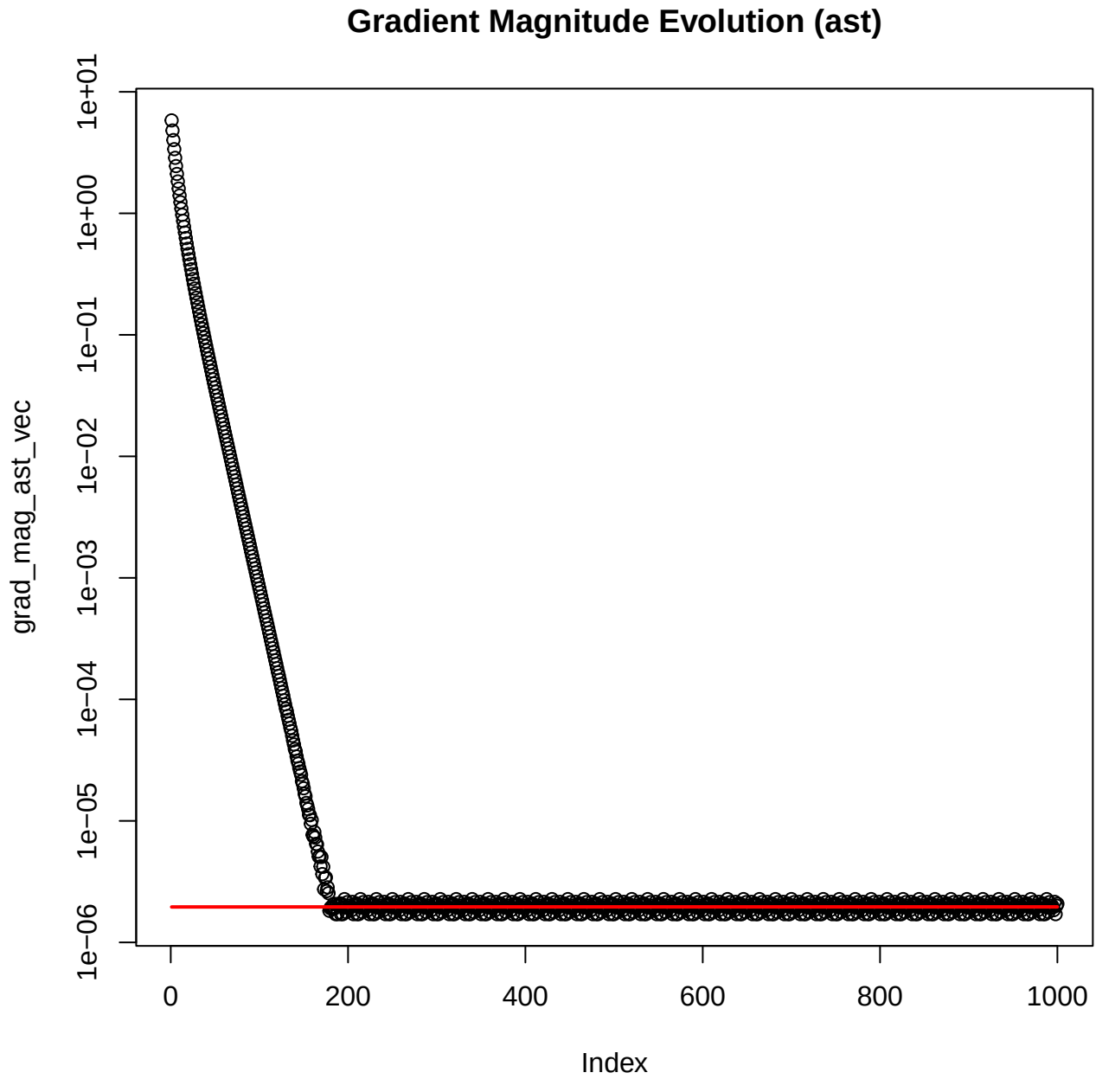

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

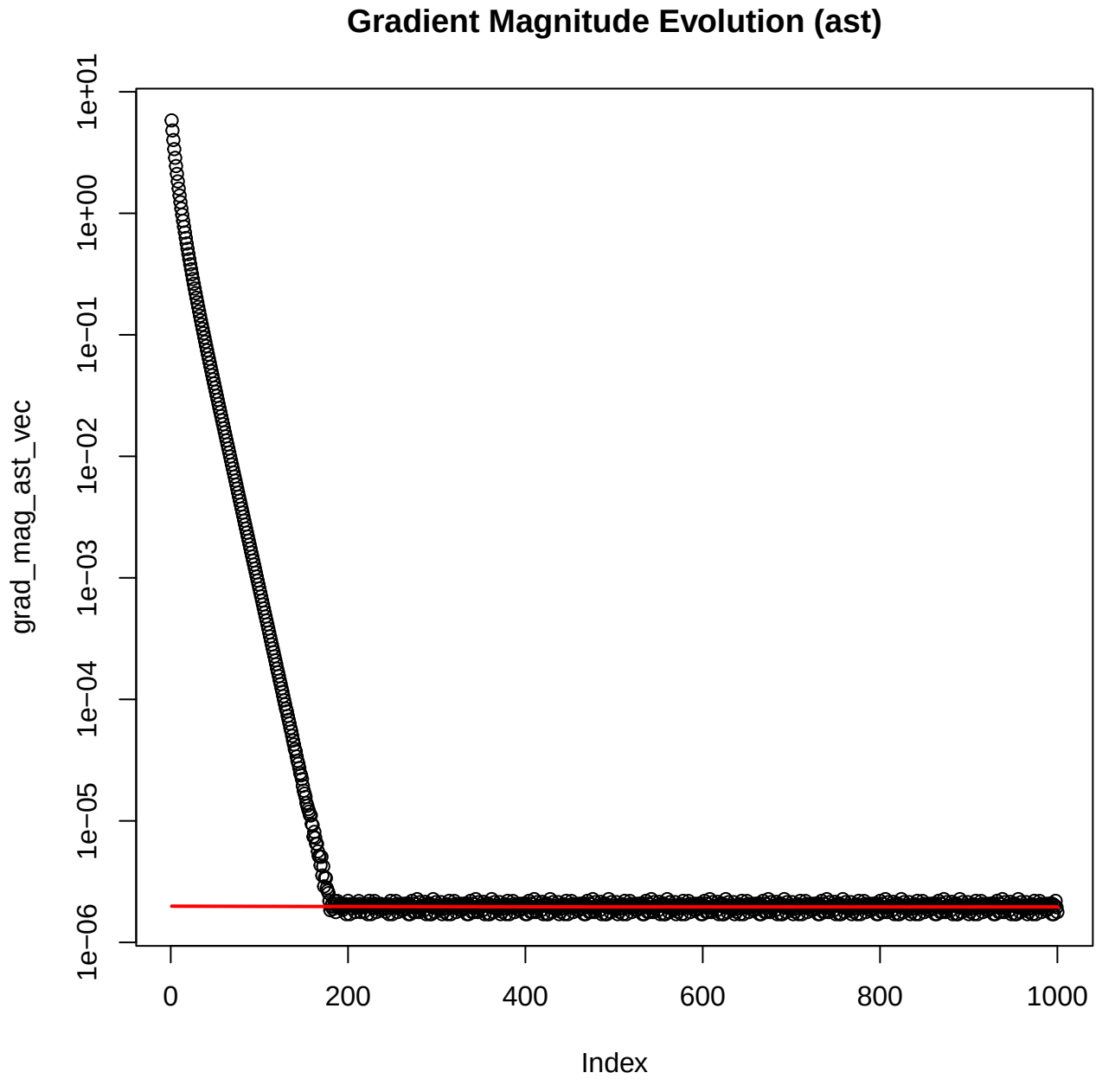

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

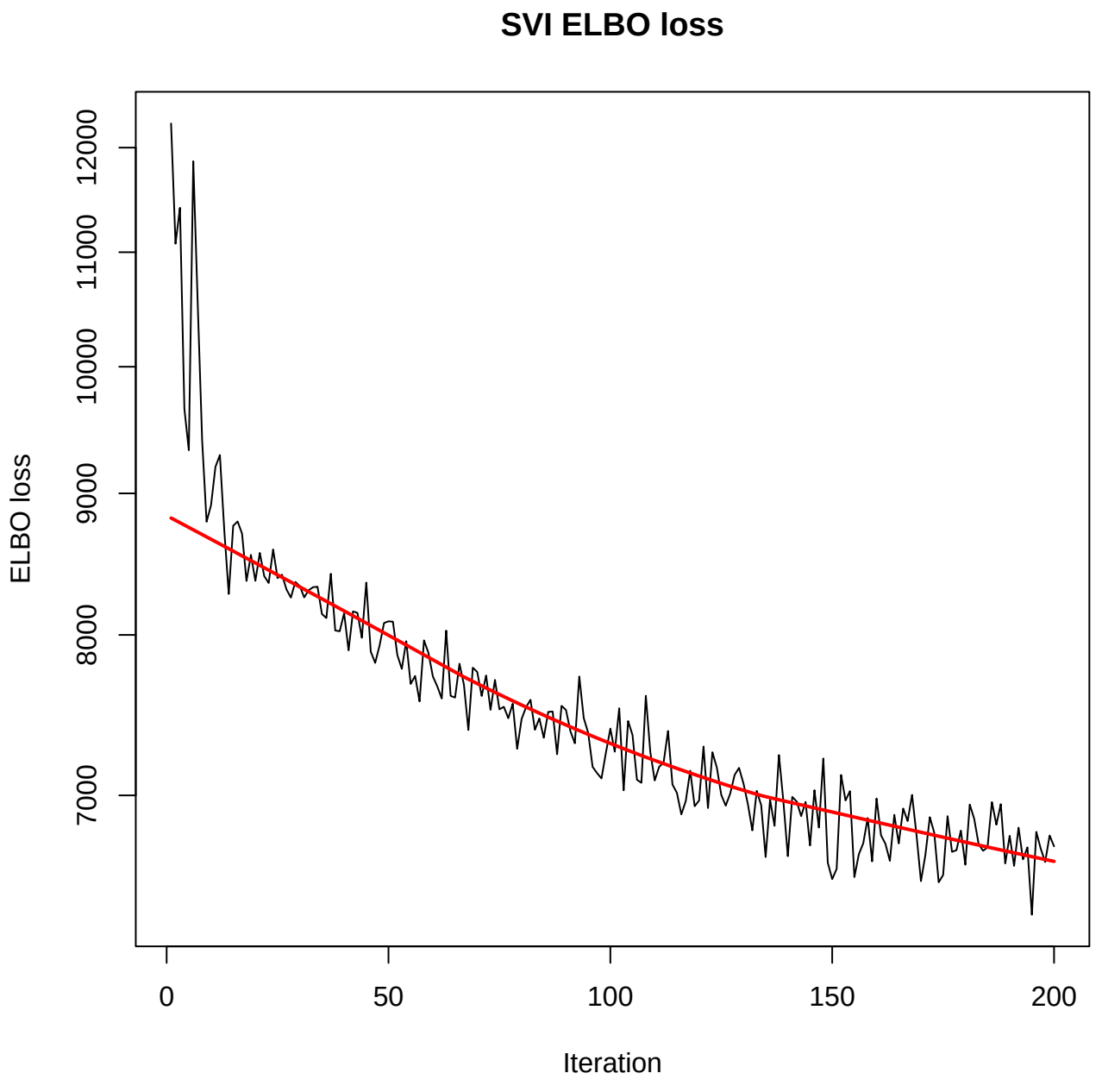

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

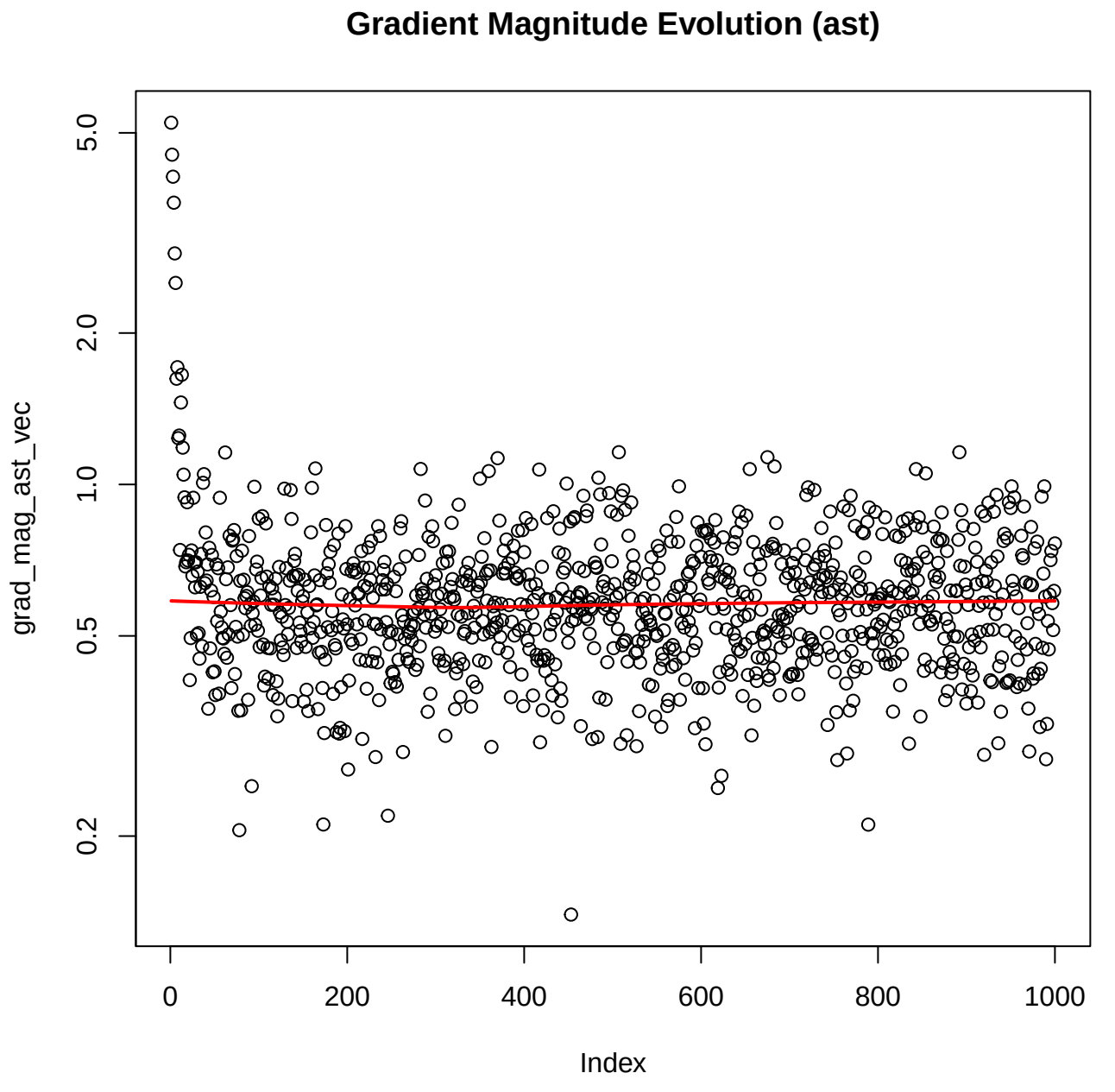

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

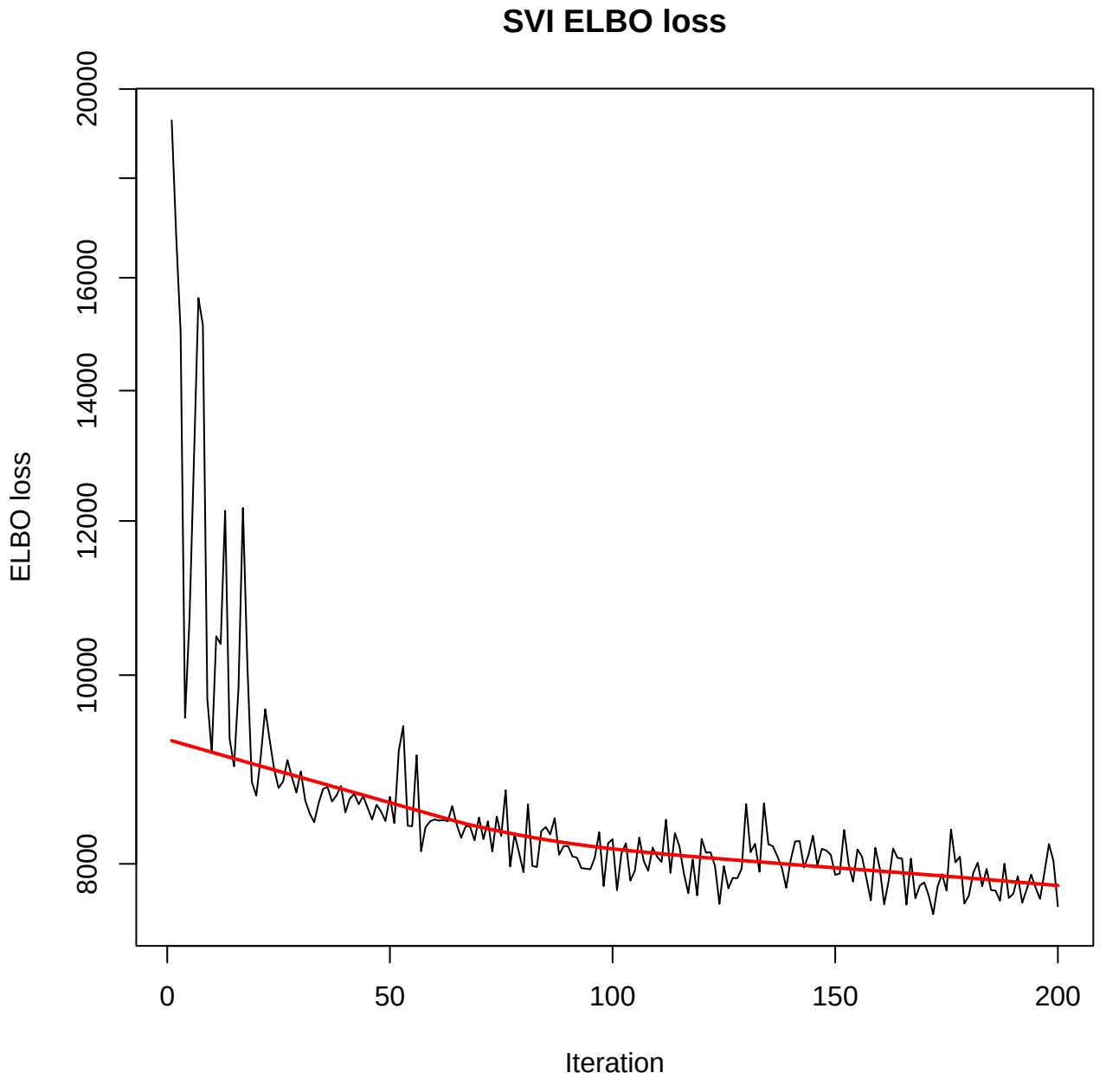

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

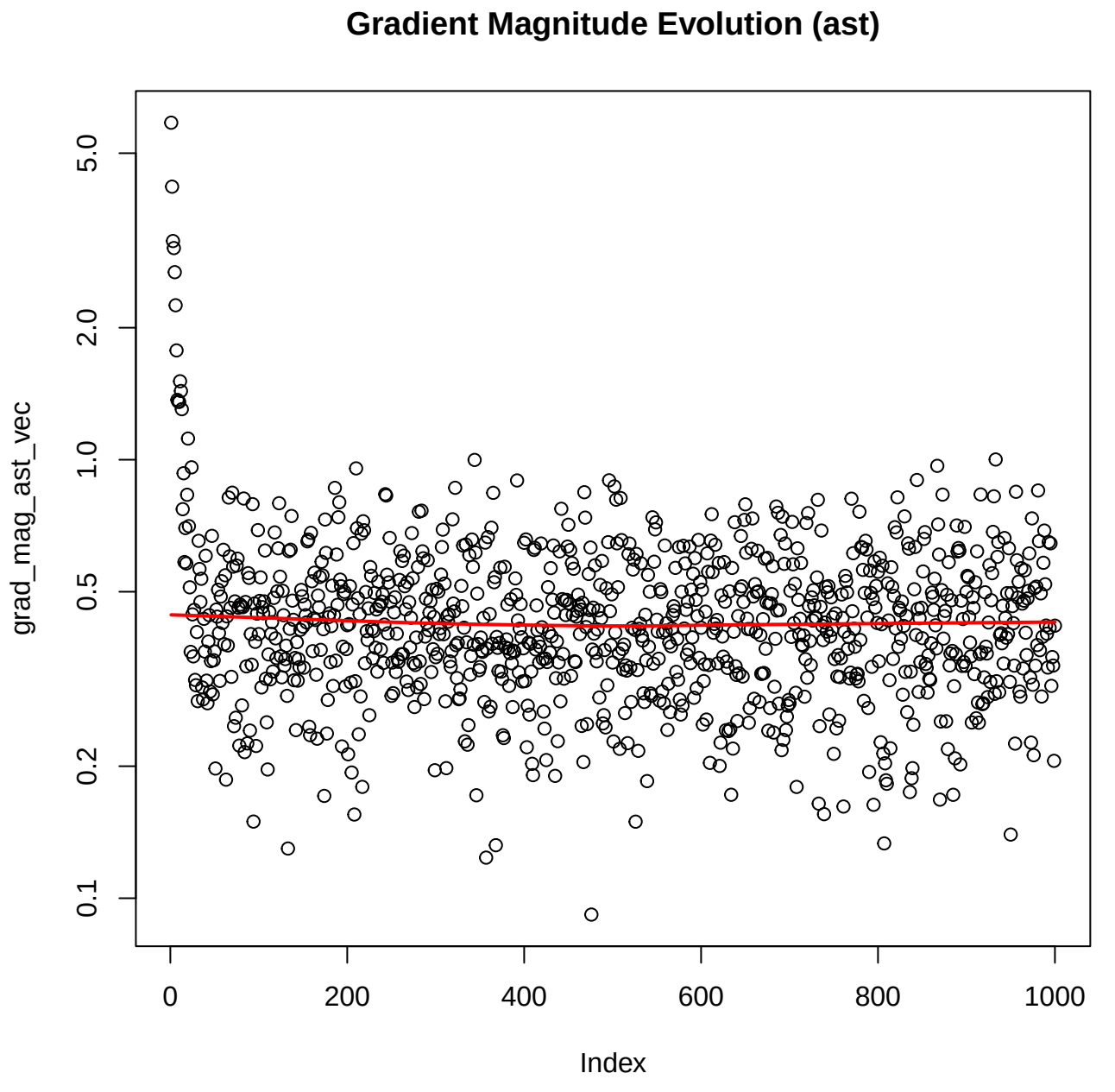

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

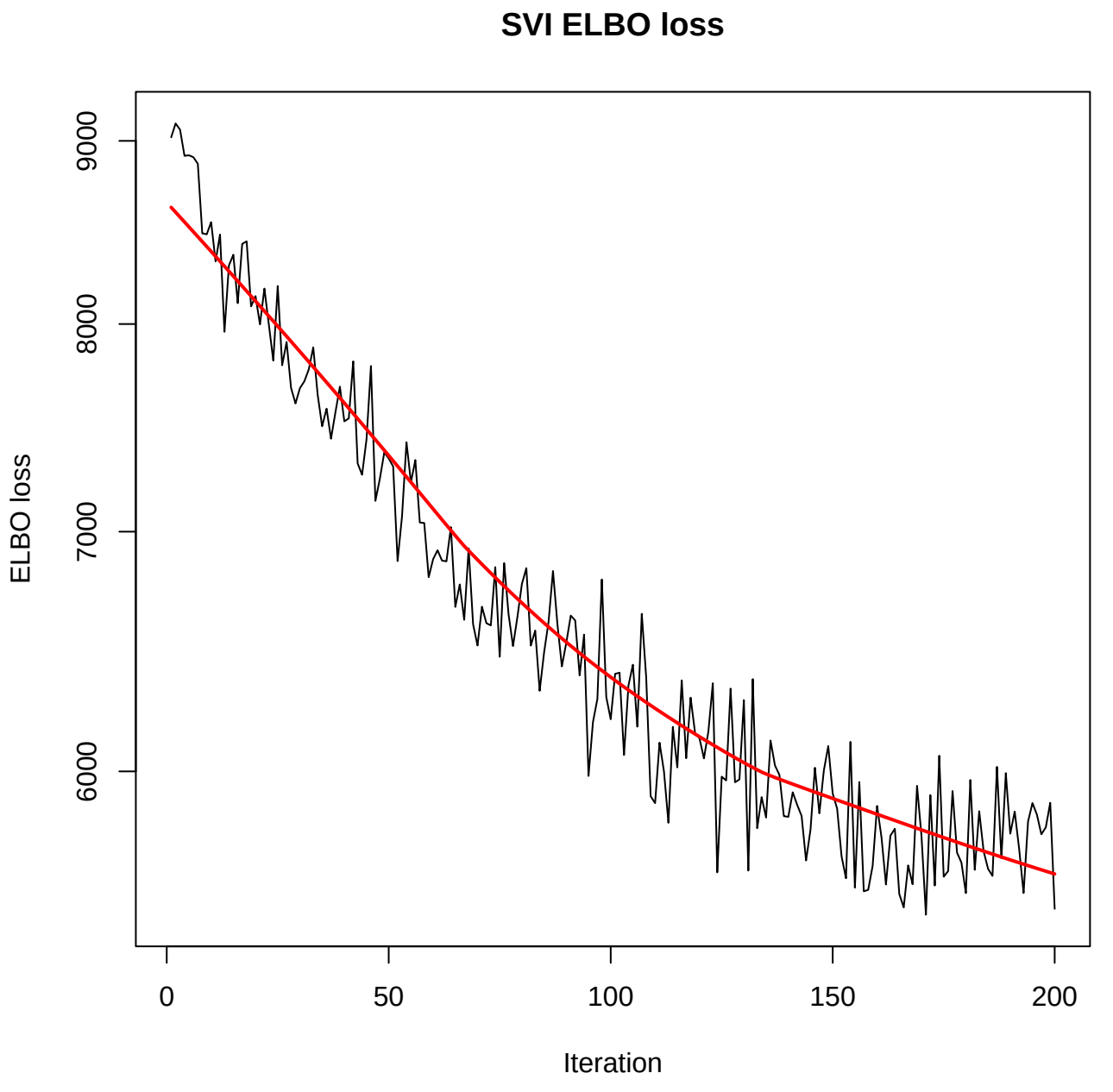

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

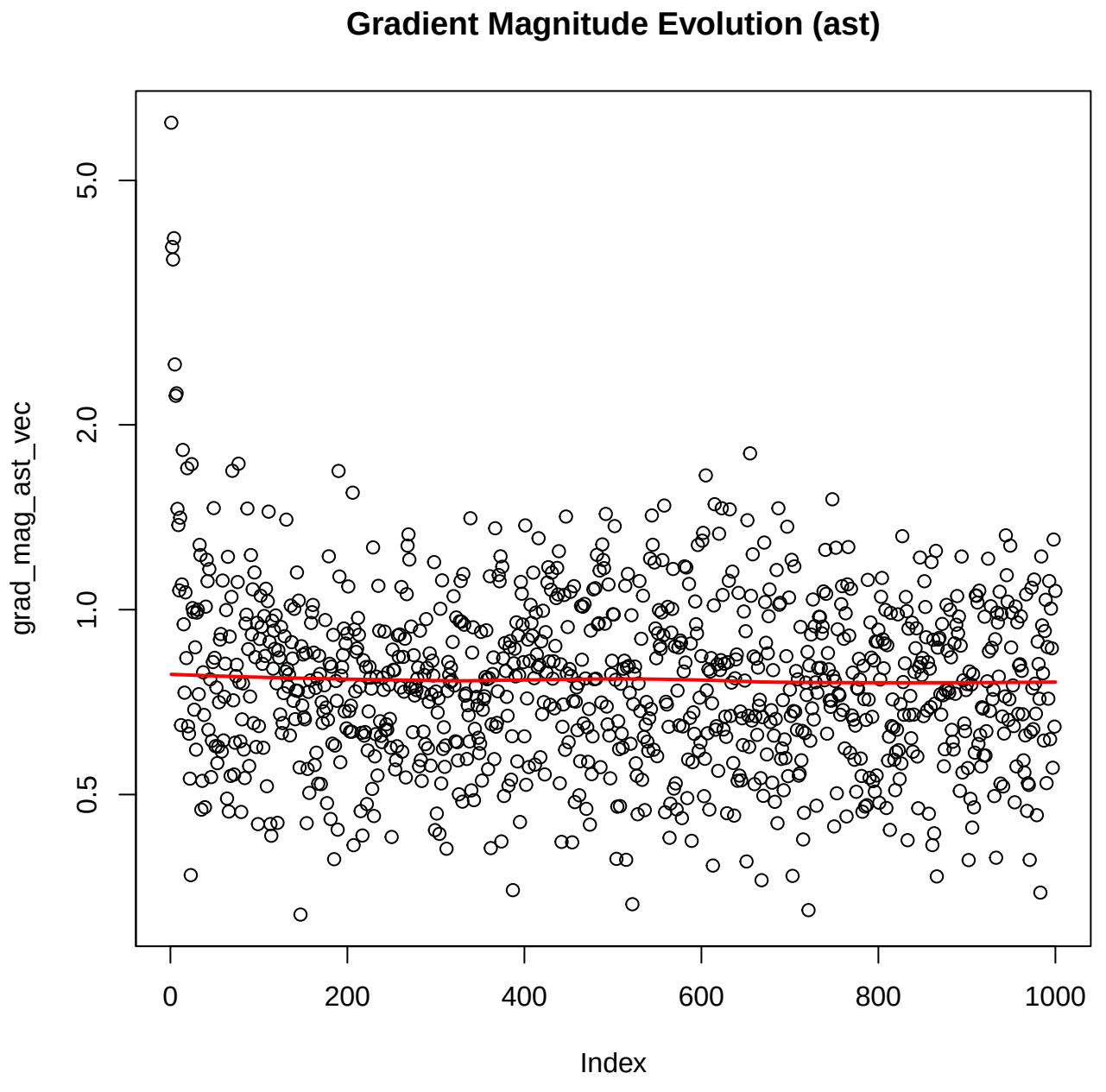

SVI ELBO loss
ELBO loss
12000
11000
10000
9000
8000
7000
0
50
100
150
200
Iteration

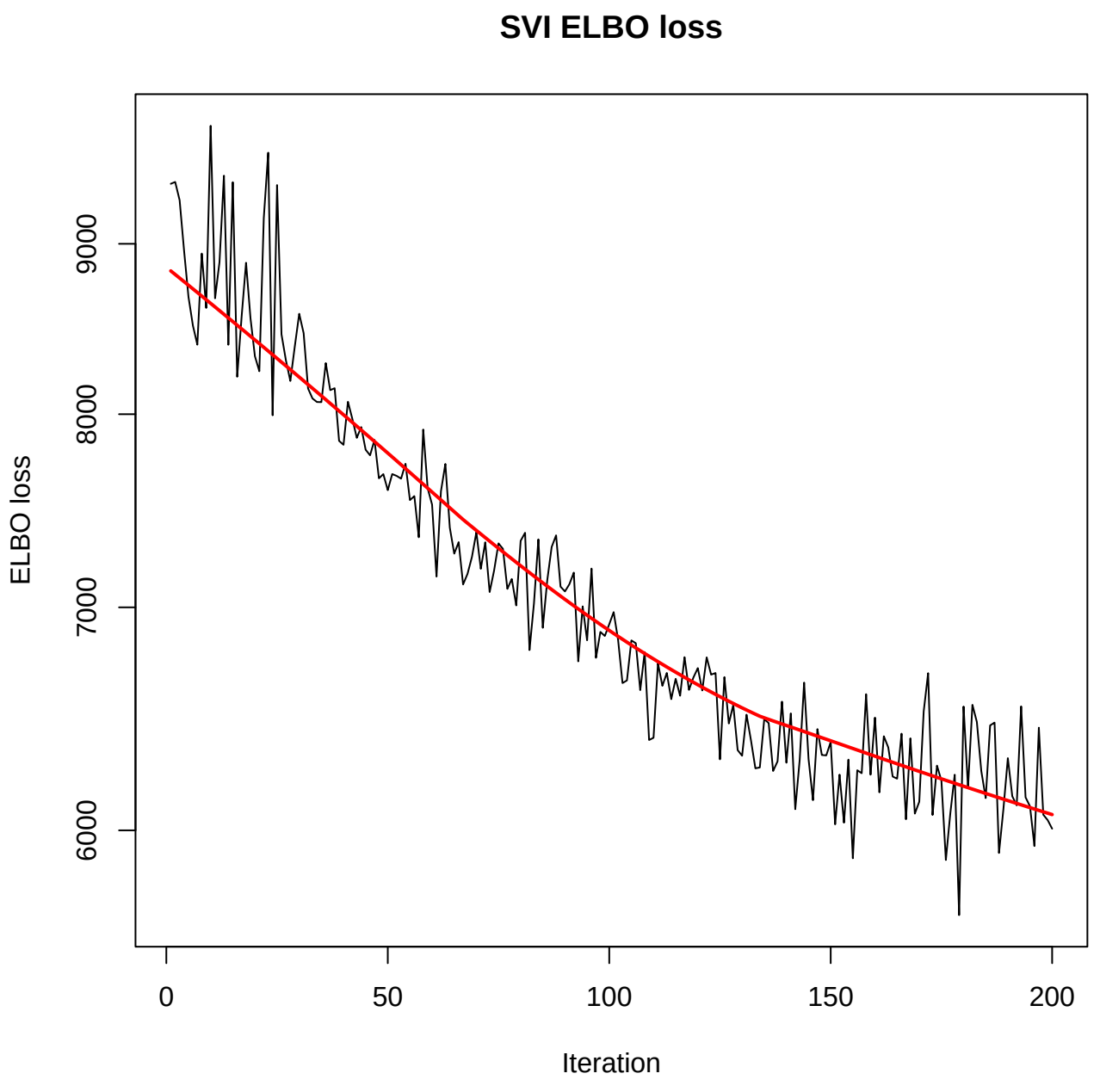

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
20000
16000
14000
12000
10000
8000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5.0
2.0
1.0
0.5
0.2
0.1
0
200
400
600
800
1000
Index


SVI ELBO loss
ELBO loss
9000
8000
7000
6000
0
50
100
150
200
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.5
1.0
2.0
5.0
0
200
400
600
800
1000
Index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ELBO loss
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7000
6000
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100
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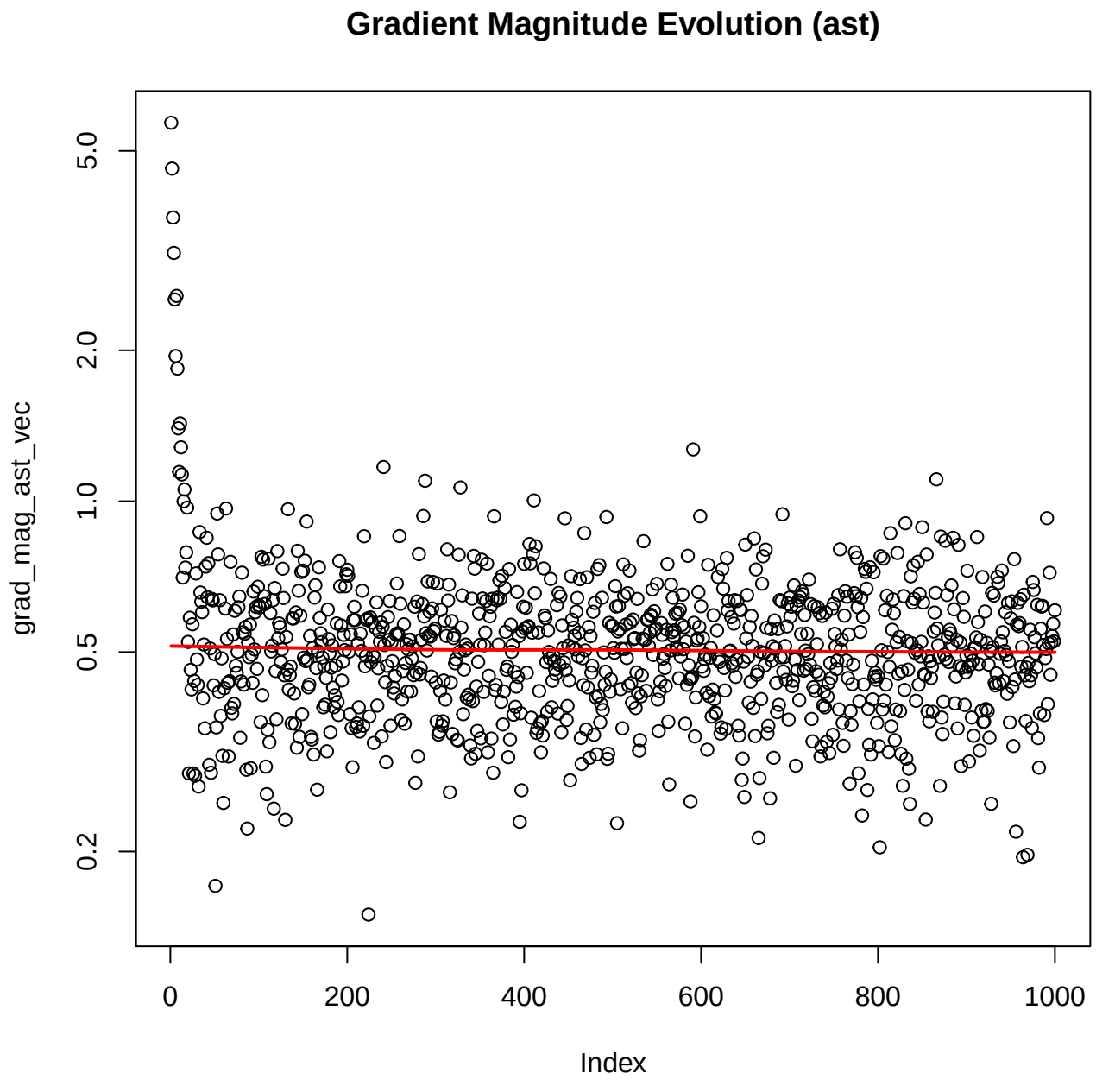
200
Iteration


Gradient Magnitude Evolution (ast)
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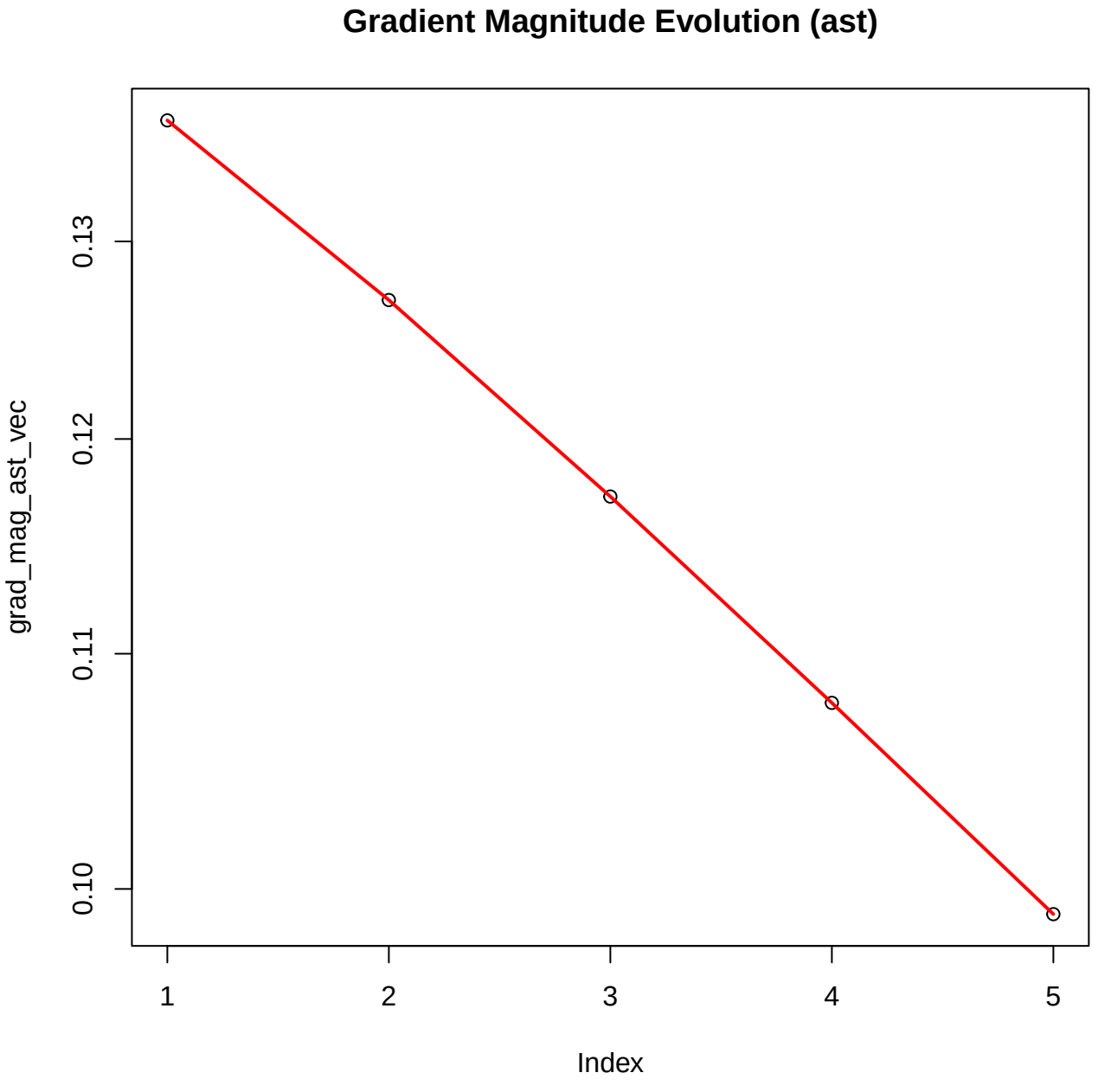
Gradient Magnitude Evolution (ast)
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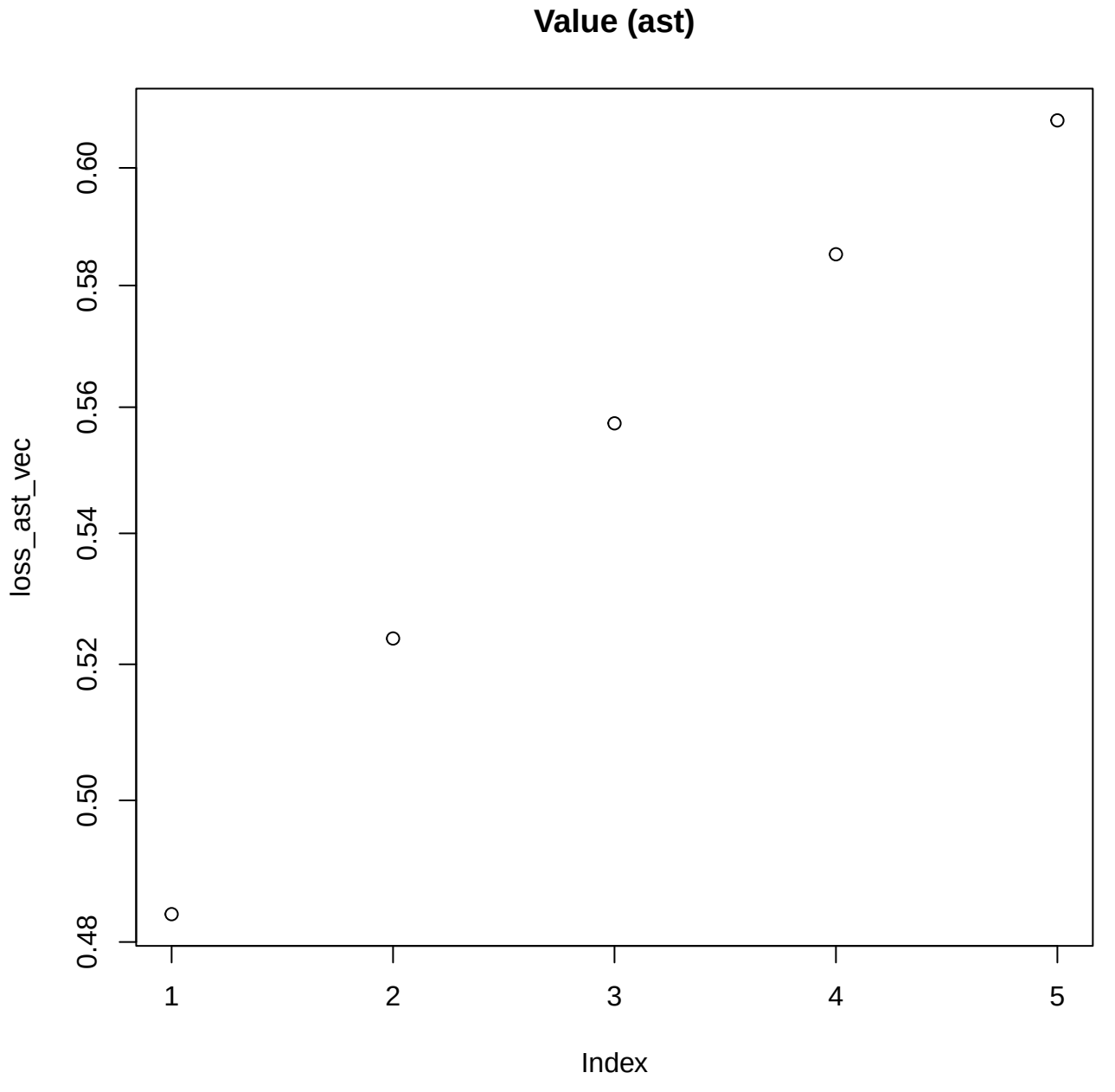
Value (ast)
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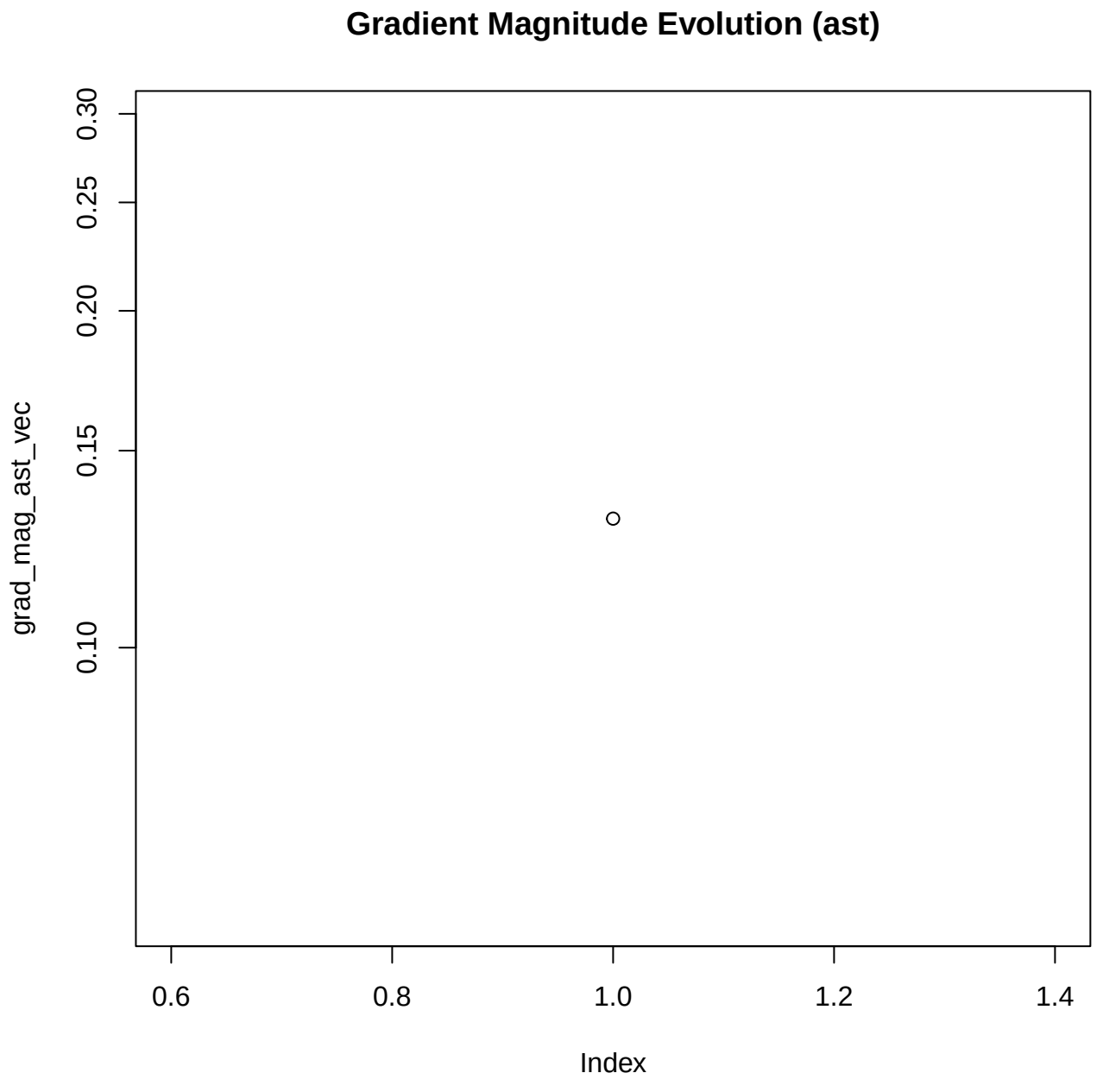
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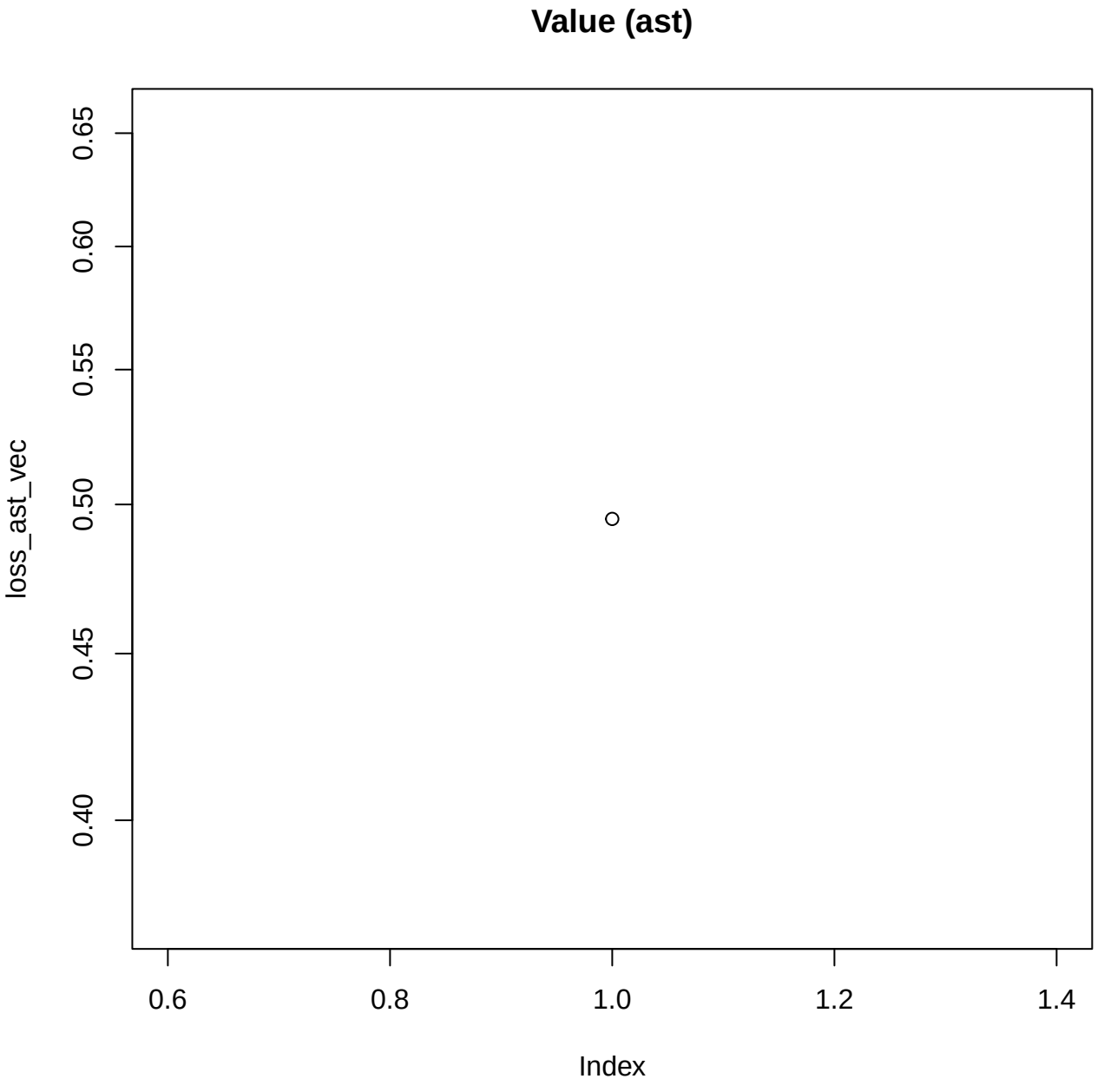
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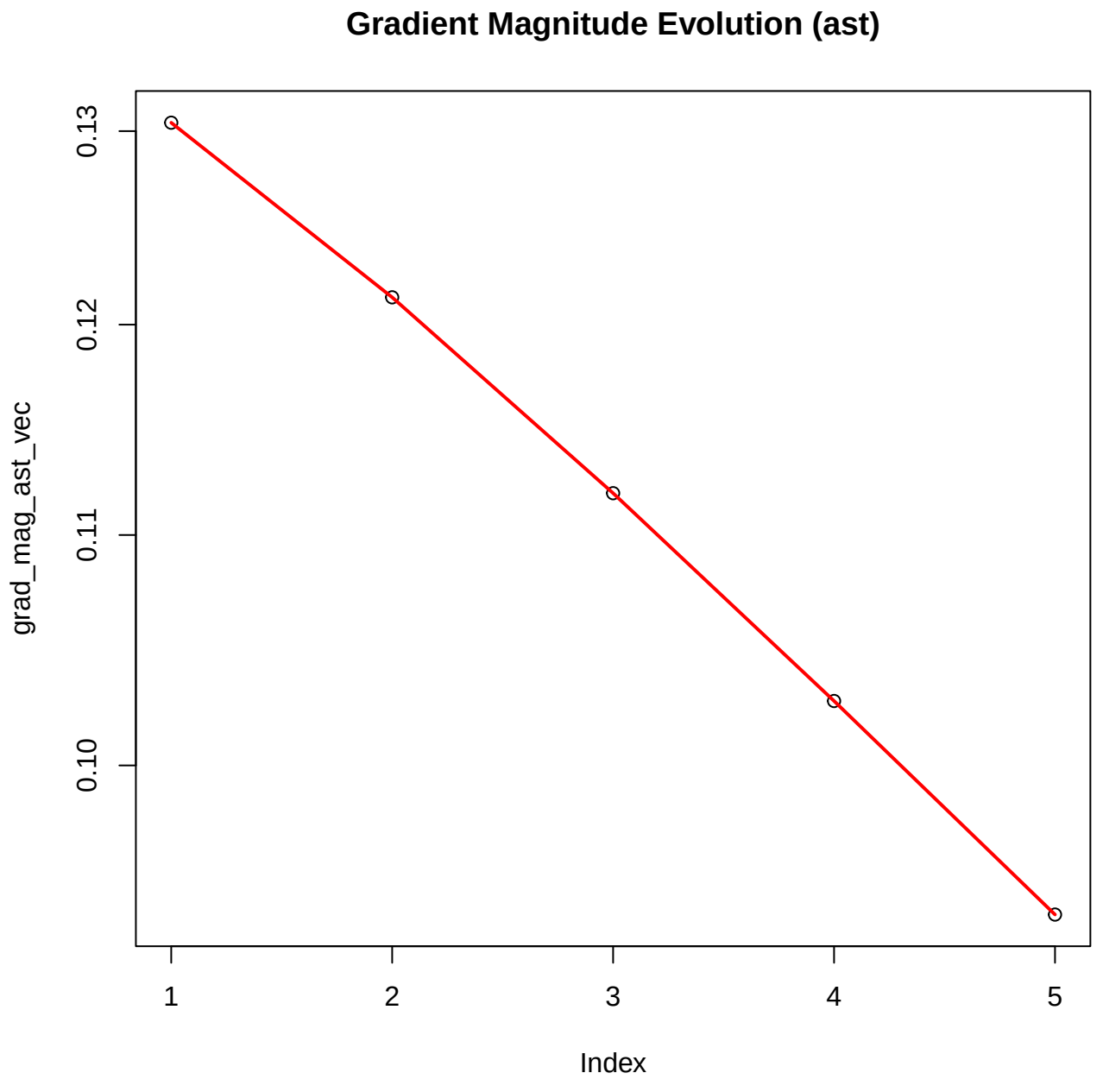
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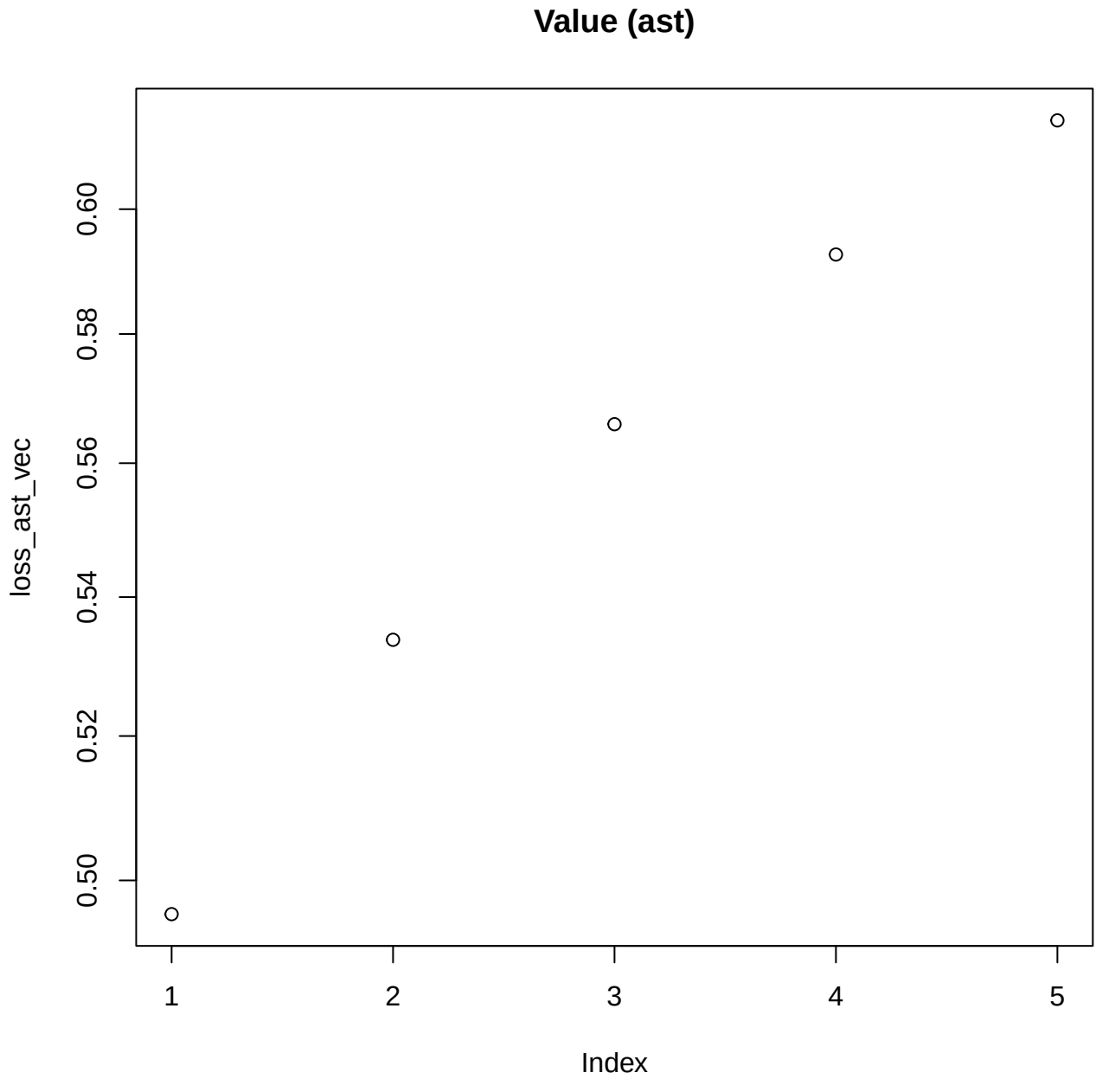
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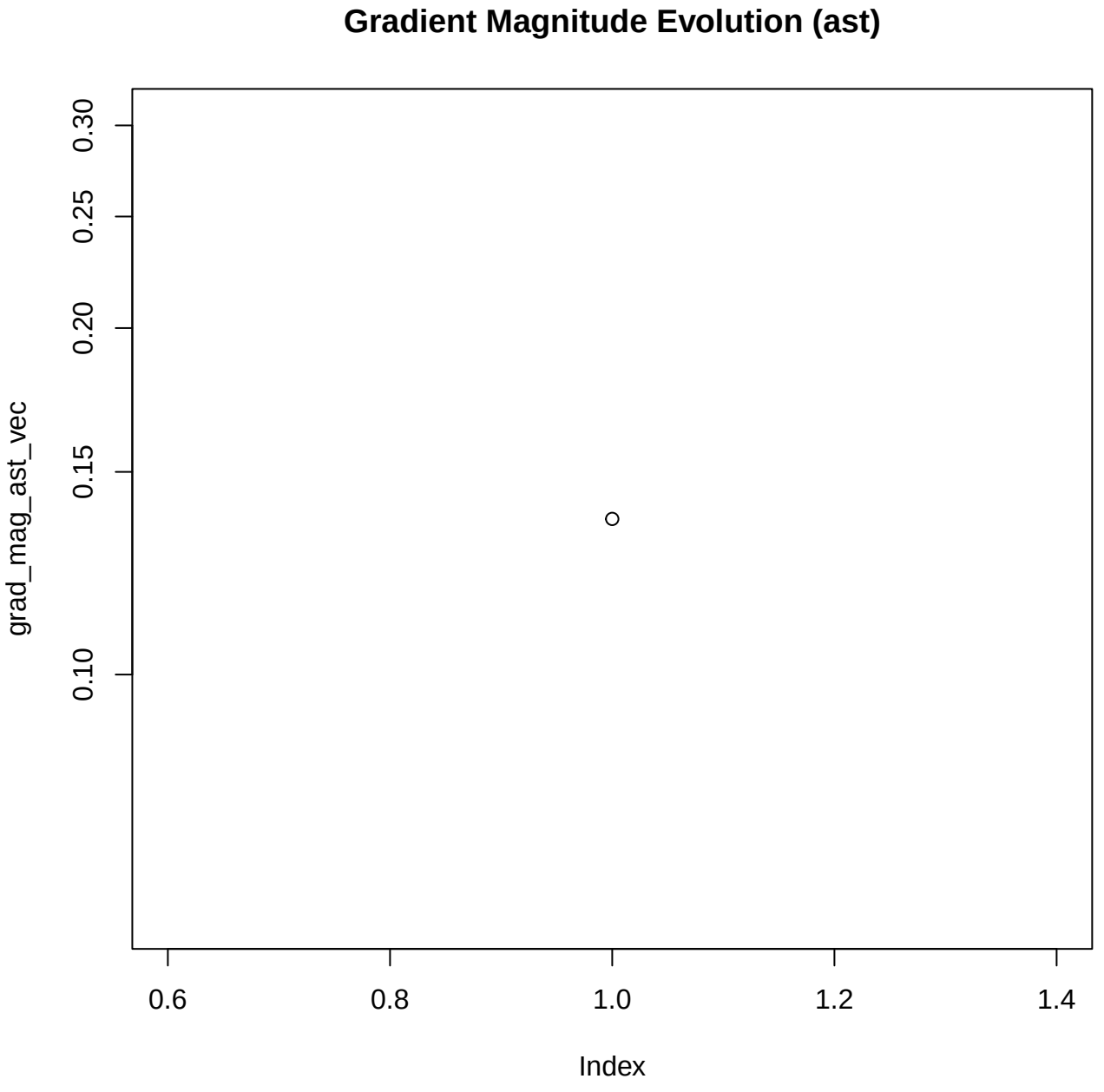
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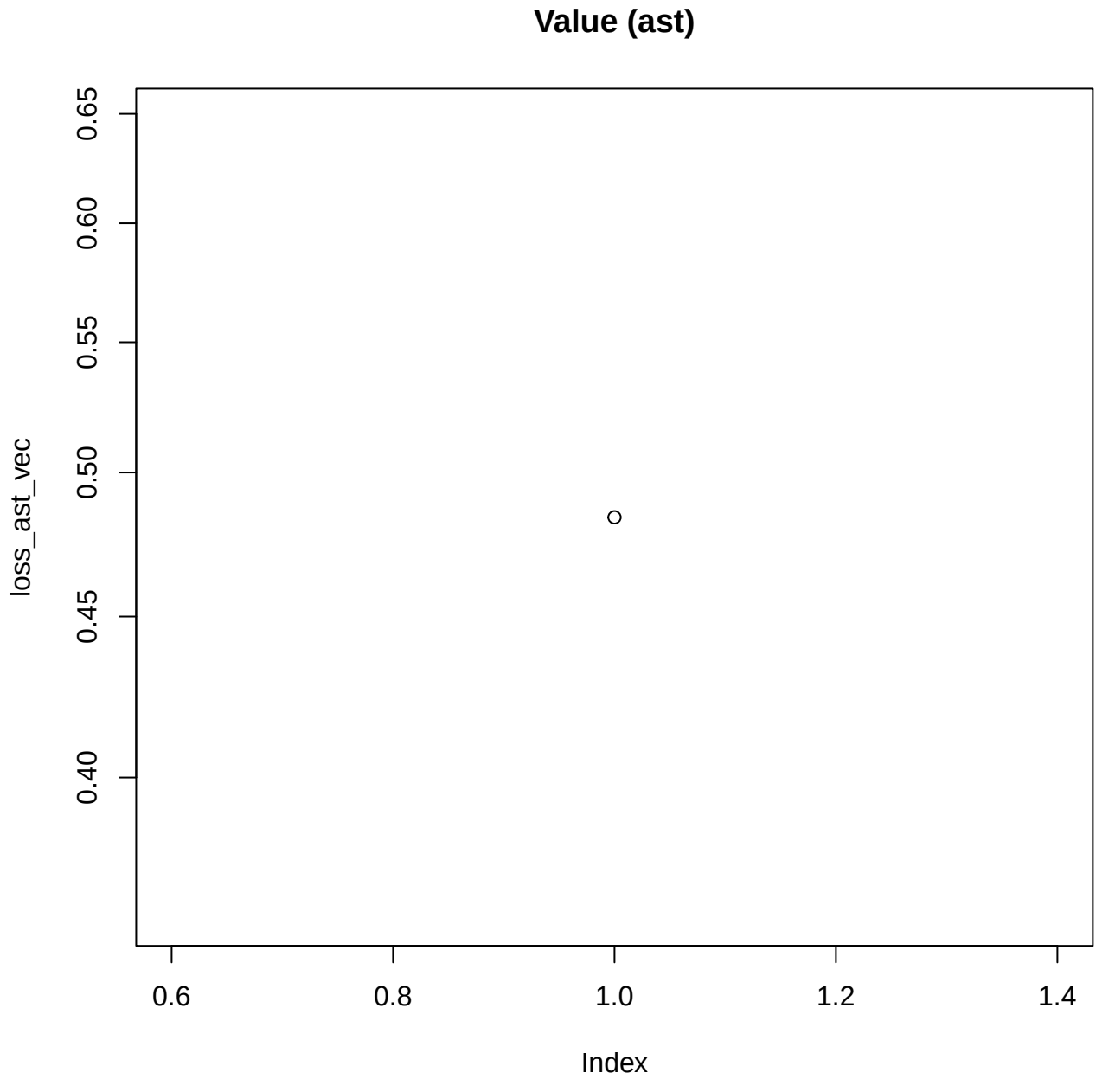
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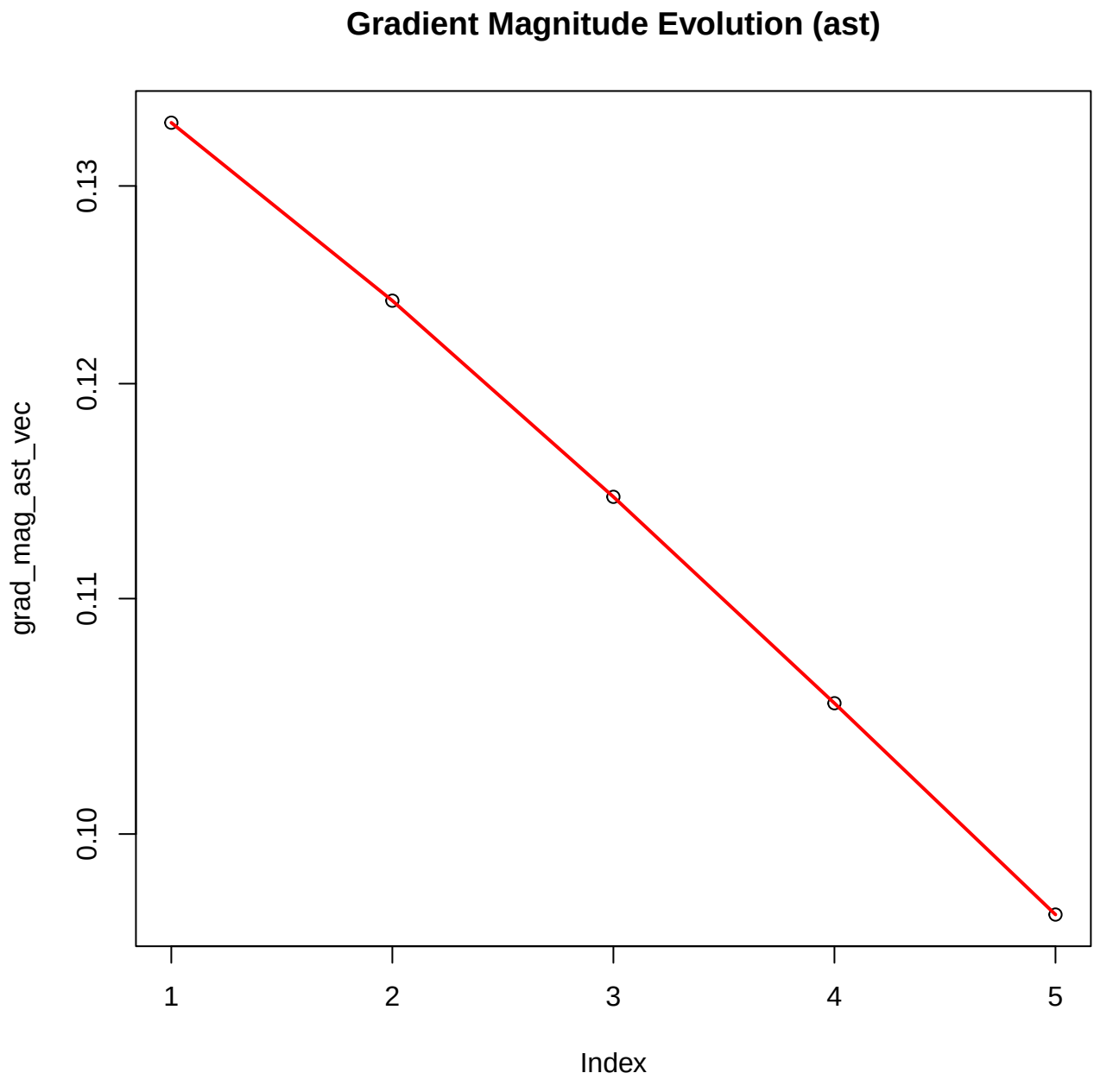
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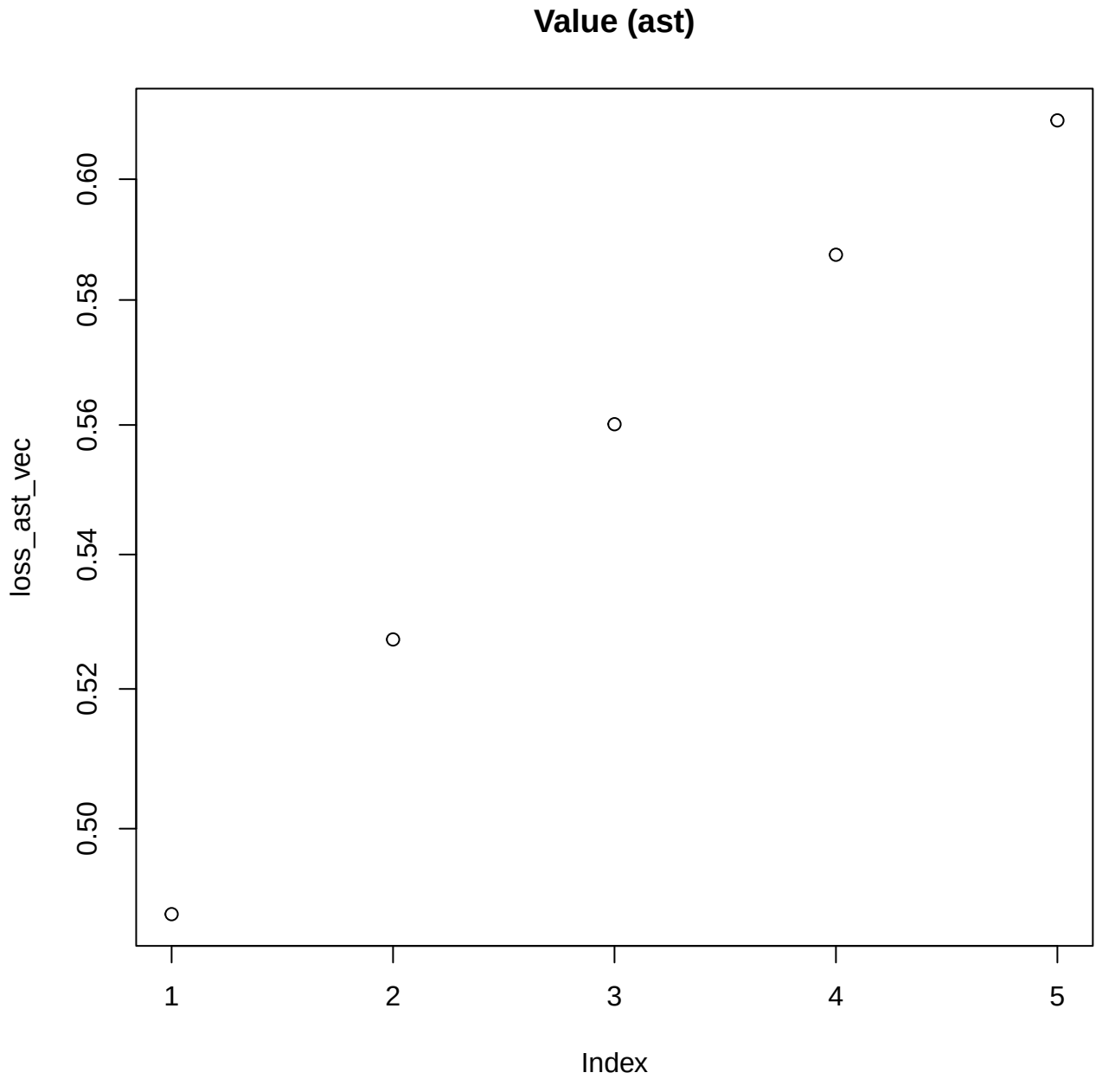
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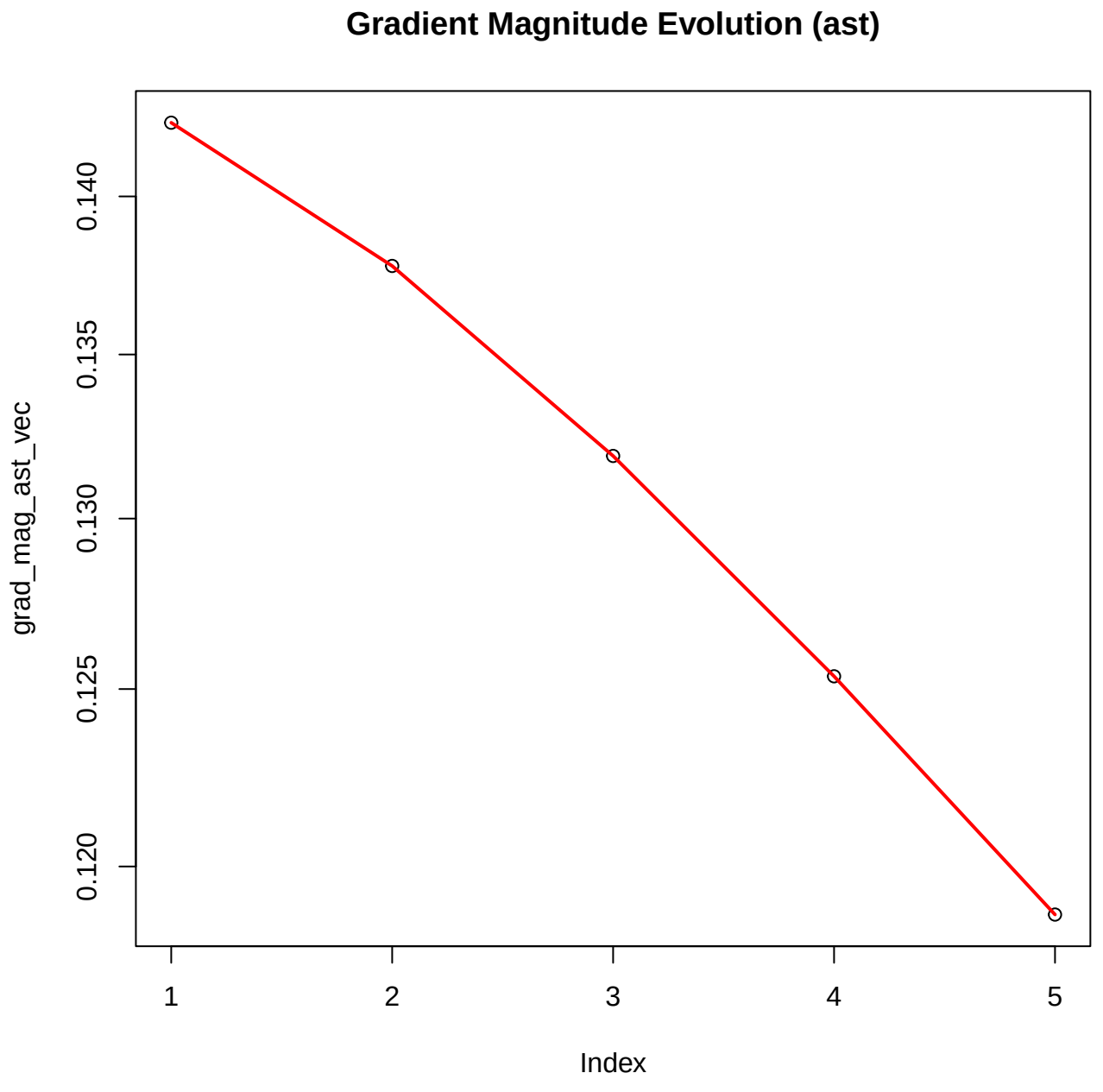
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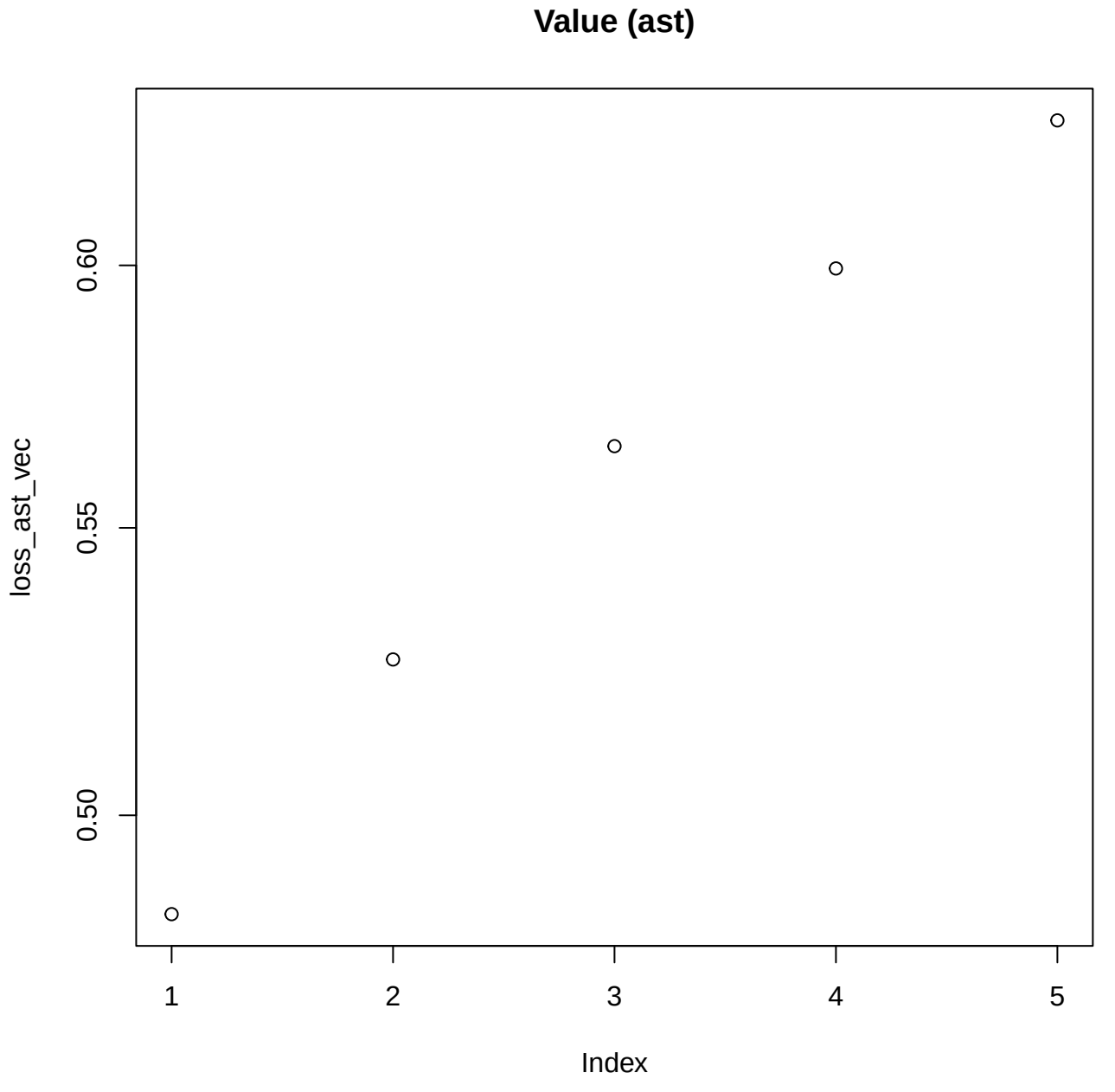
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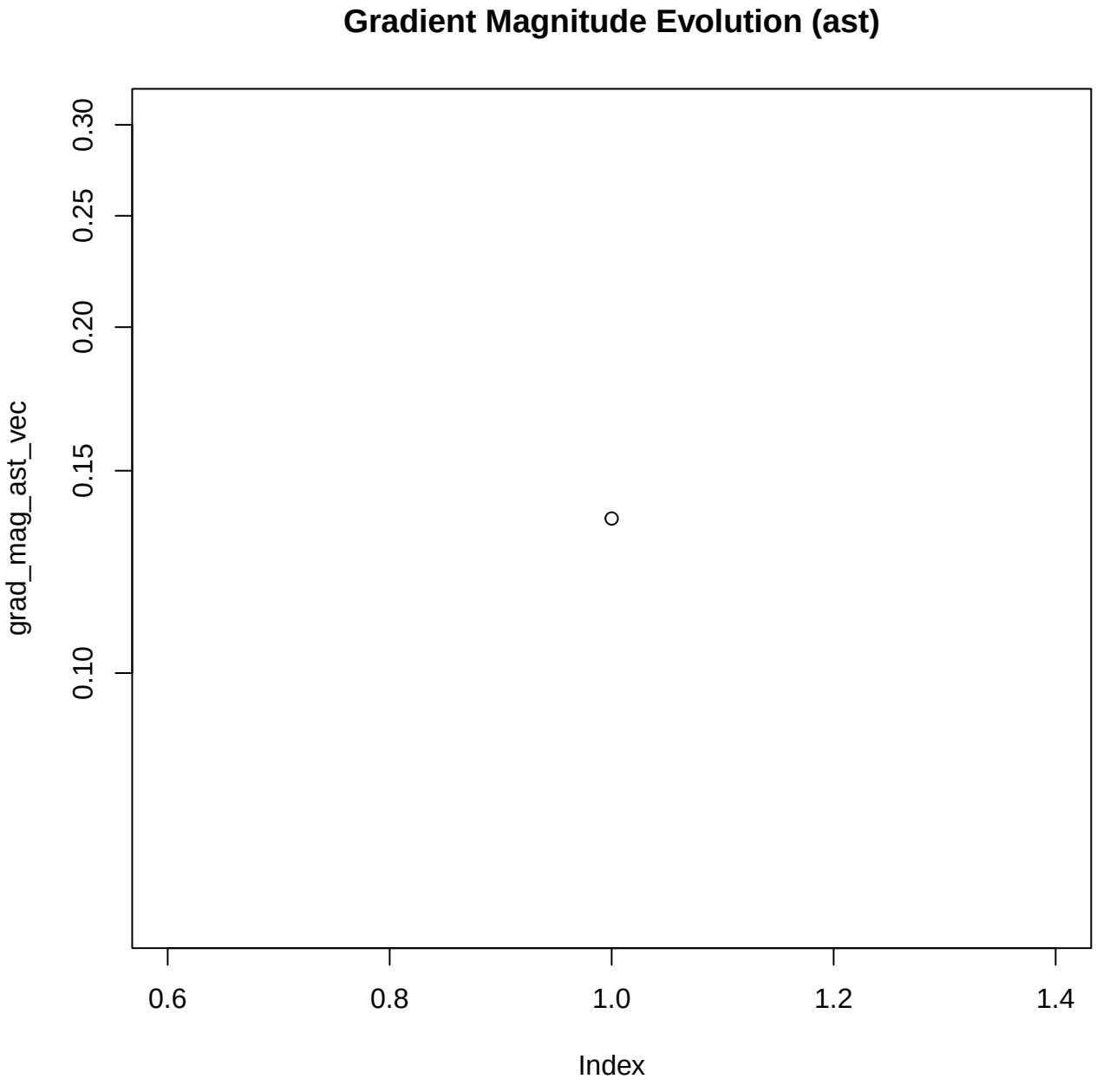
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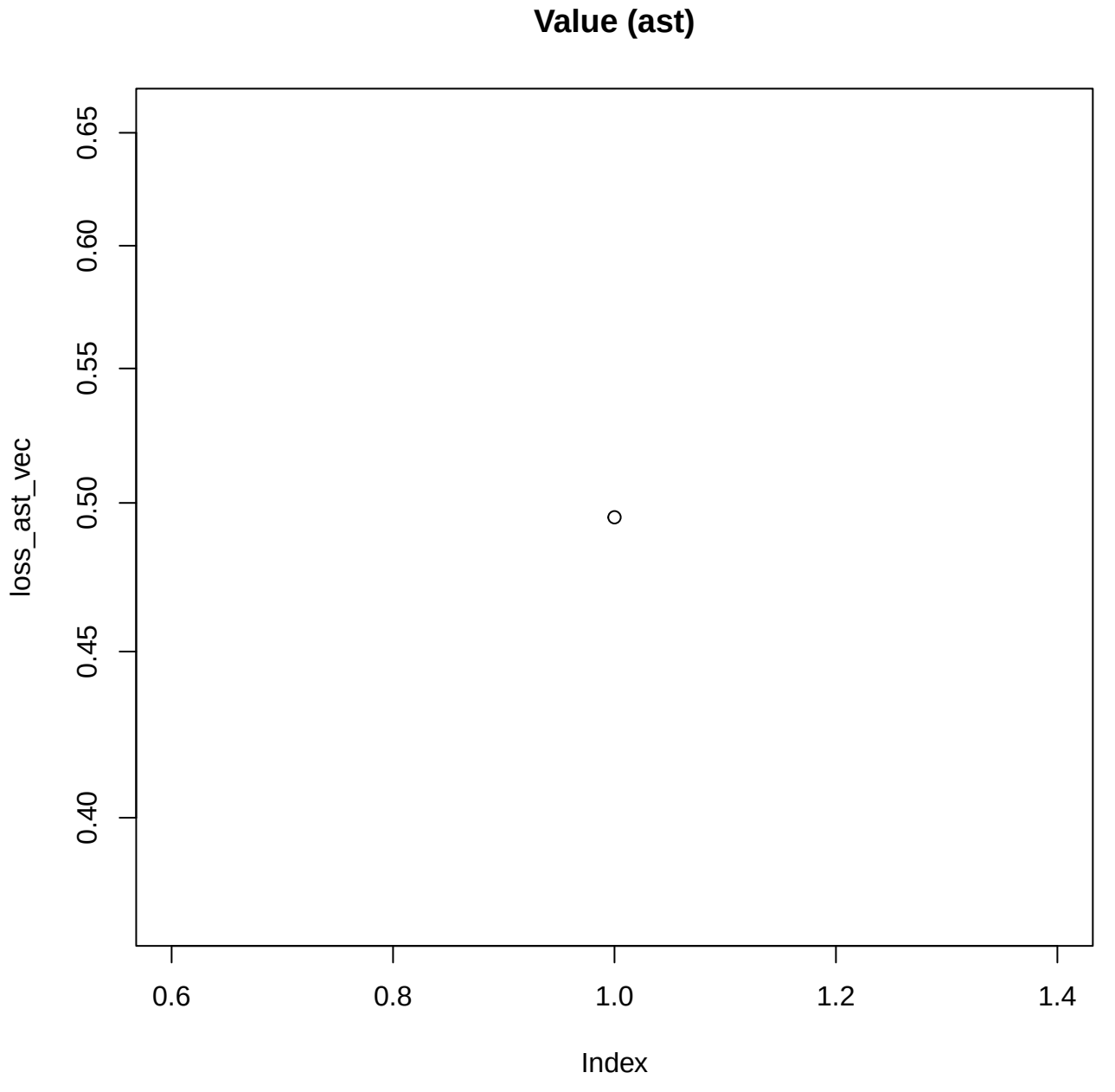
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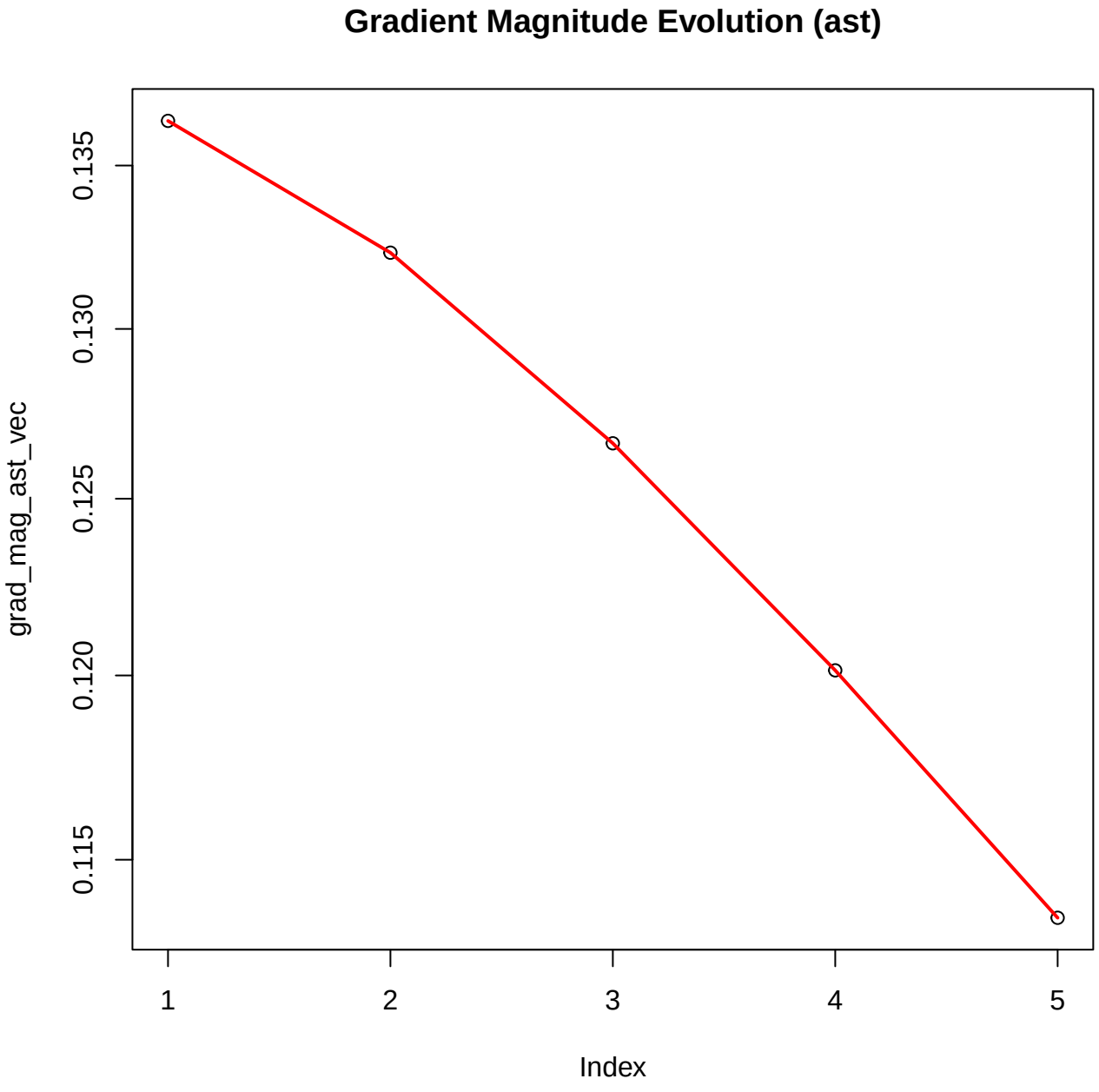
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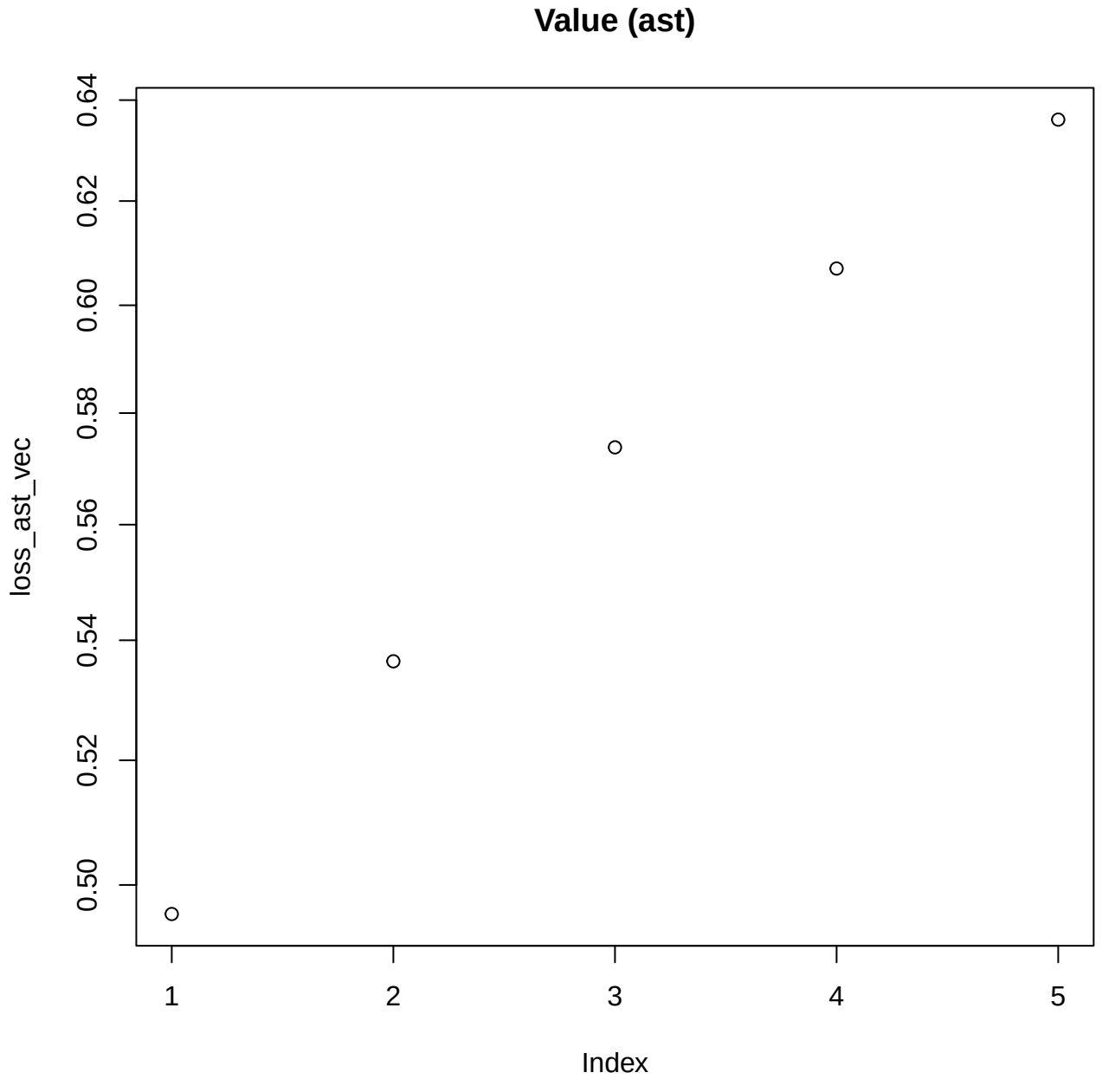
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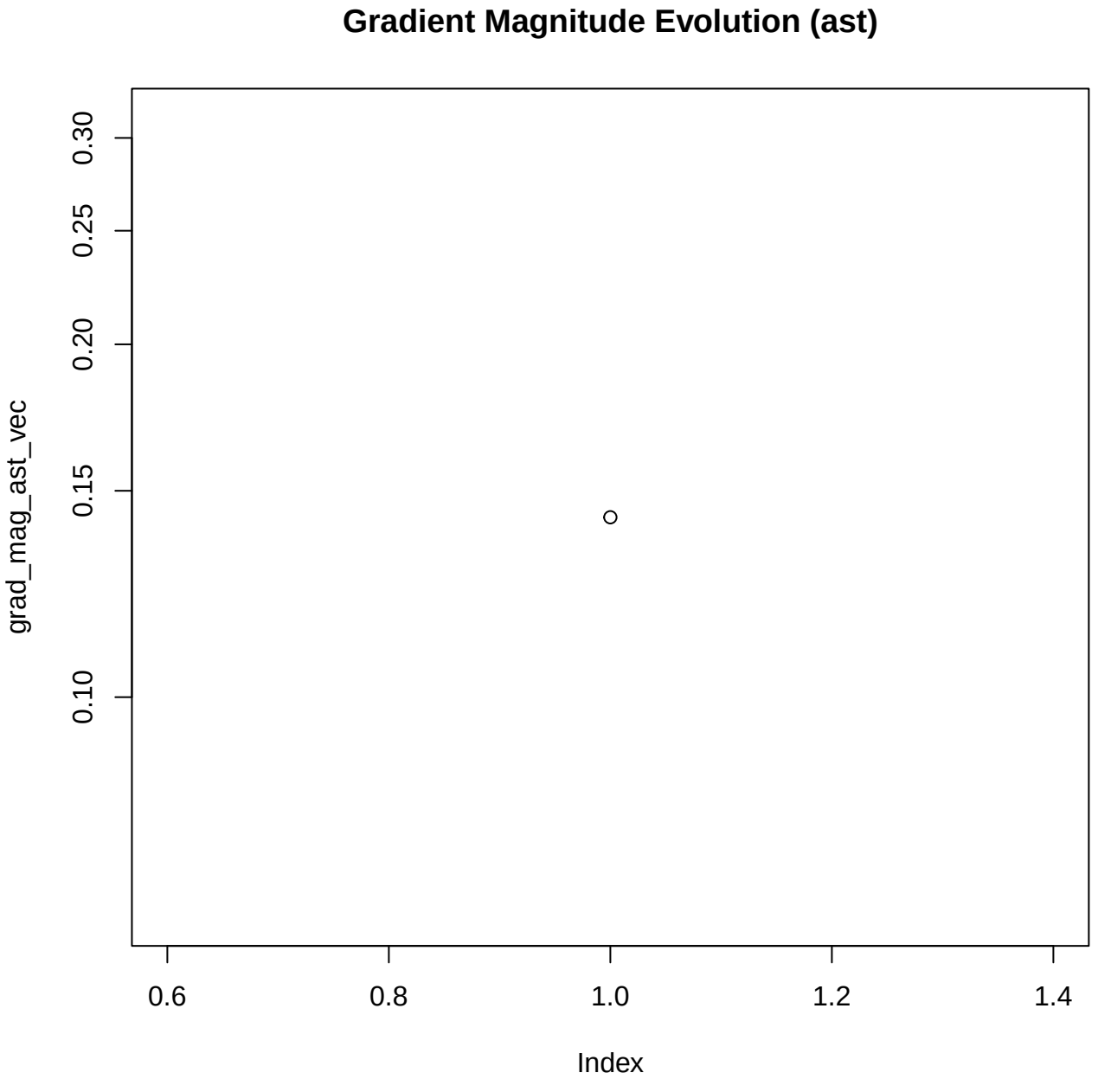
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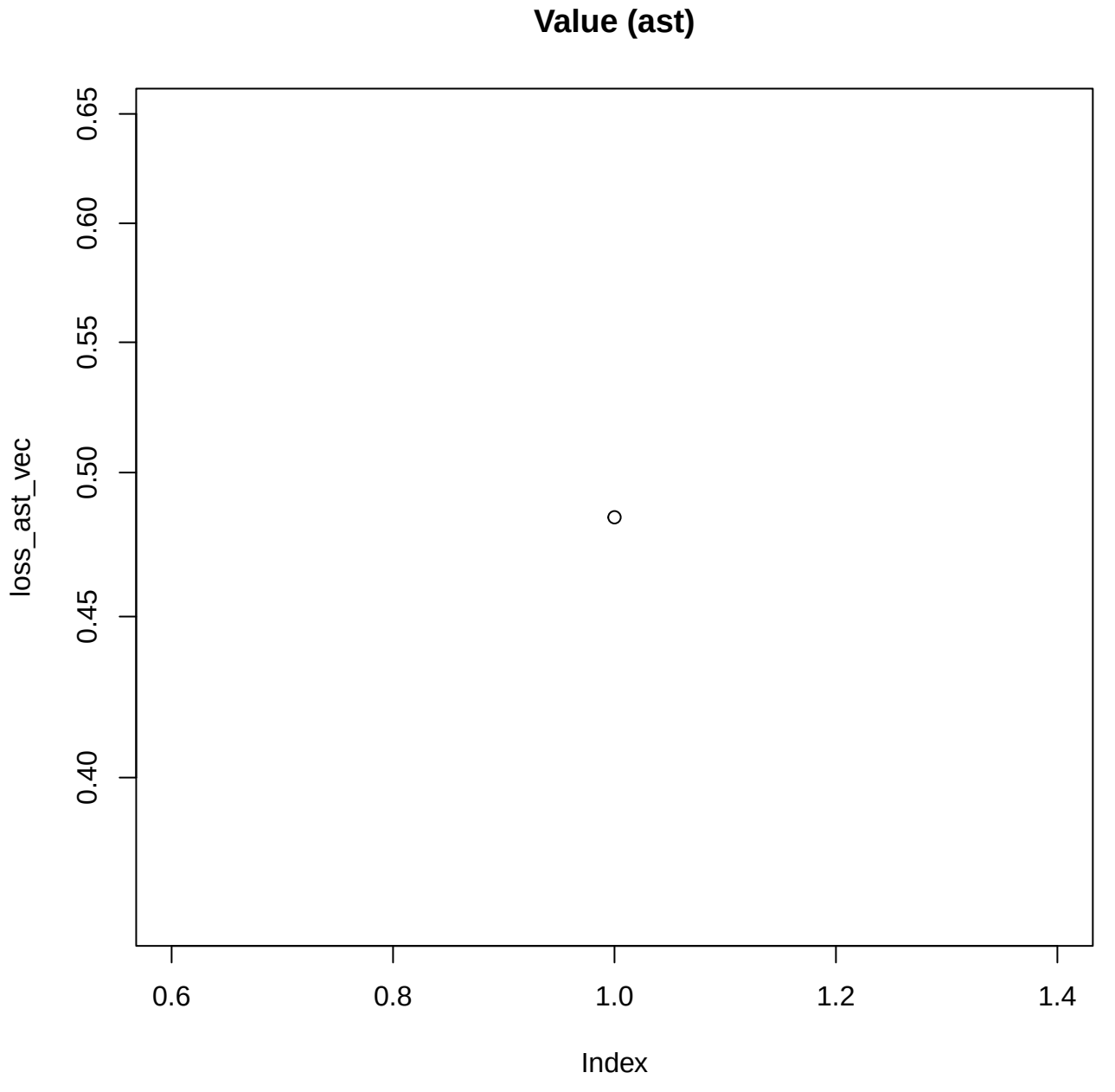
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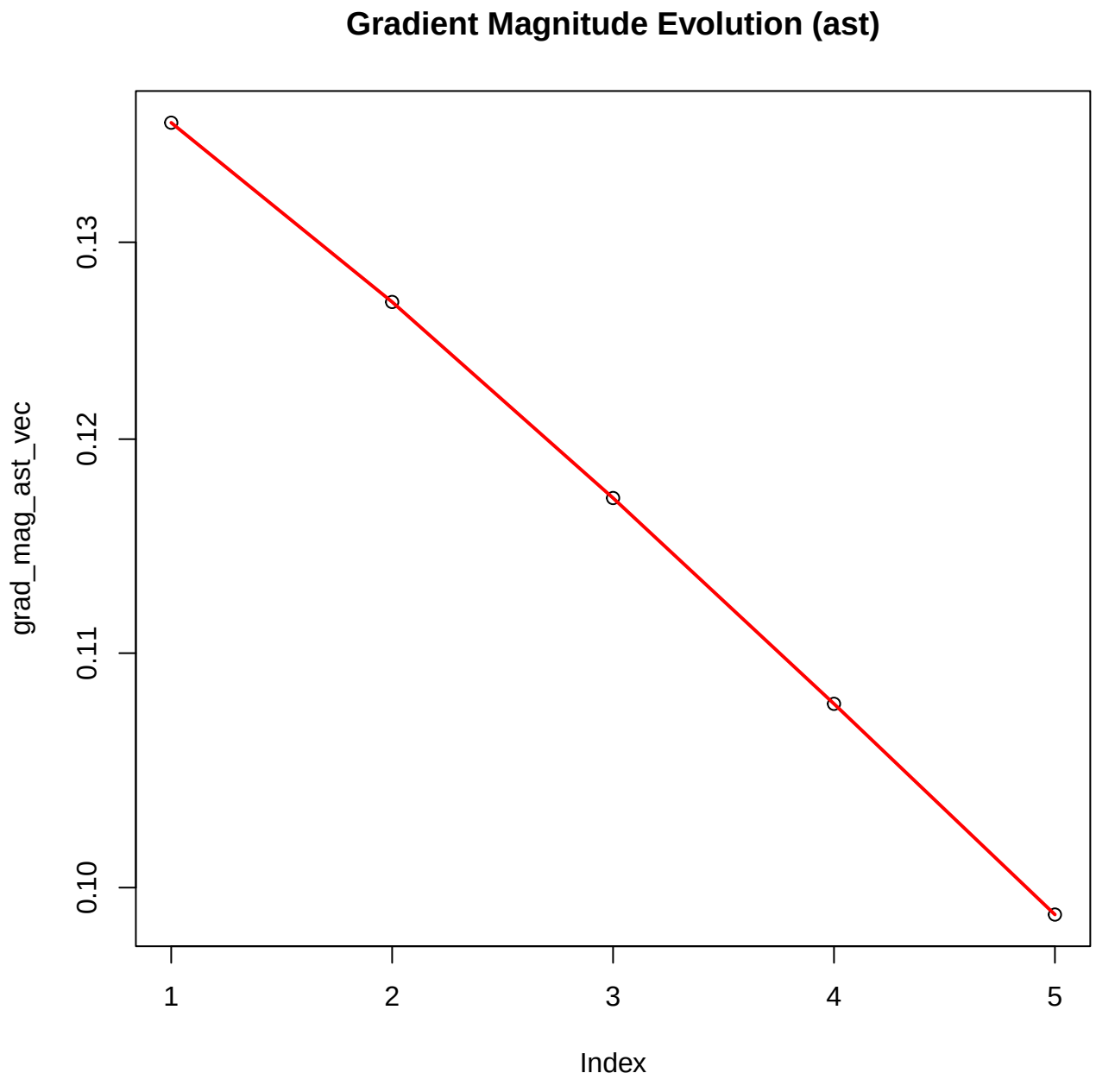
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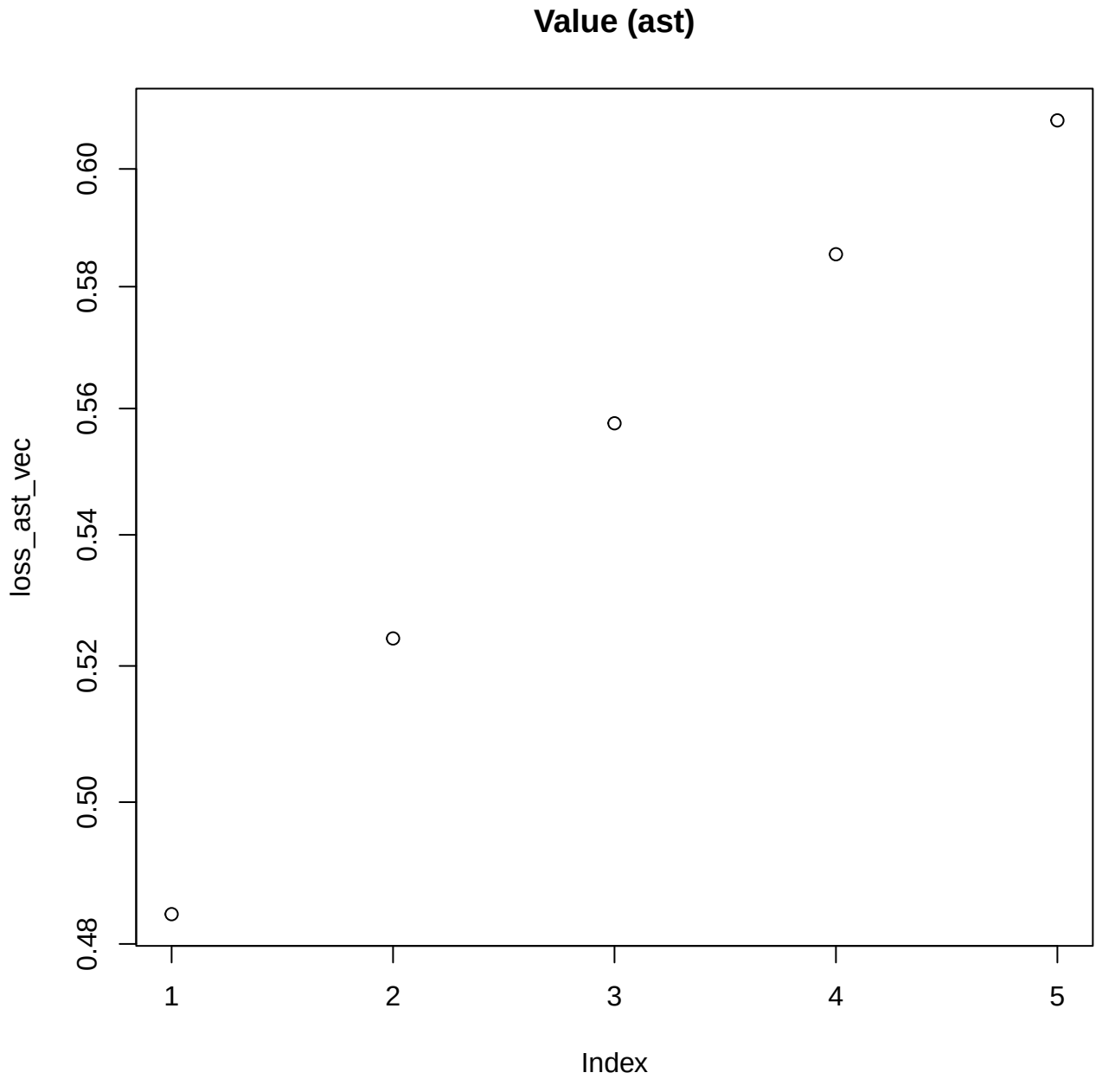
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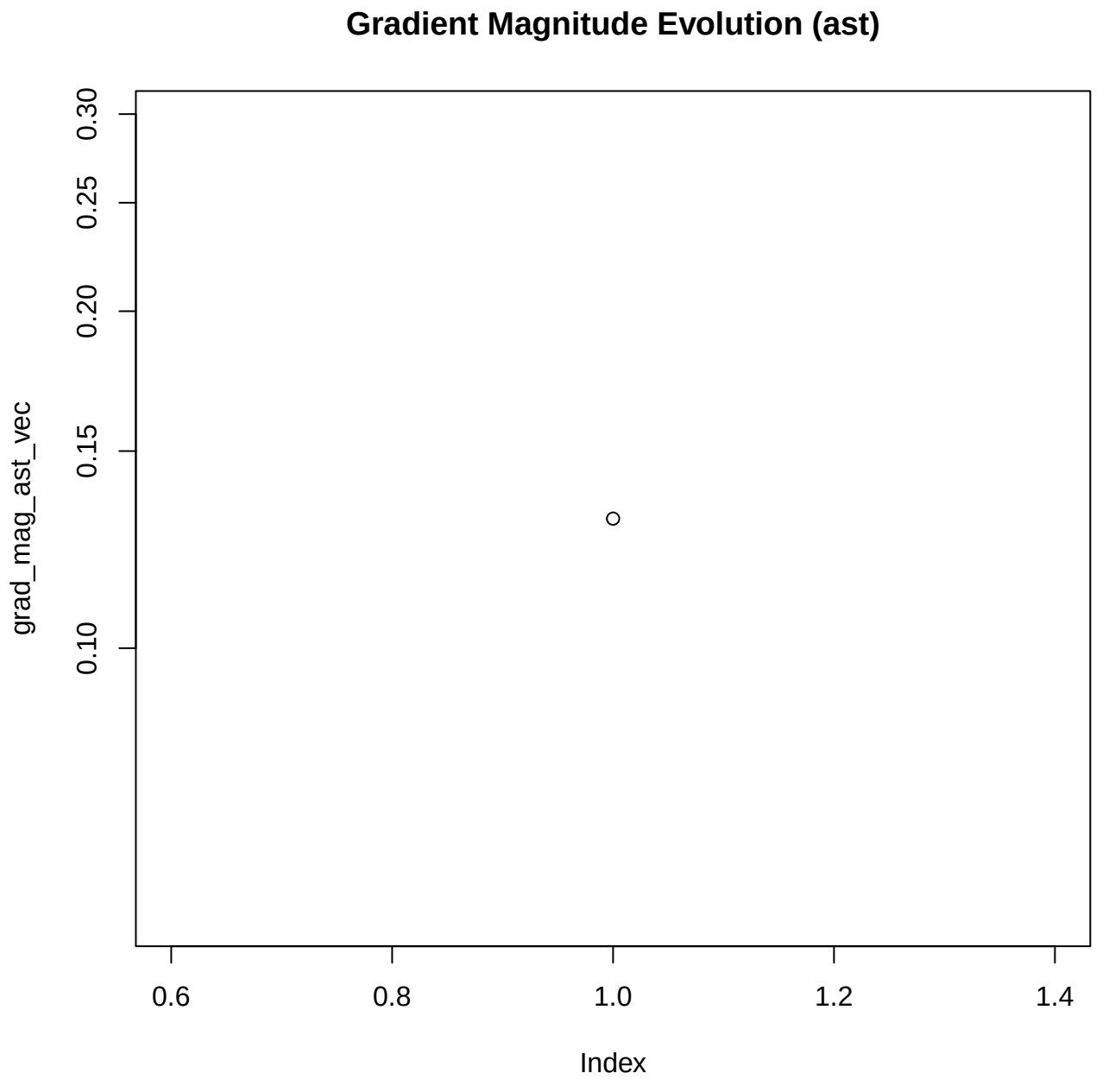
Gradient Magnitude Evolution (ast)
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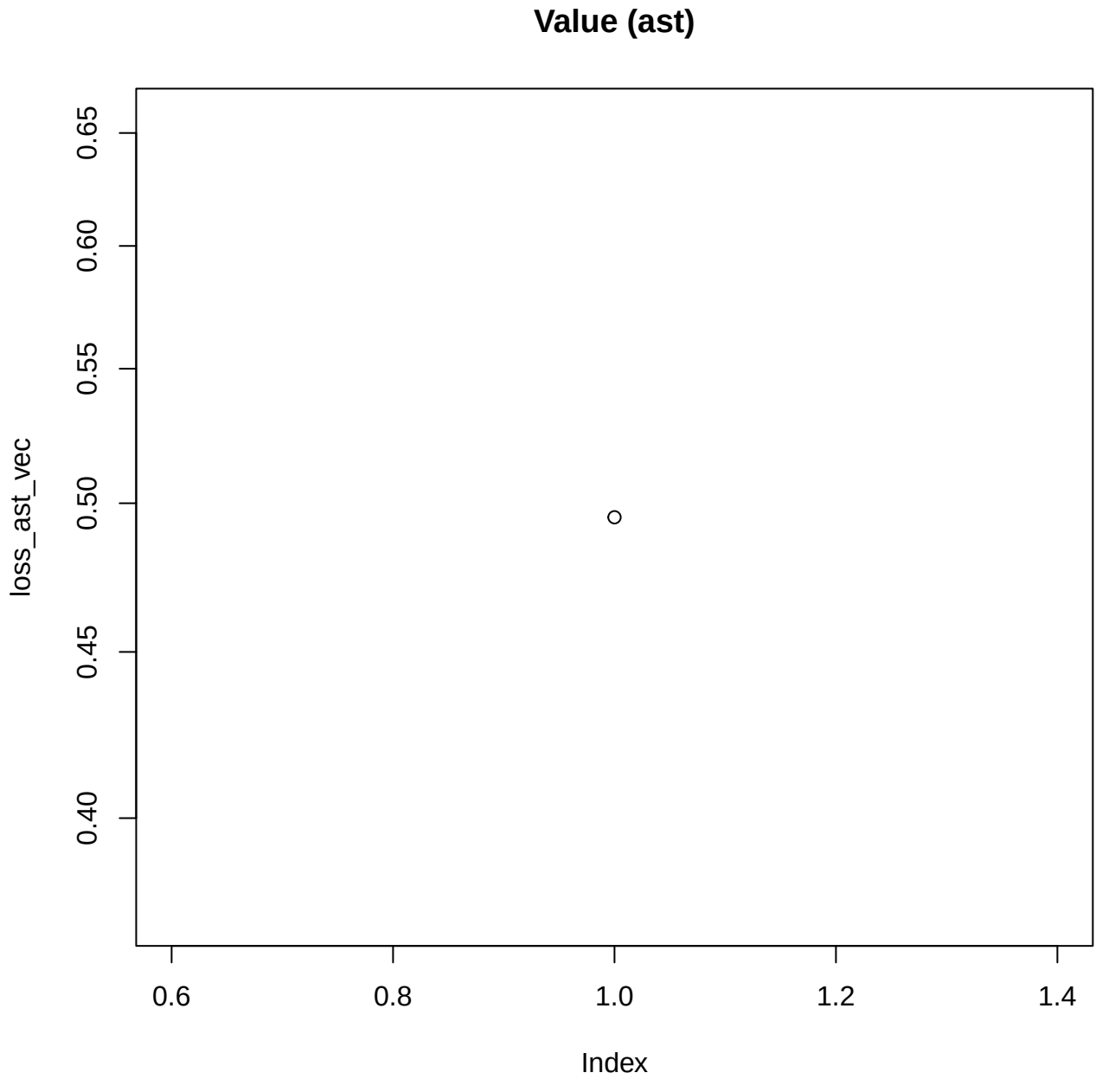
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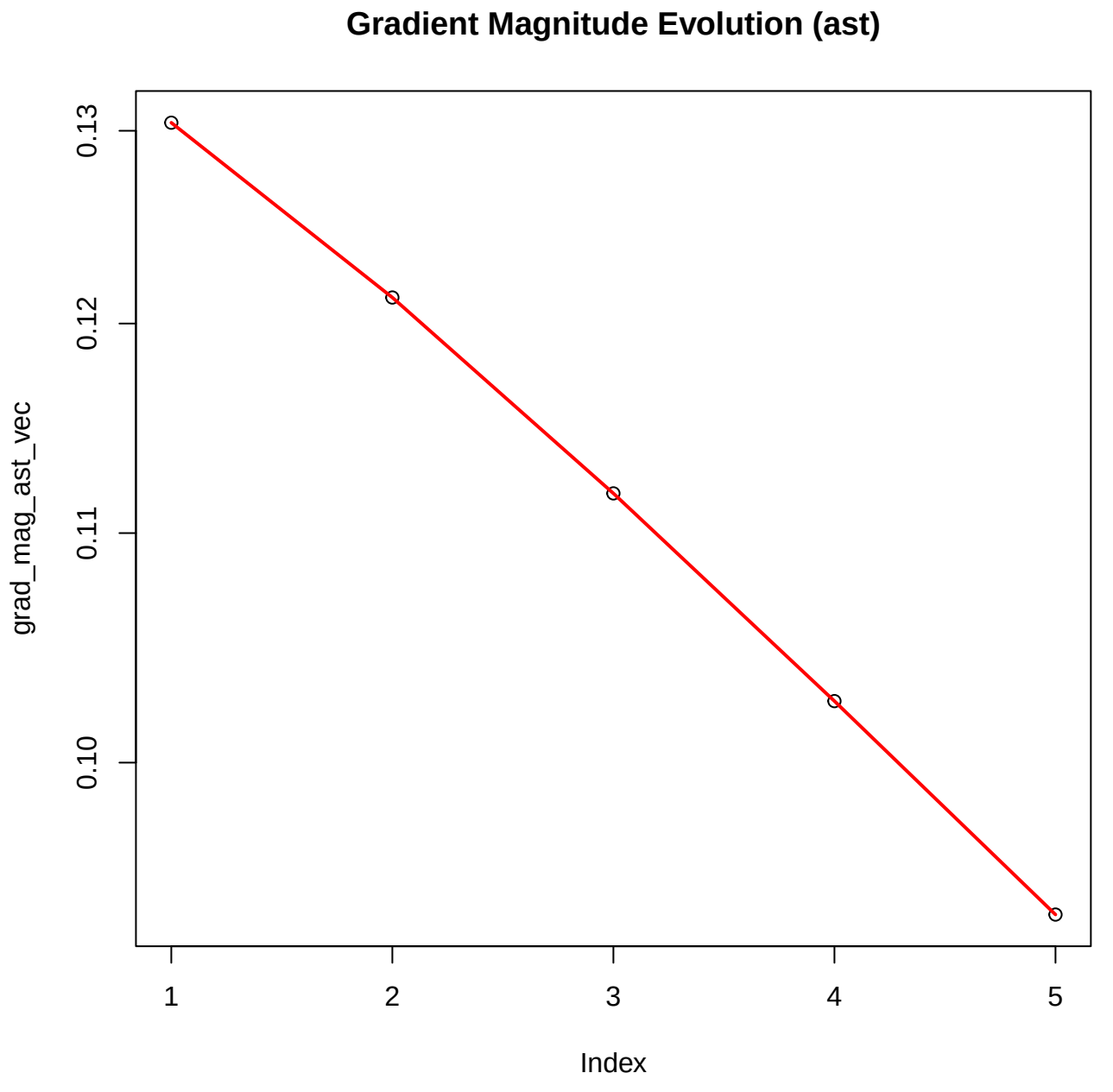
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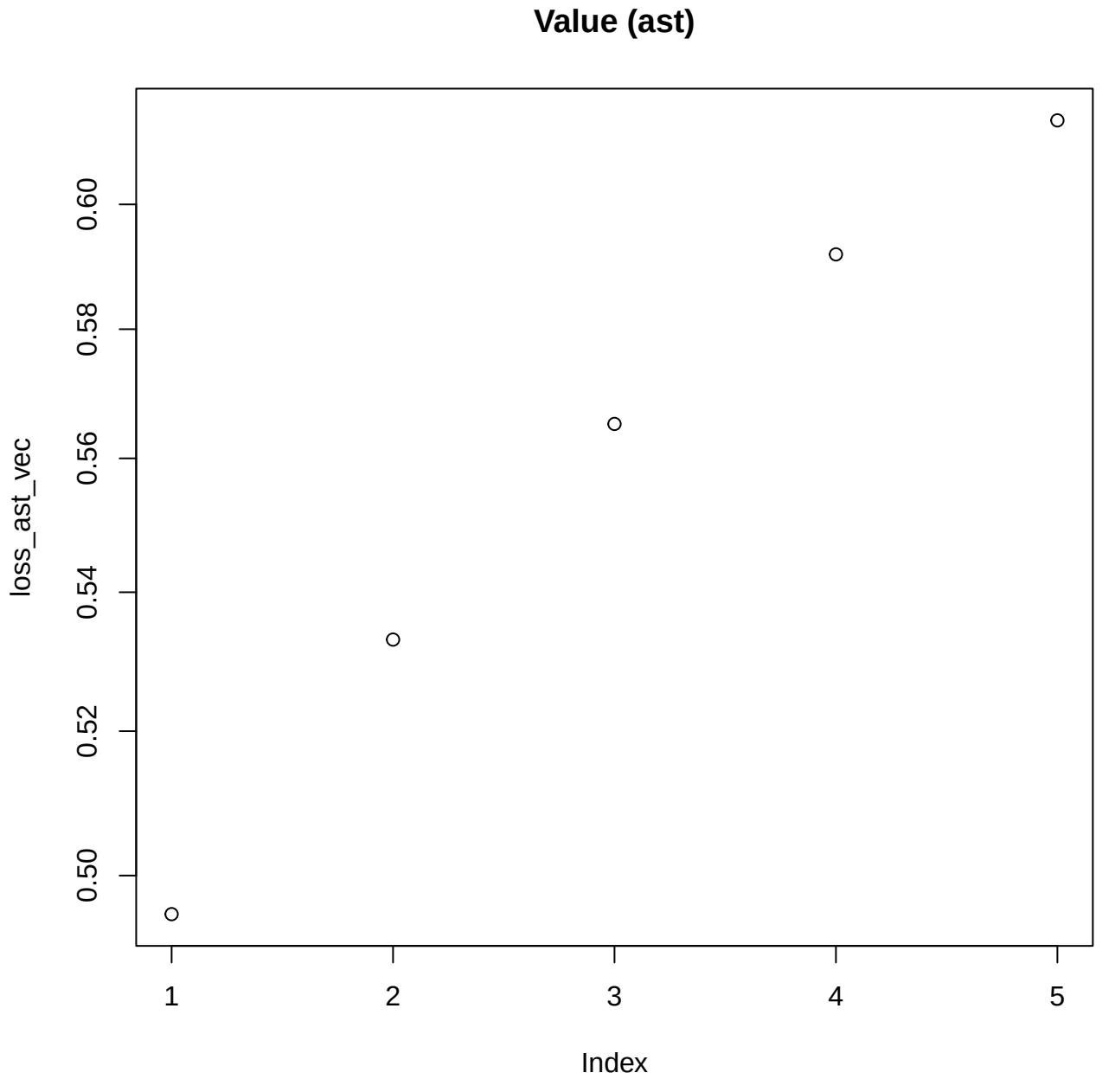
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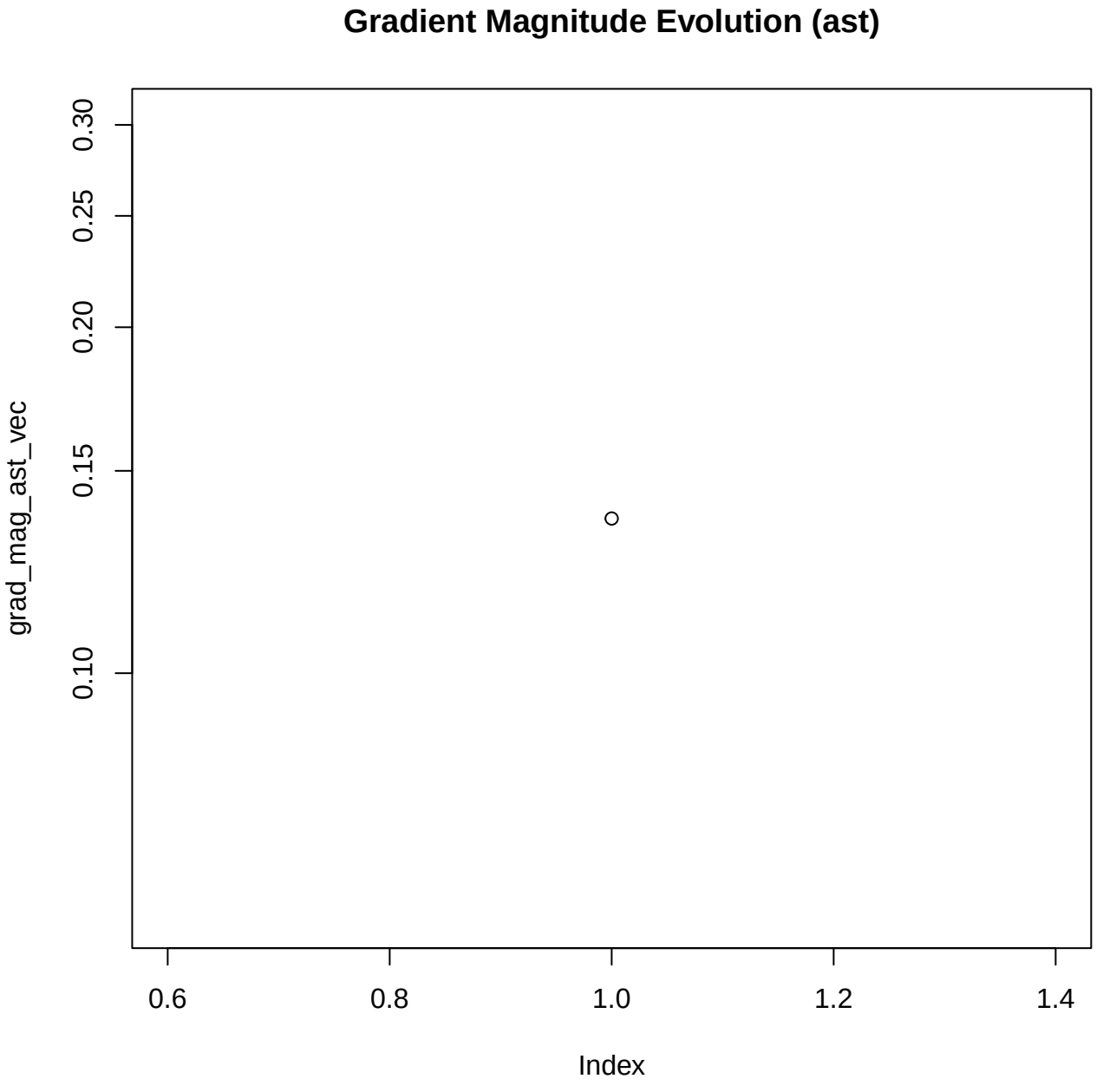
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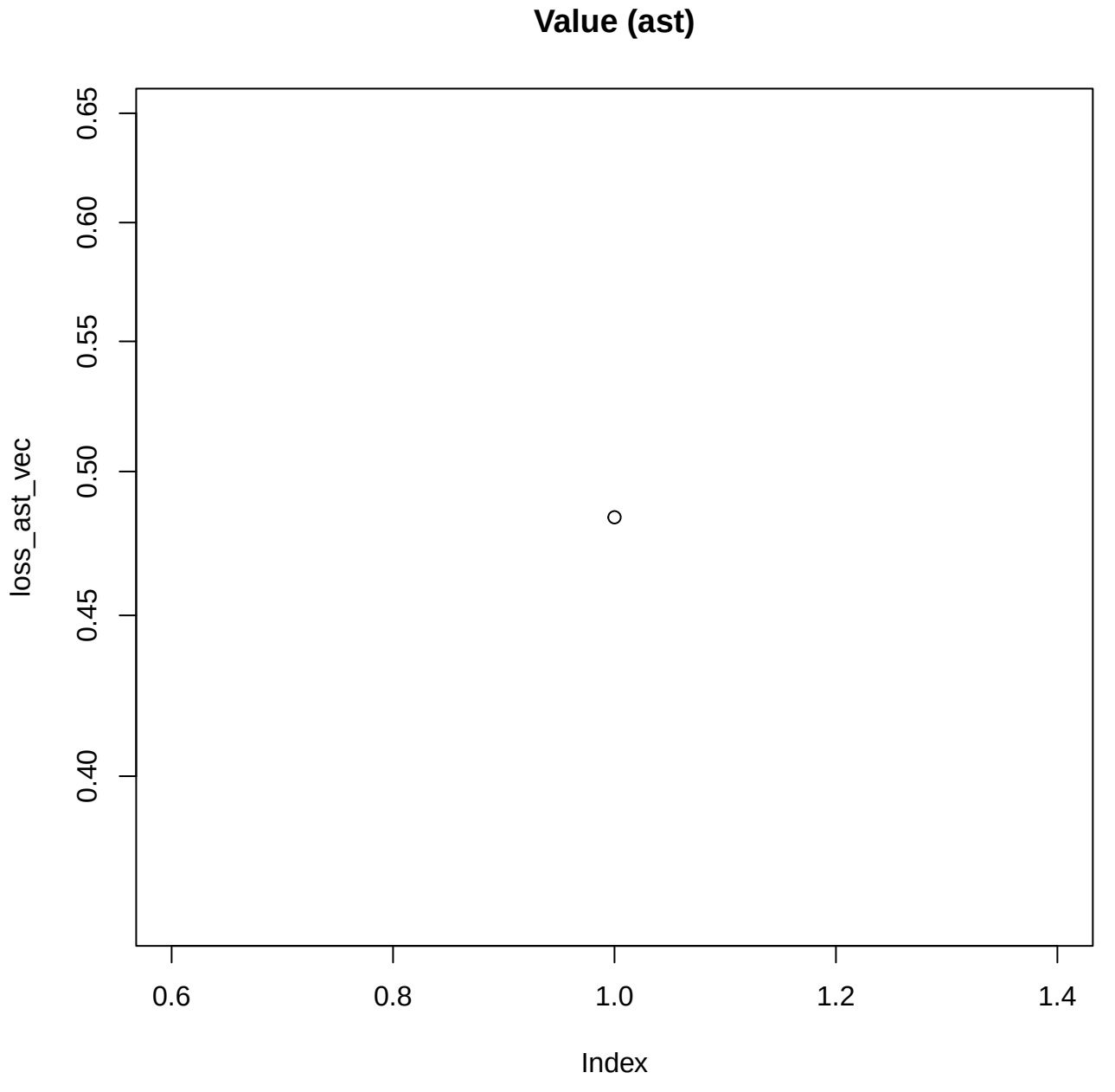
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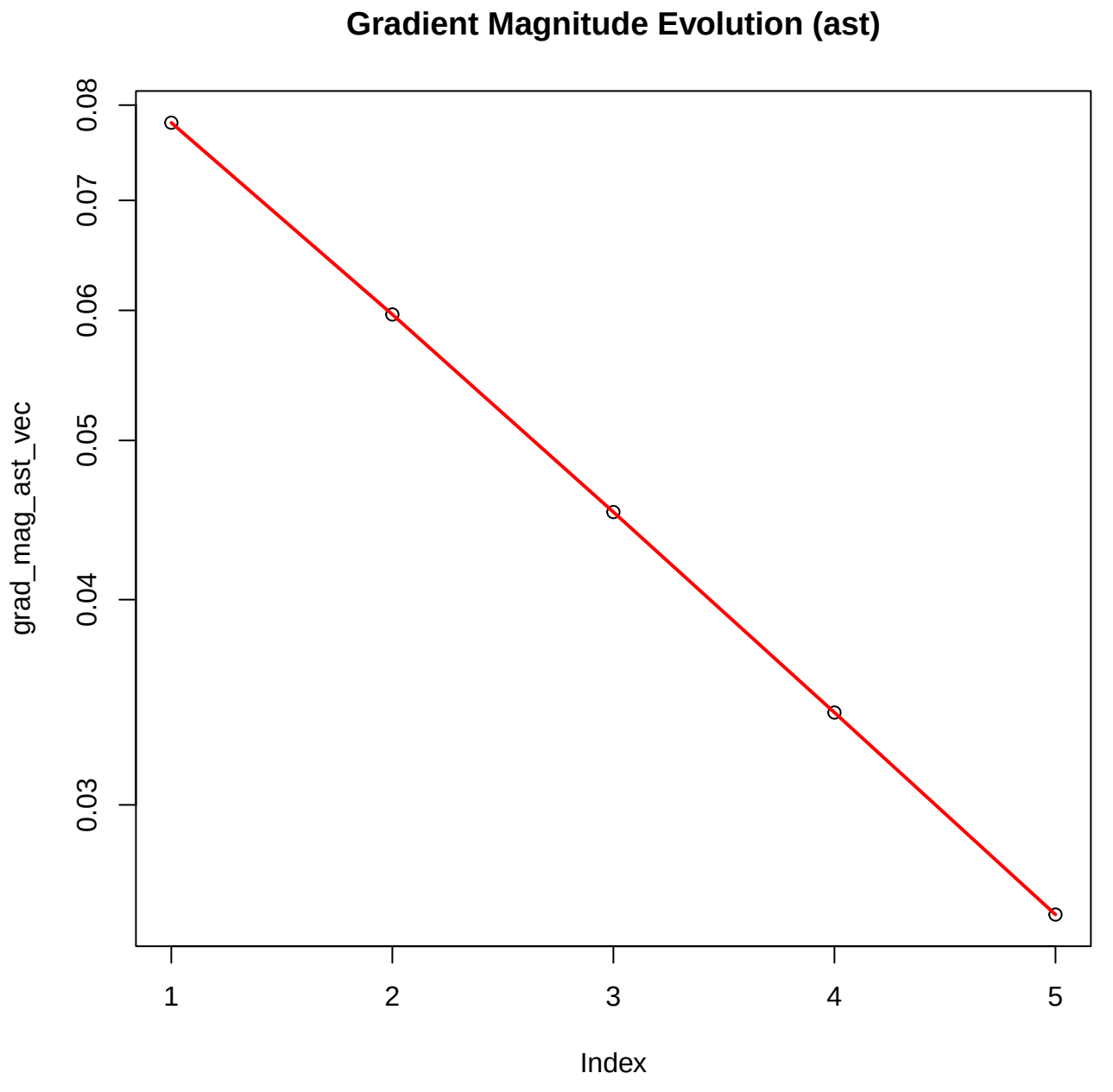
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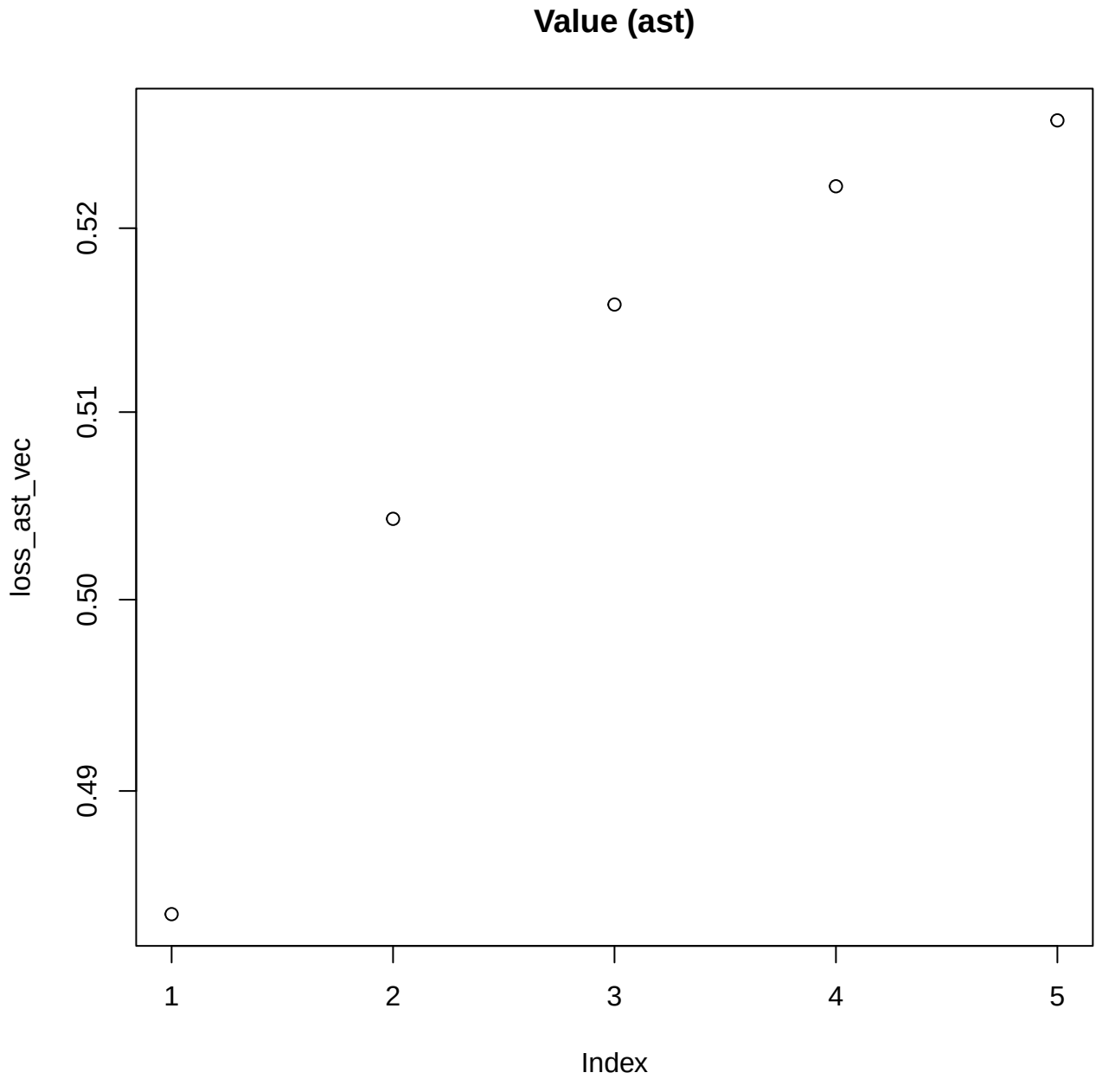
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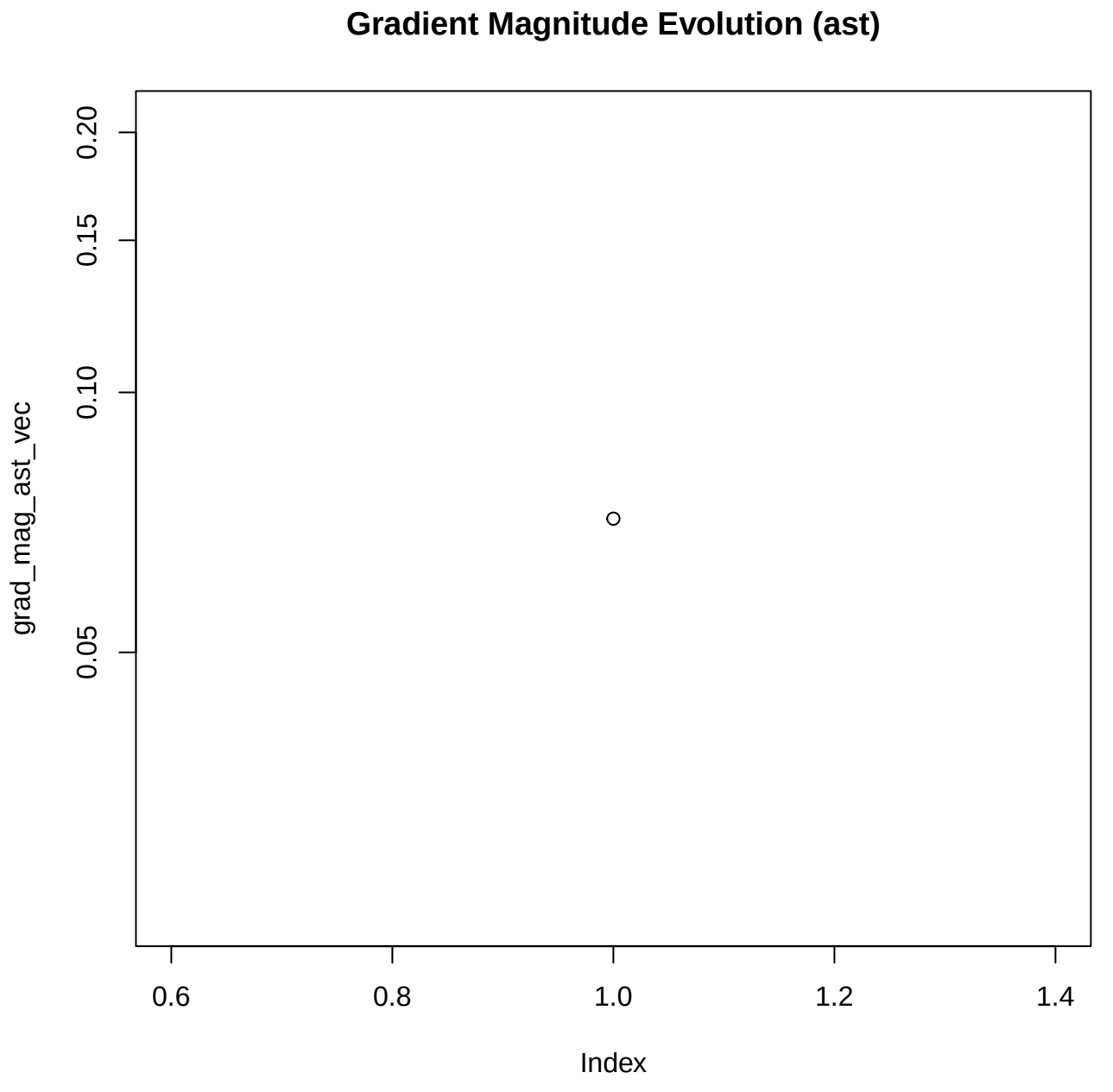
Gradient Magnitude Evolution (ast)
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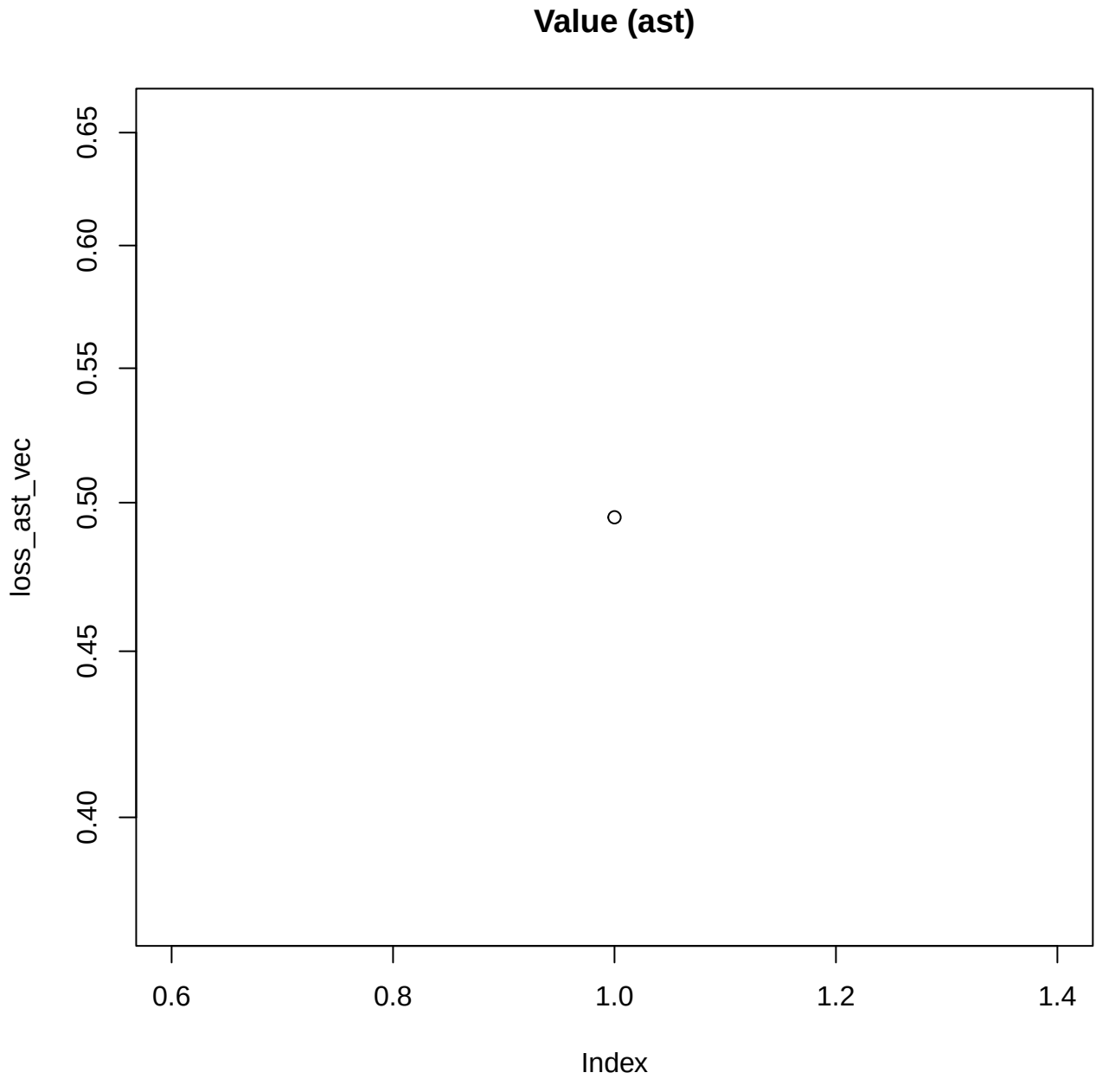
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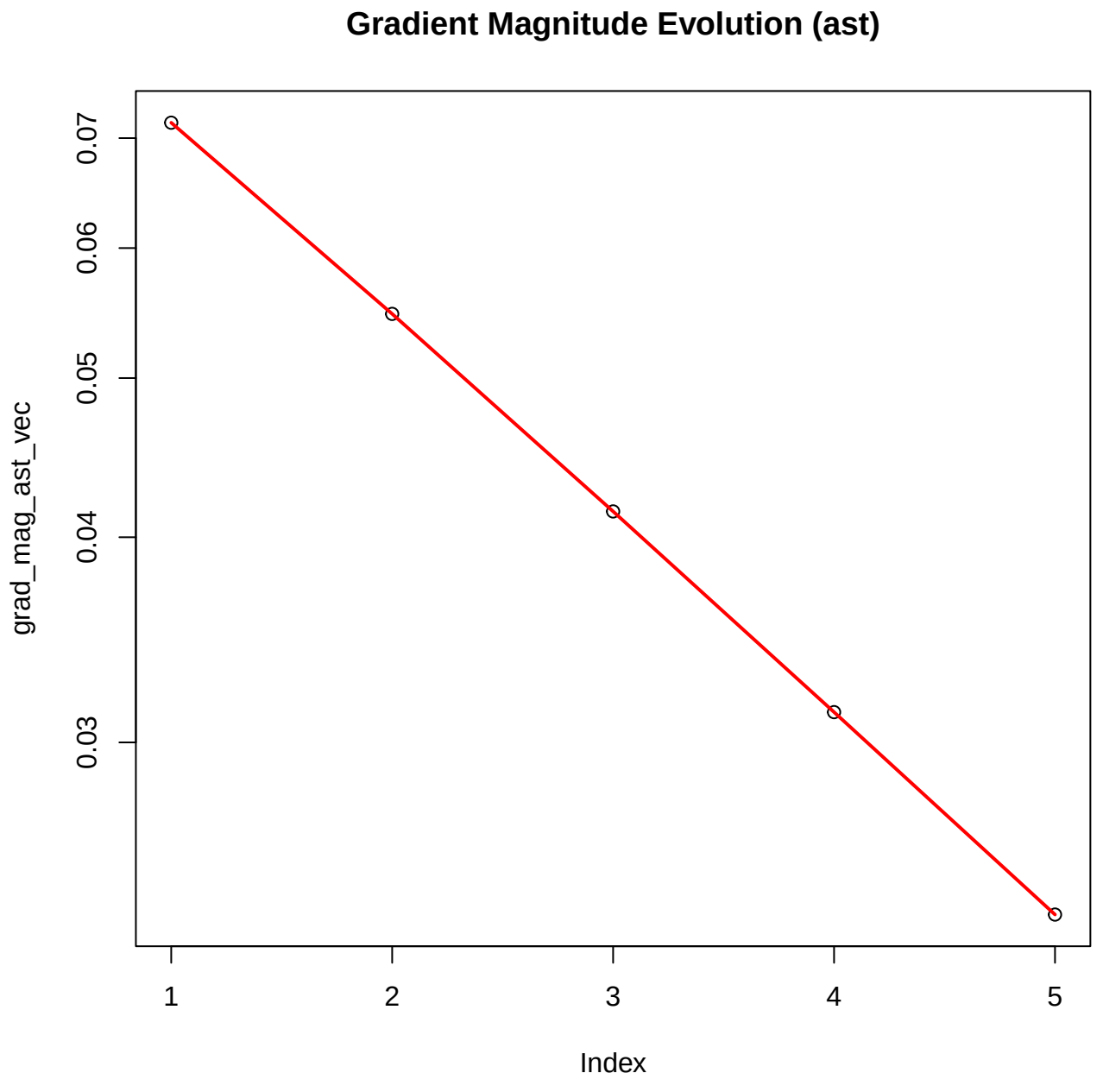
Gradient Magnitude Evolution (ast)
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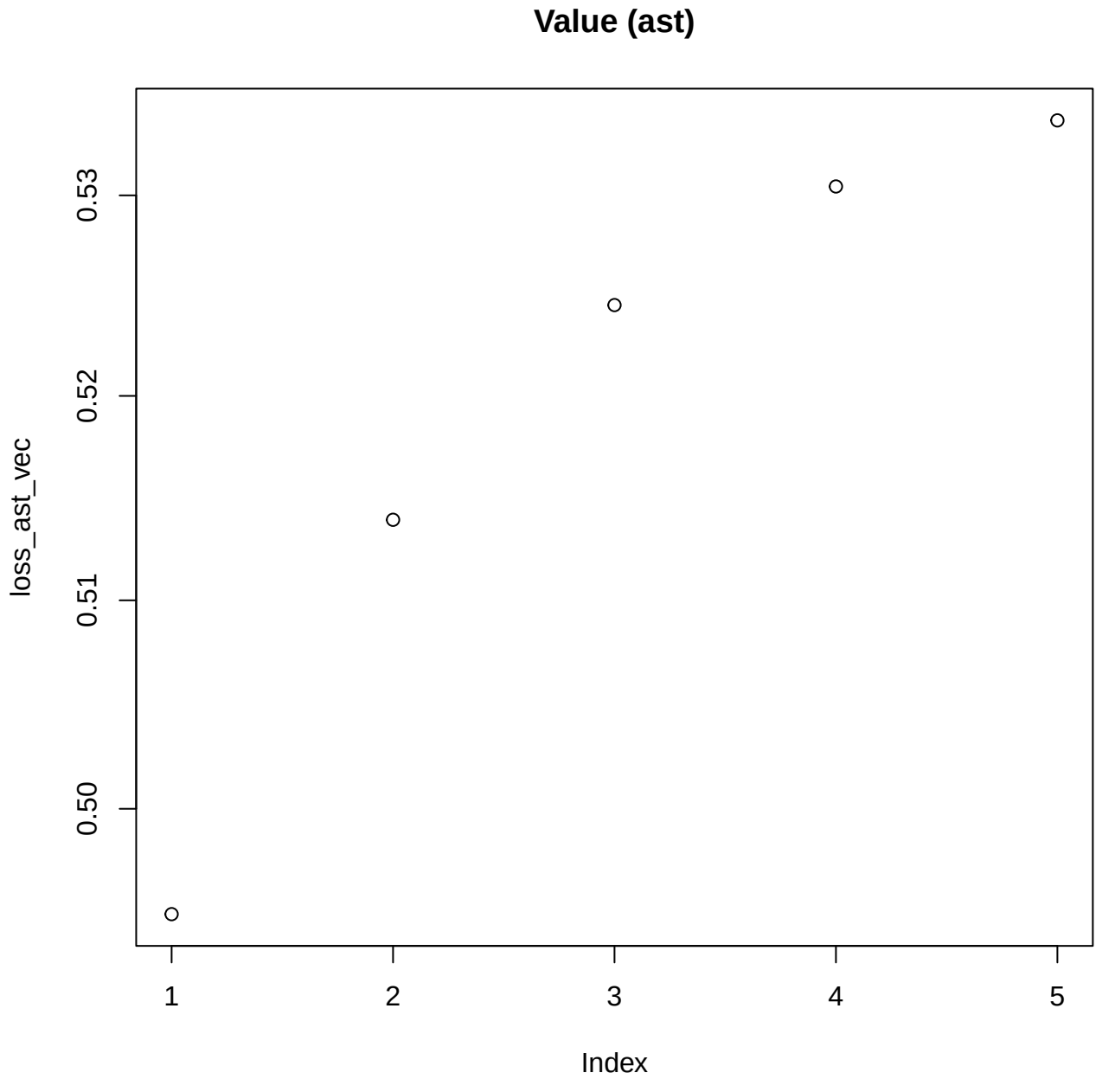
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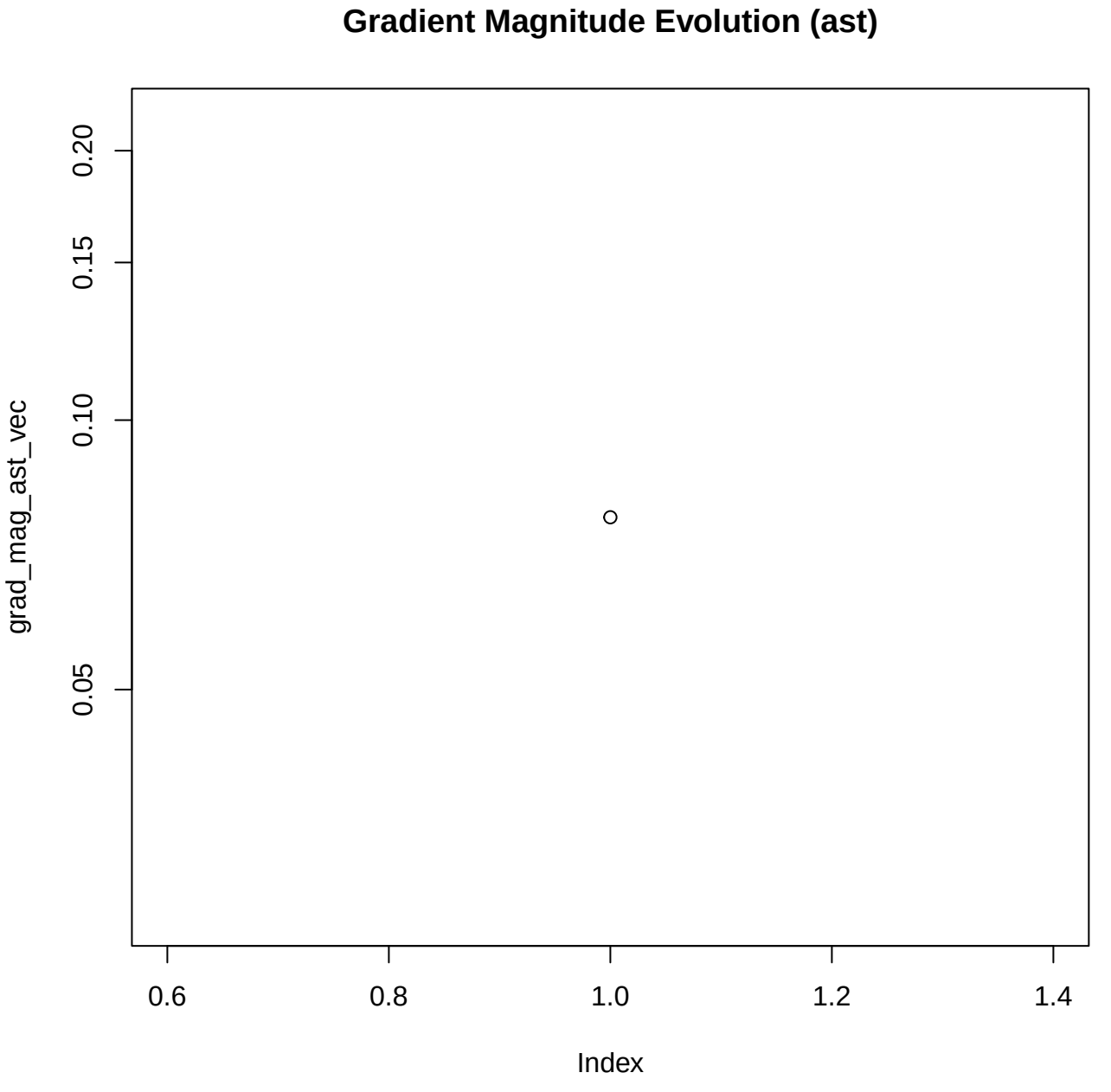
Gradient Magnitude Evolution (ast)
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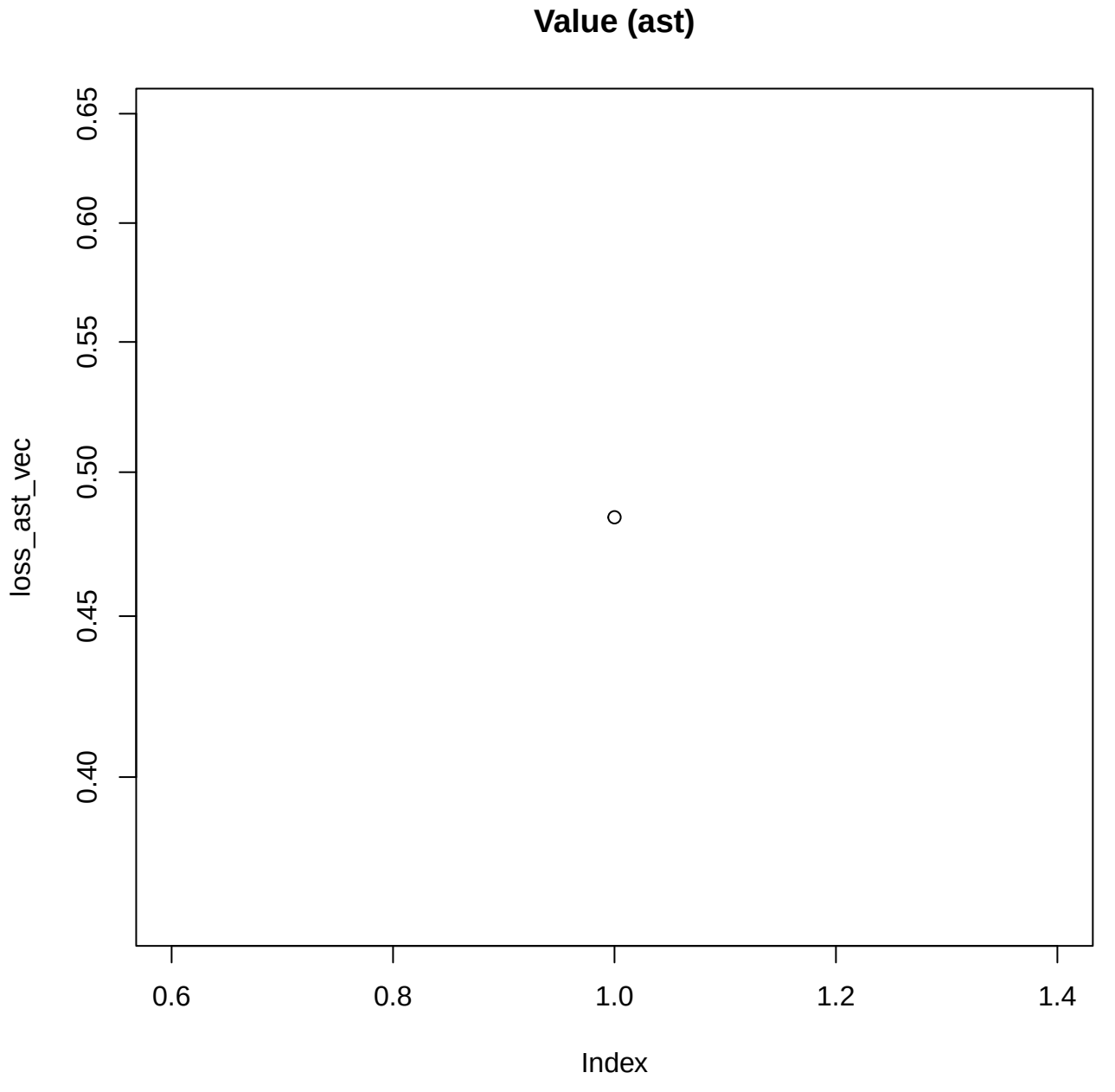
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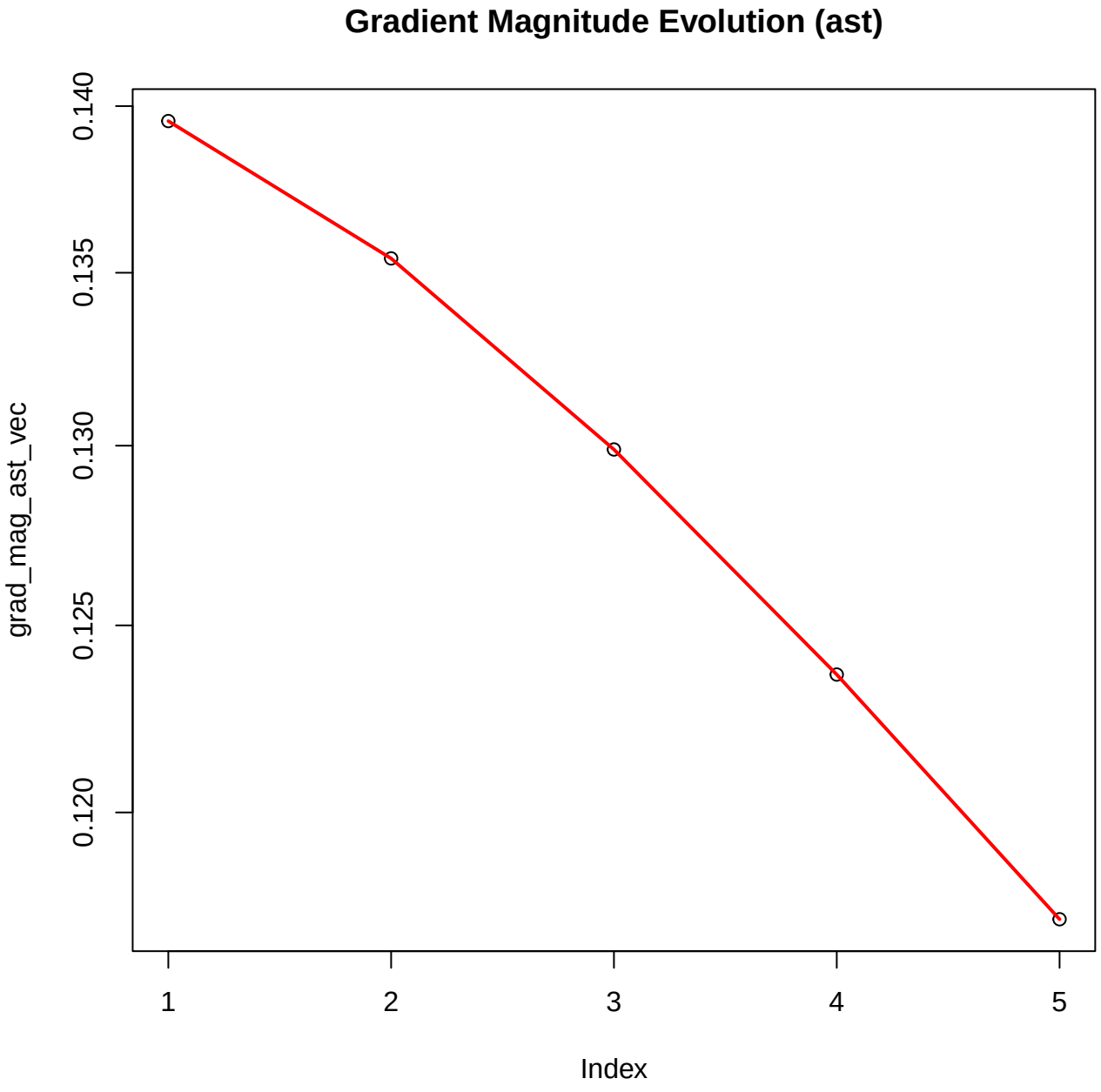
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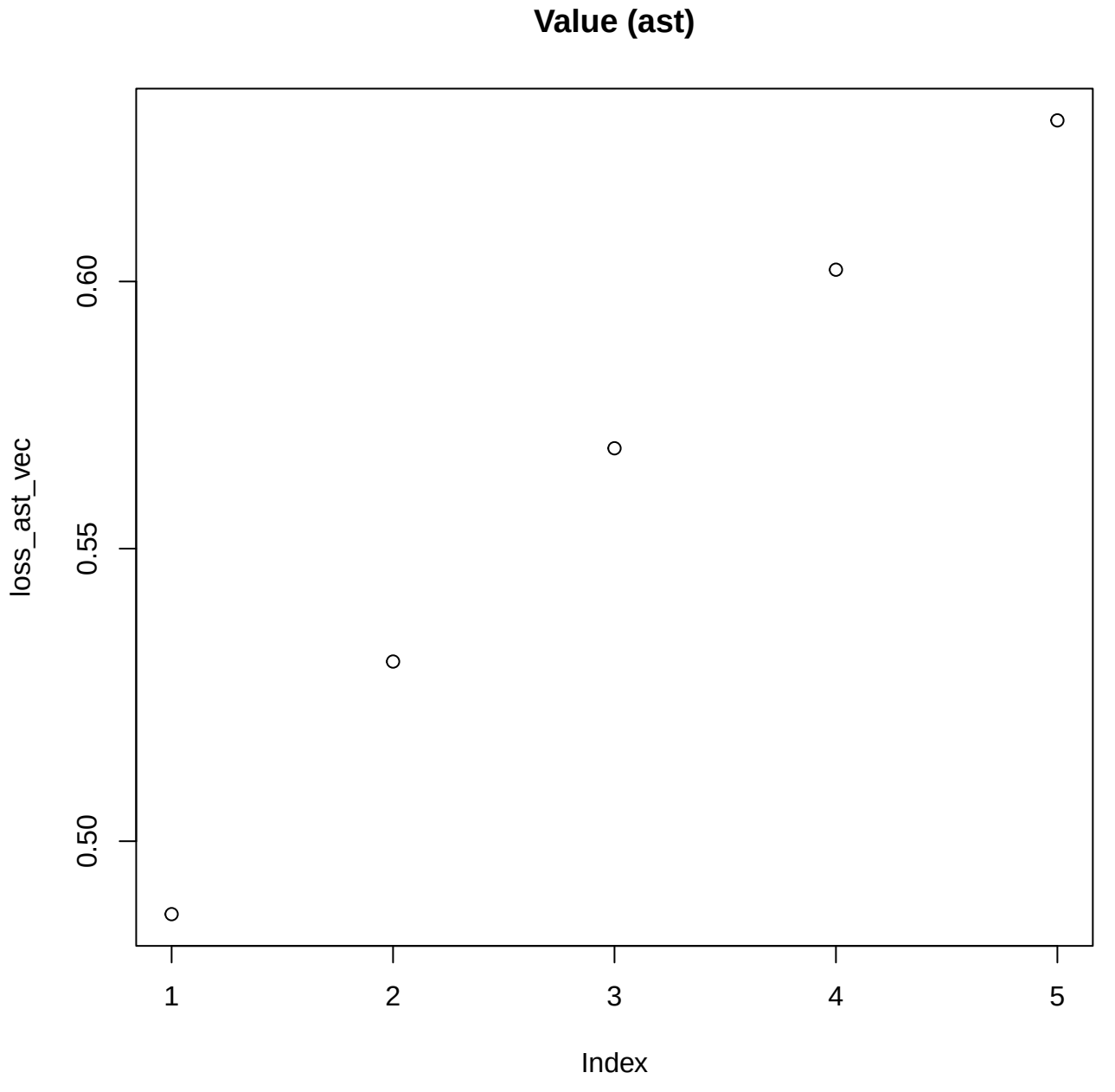
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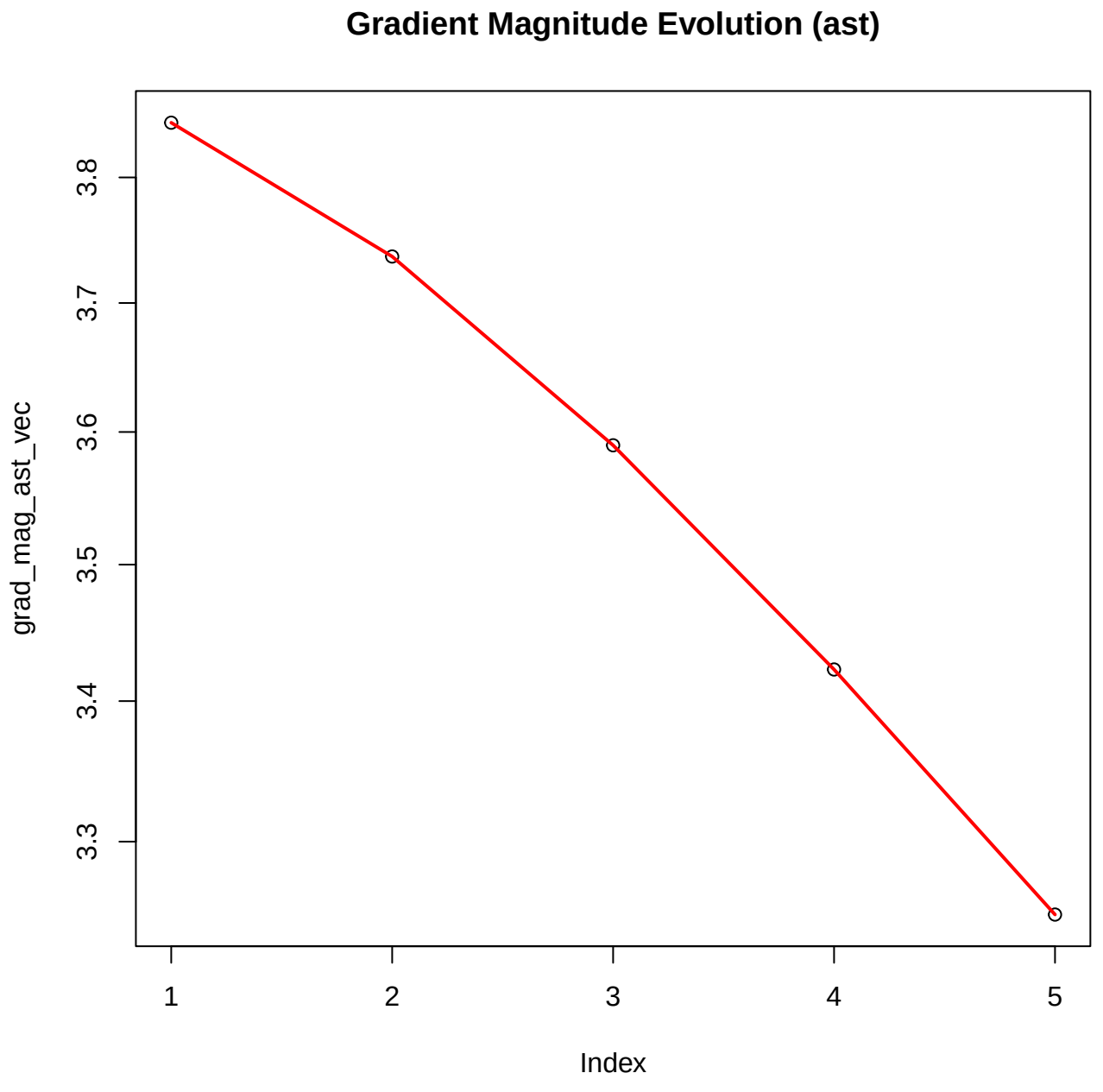
Gradient Magnitude Evolution (ast)
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2
3
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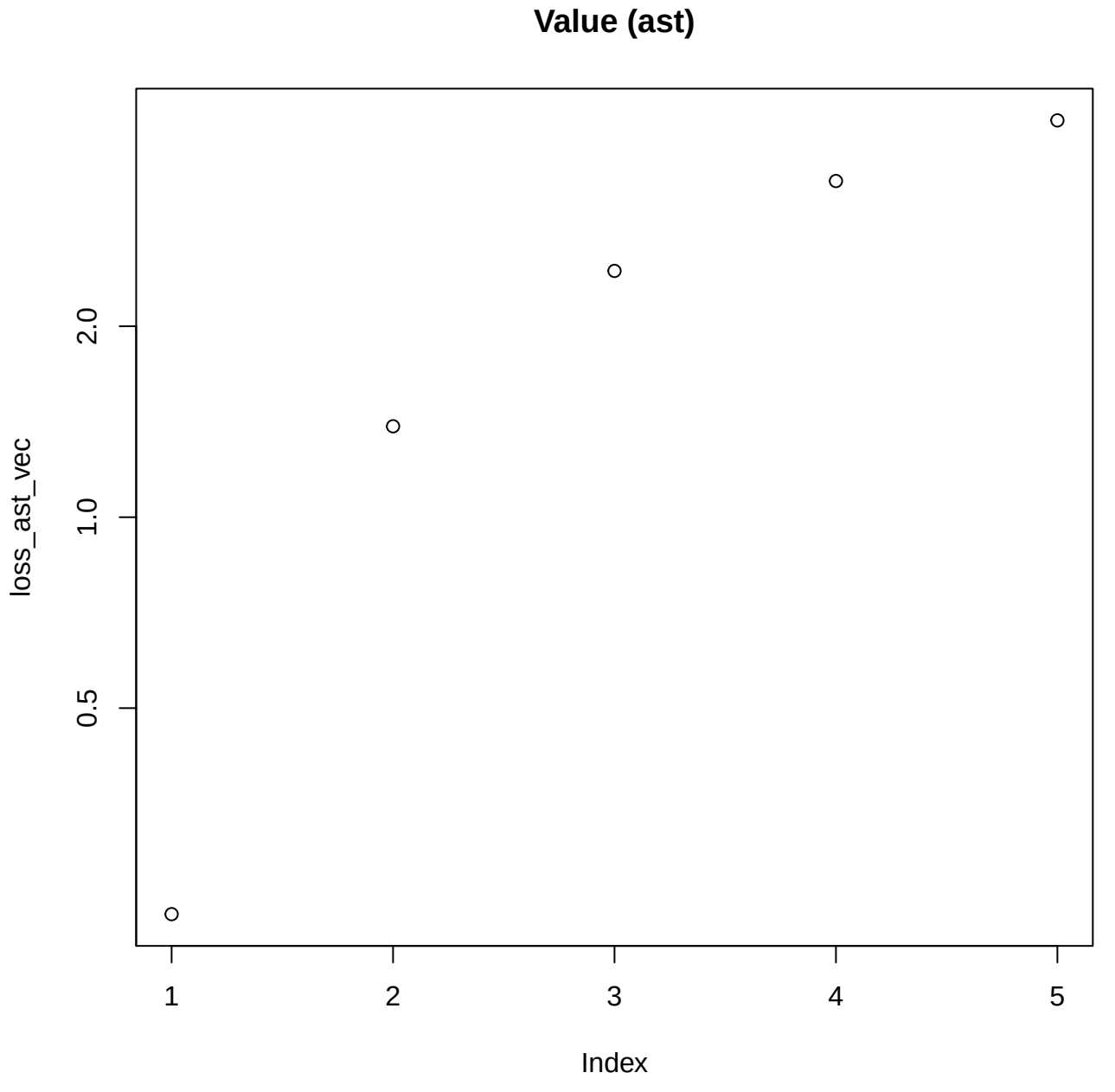
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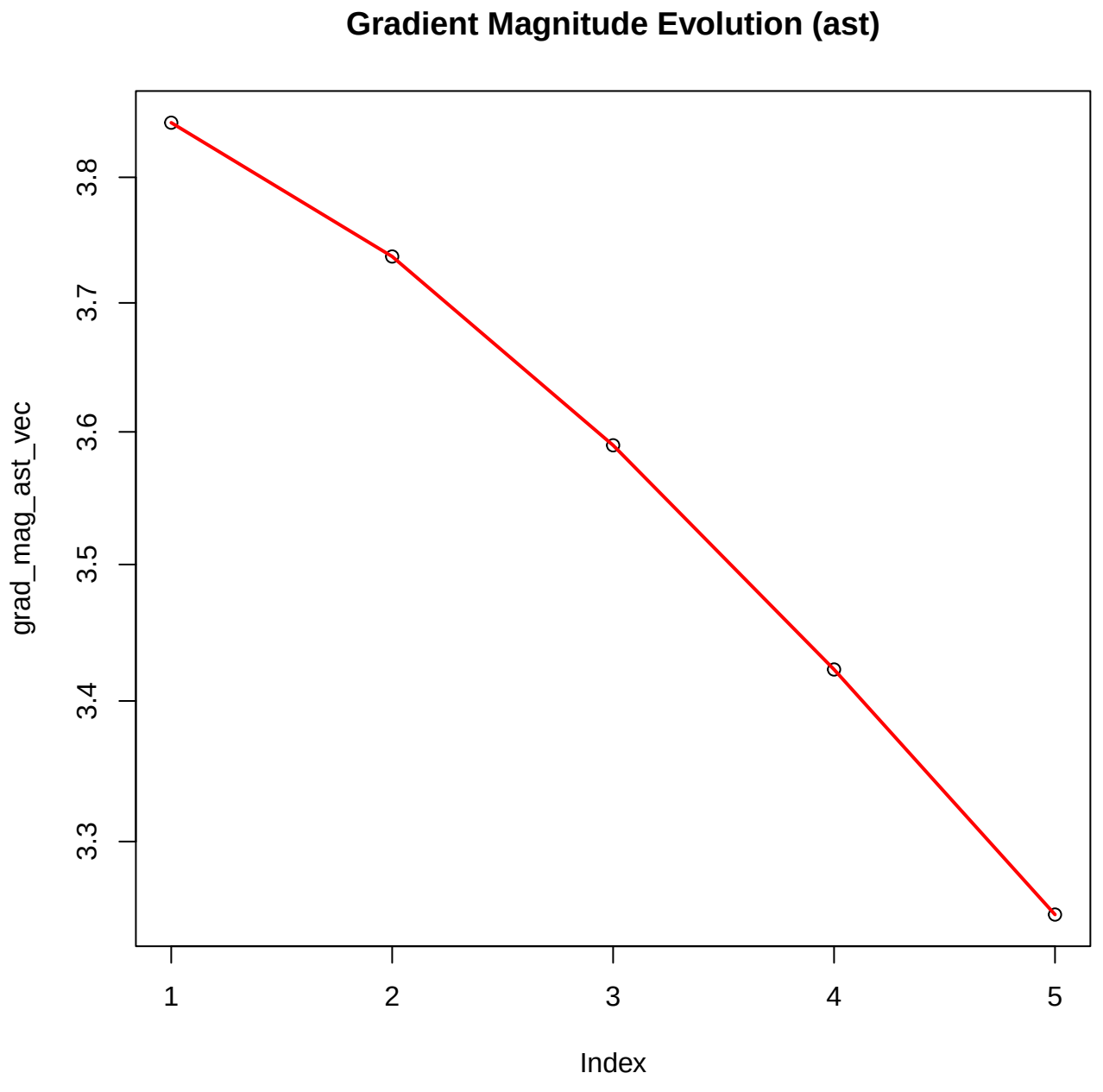
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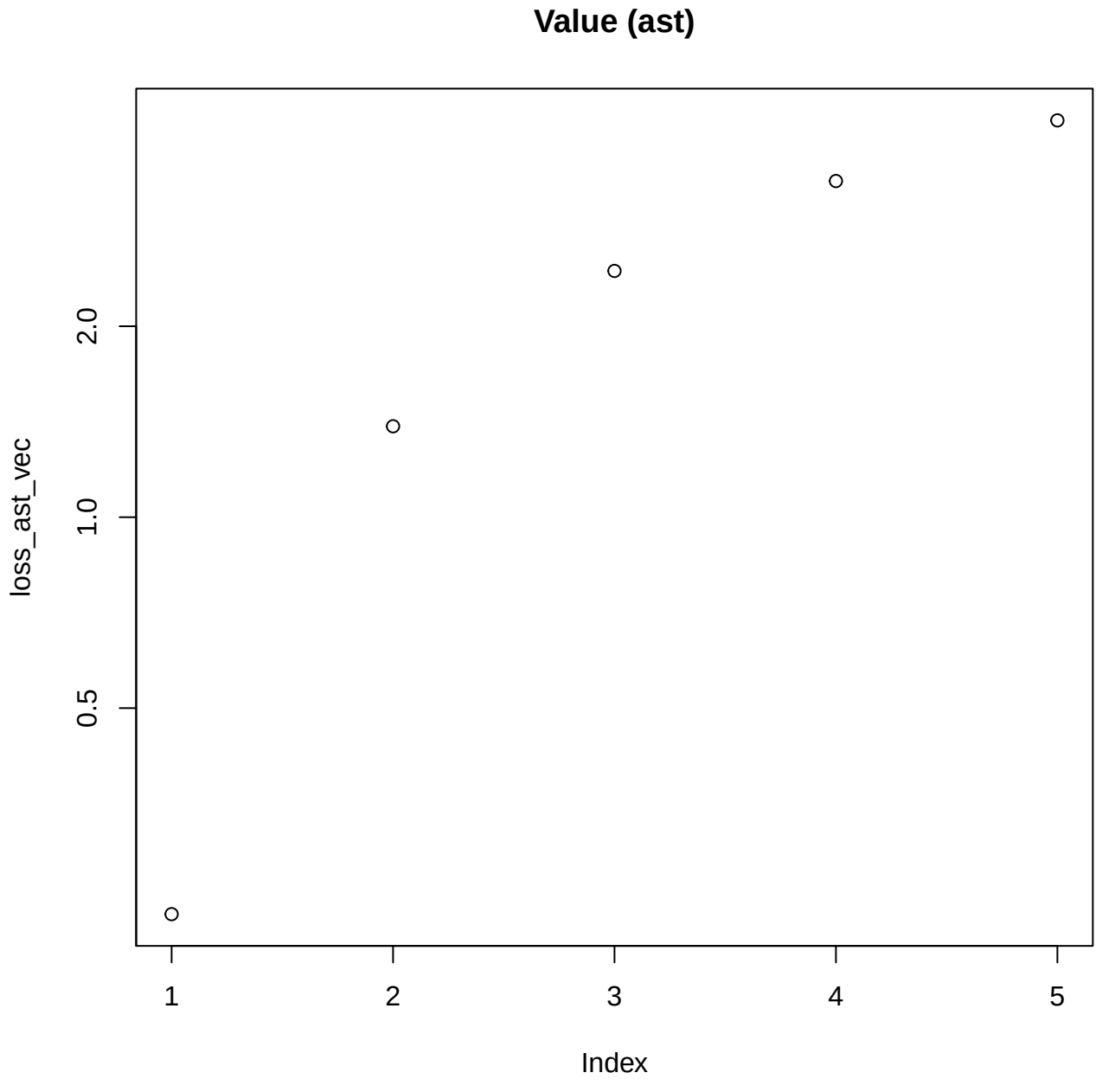
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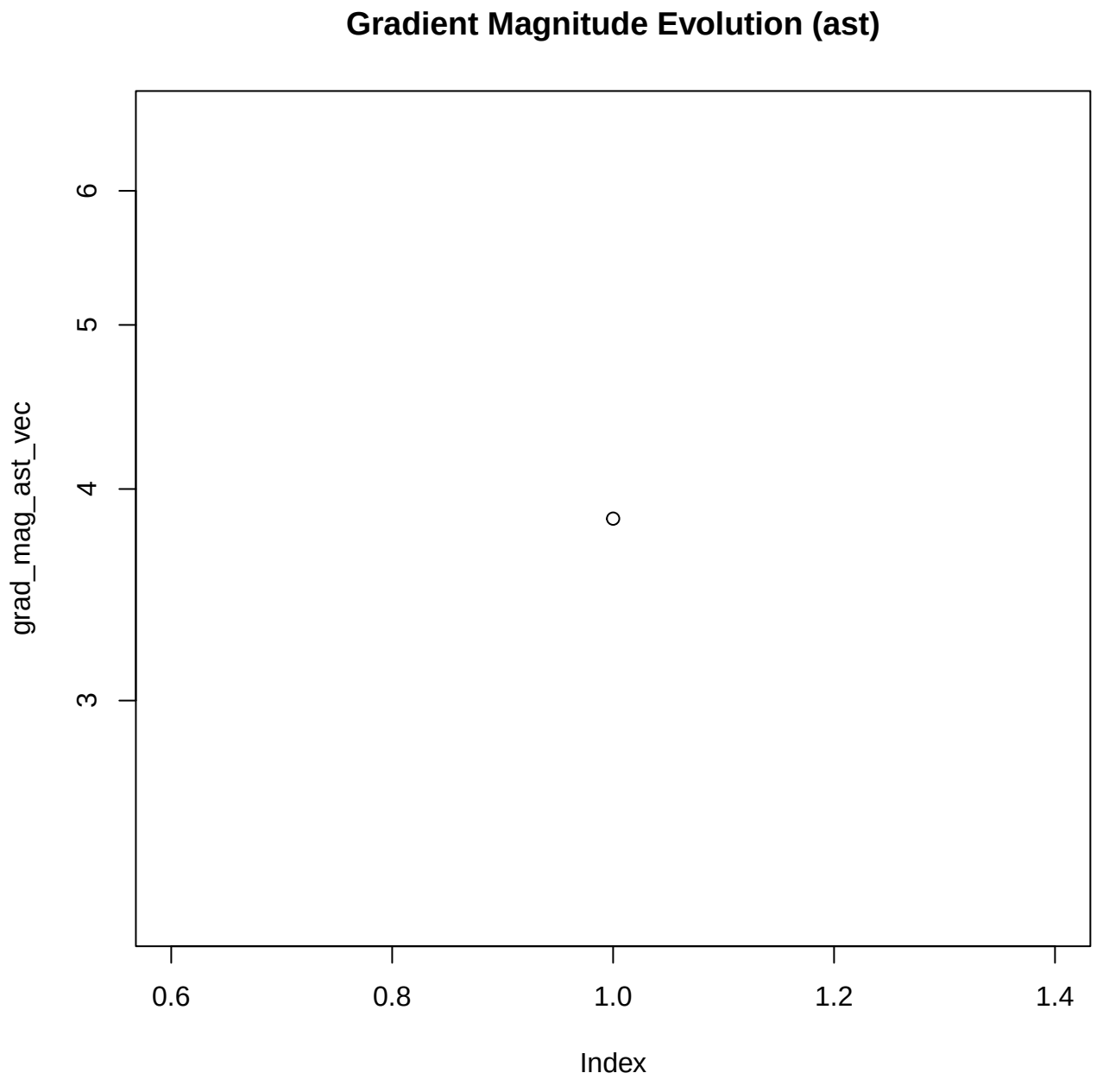
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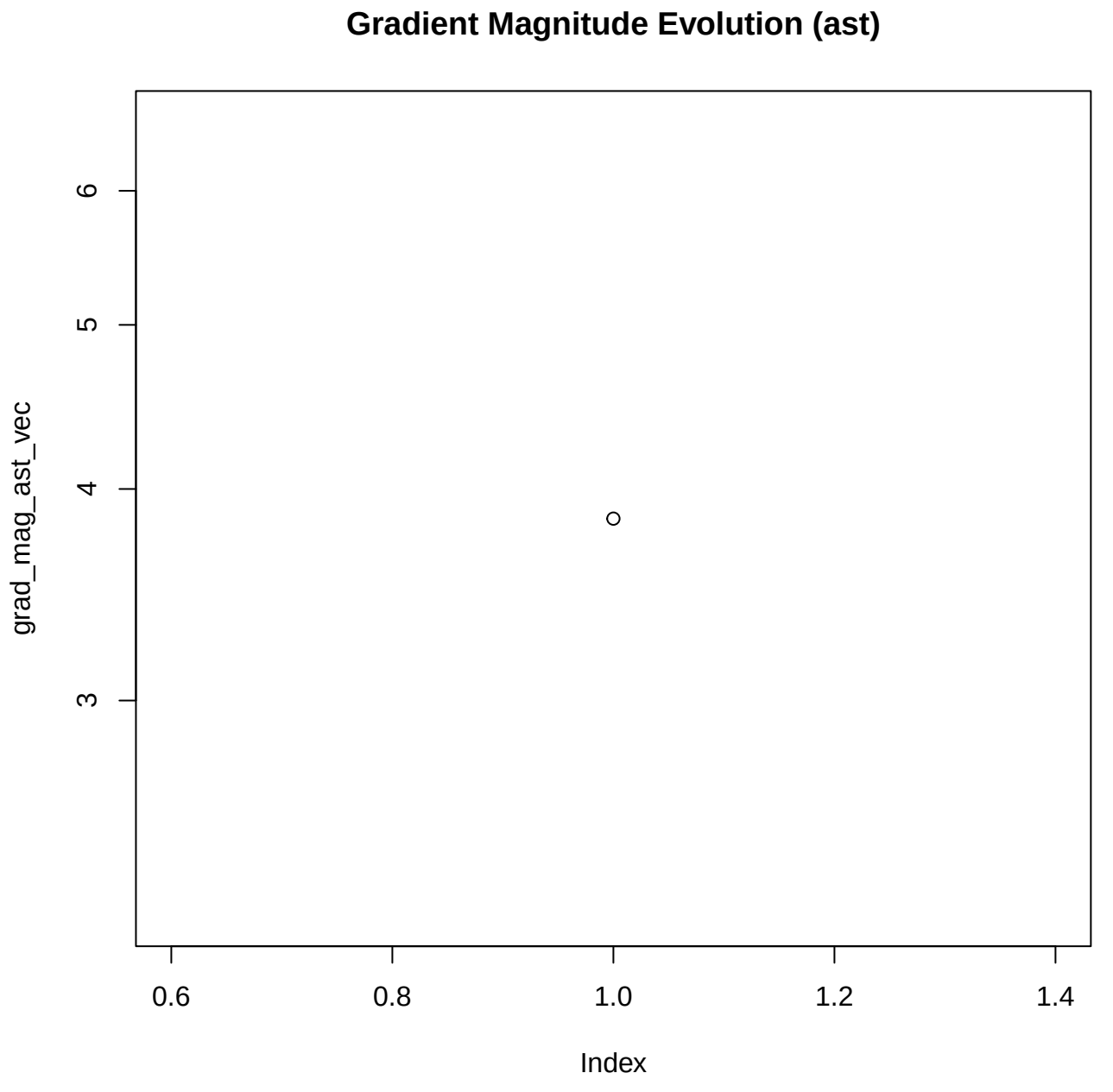
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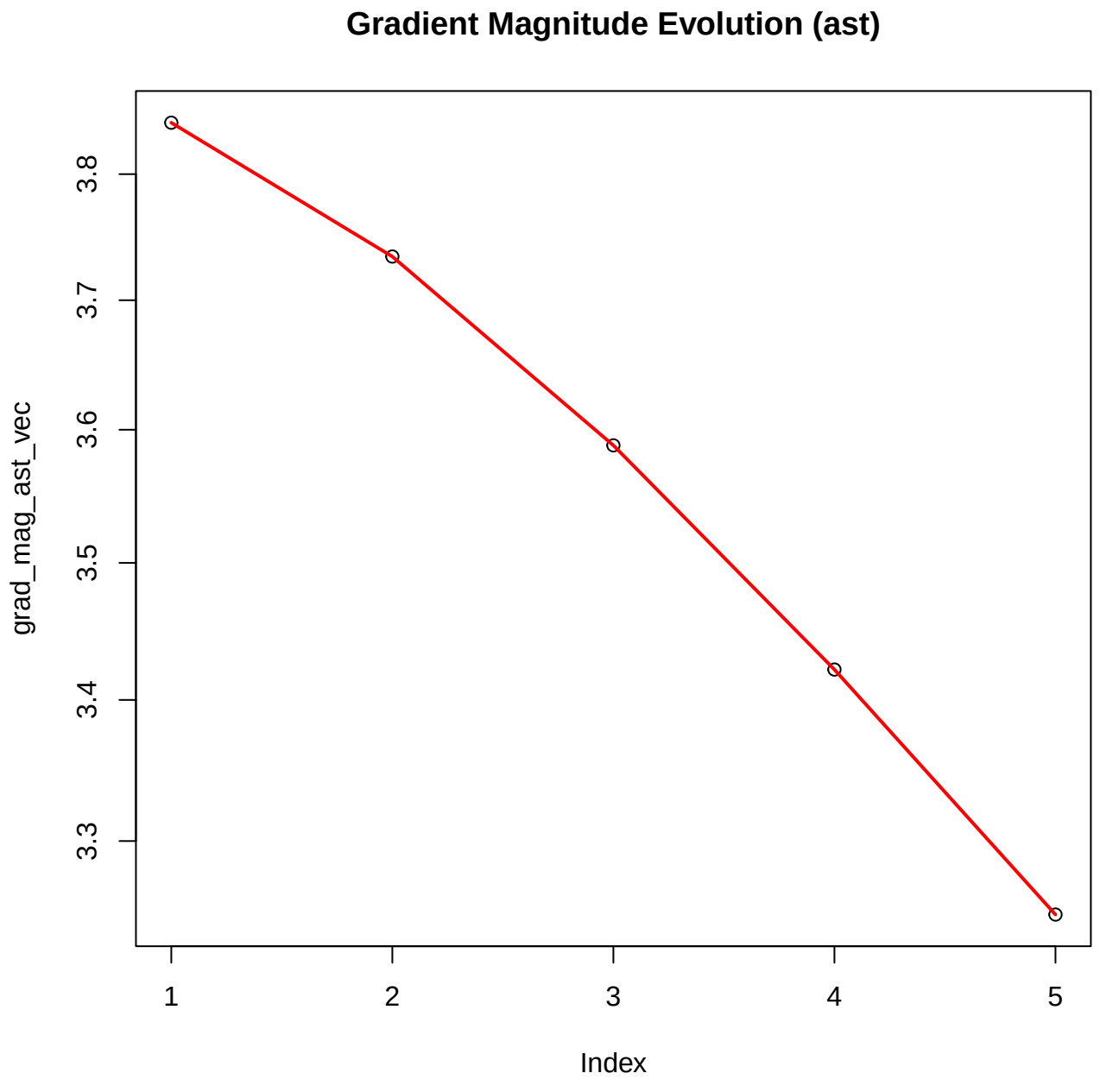
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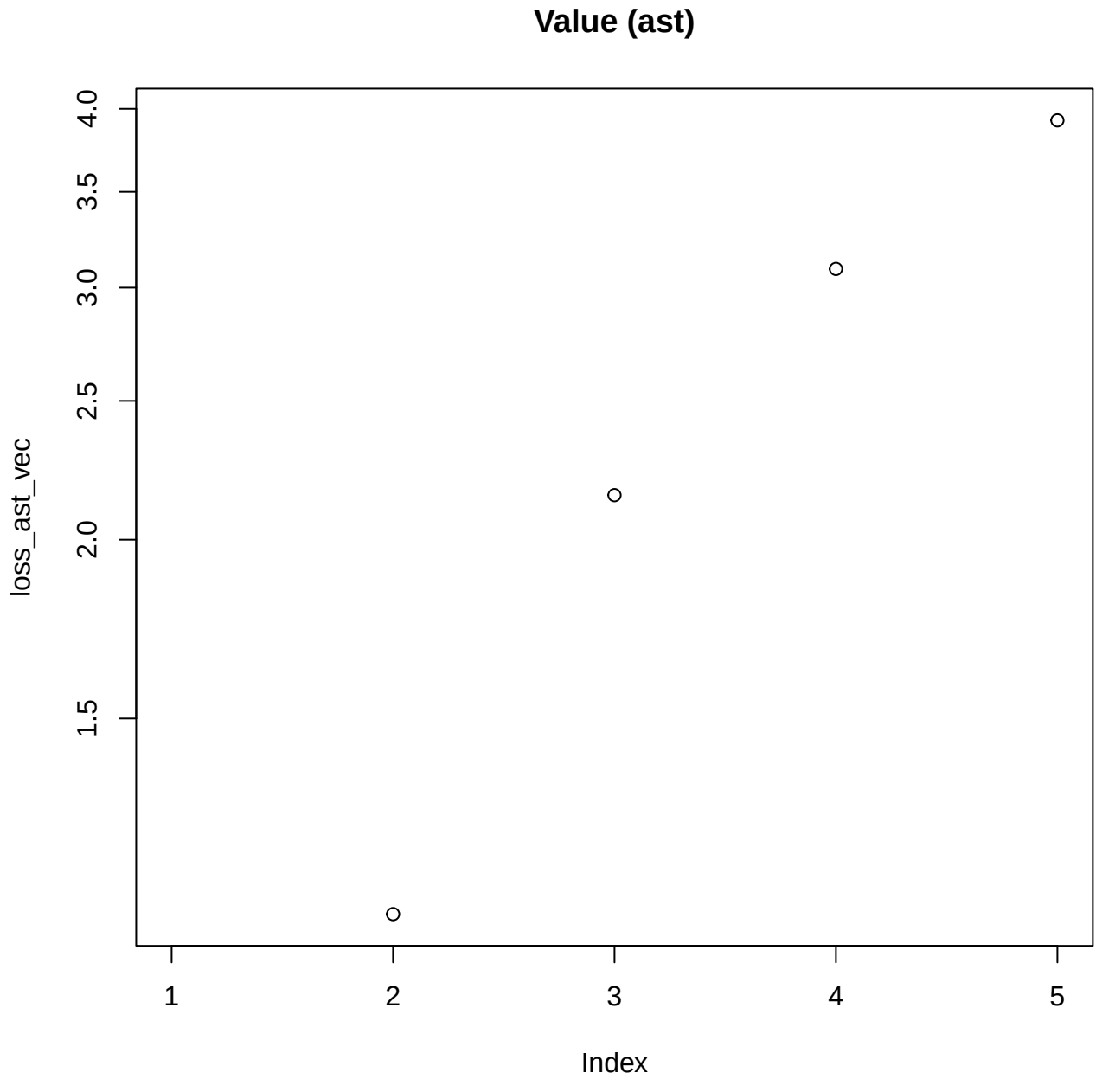
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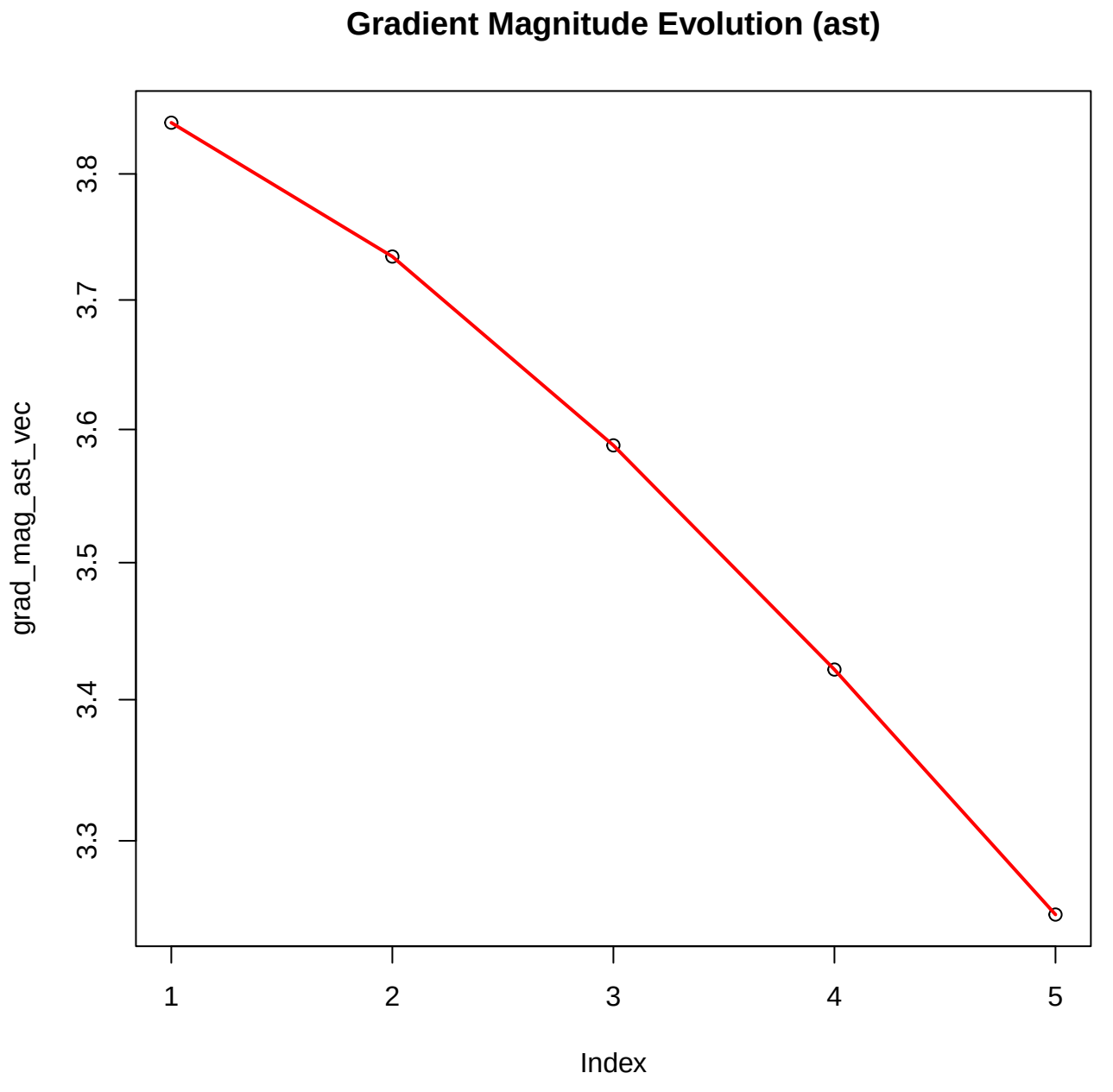
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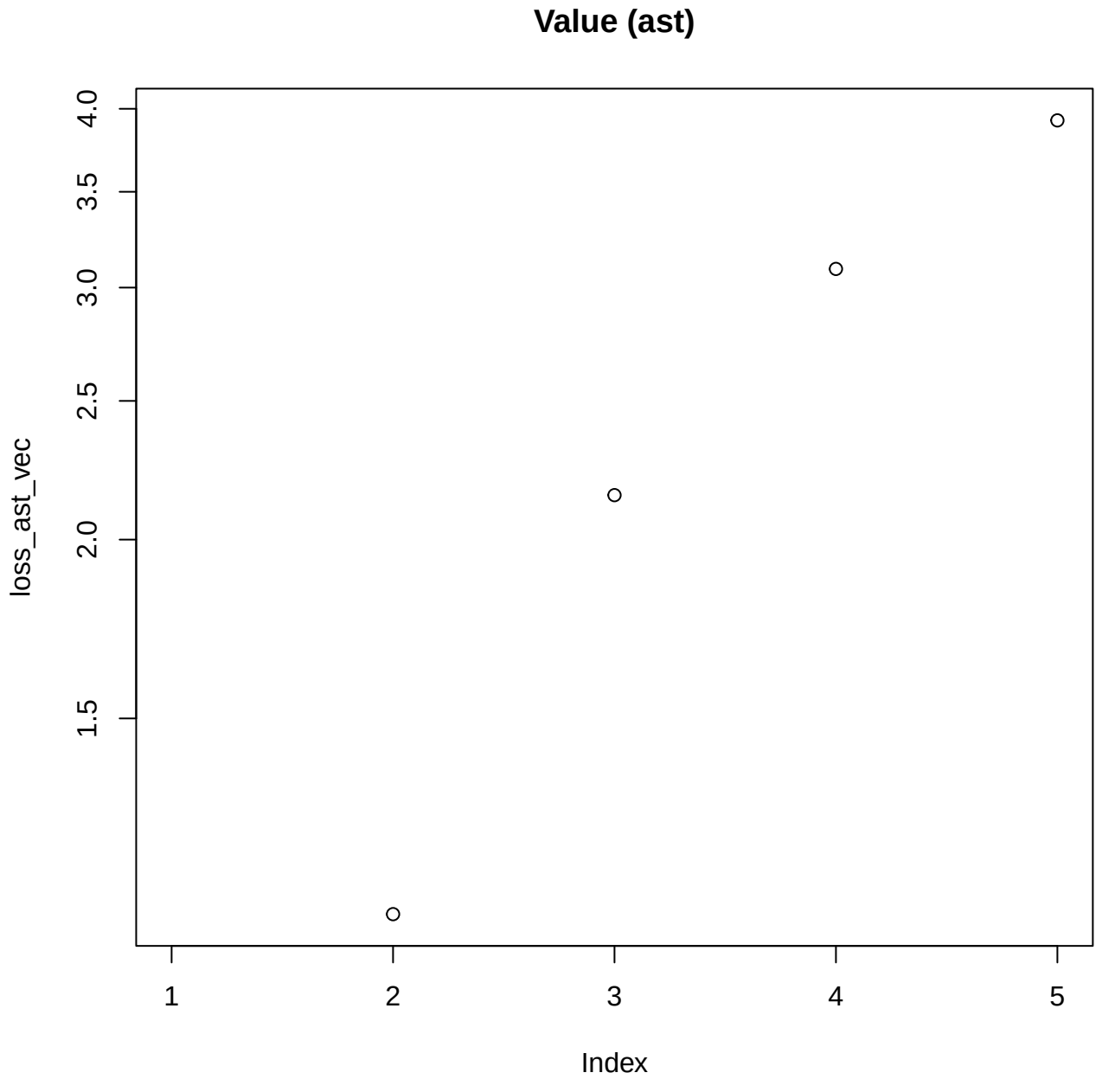
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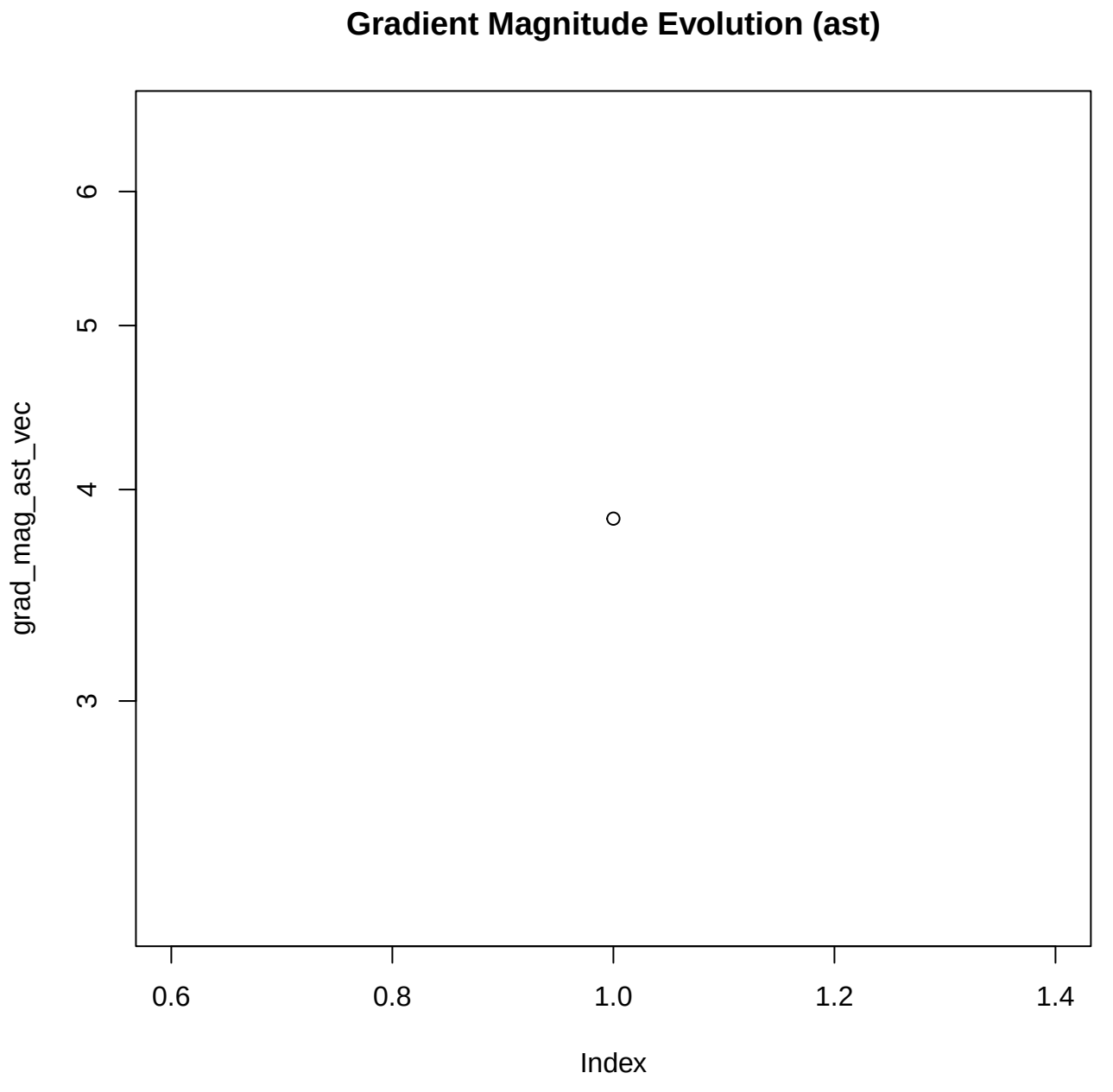
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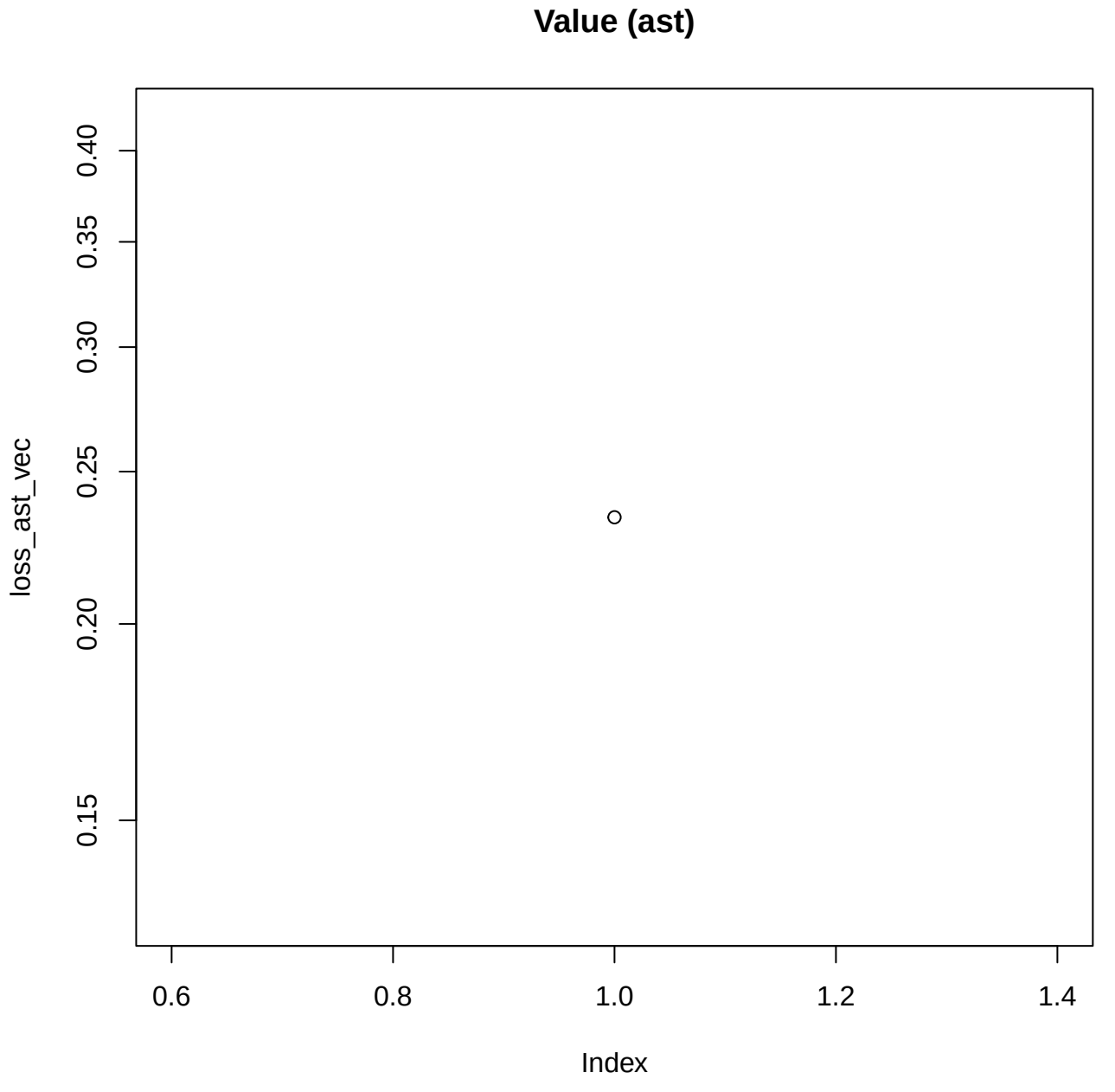
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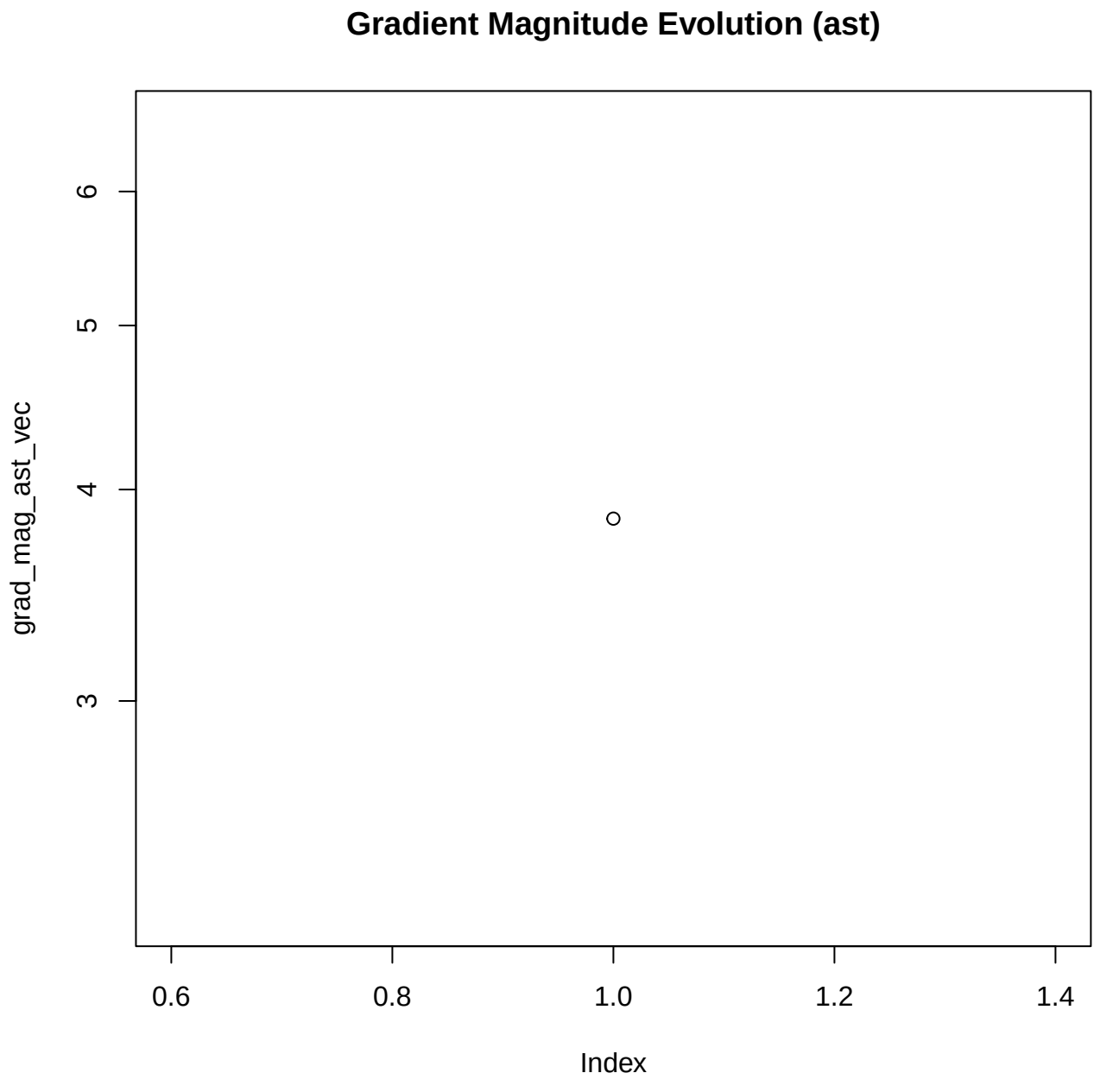
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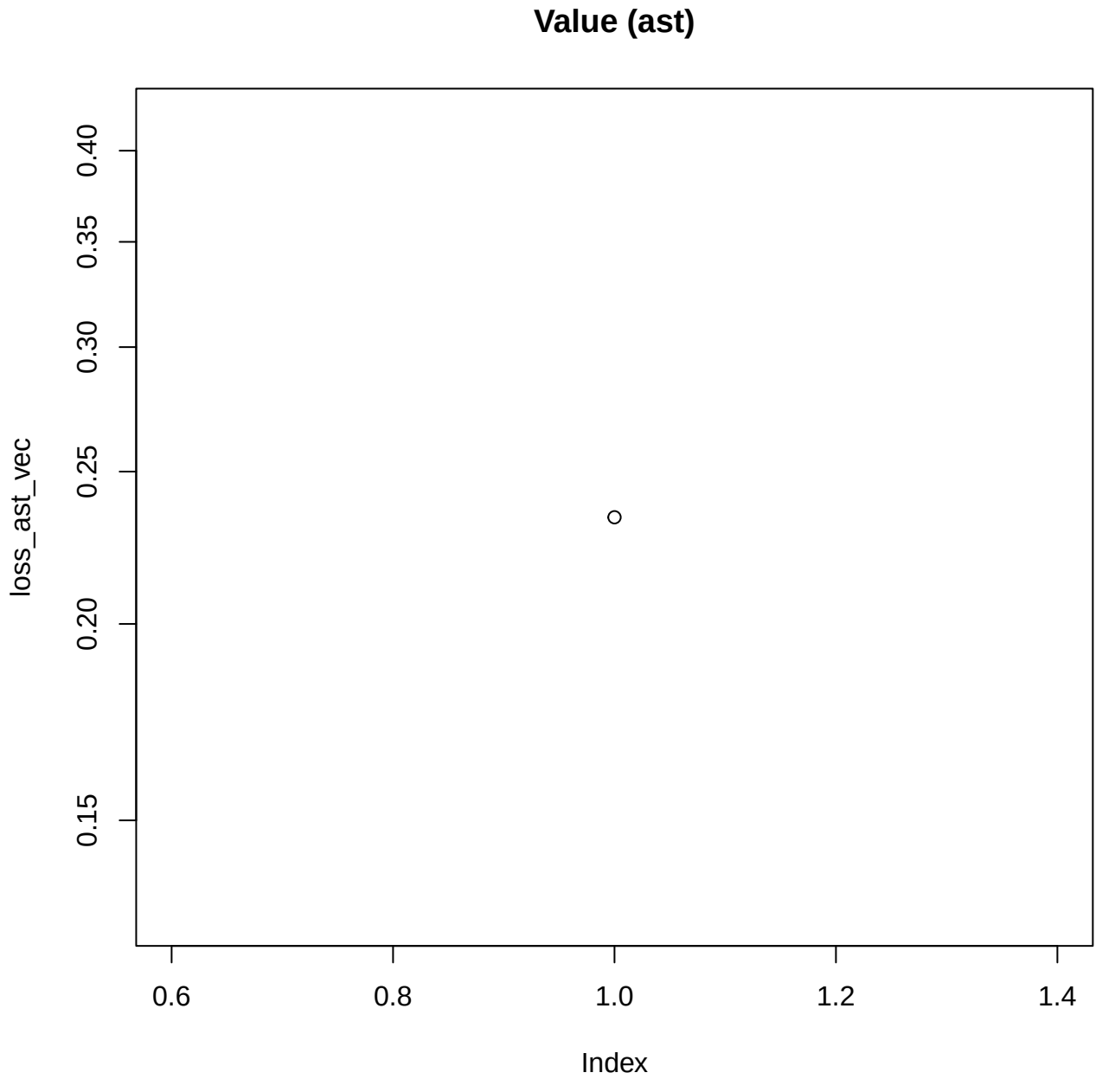
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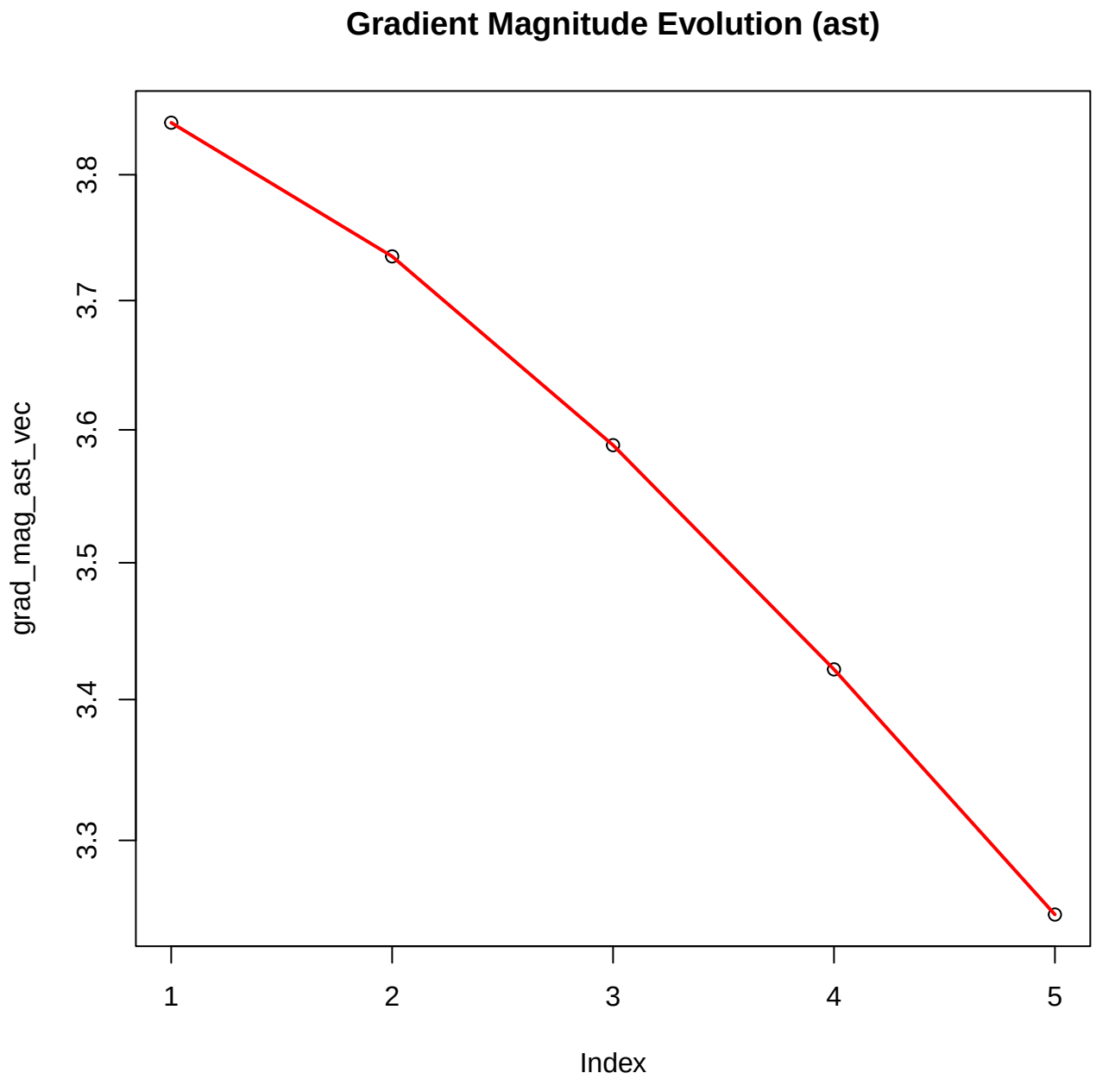
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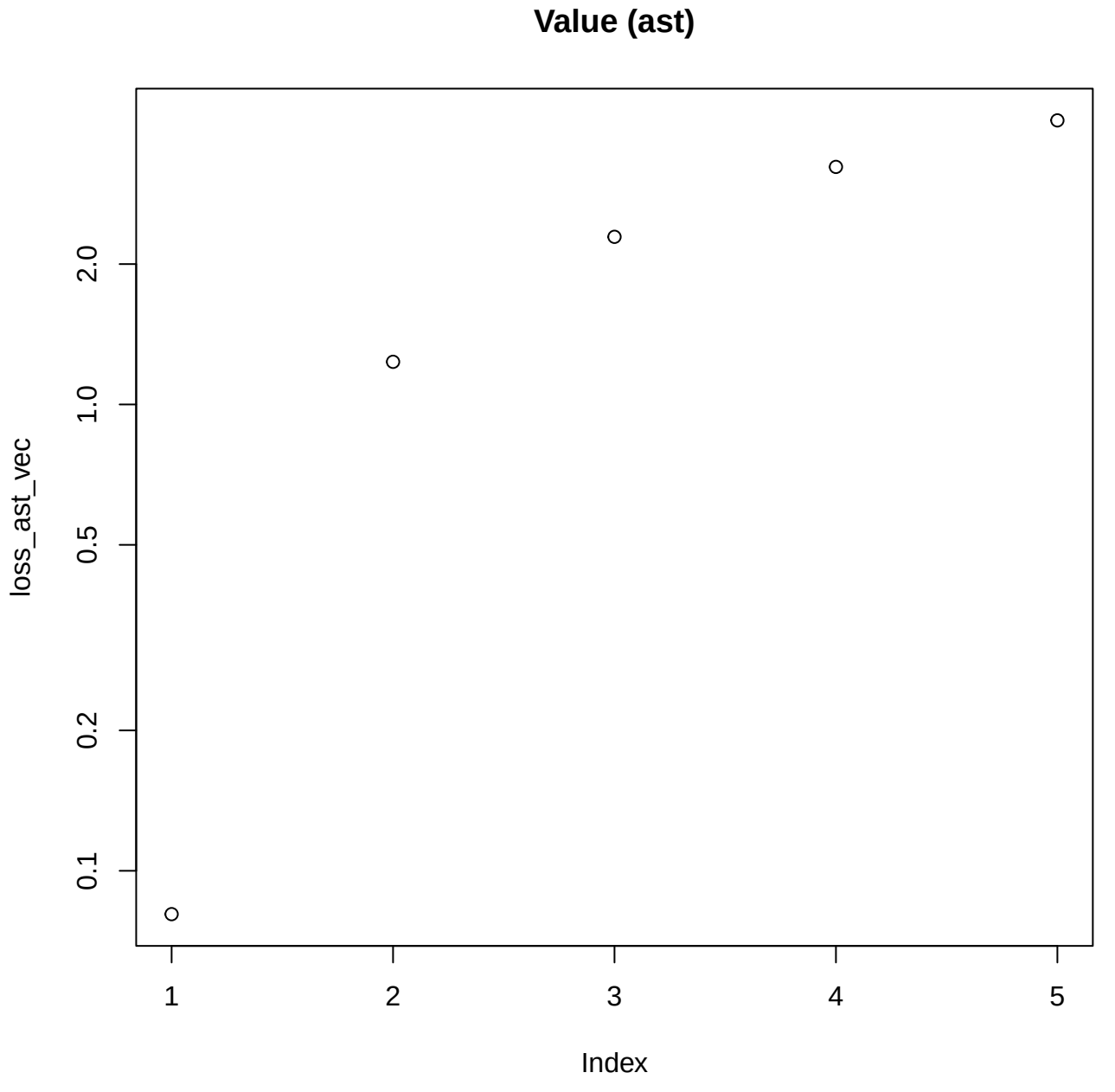
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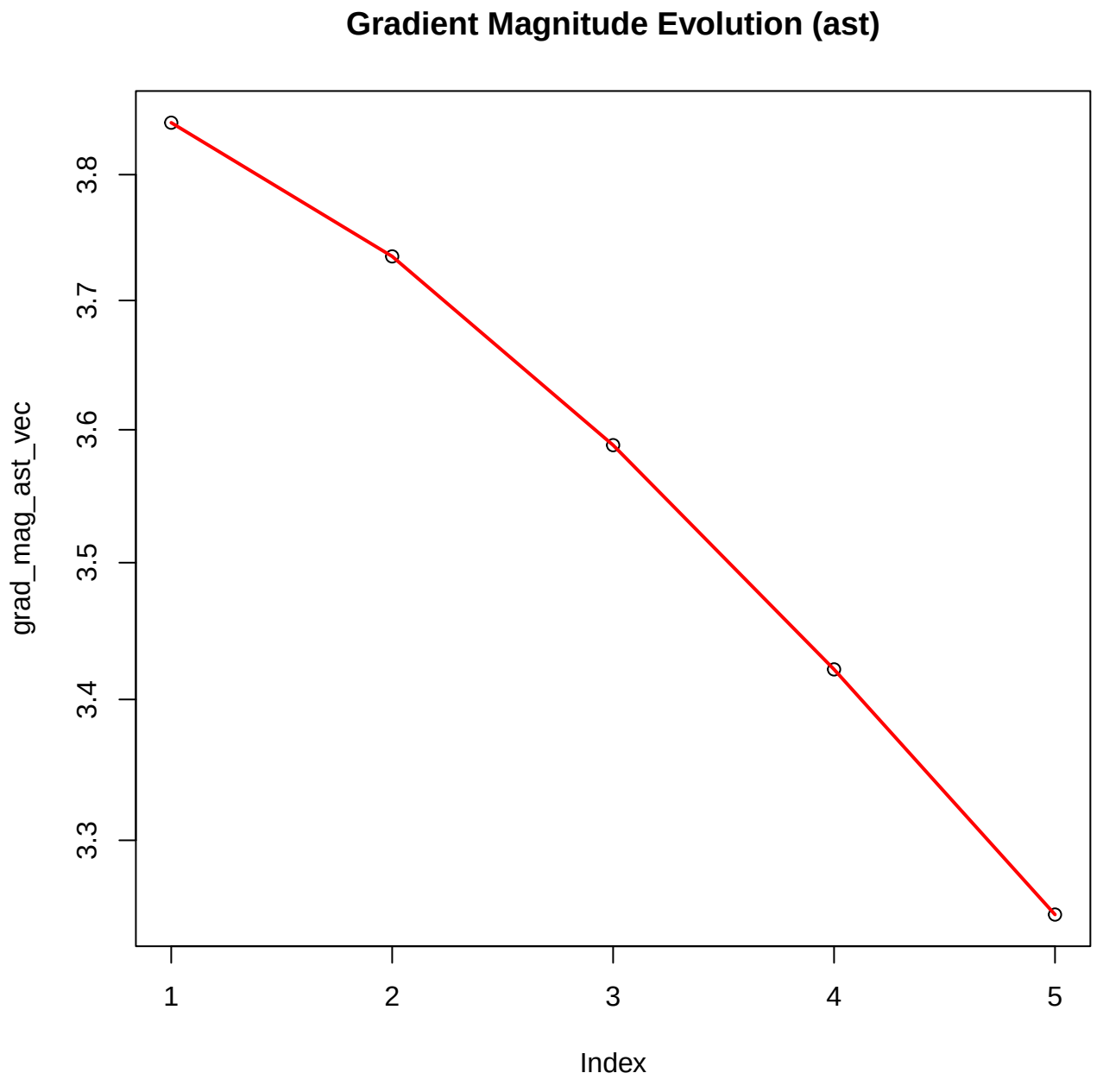
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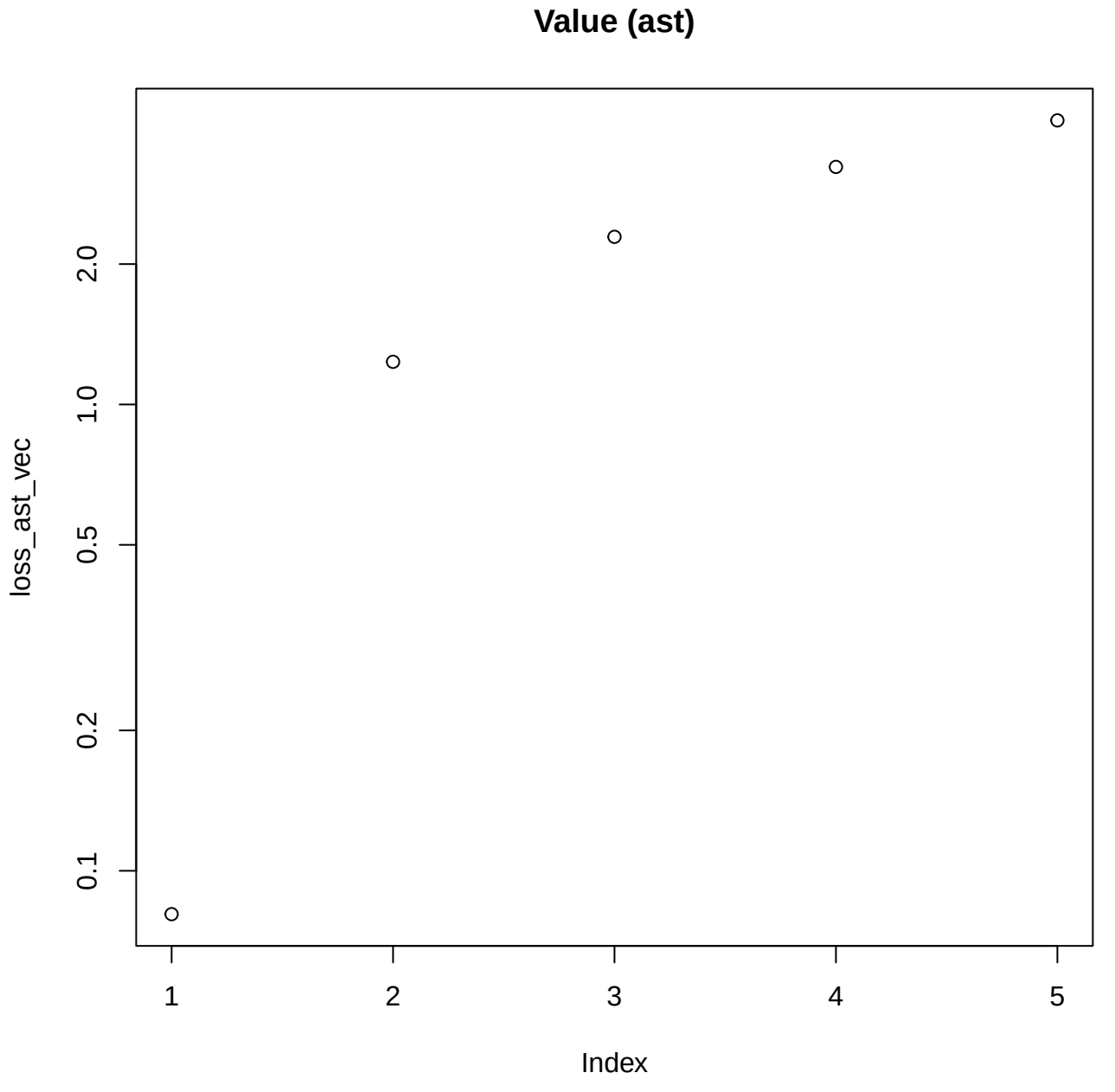
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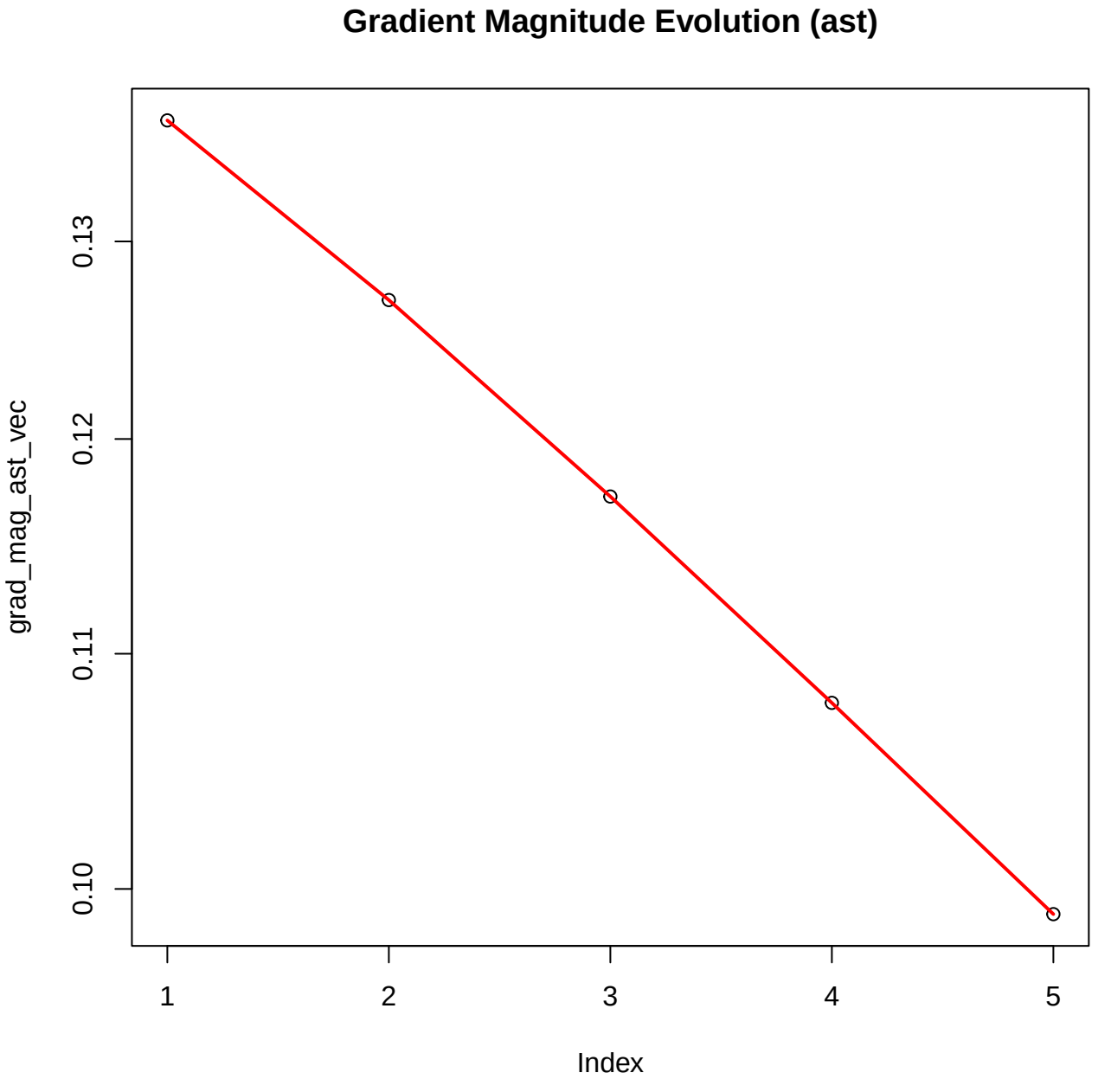
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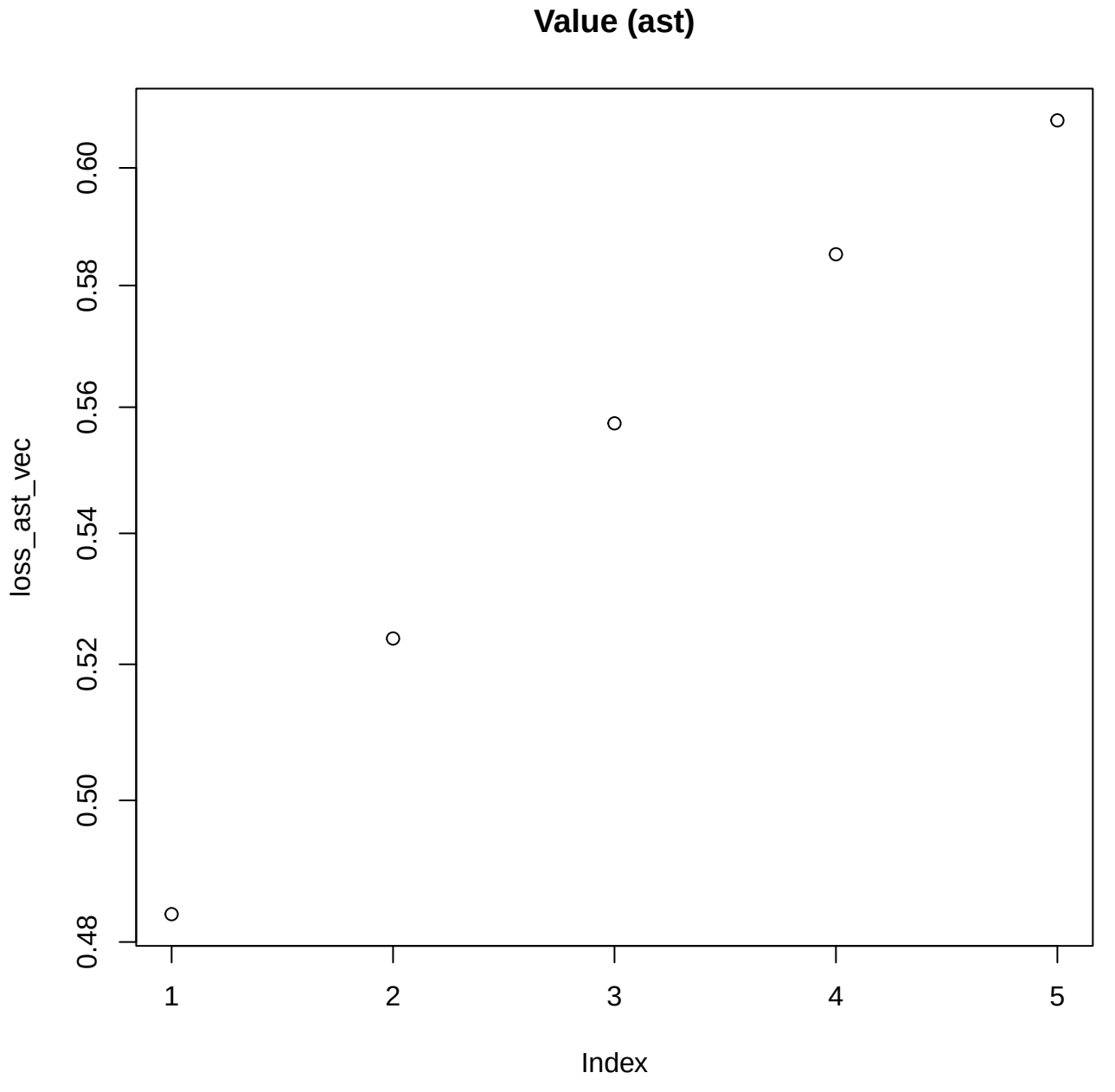
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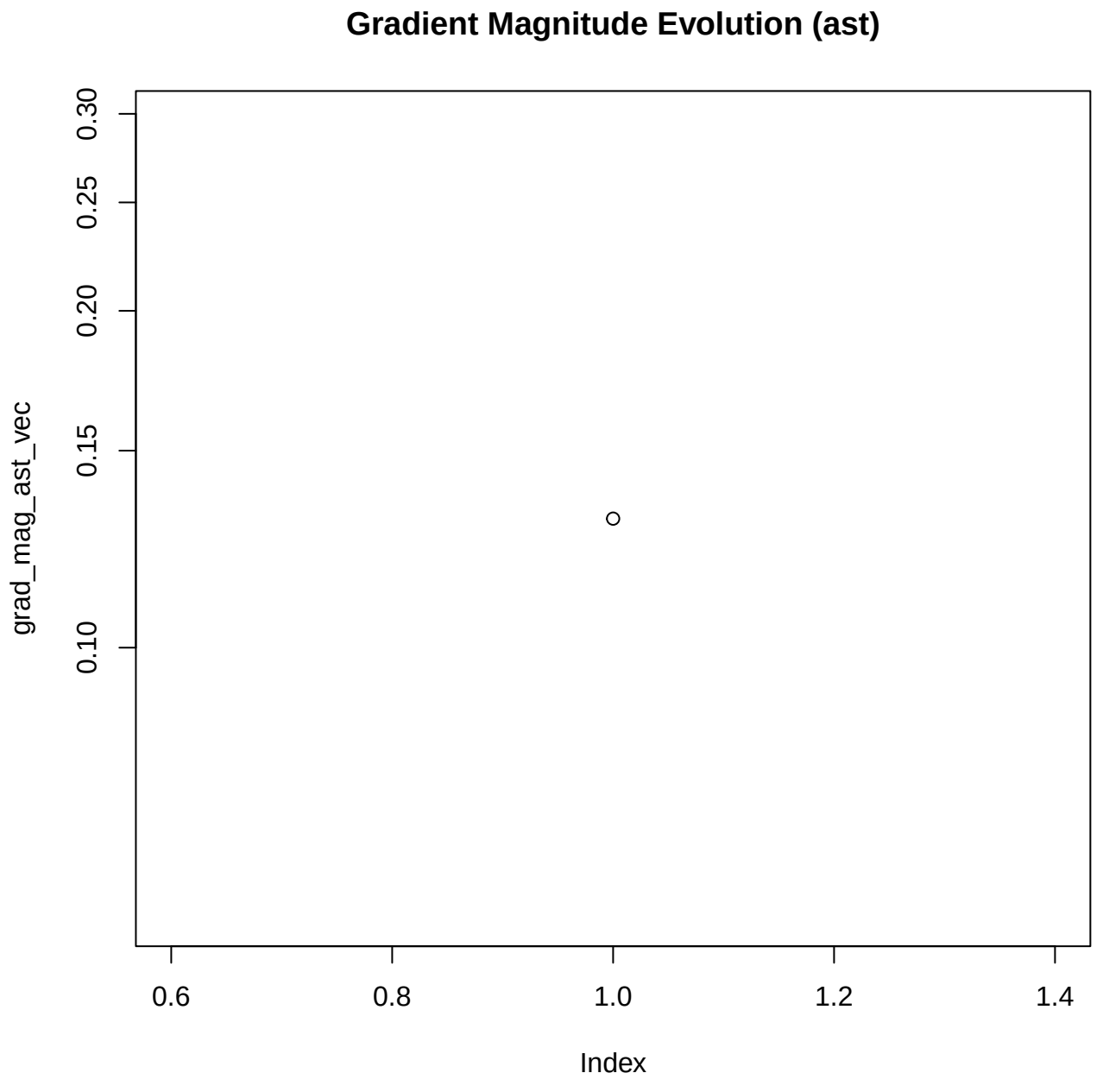
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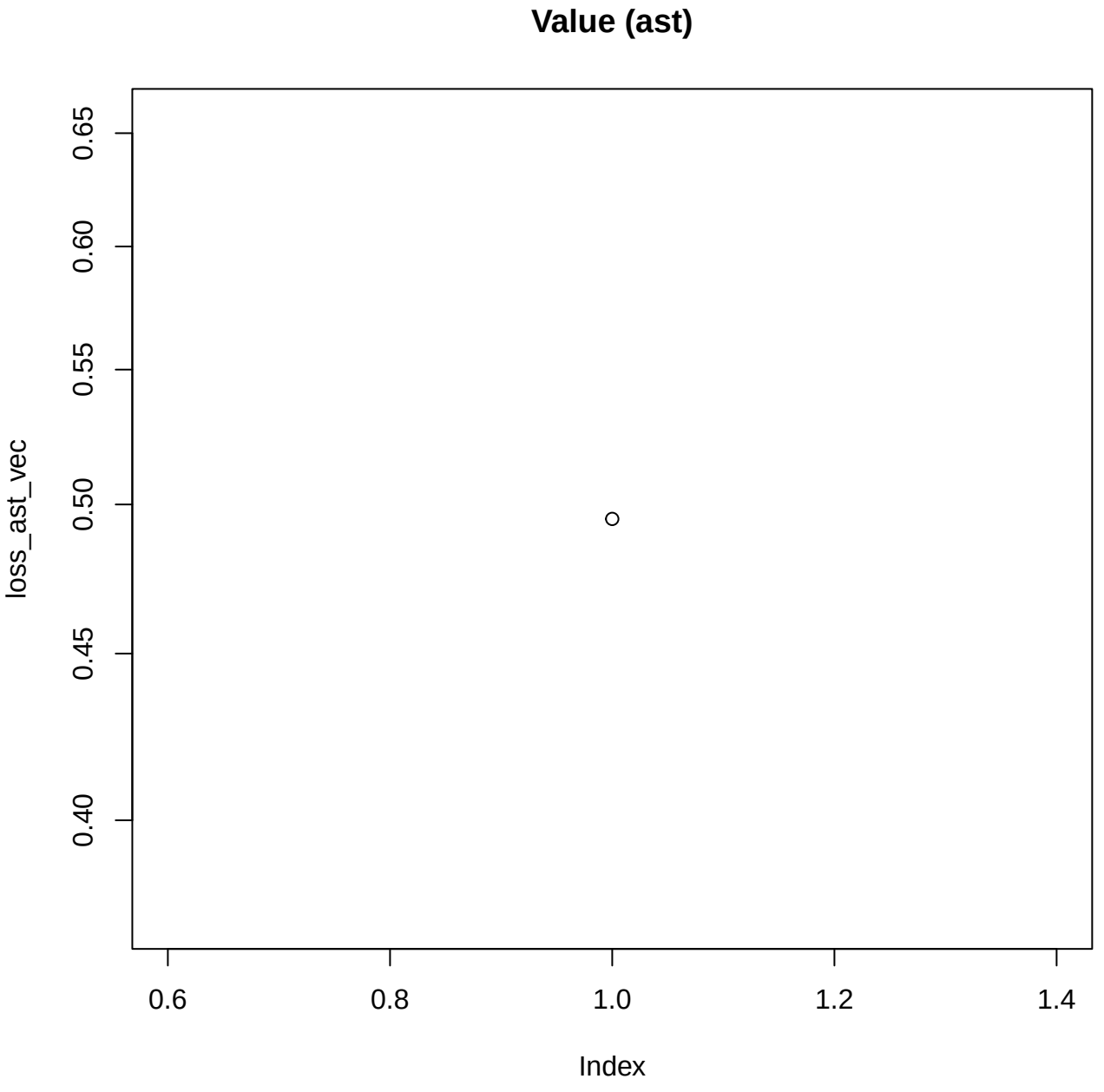
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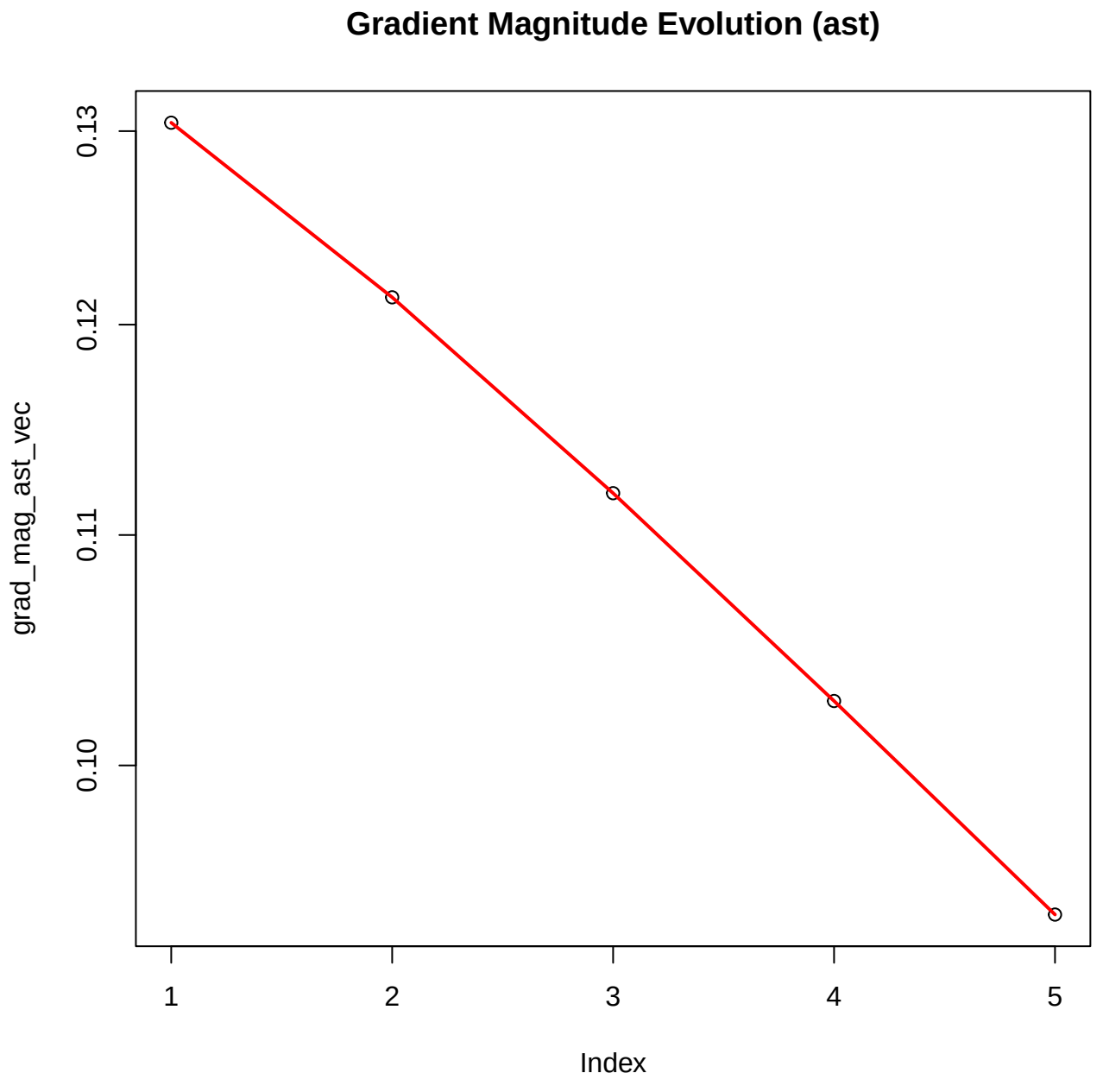
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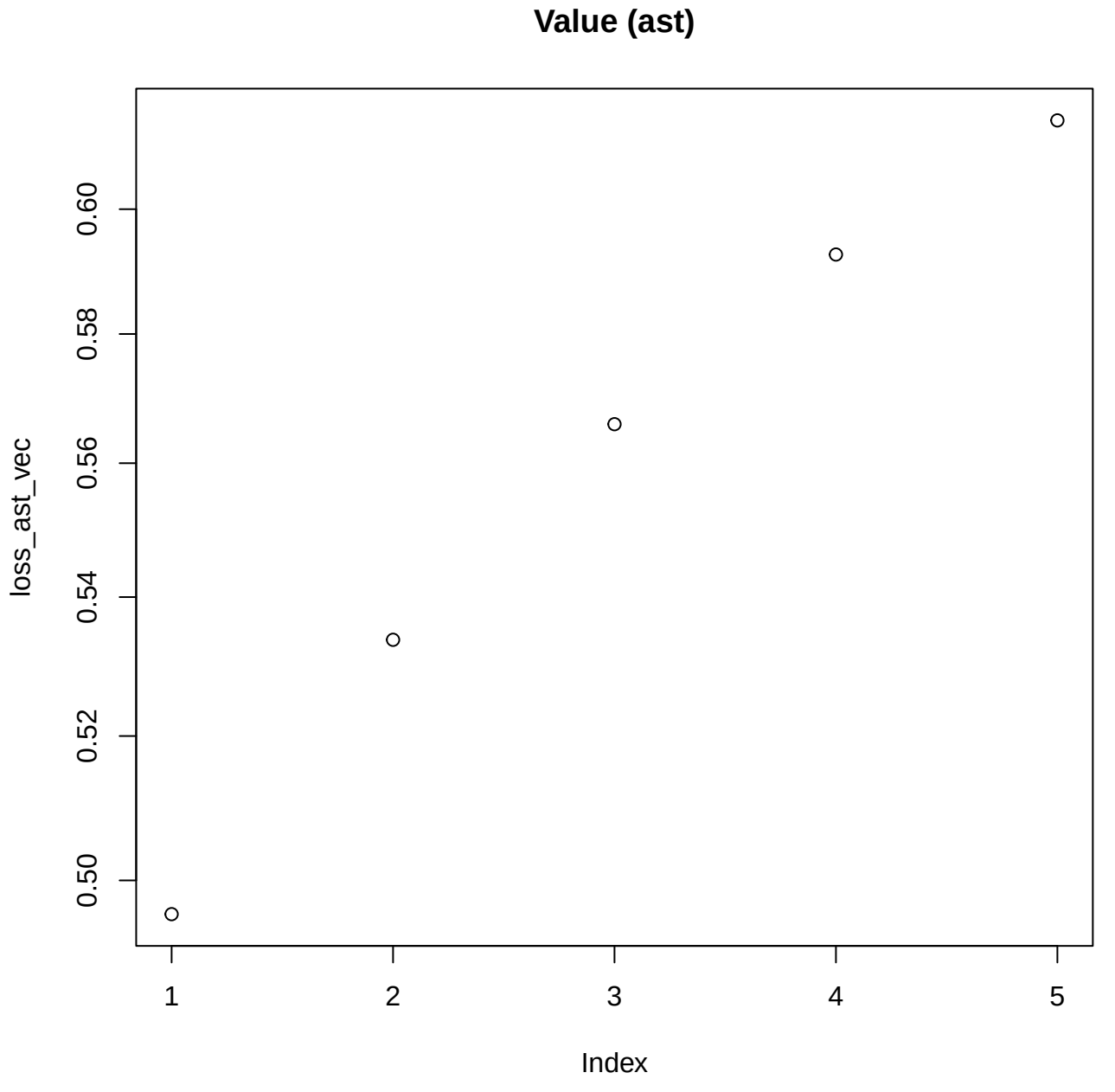
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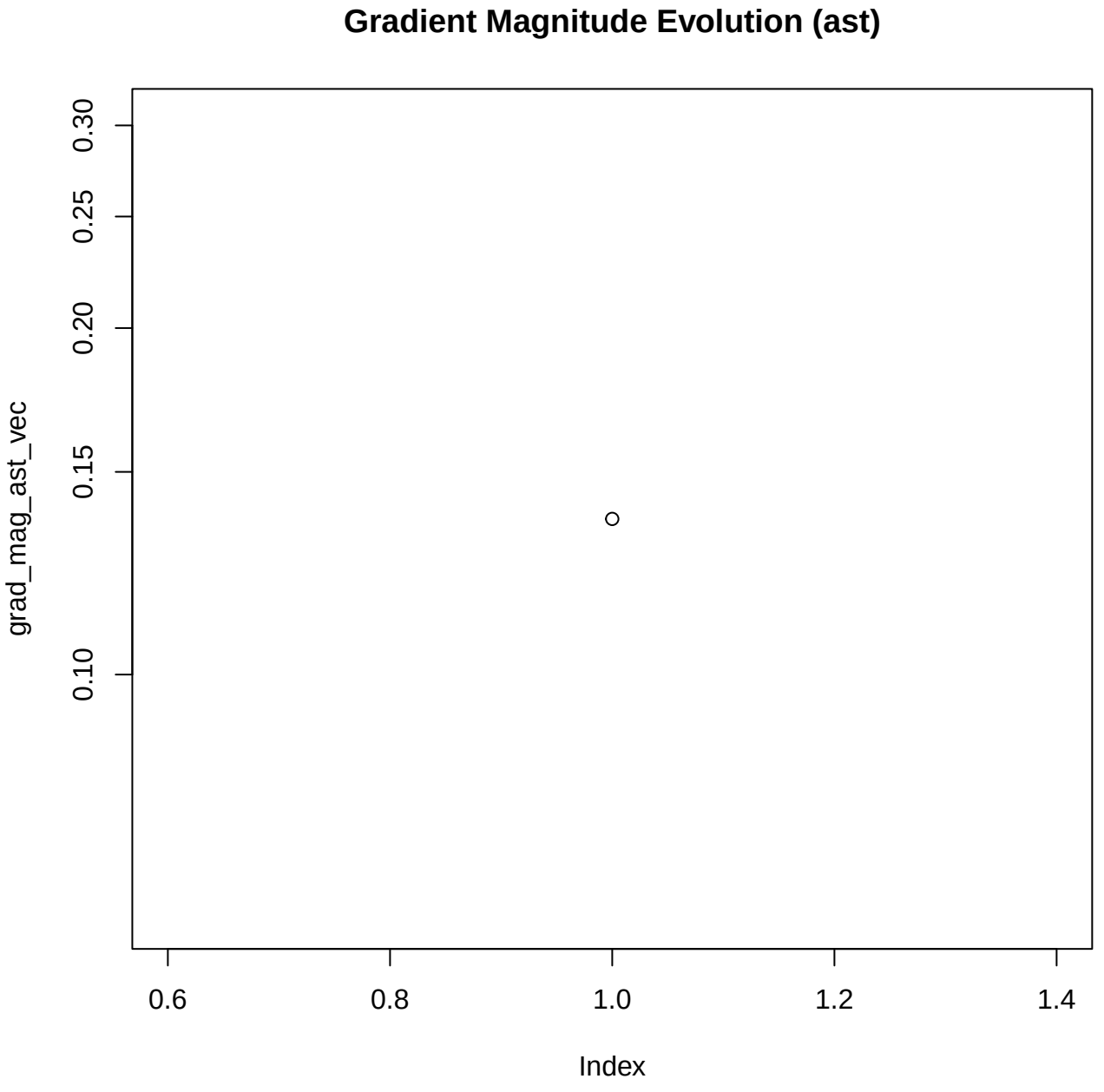
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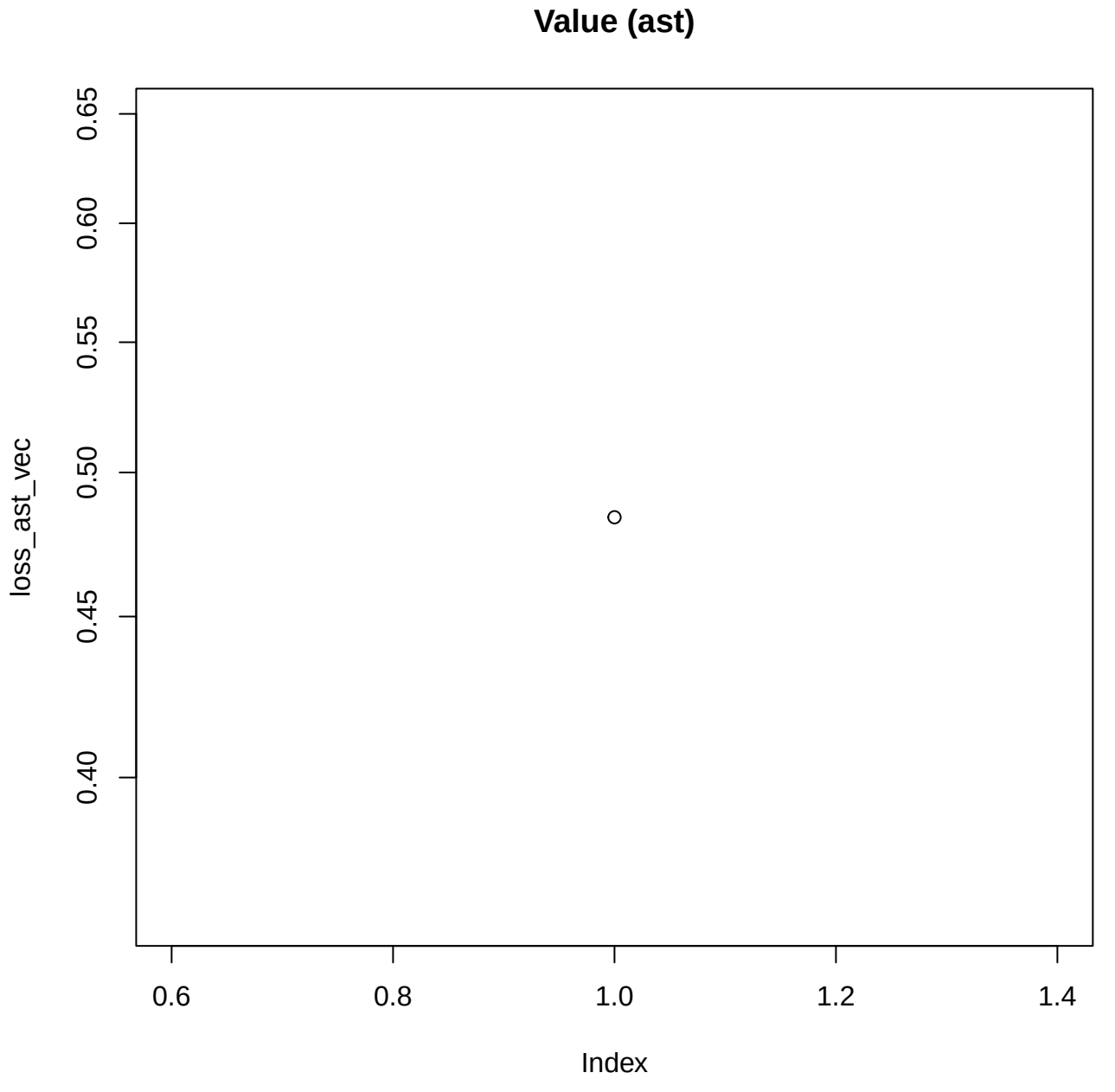
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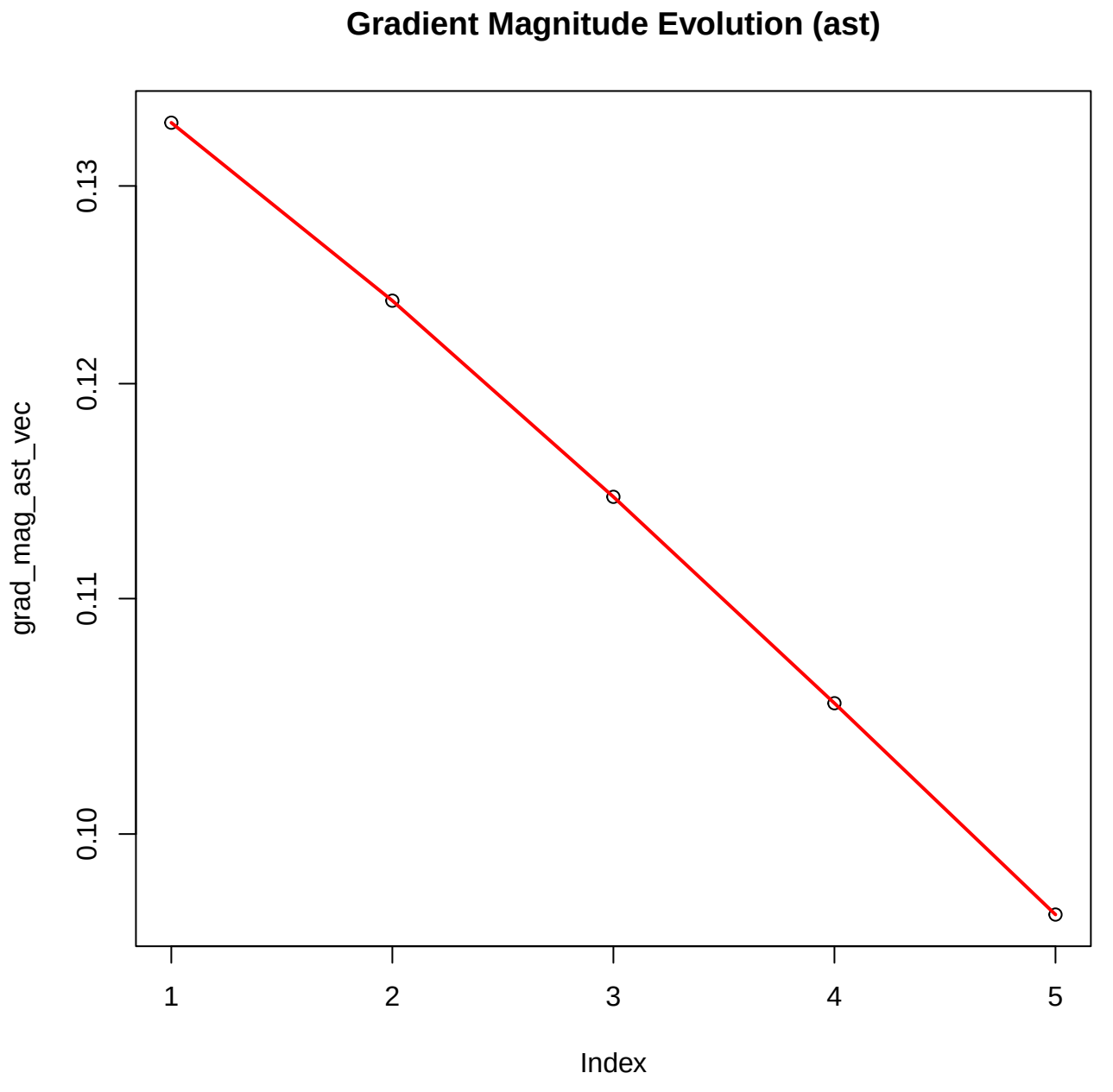
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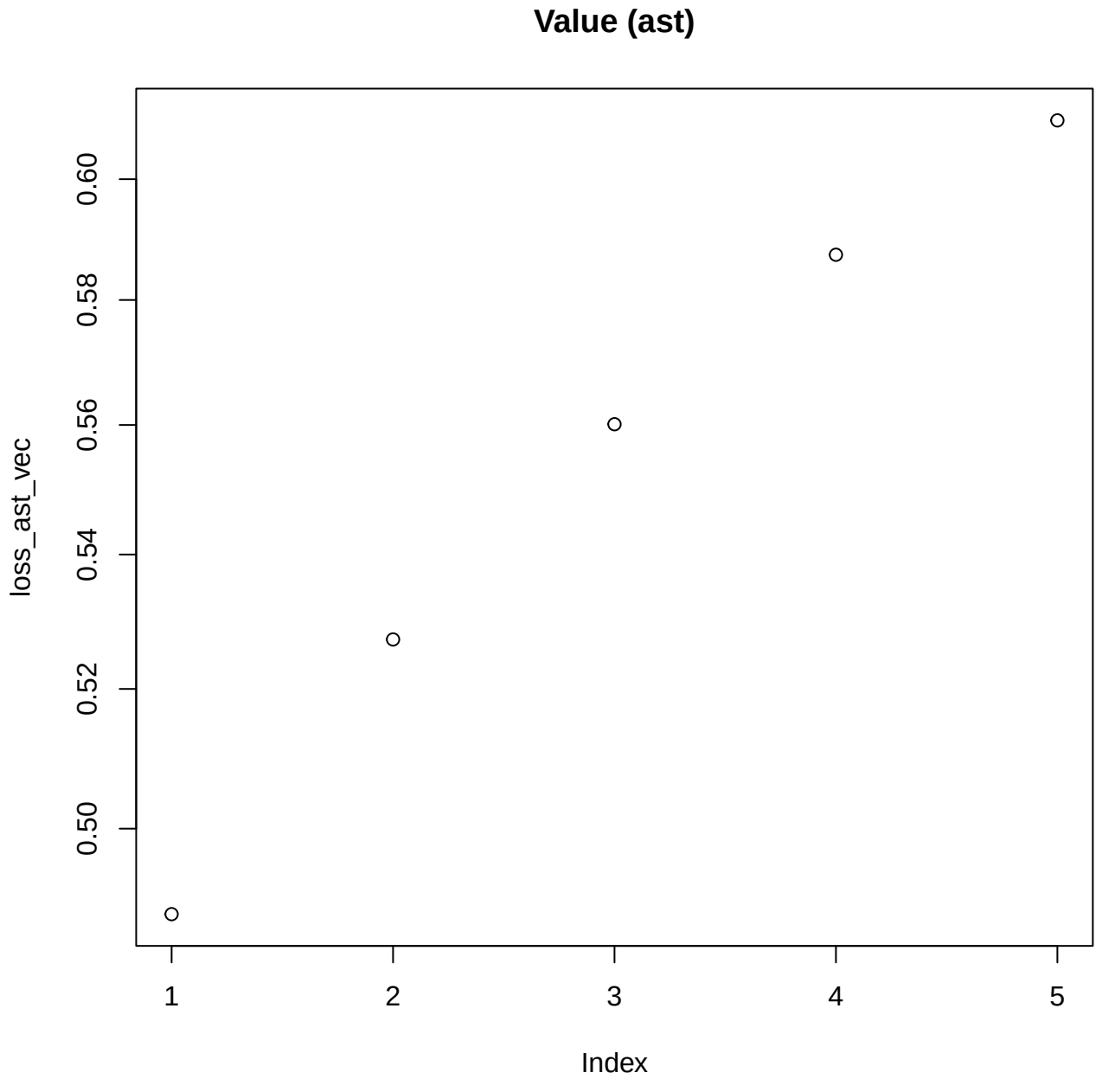
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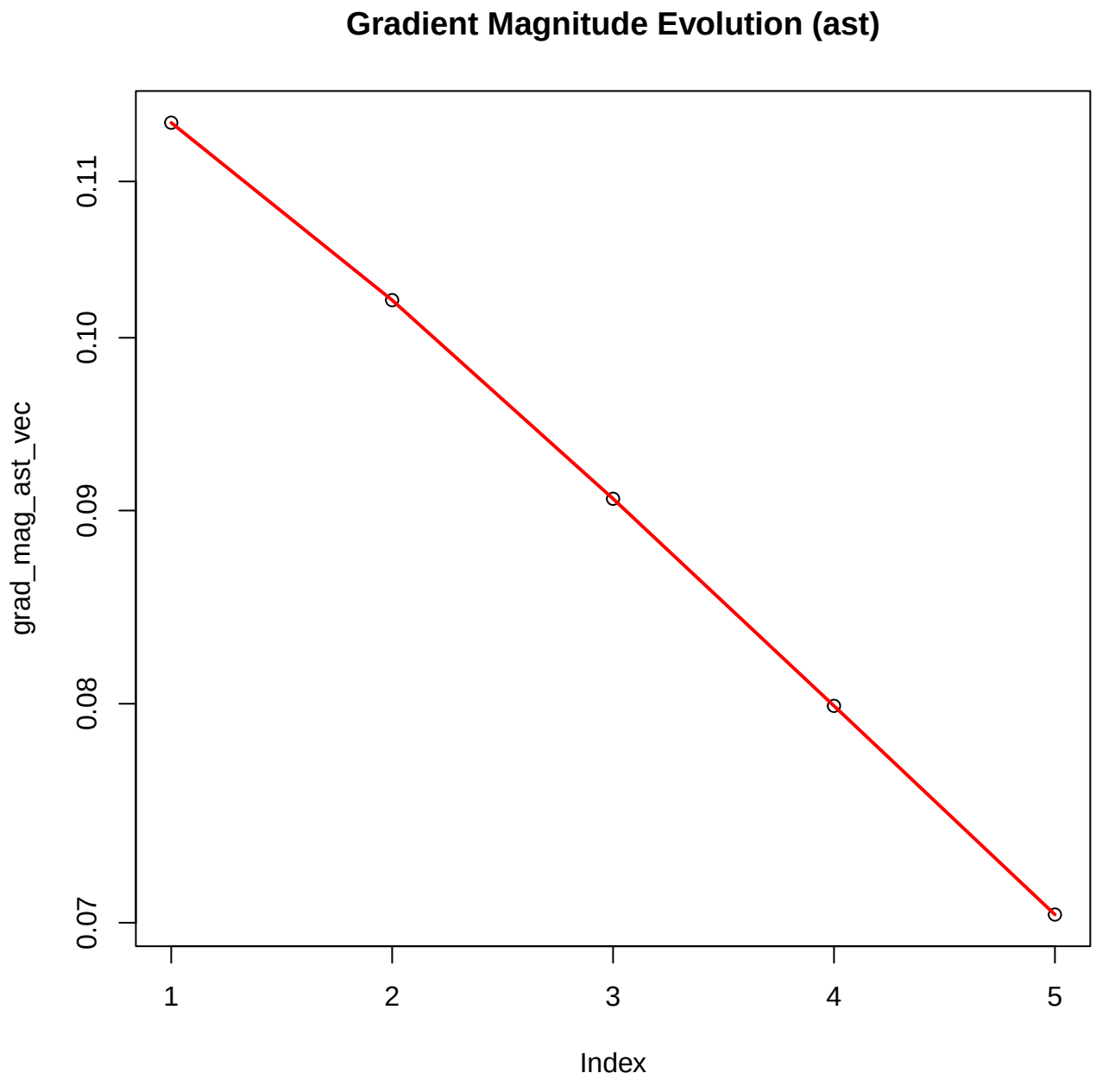
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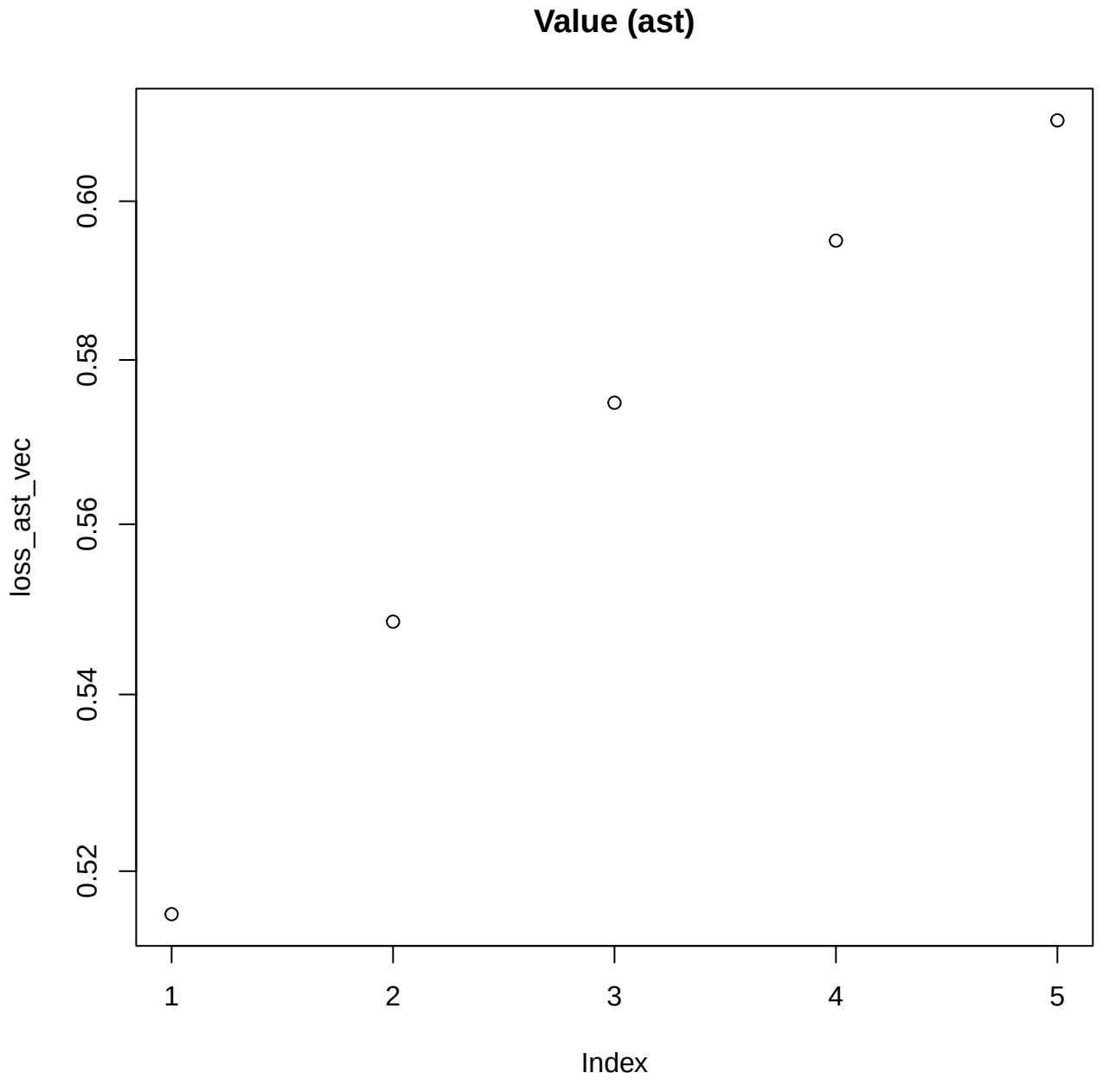
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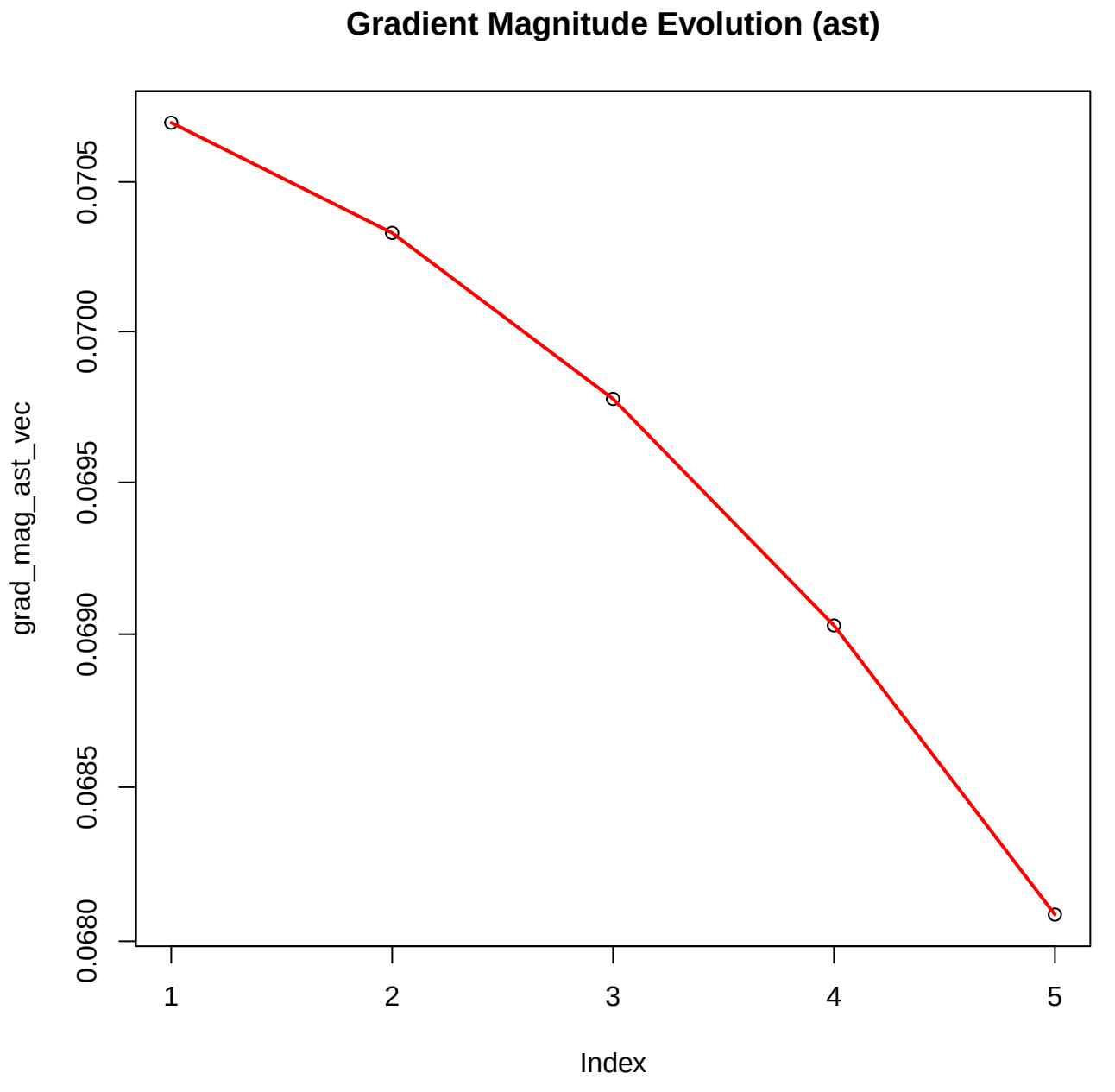
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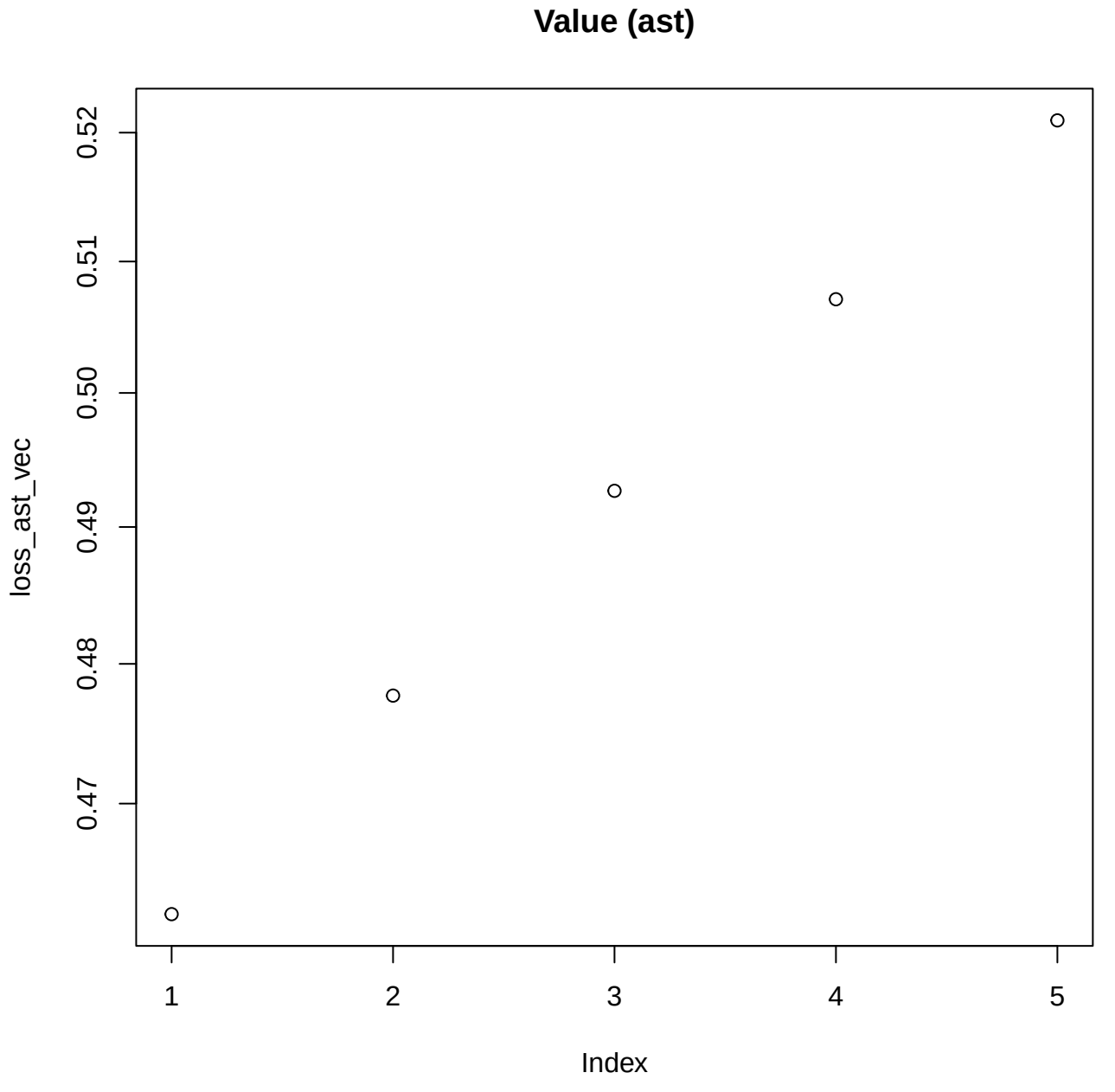
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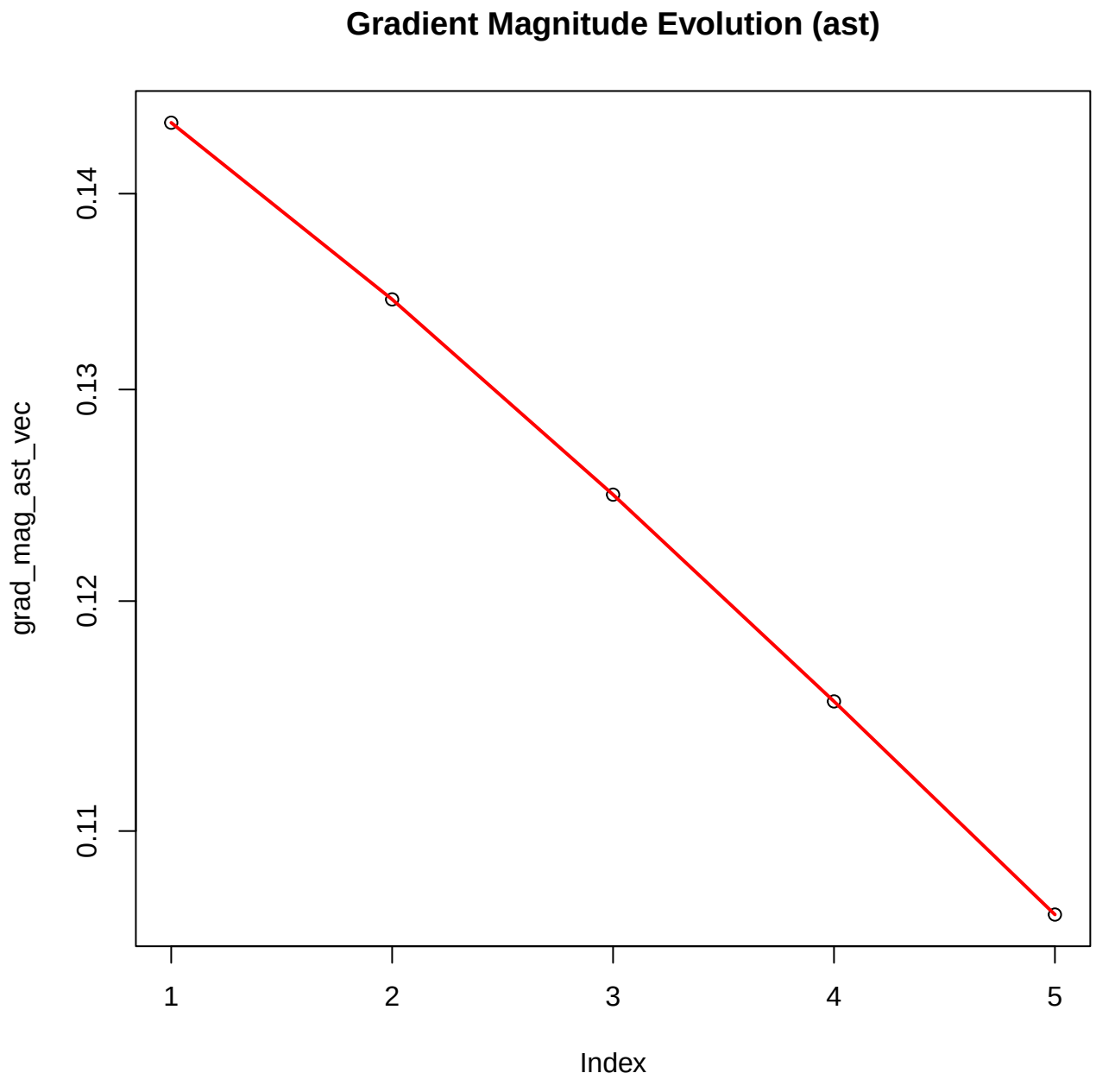
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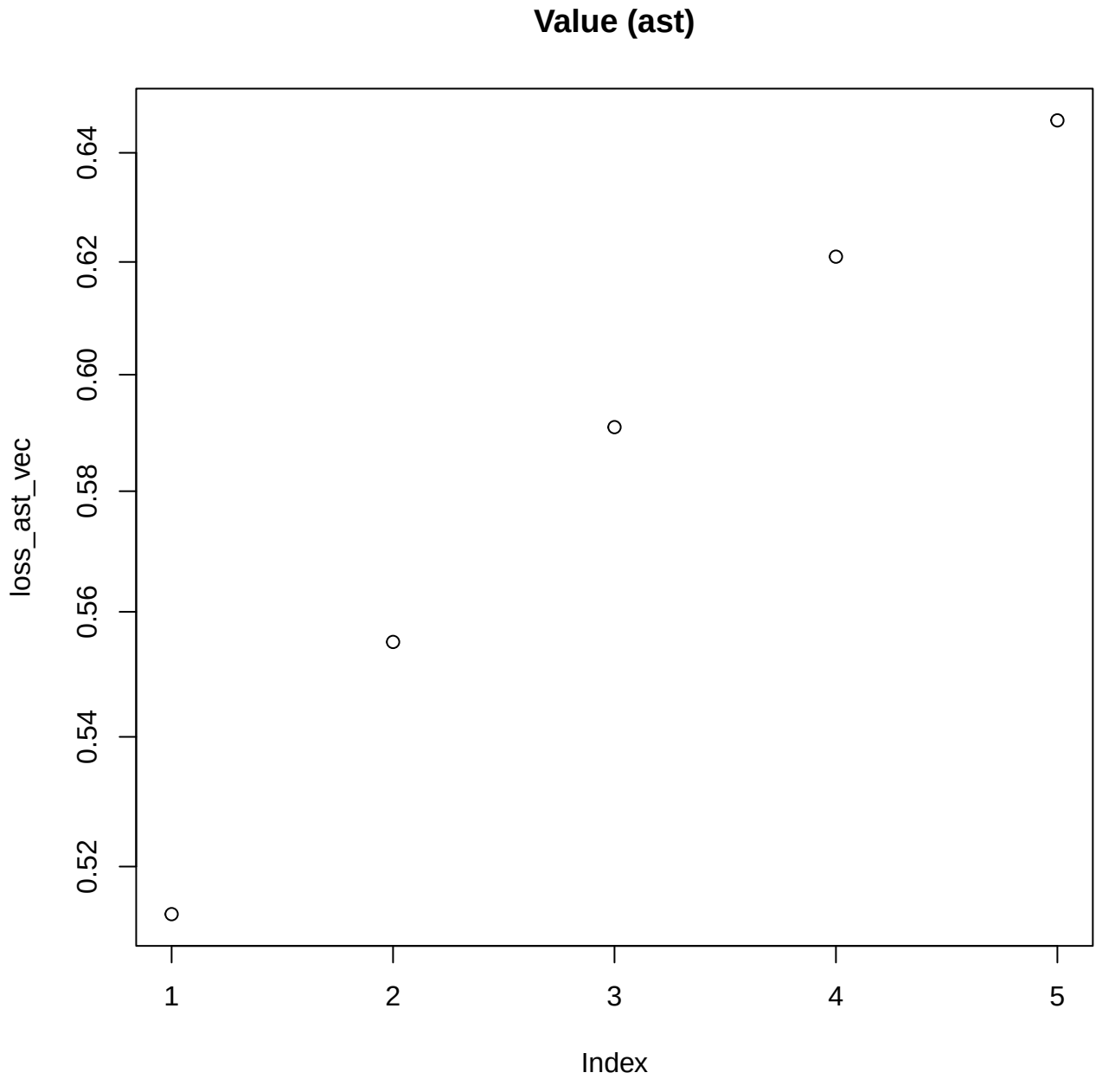
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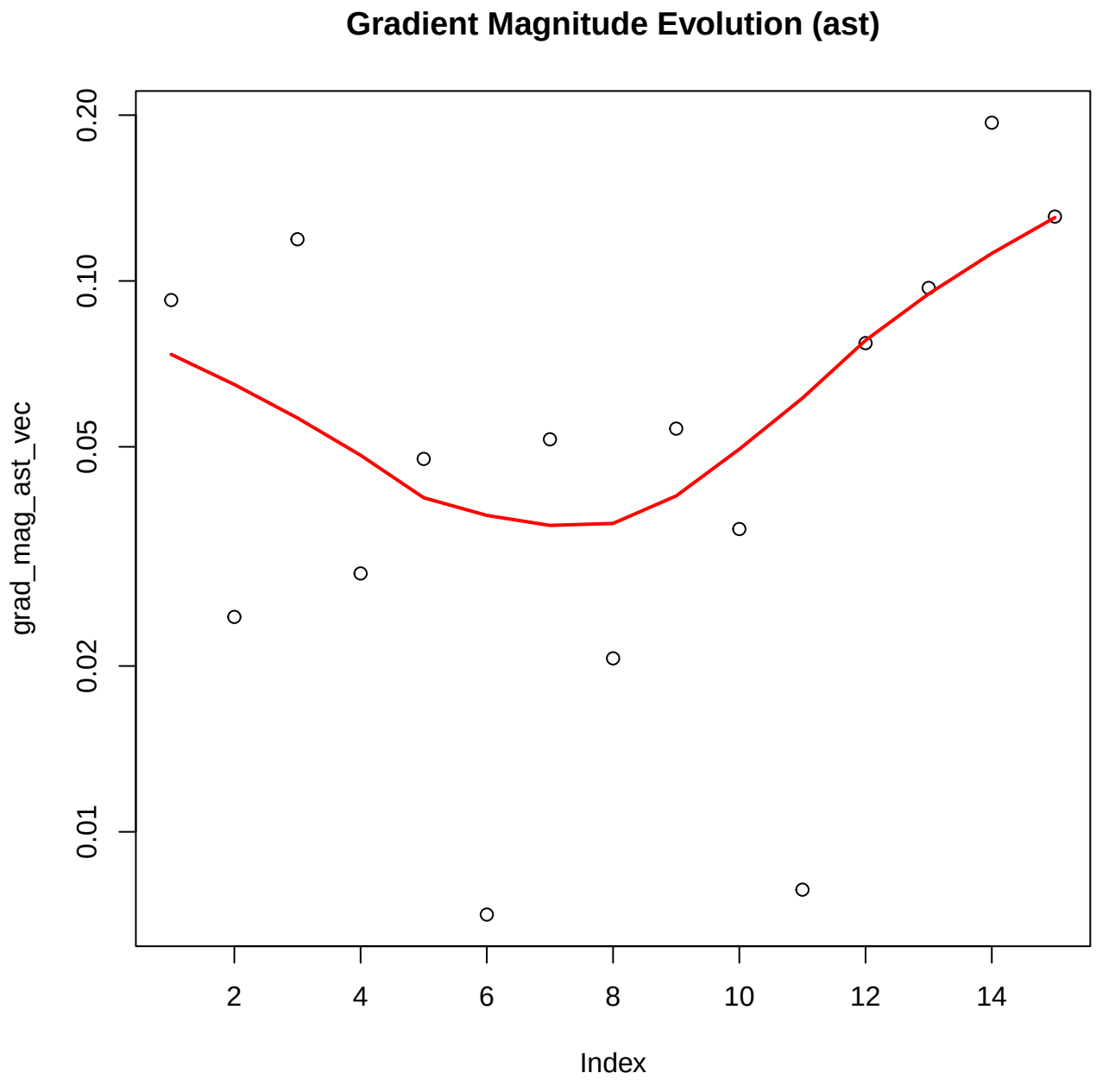
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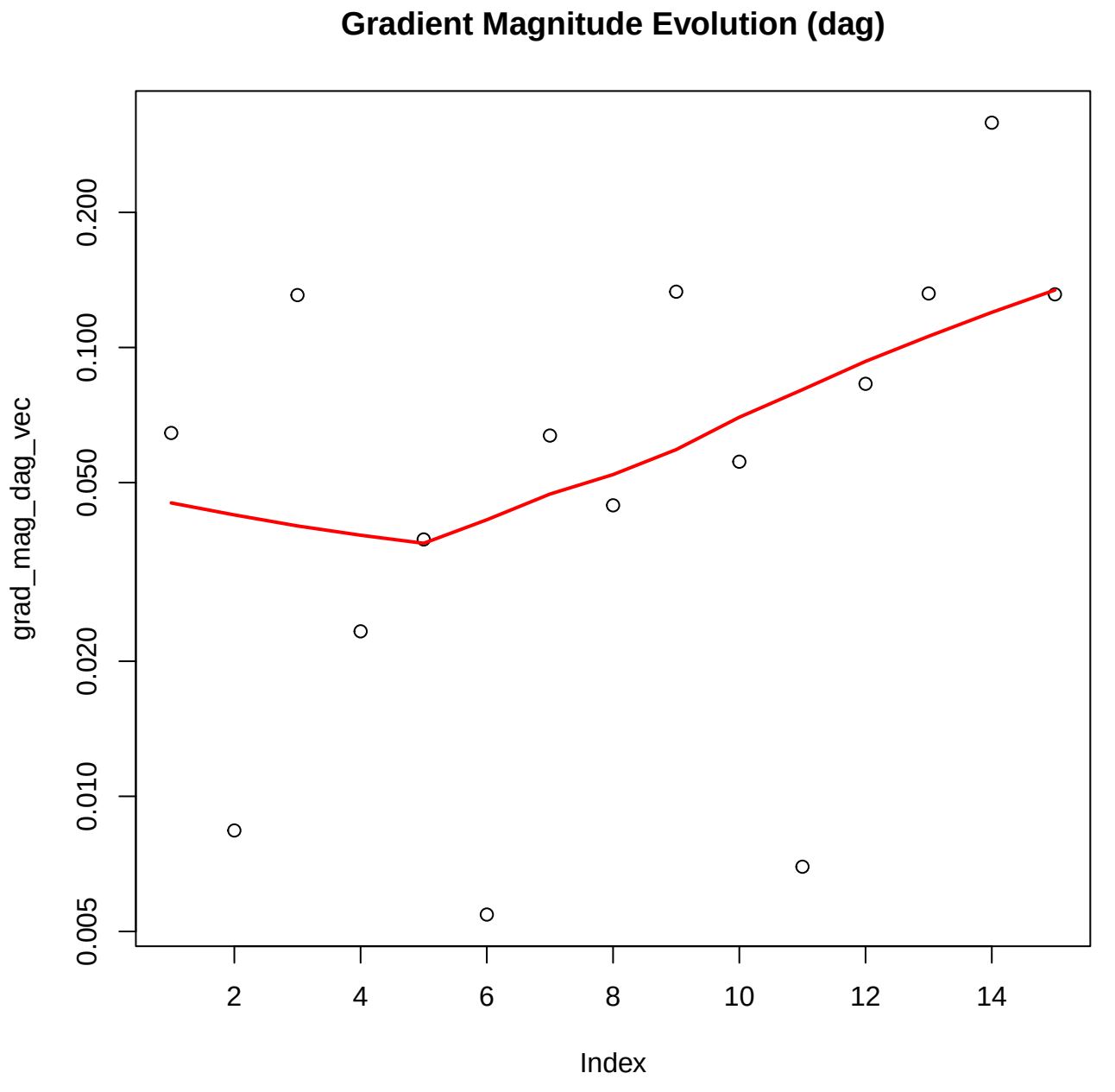
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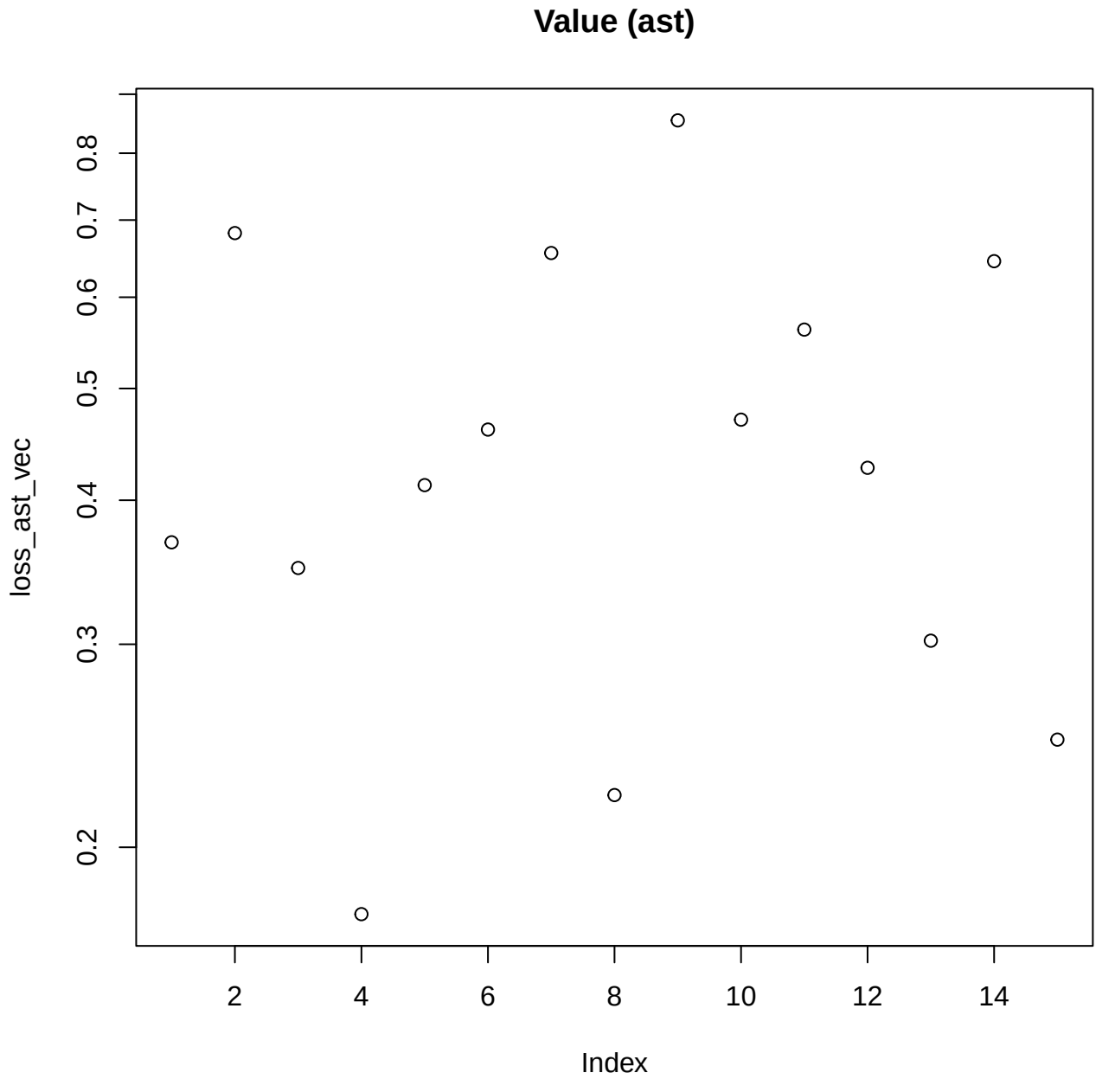
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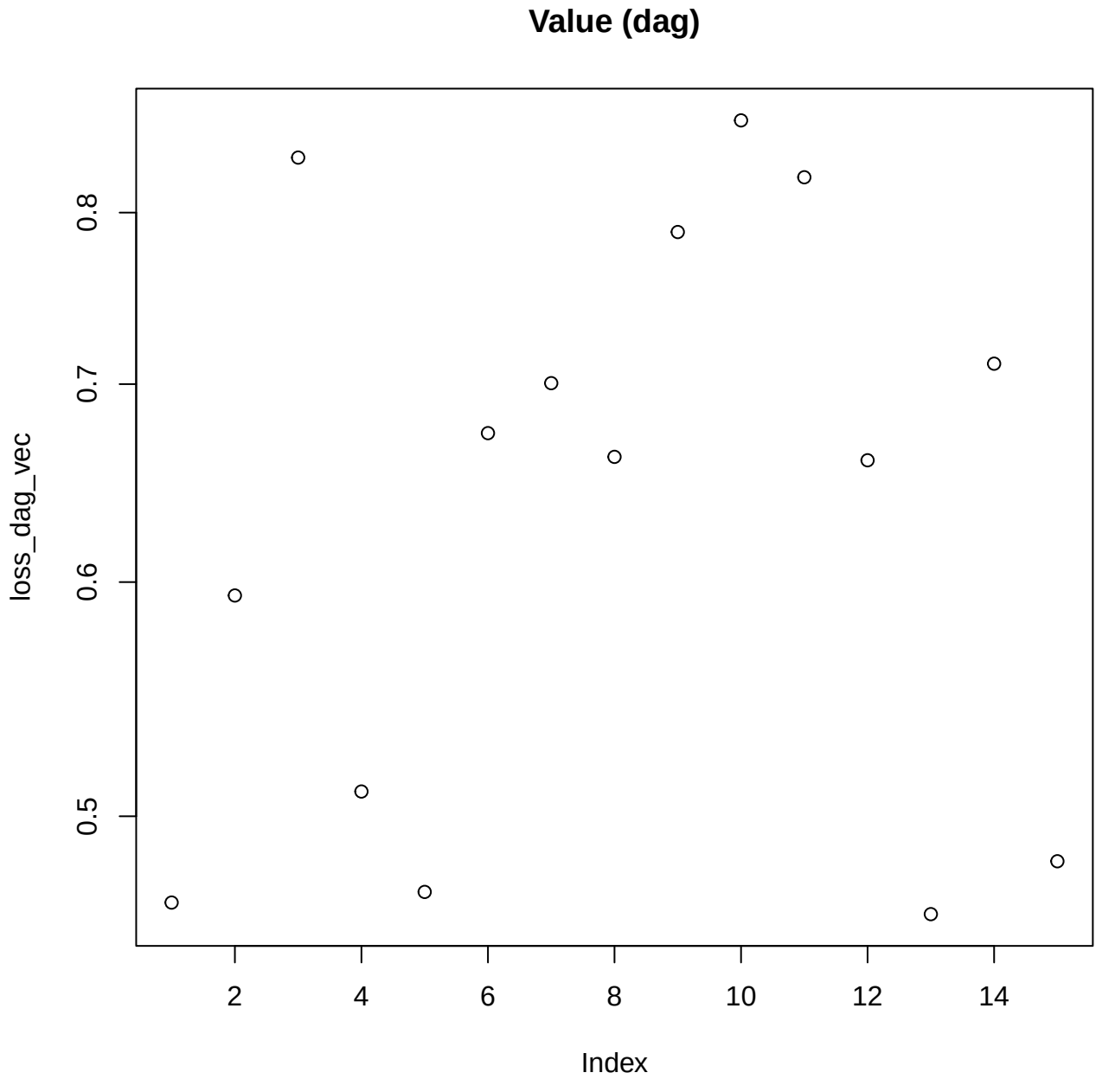
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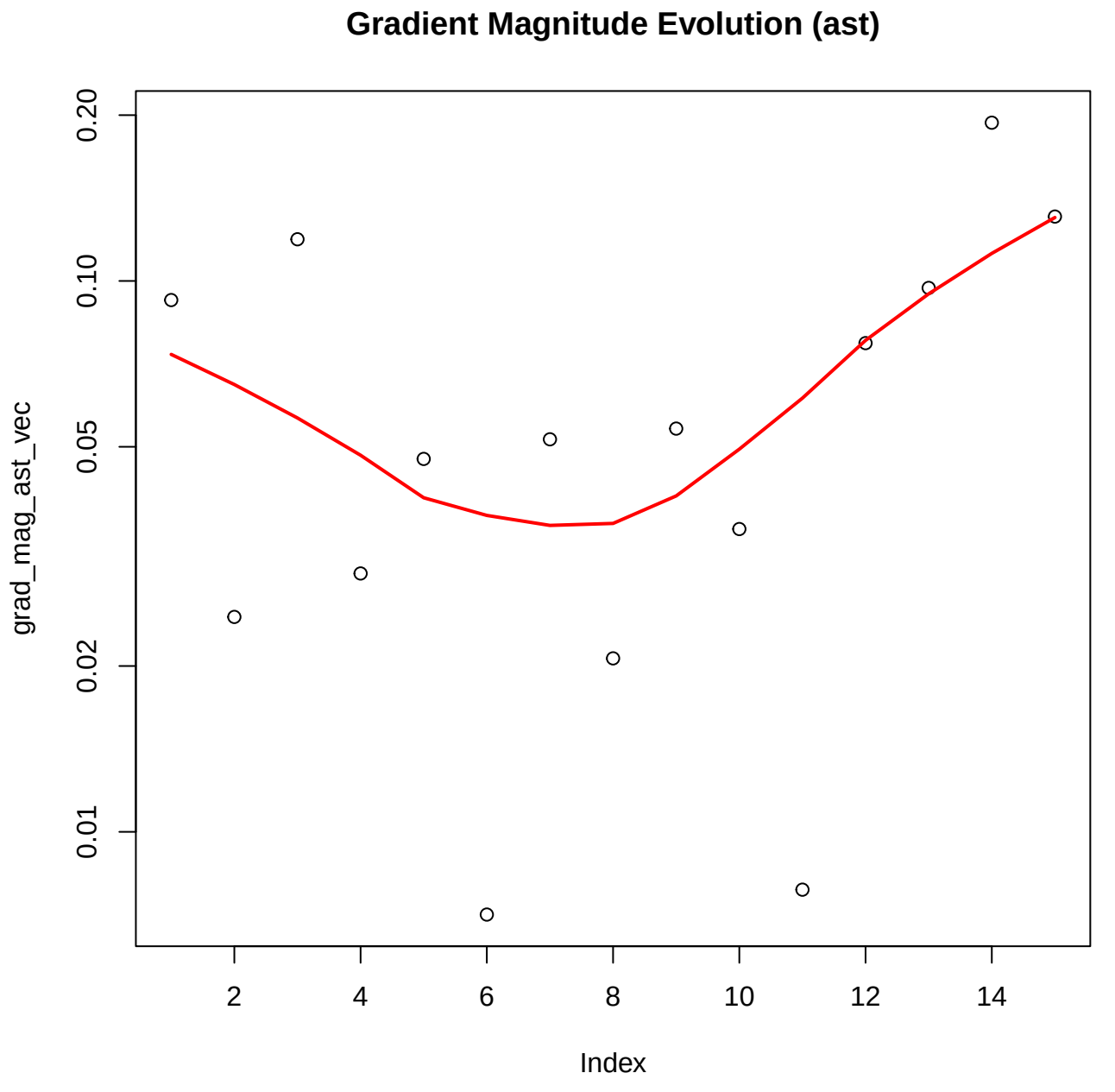
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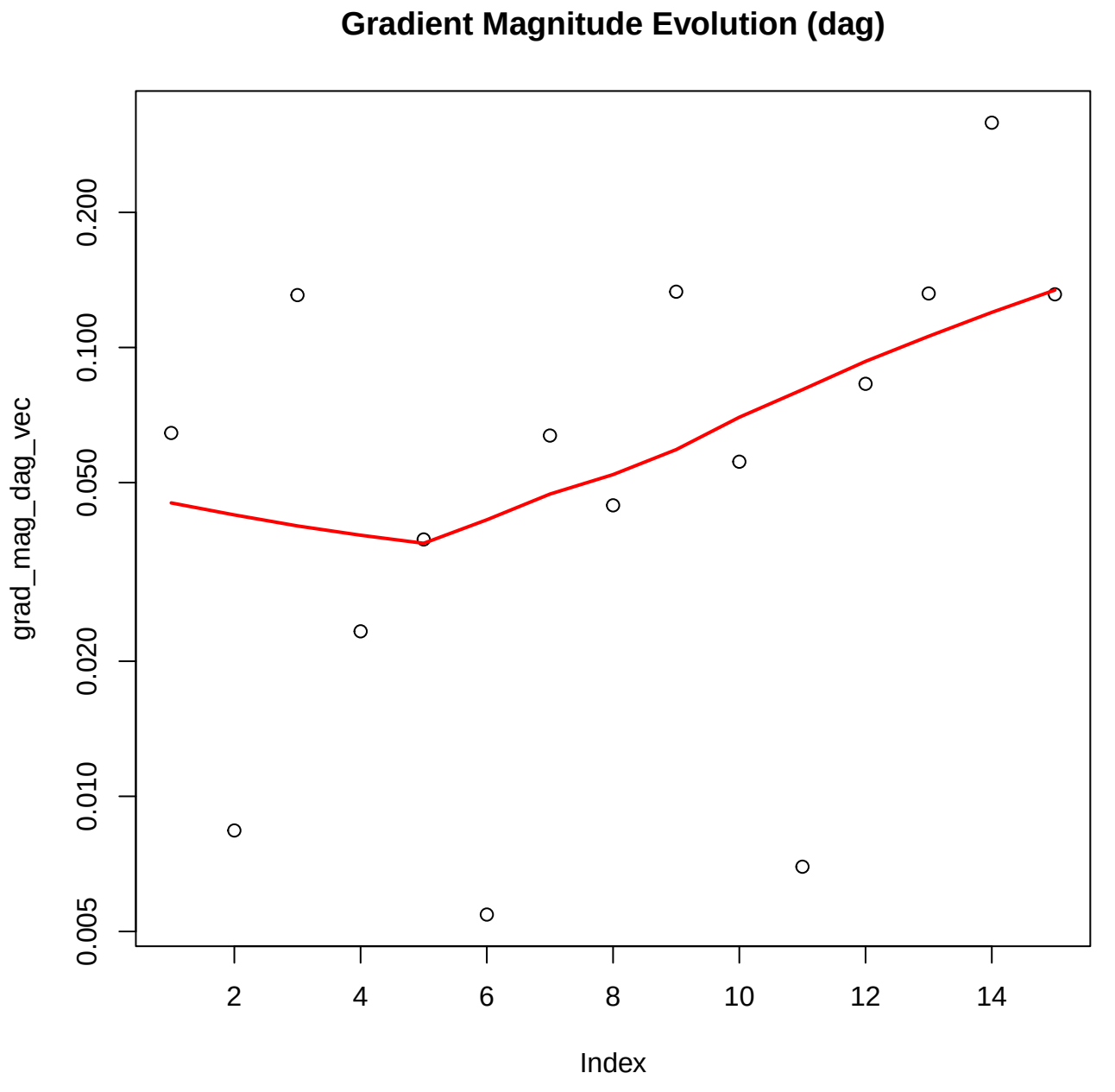
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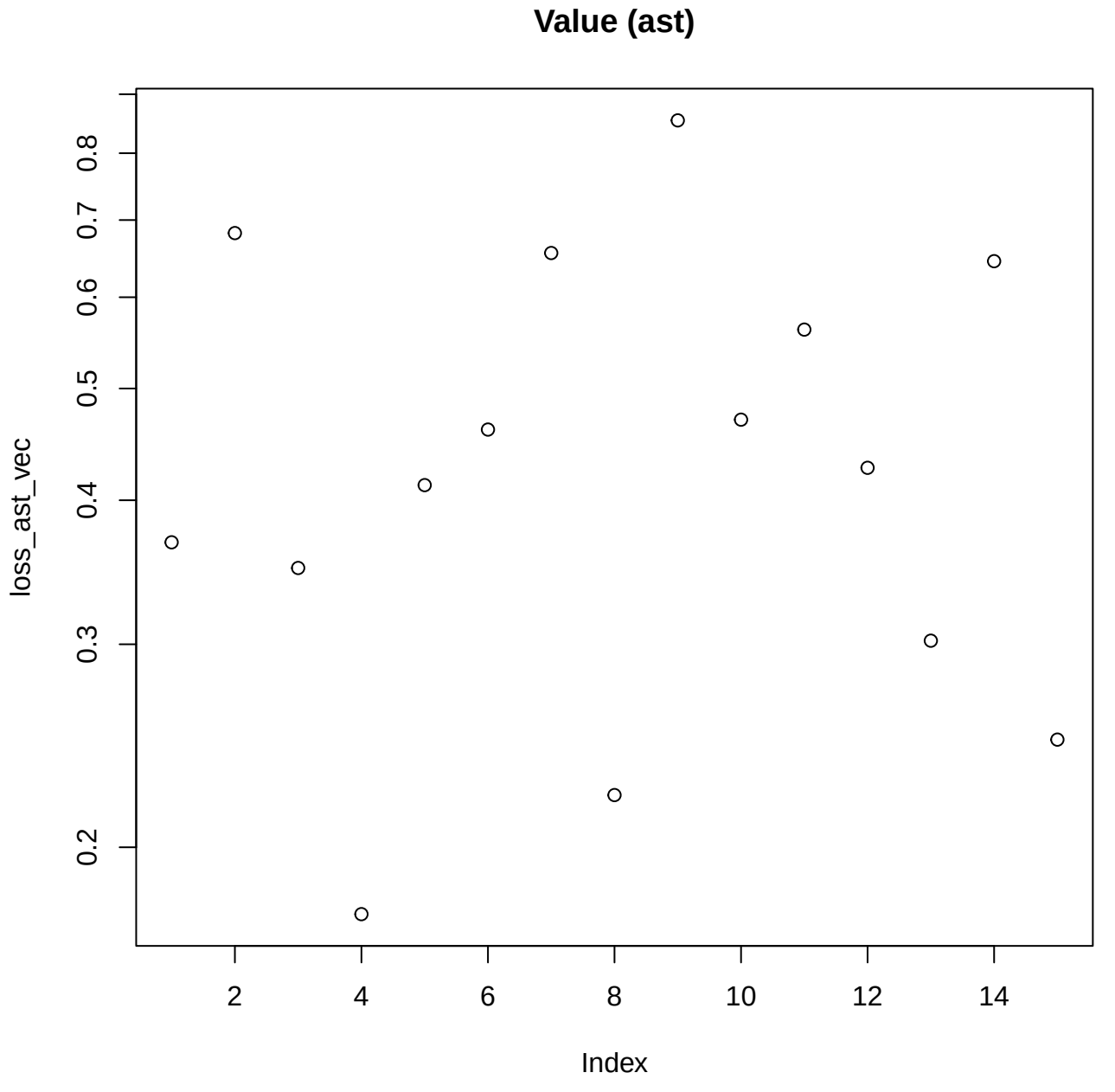
Value (ast)
loss_ast_vec
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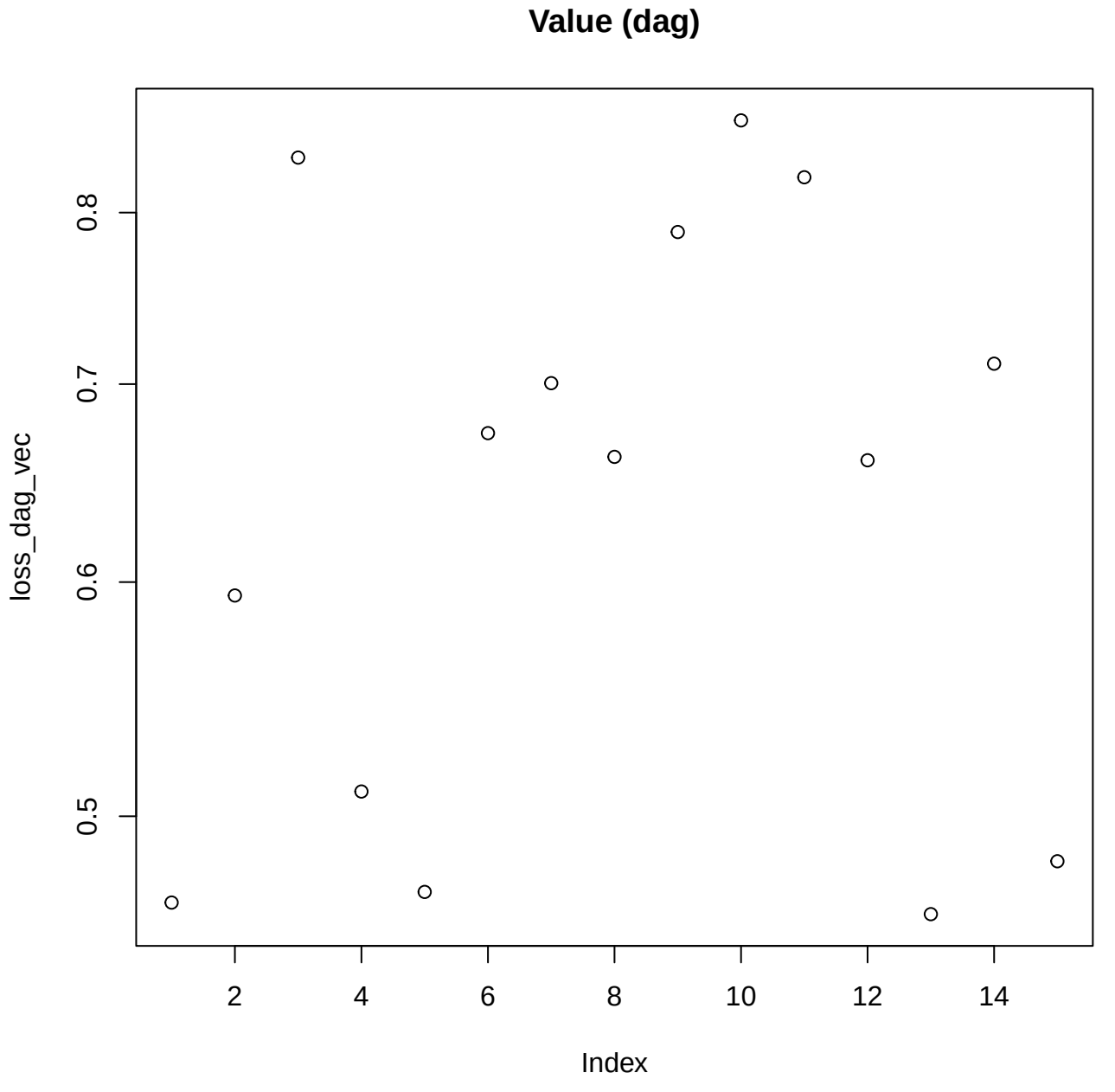
12
14
Index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14
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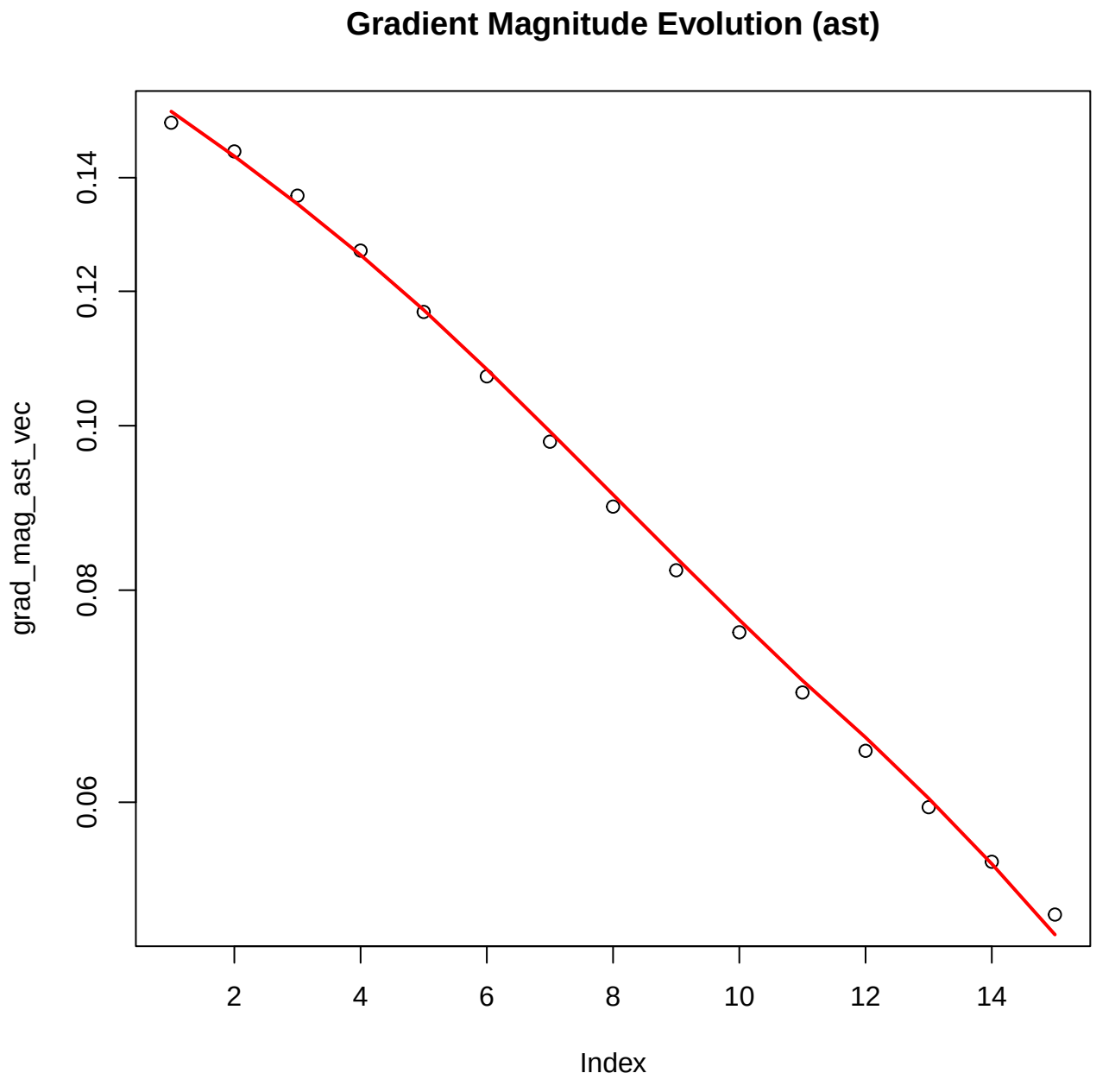
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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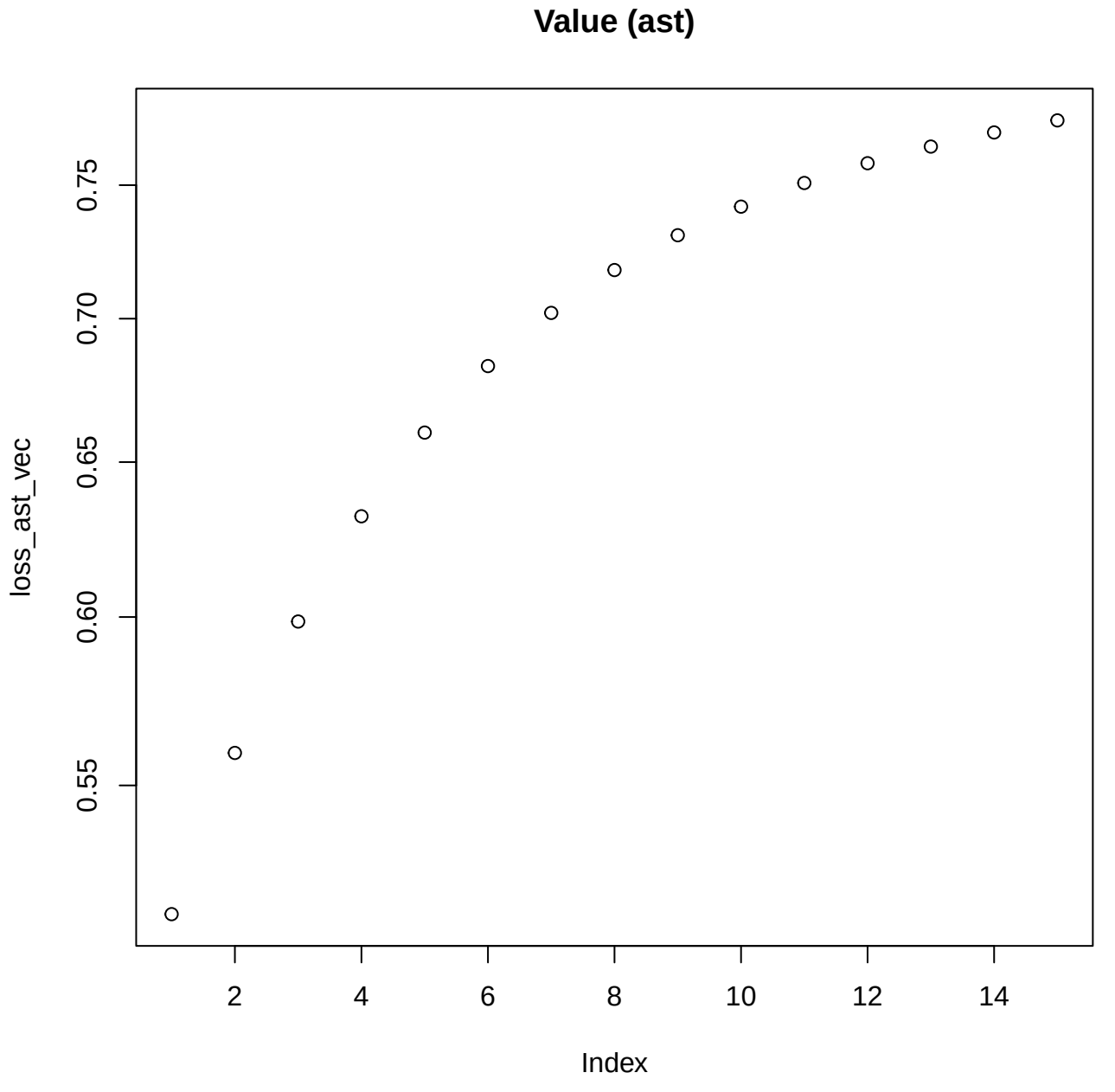
12
14
Index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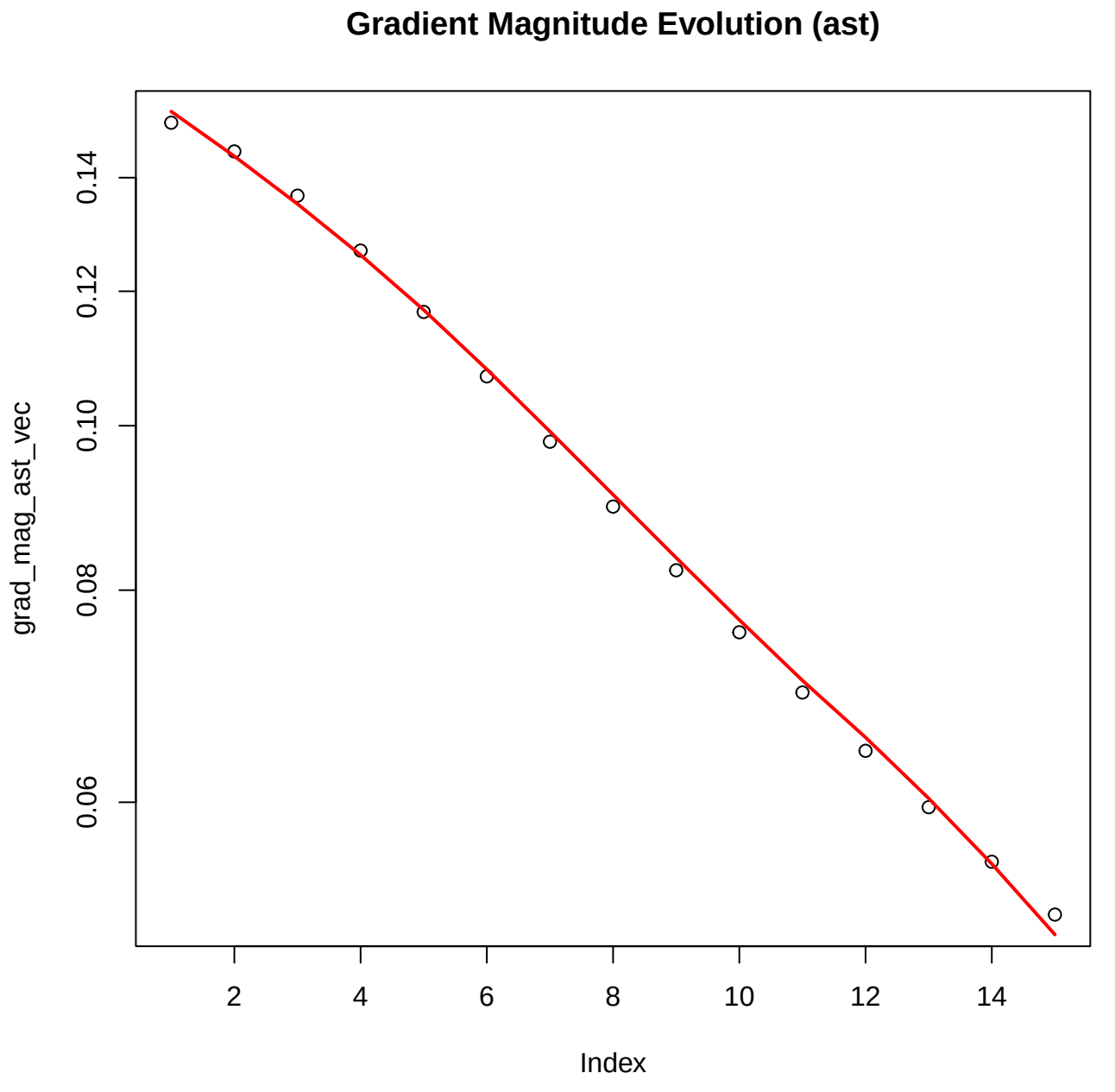
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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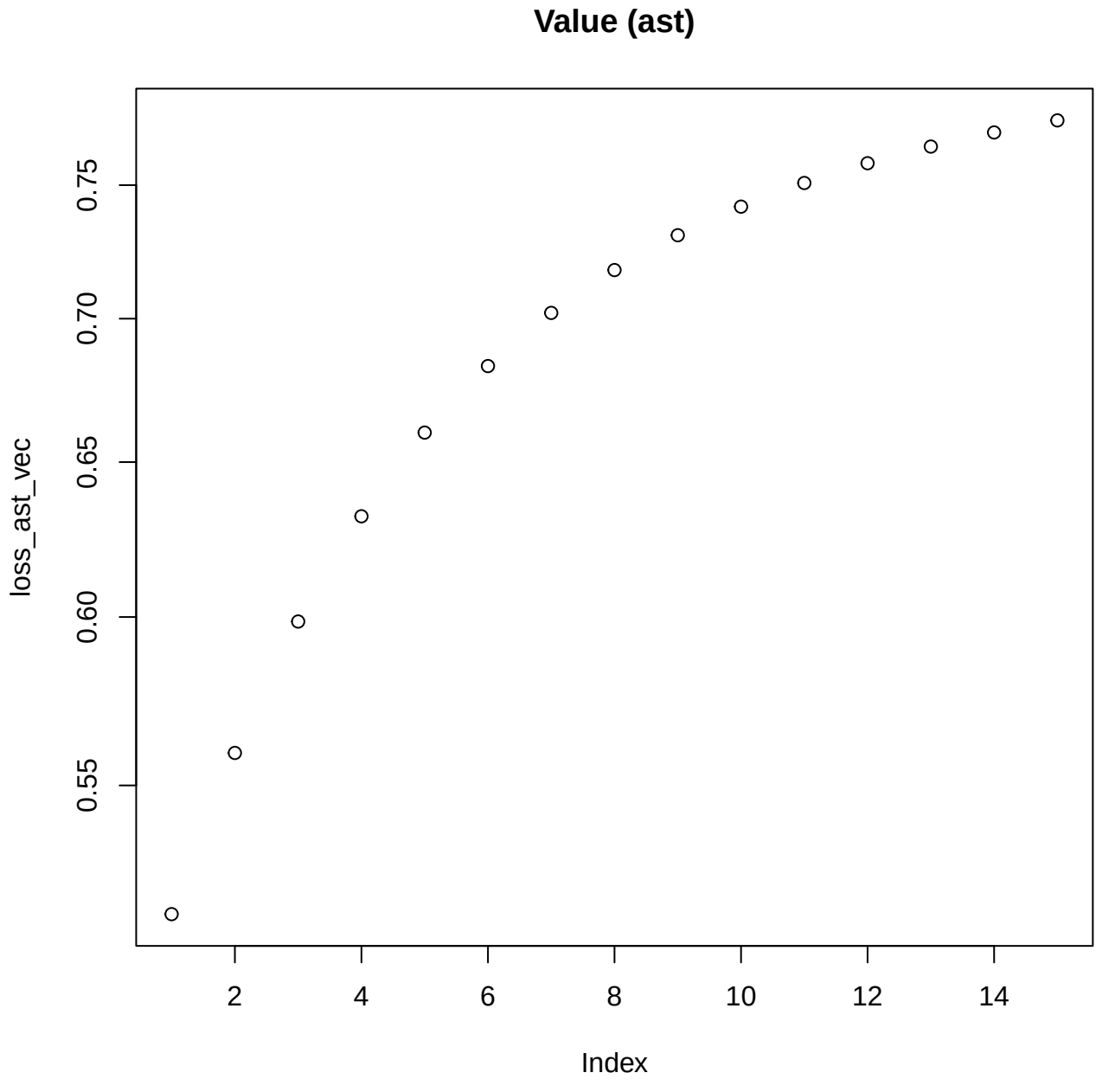
Value (ast)
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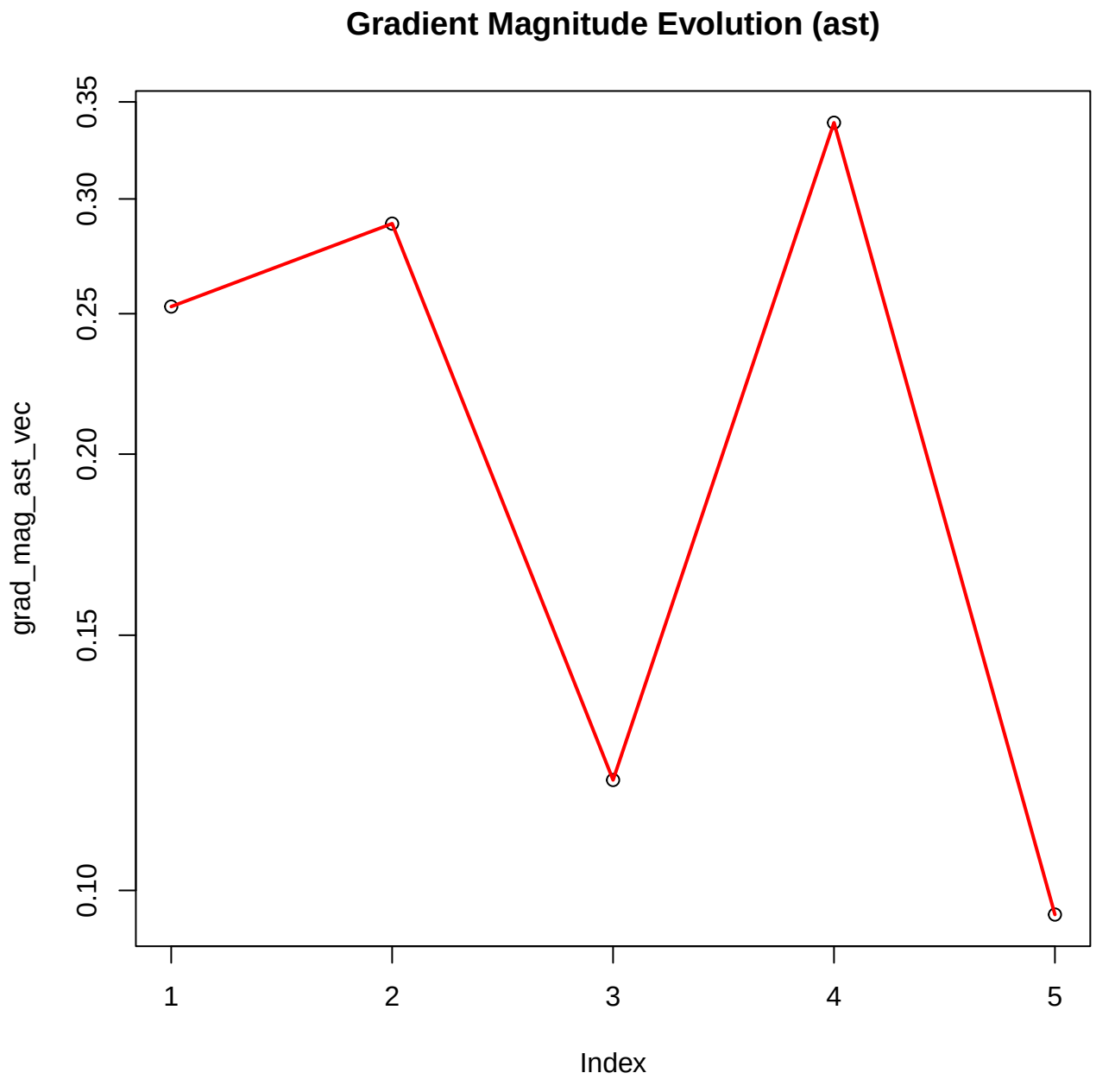
Gradient Magnitude Evolution (ast)
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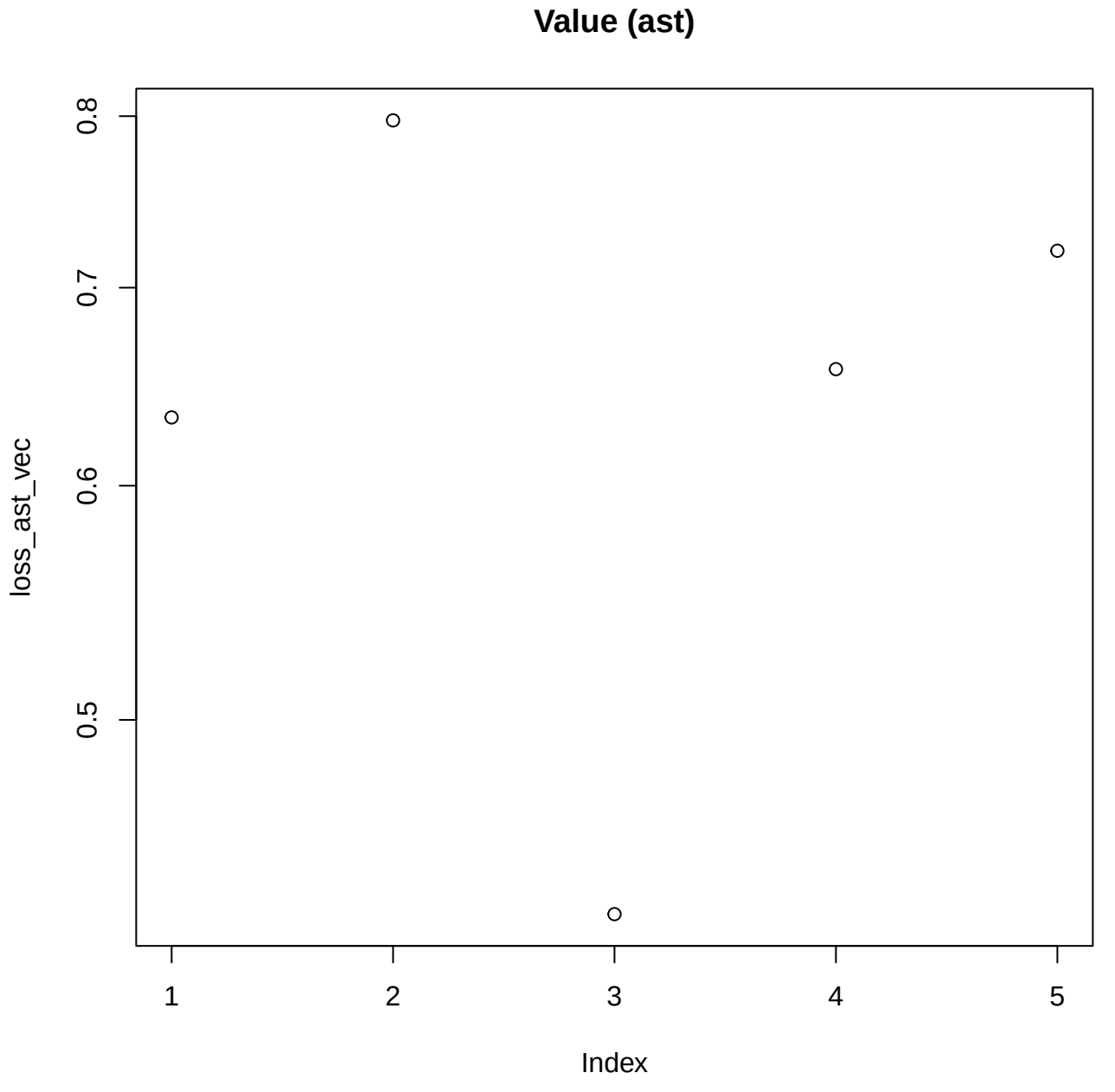
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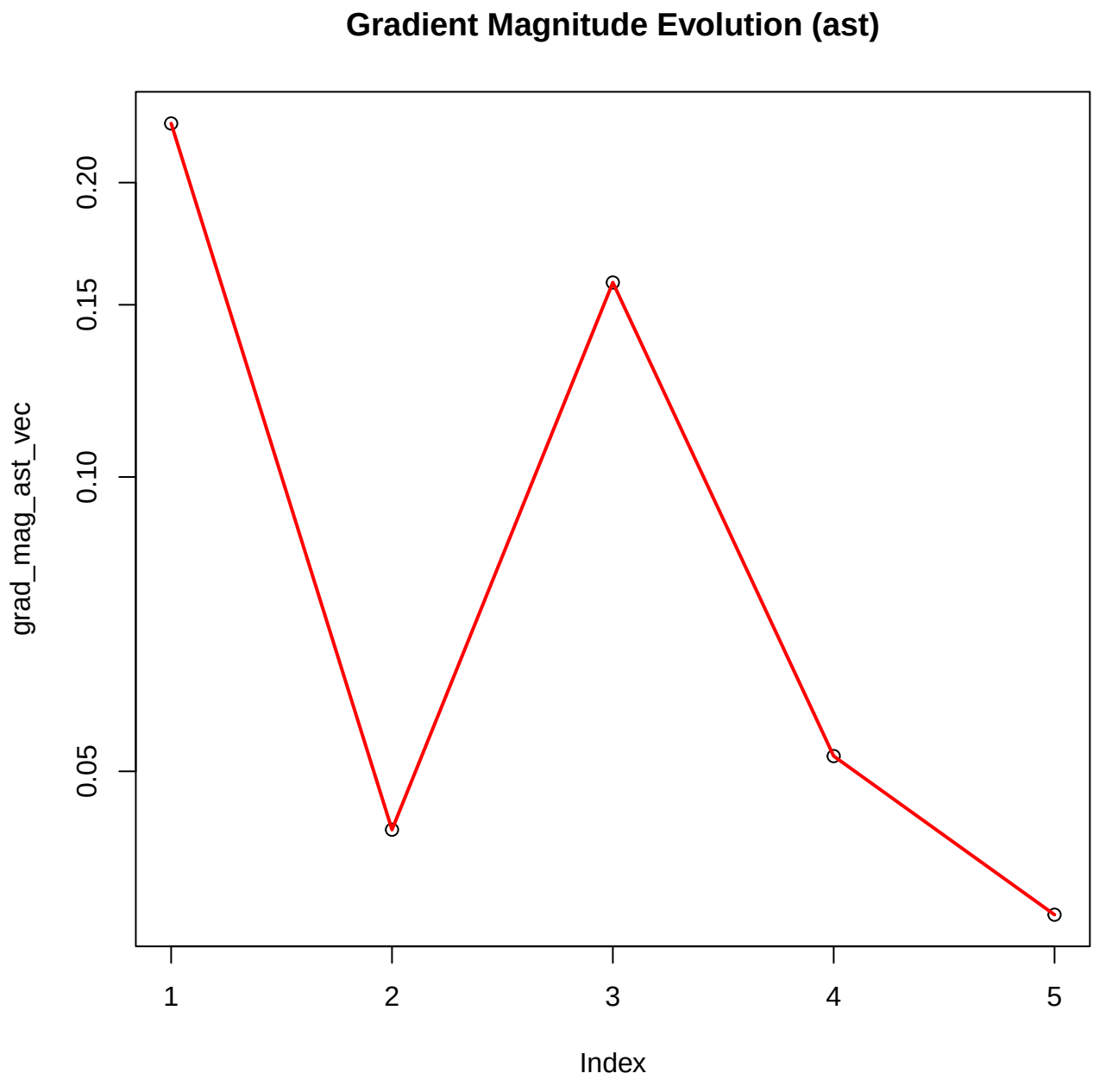
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grad_mag_ast_vec
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0.20
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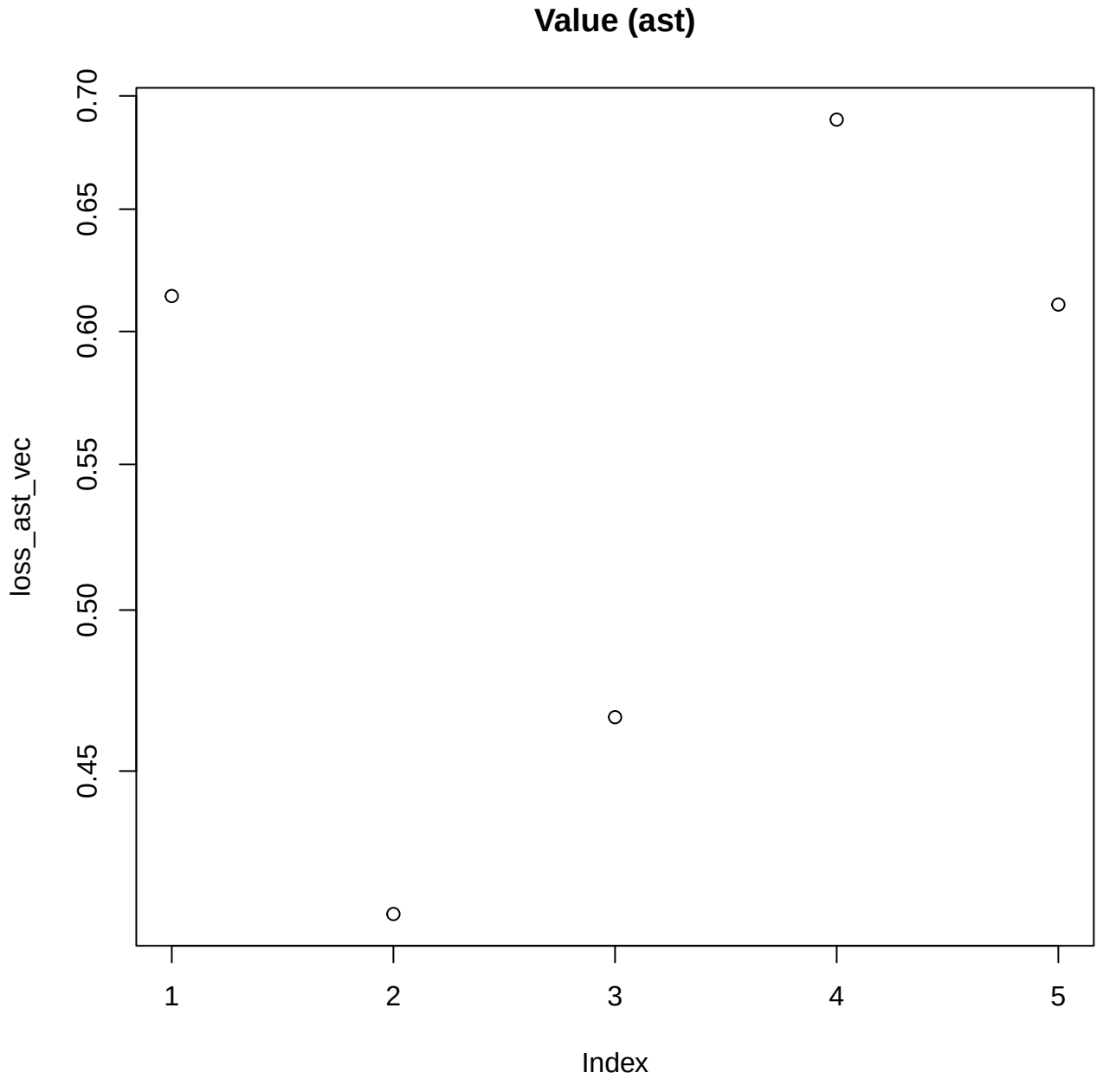
5
Index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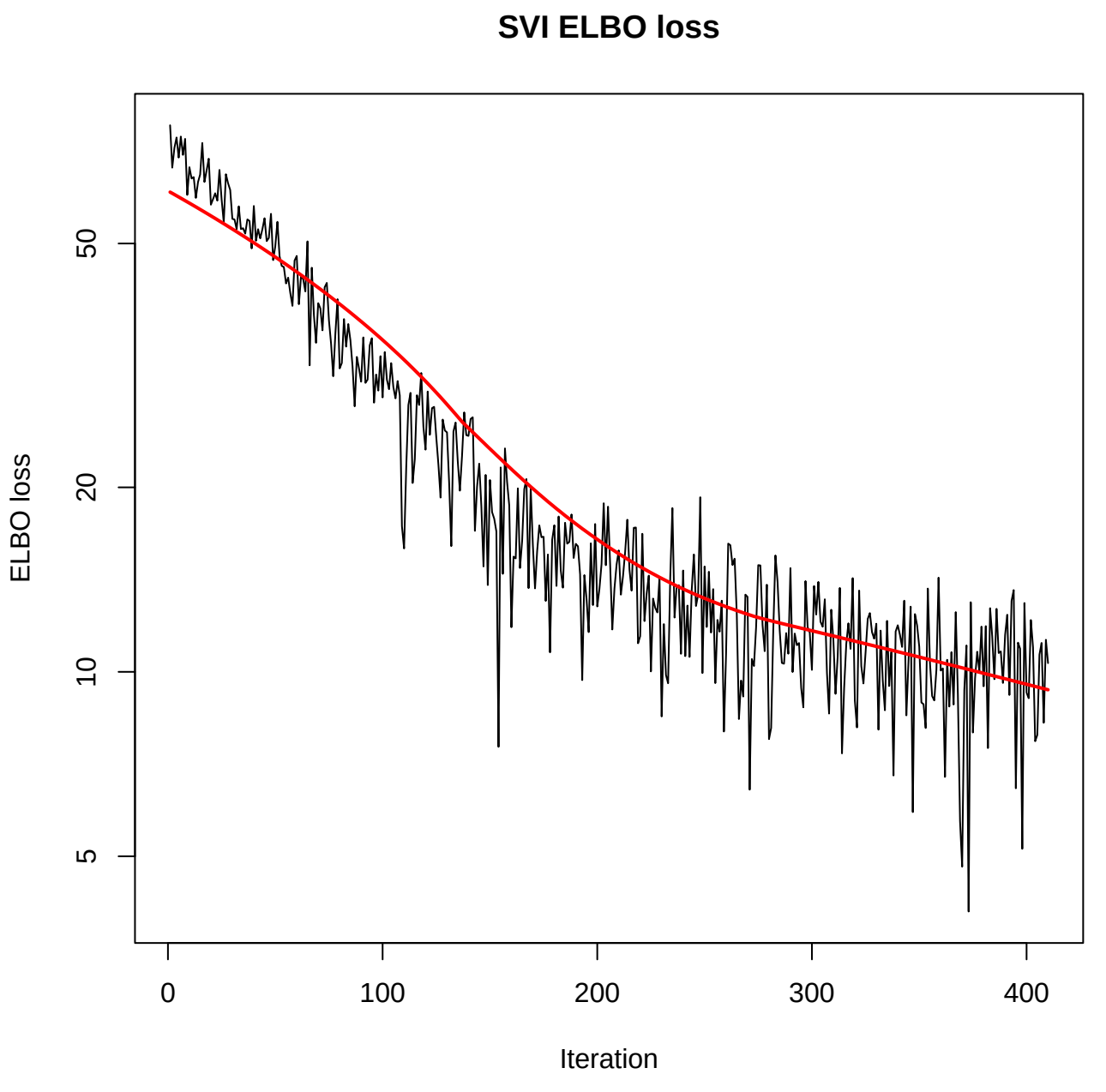
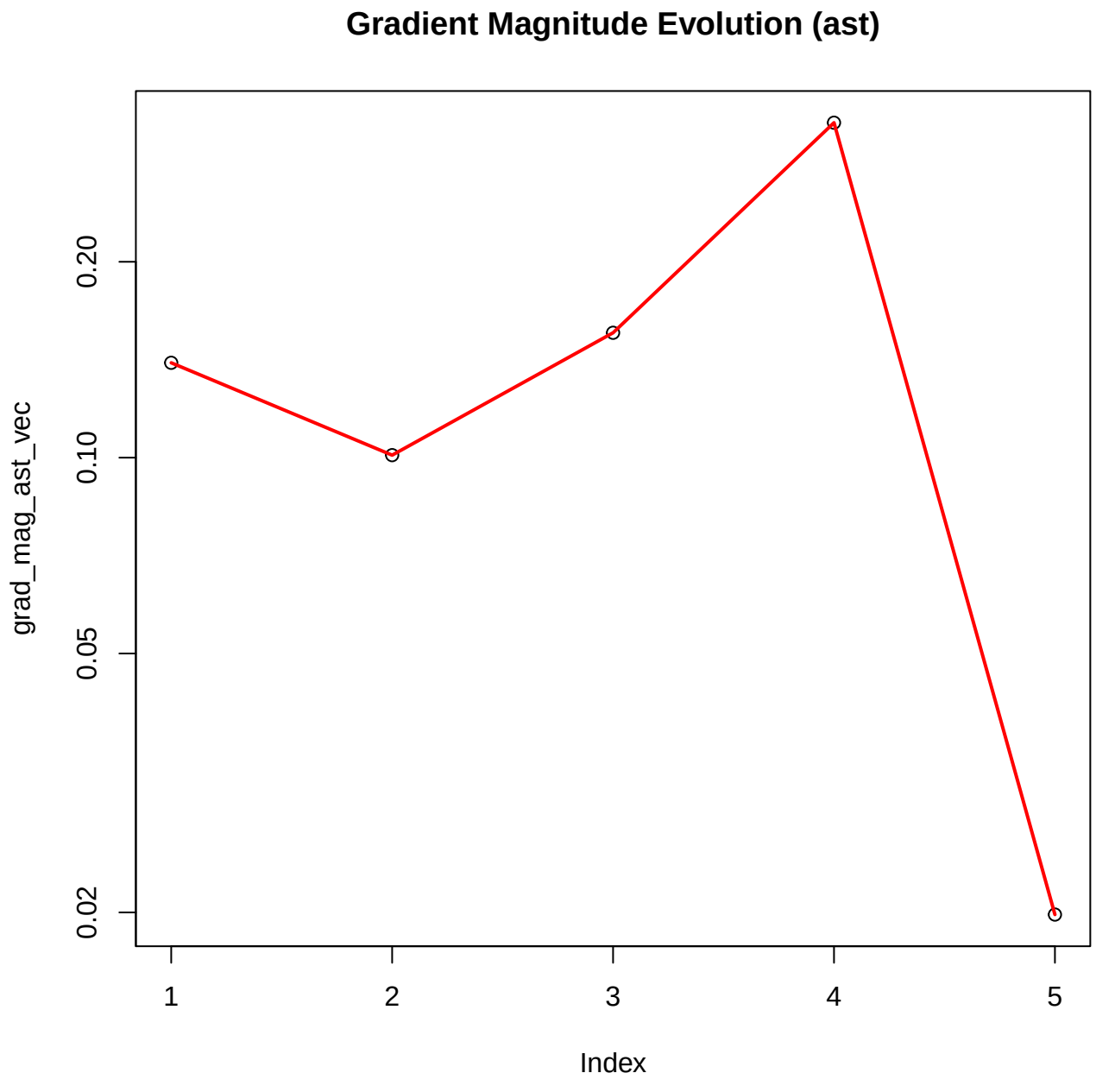
Index


SVI ELBO loss
ELBO loss
50
20
10
5
0
100
200
300
400
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
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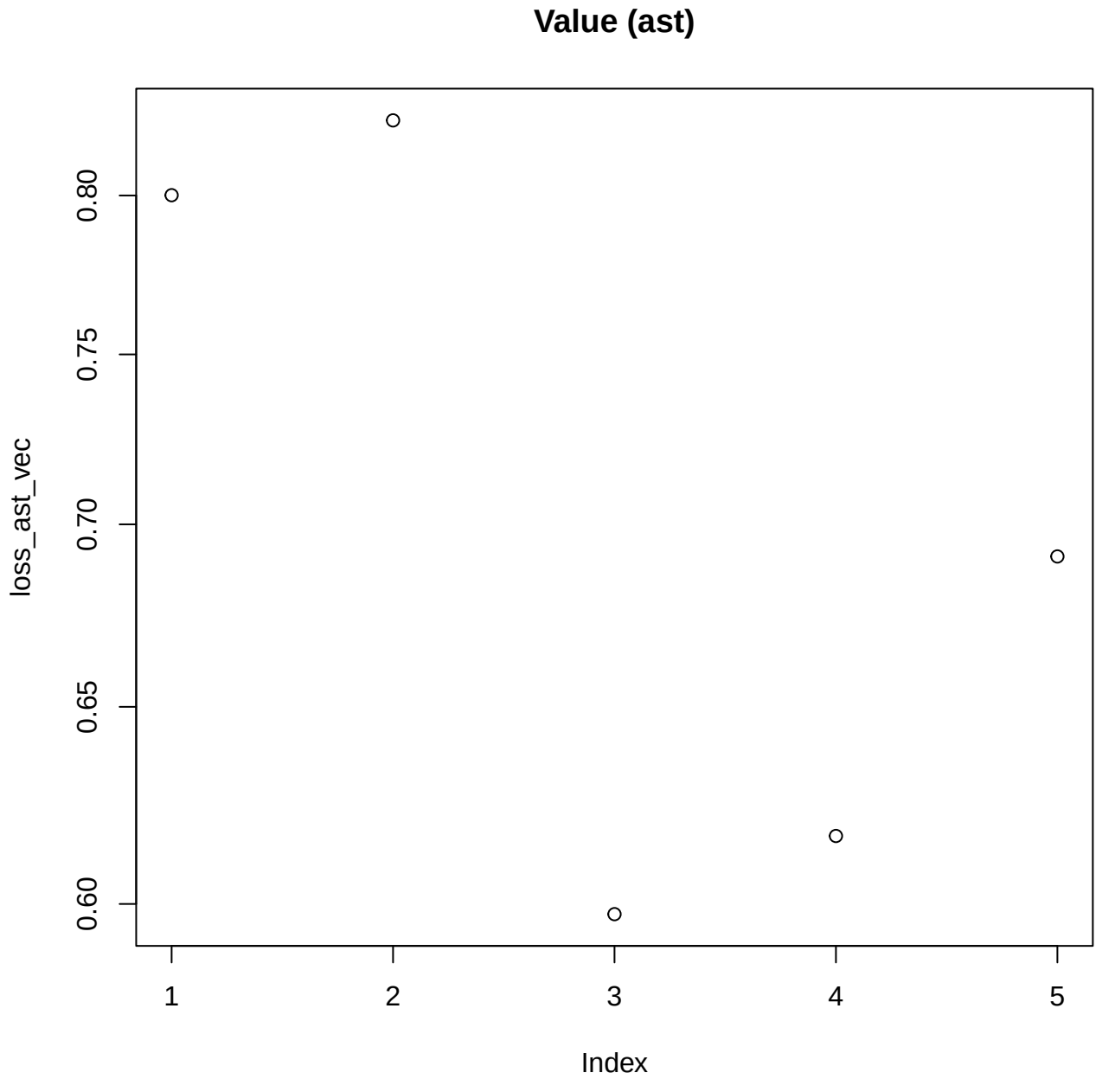
2
3
4
5
Index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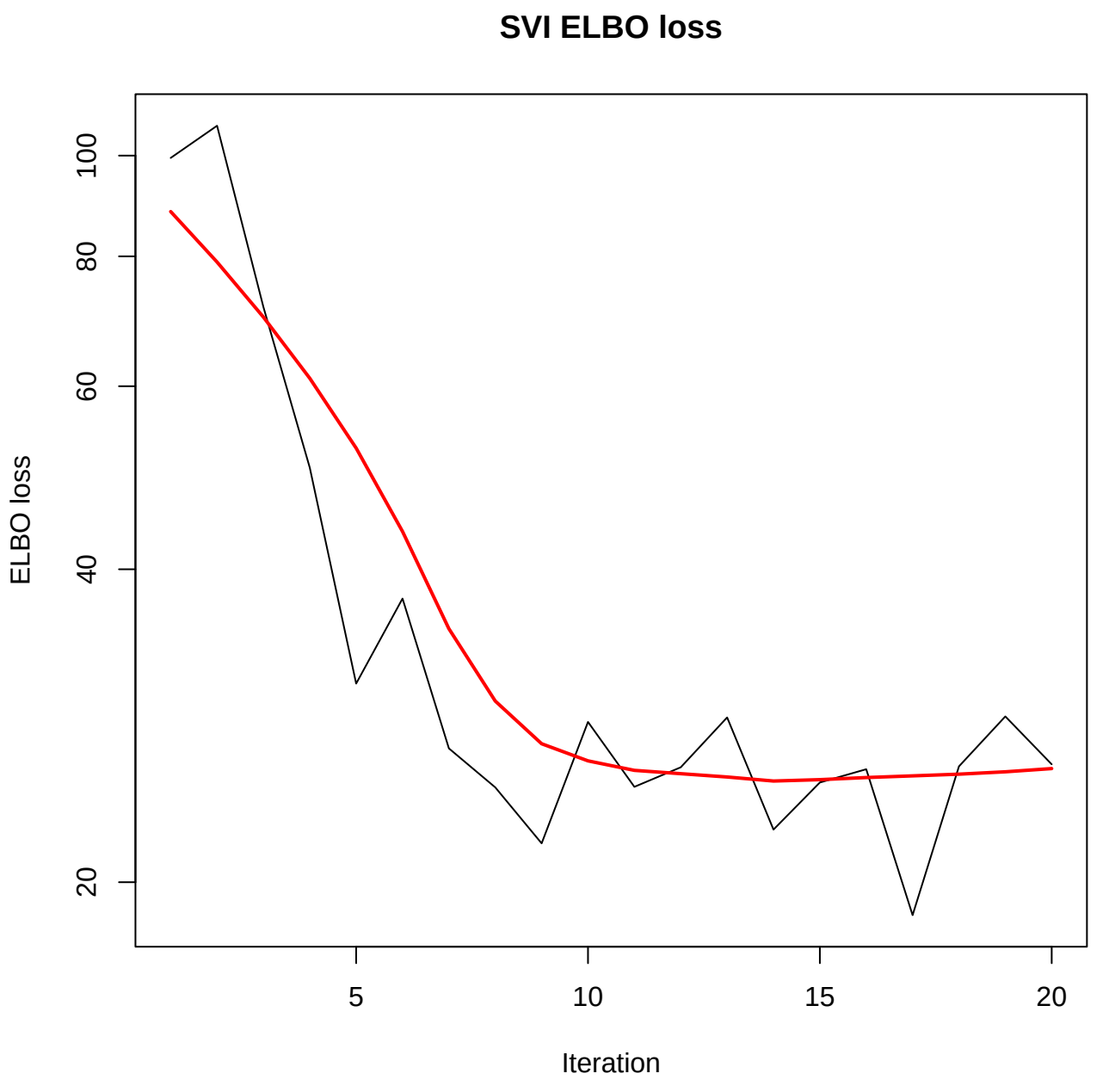
SVI ELBO loss
ELBO loss
100
80
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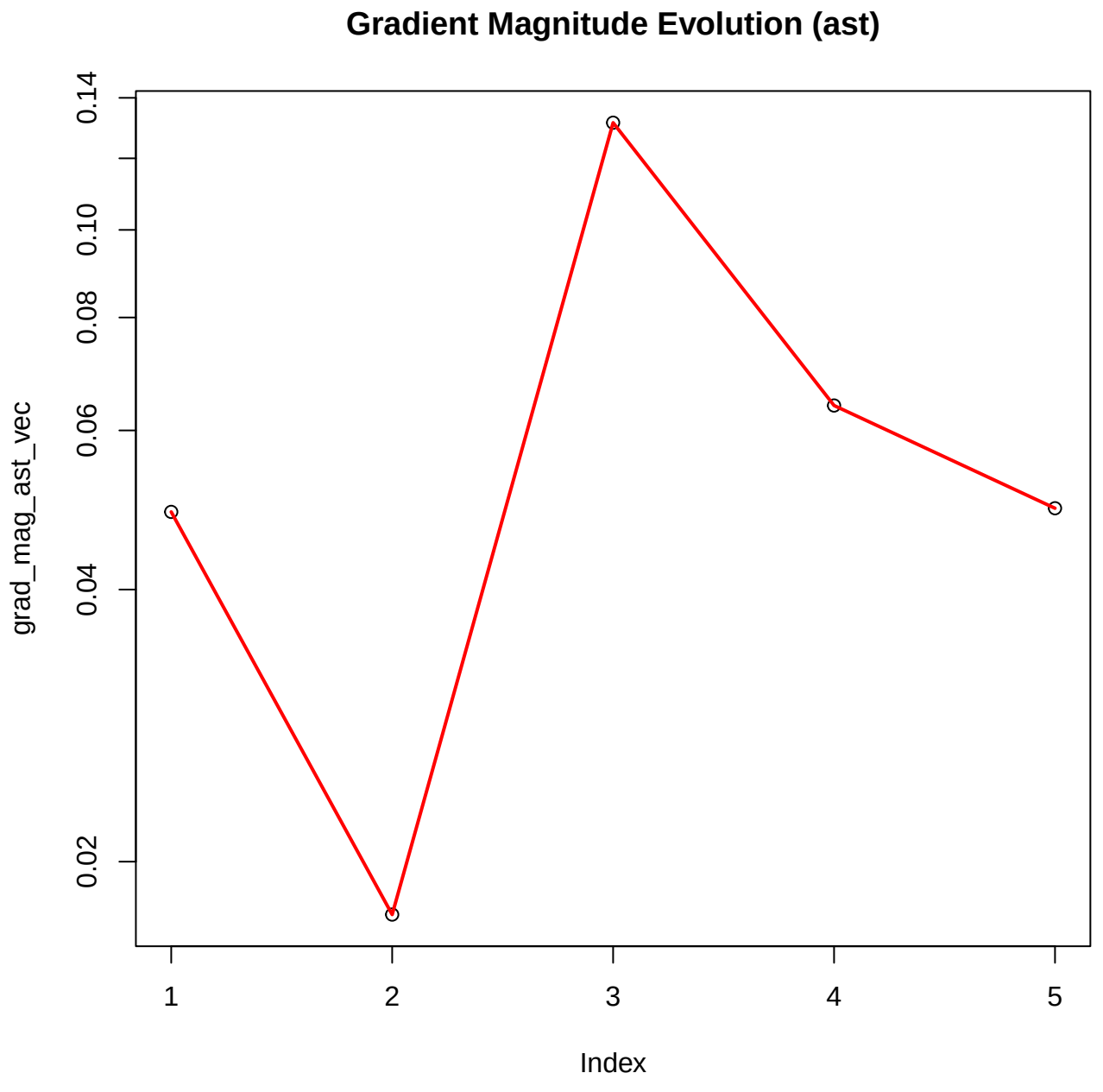
40
20
5
10
15
20
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.06
0.08
0.10
0.14
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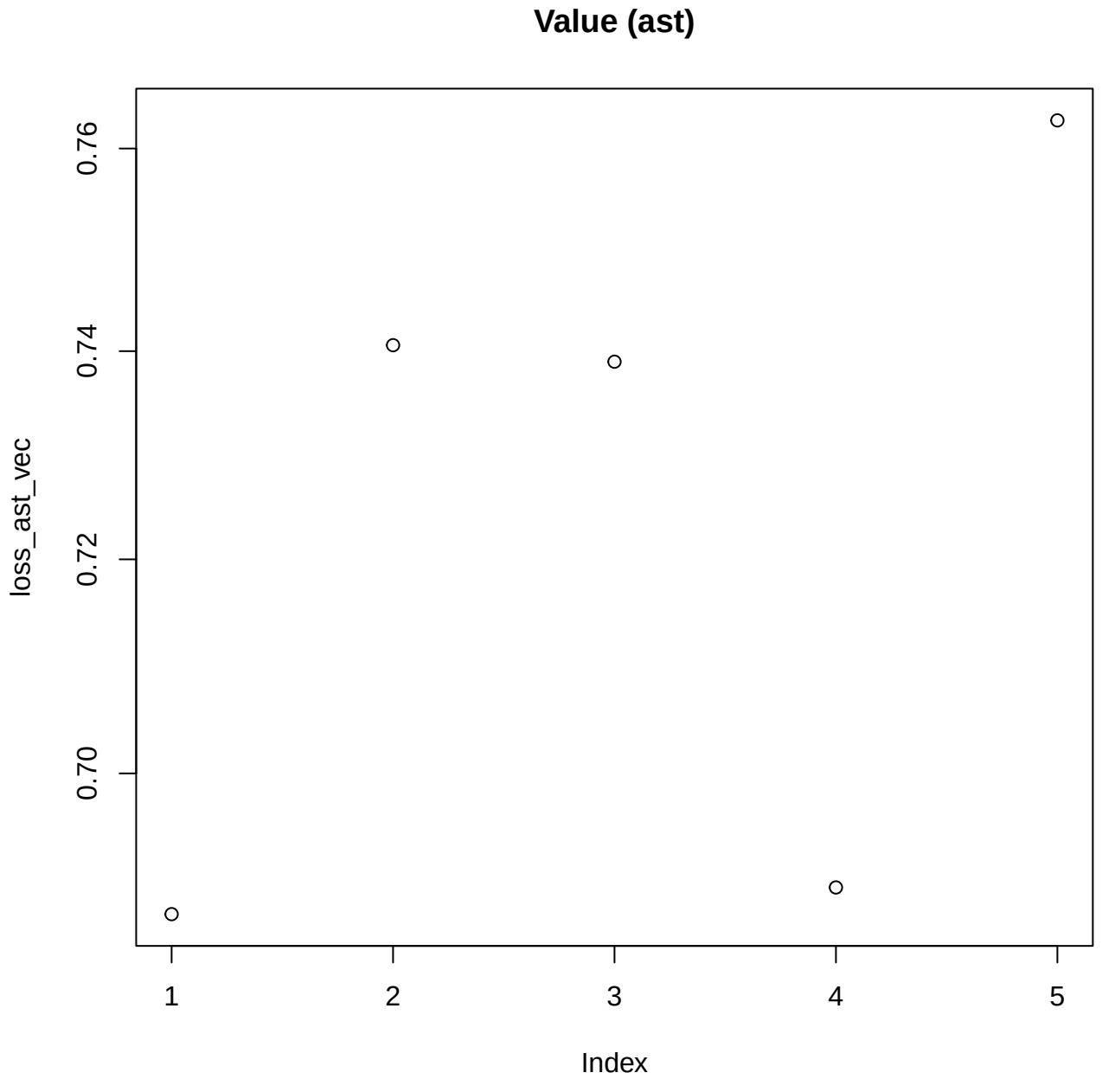
2
3
4
5
Index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5
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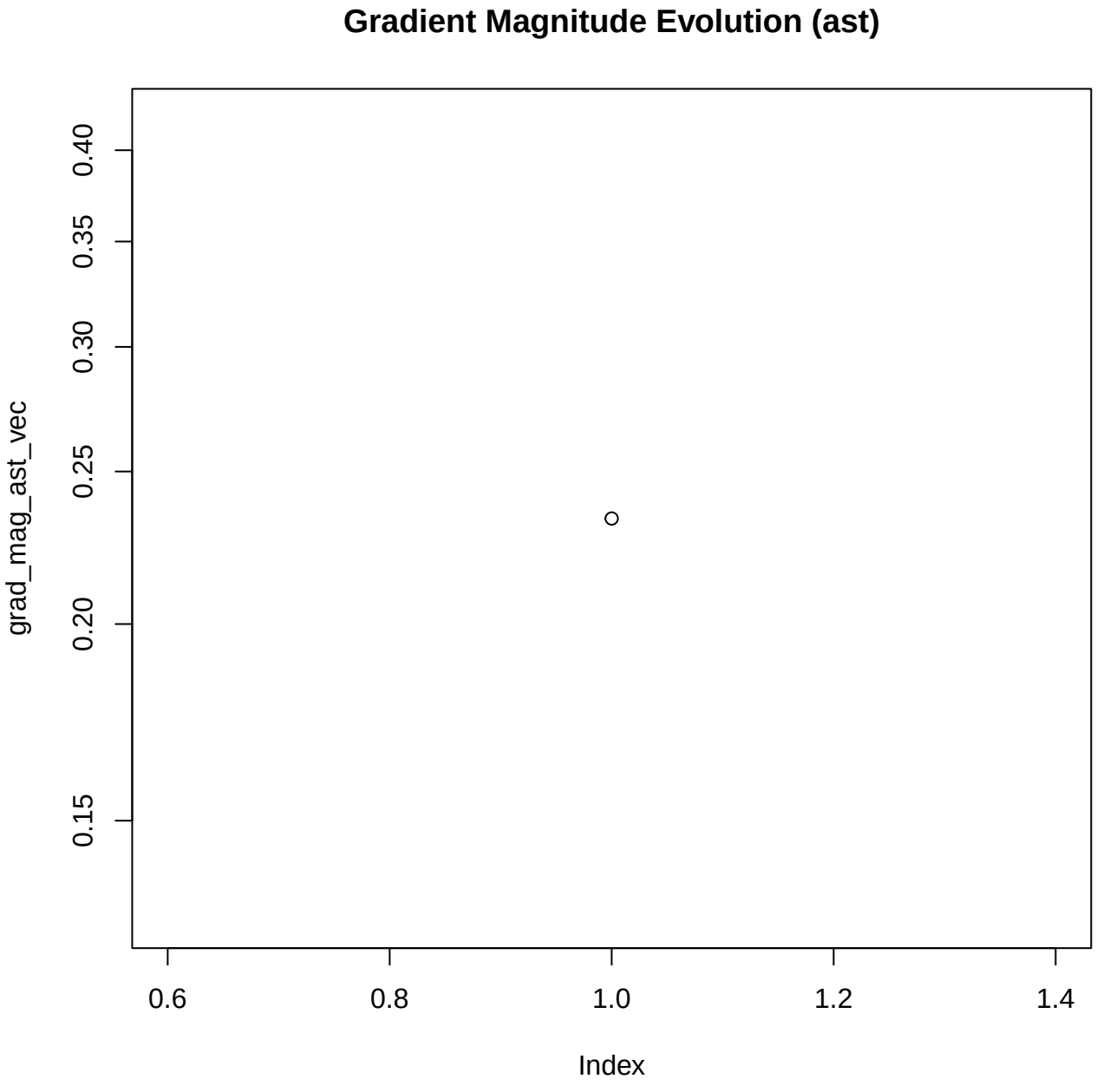
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.8
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1.2
1.4
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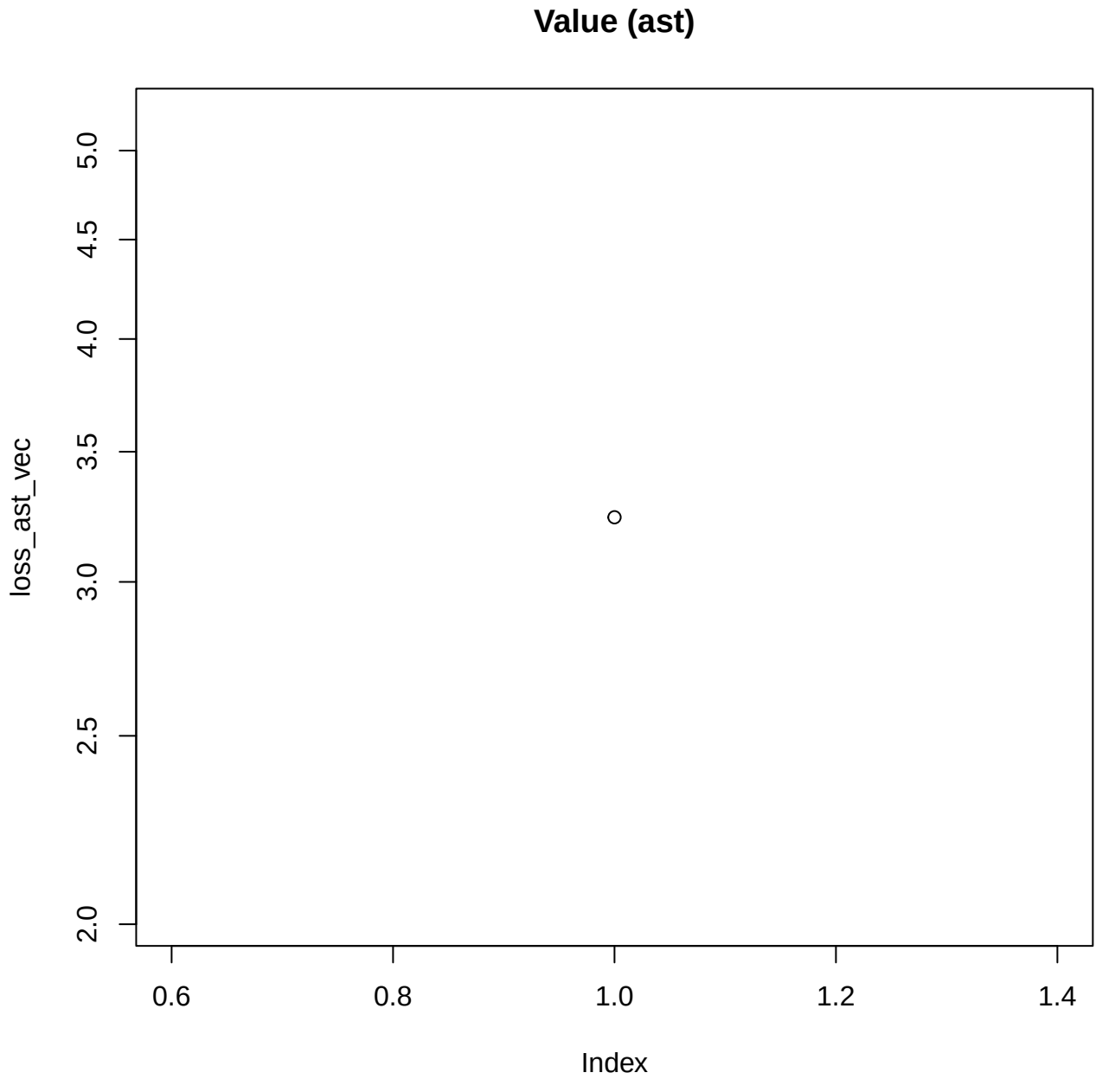
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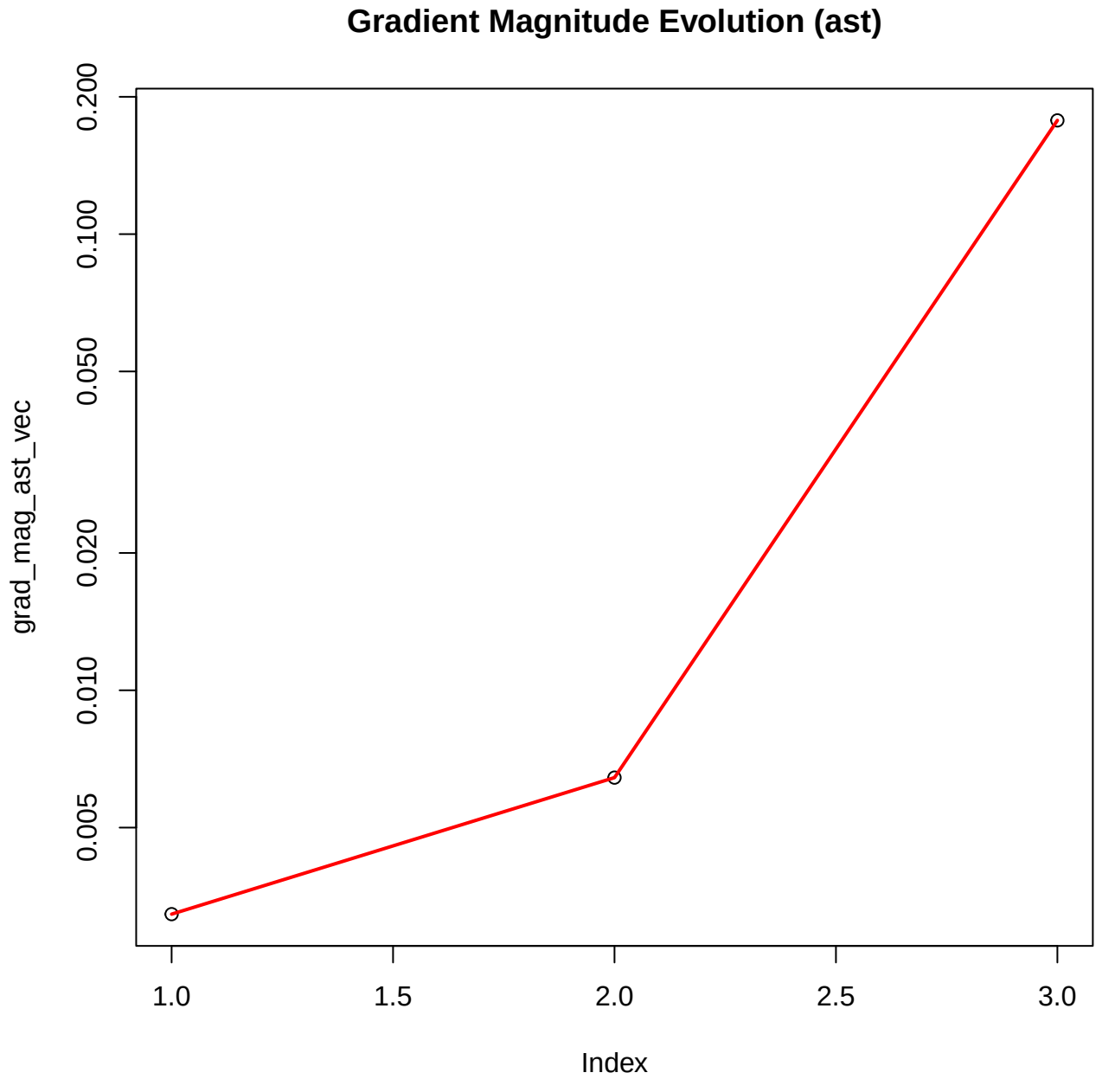
0.8
1.0
1.2
1.4
Index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grad_mag_ast_vec
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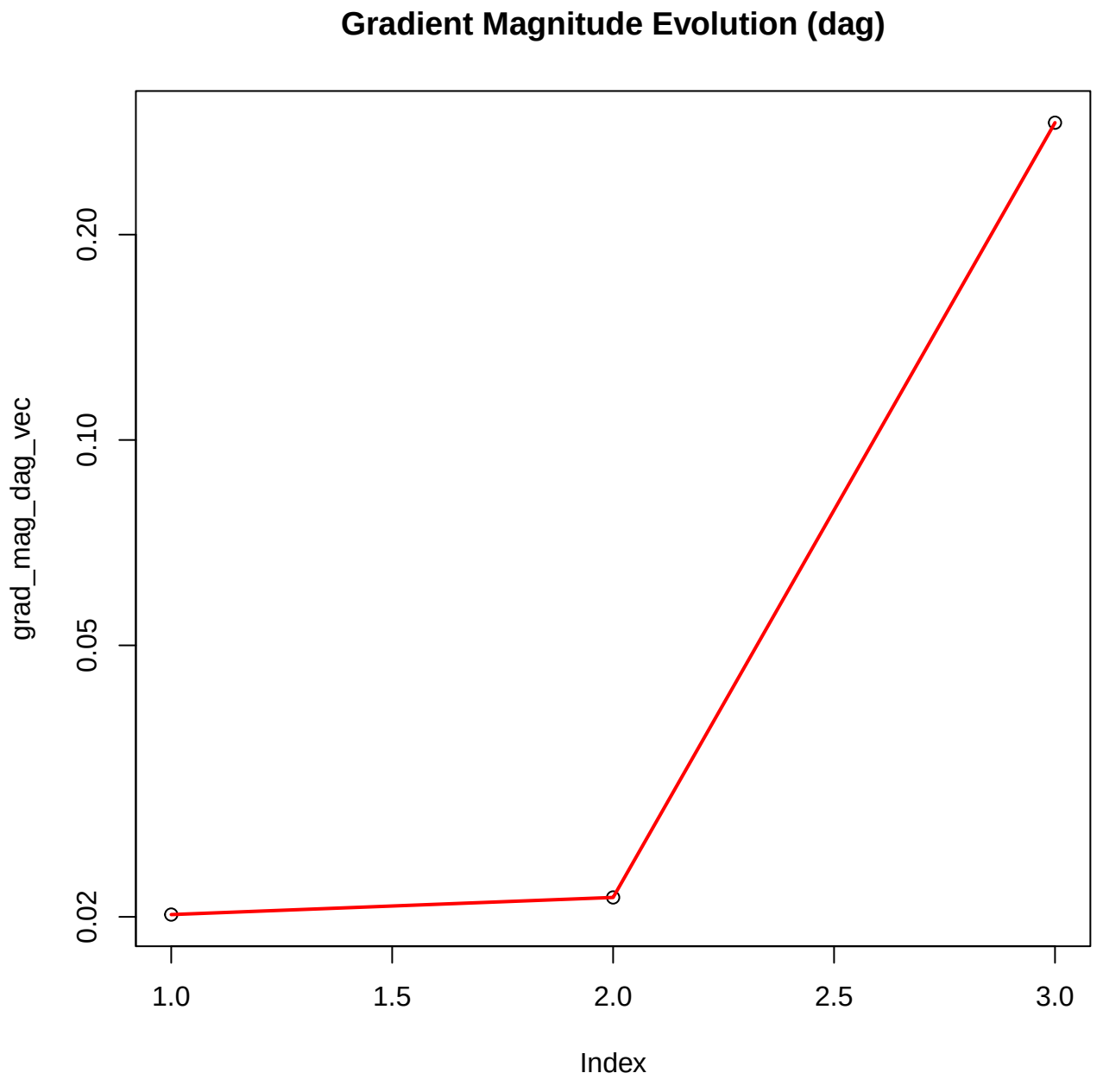
Gradient Magnitude Evolution (dag)
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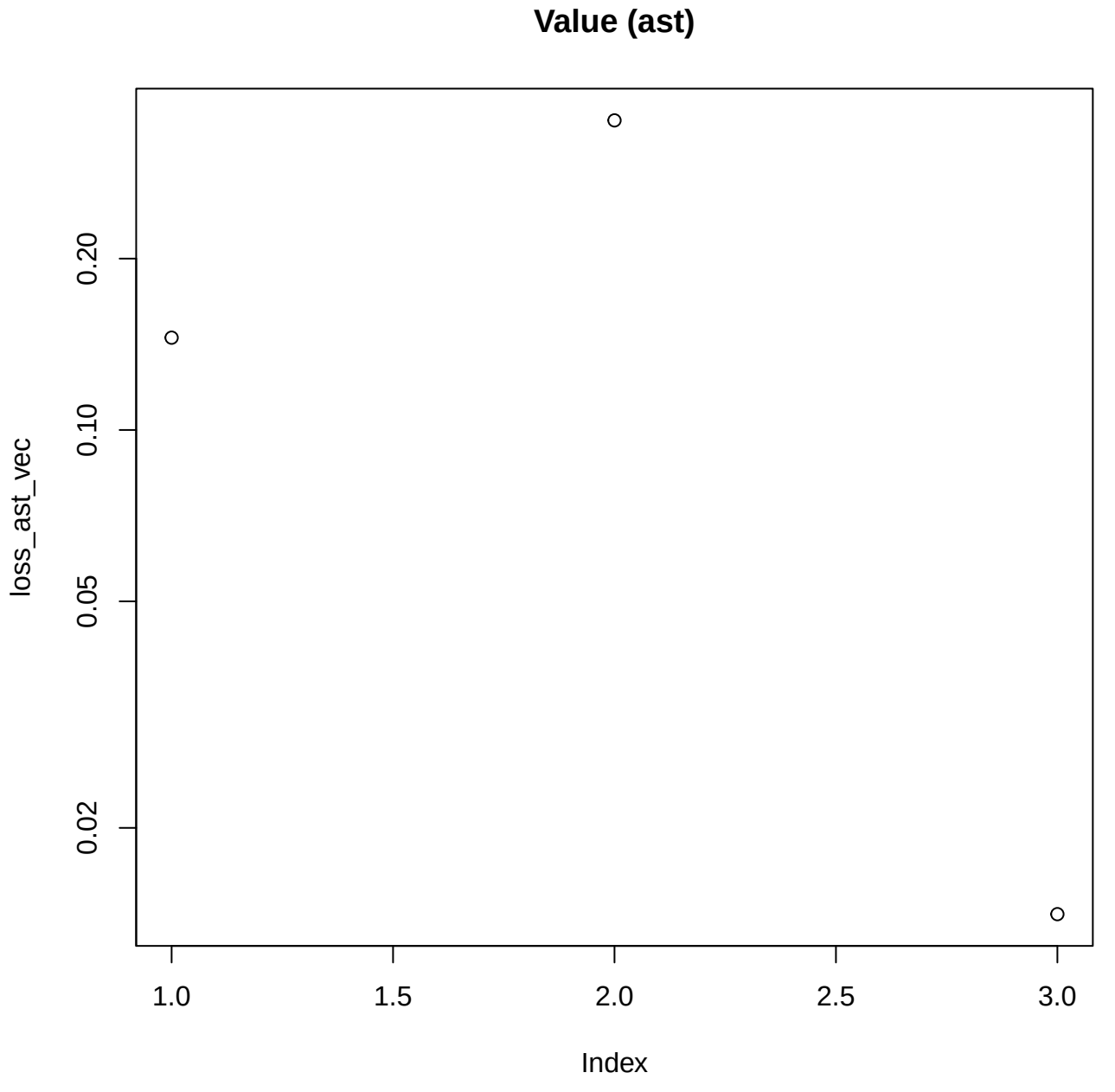
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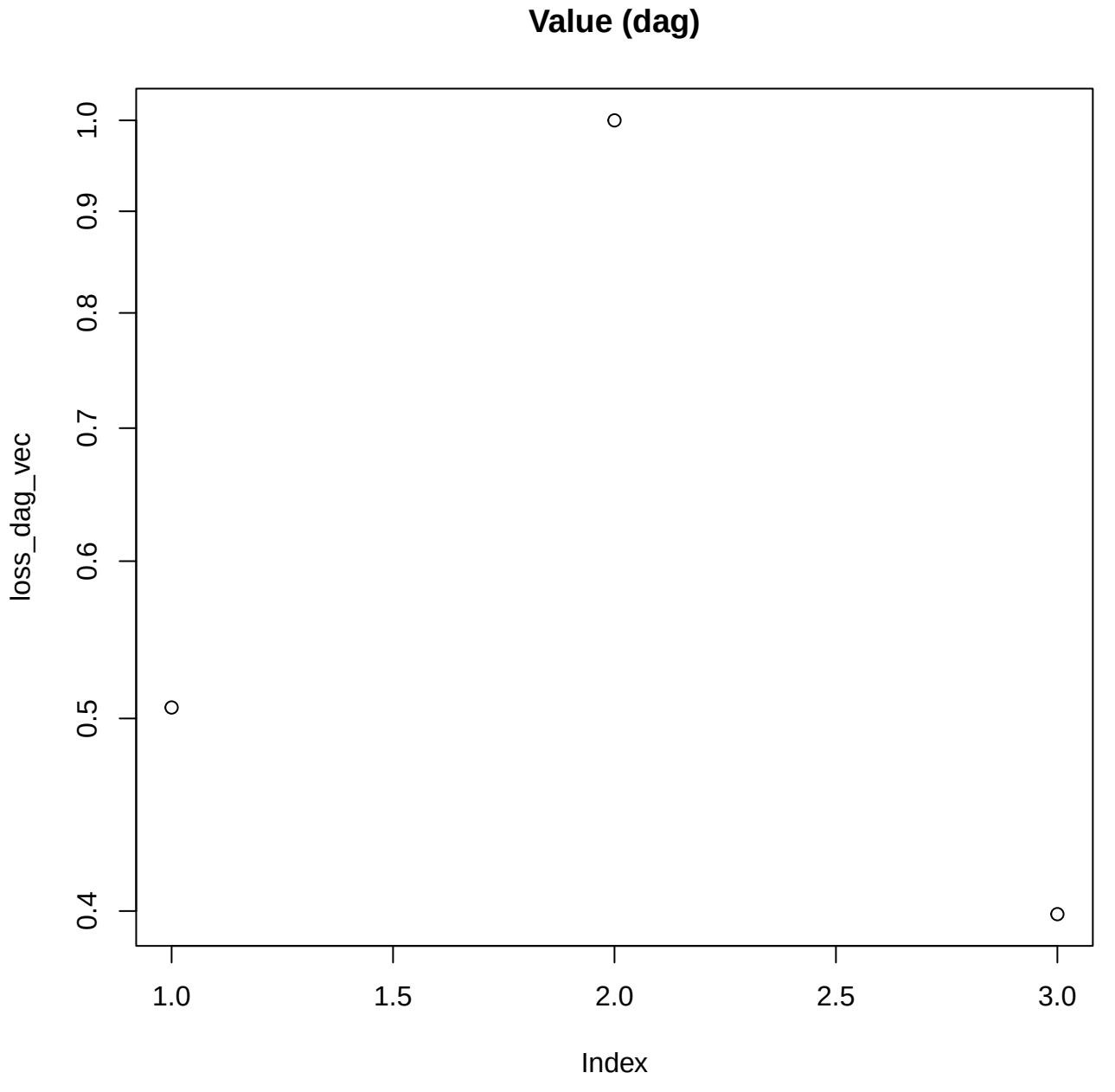
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3.0
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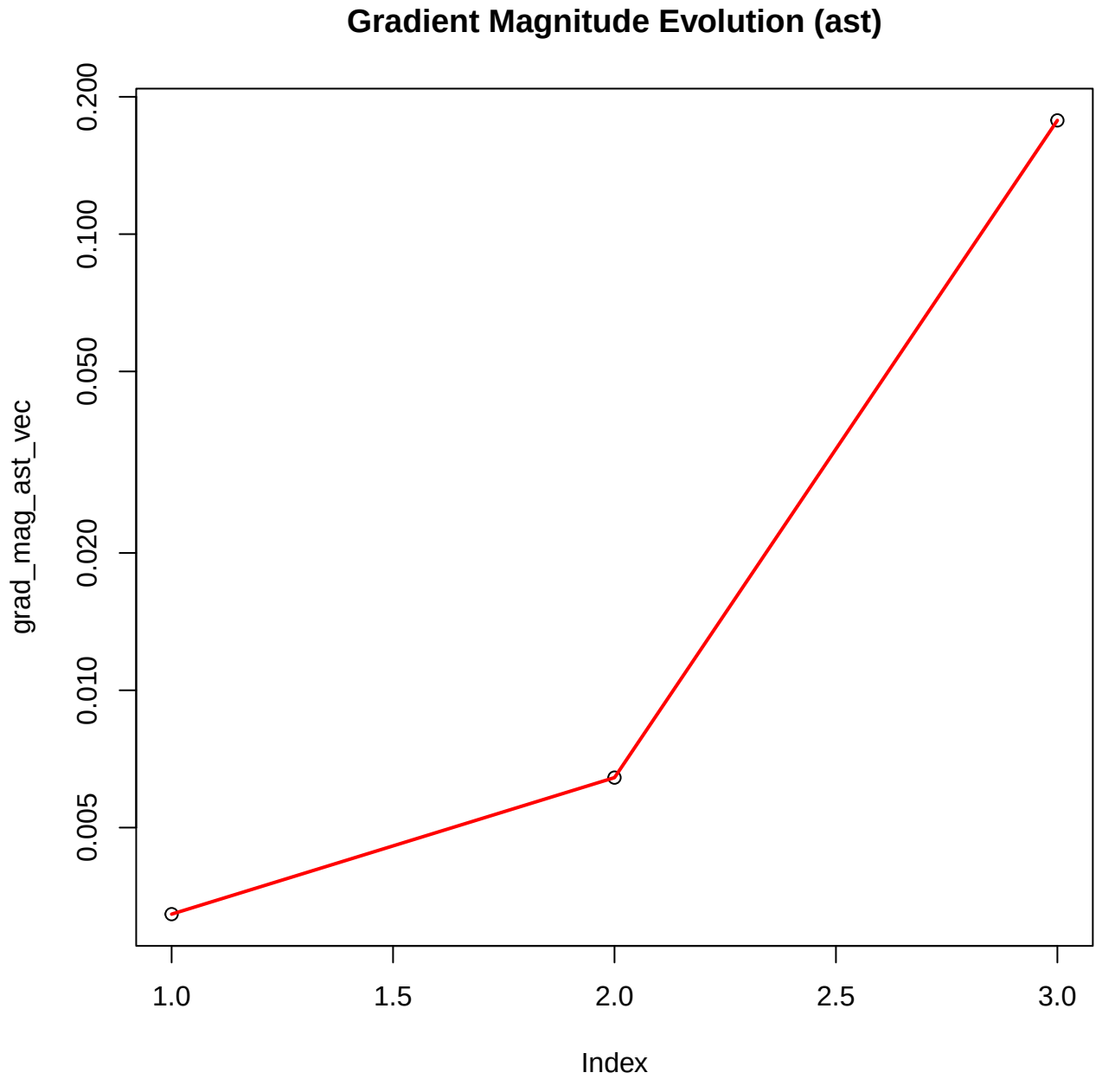
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.010
0.020
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0.200
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1.5
2.0
2.5
3.0
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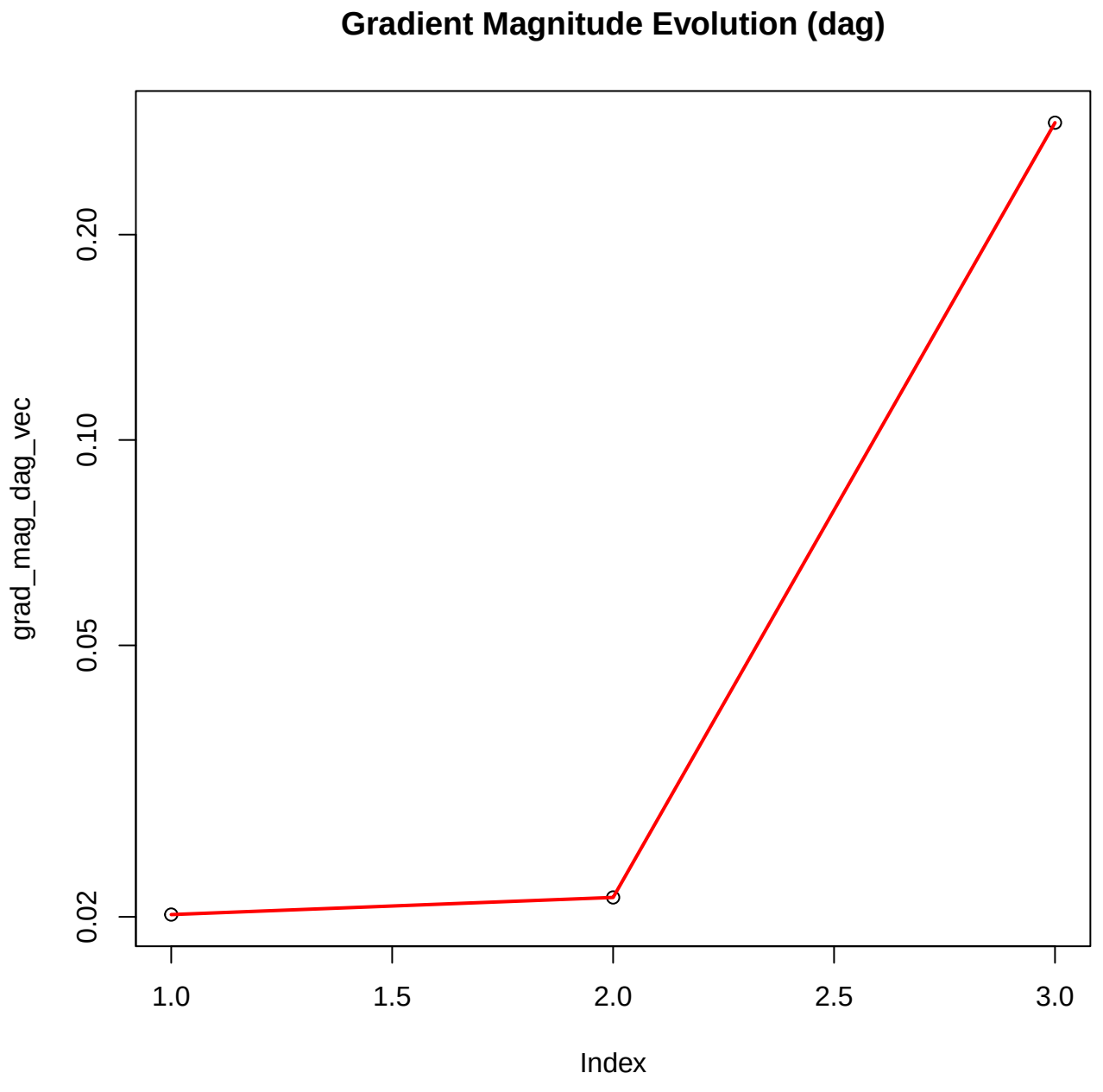
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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0.10
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2.5
3.0
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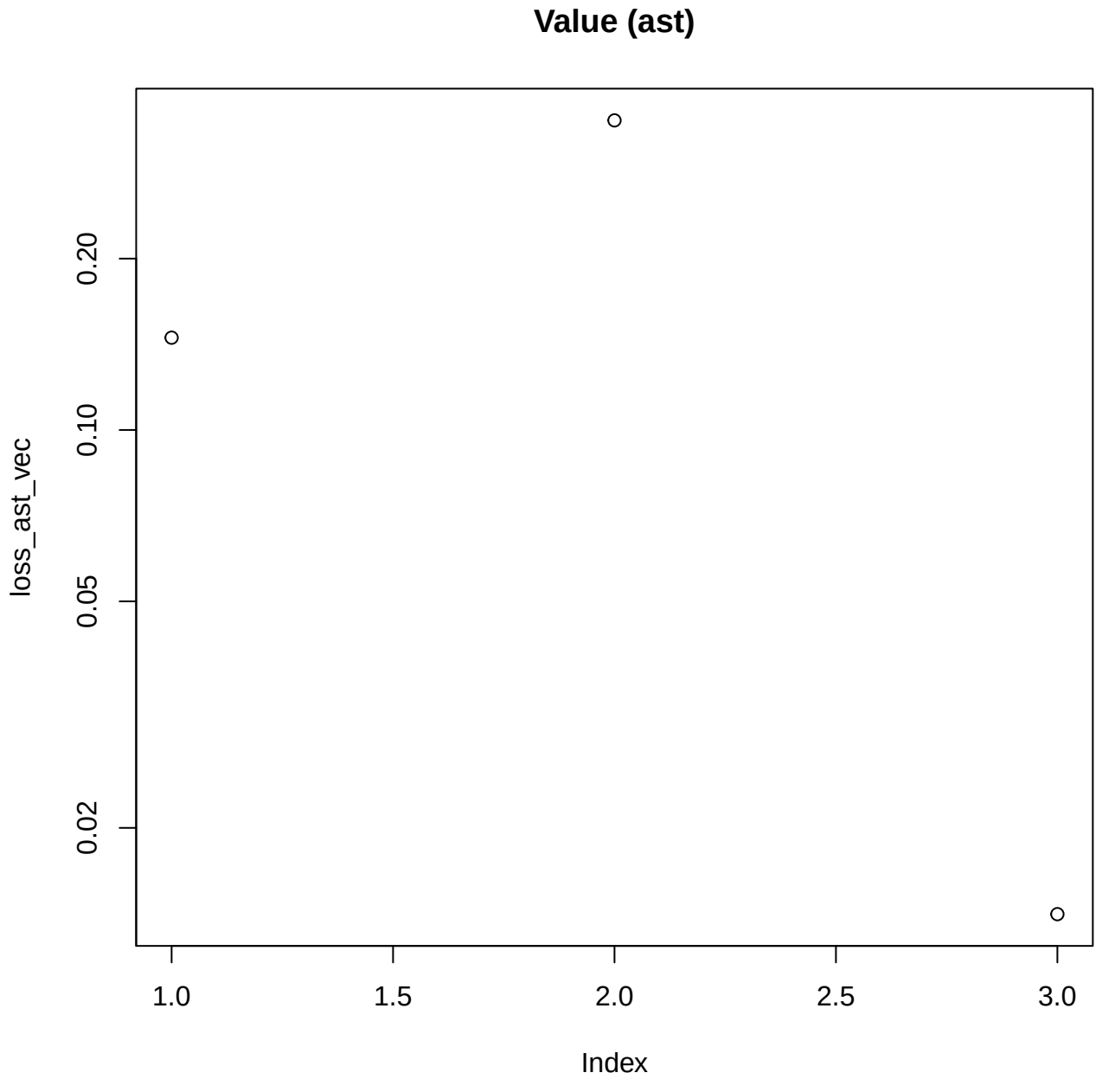
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0.20
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2.5
3.0
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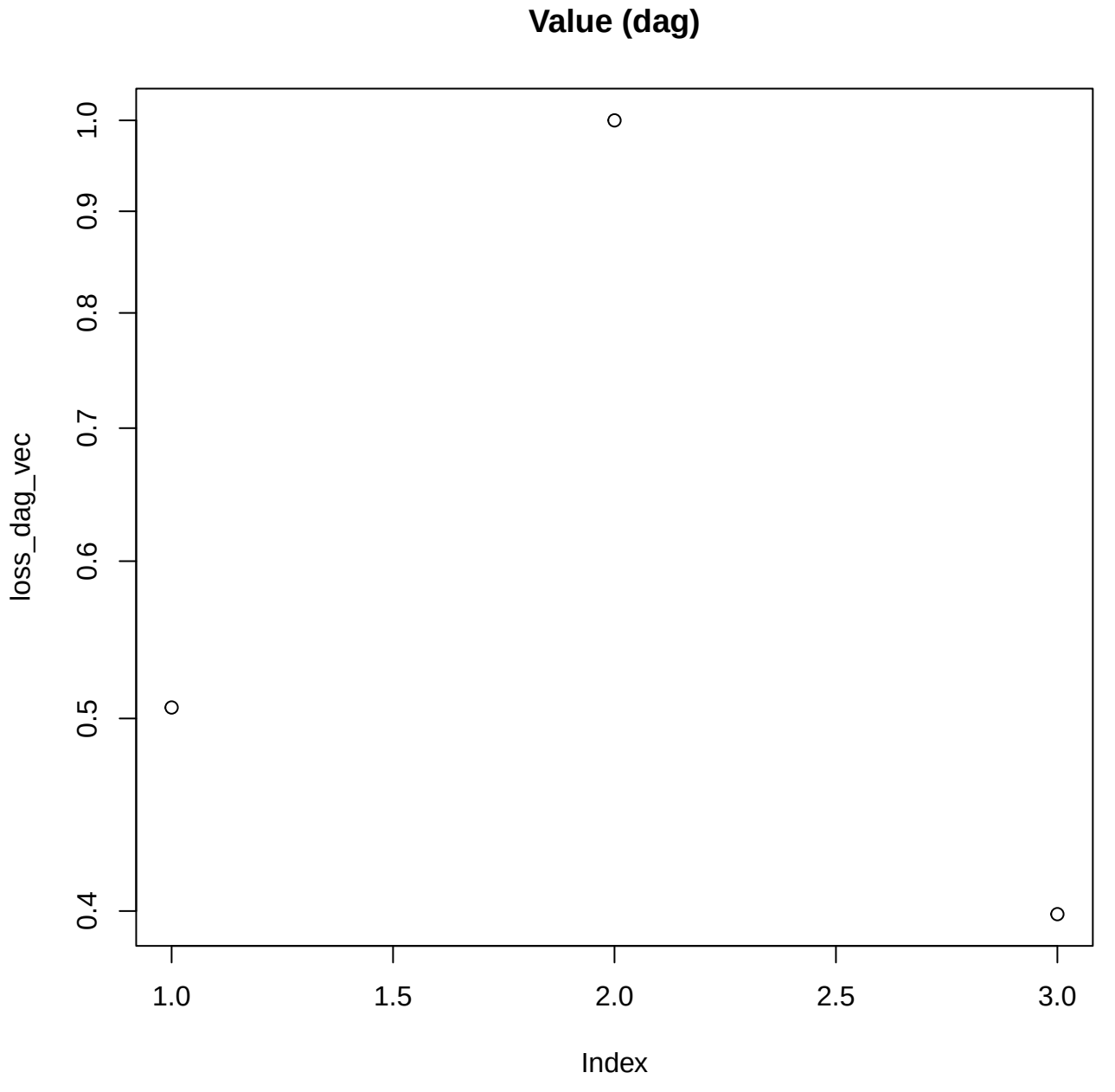
Value (dag)
loss_dag_vec
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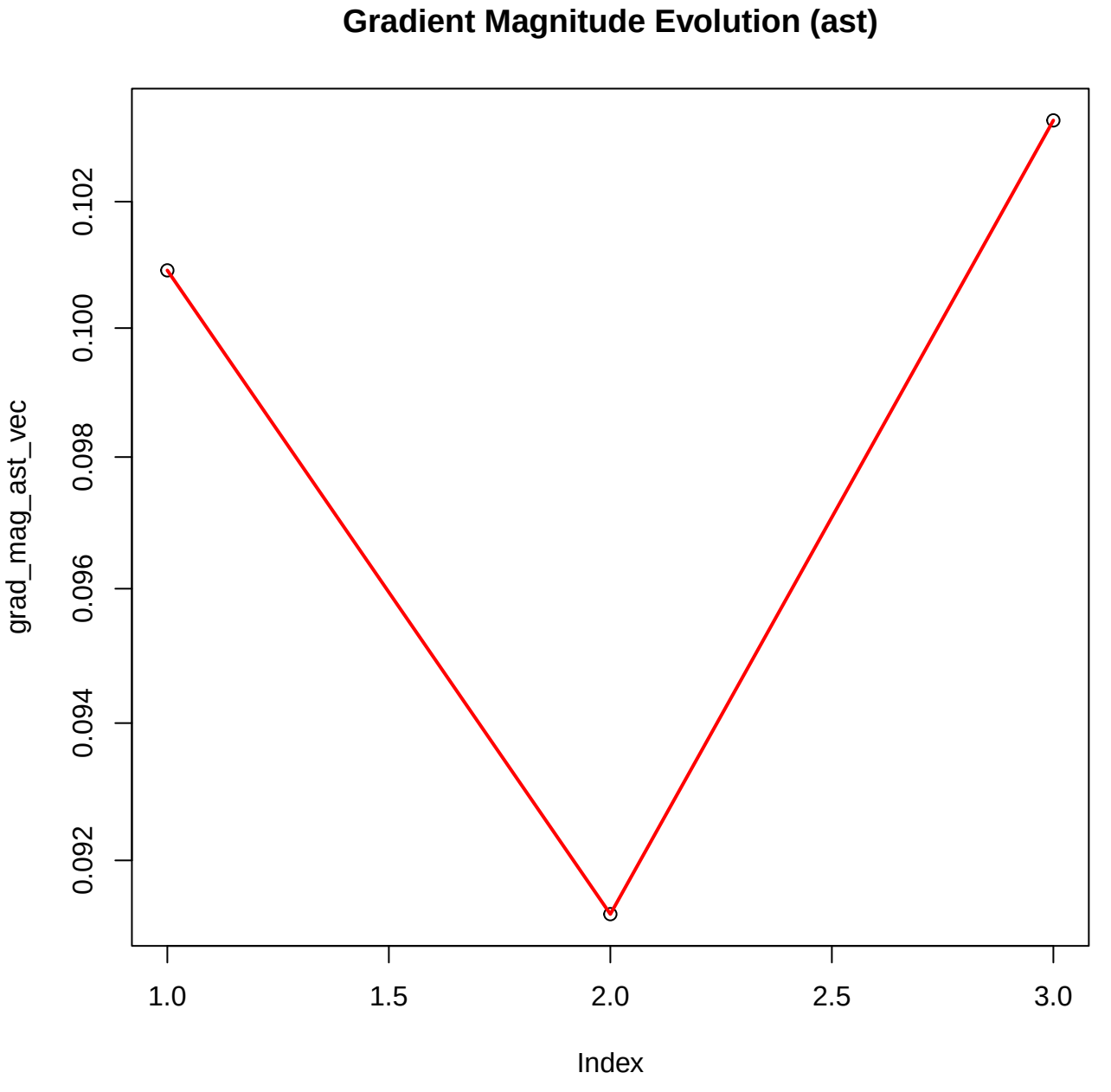
3.0
Index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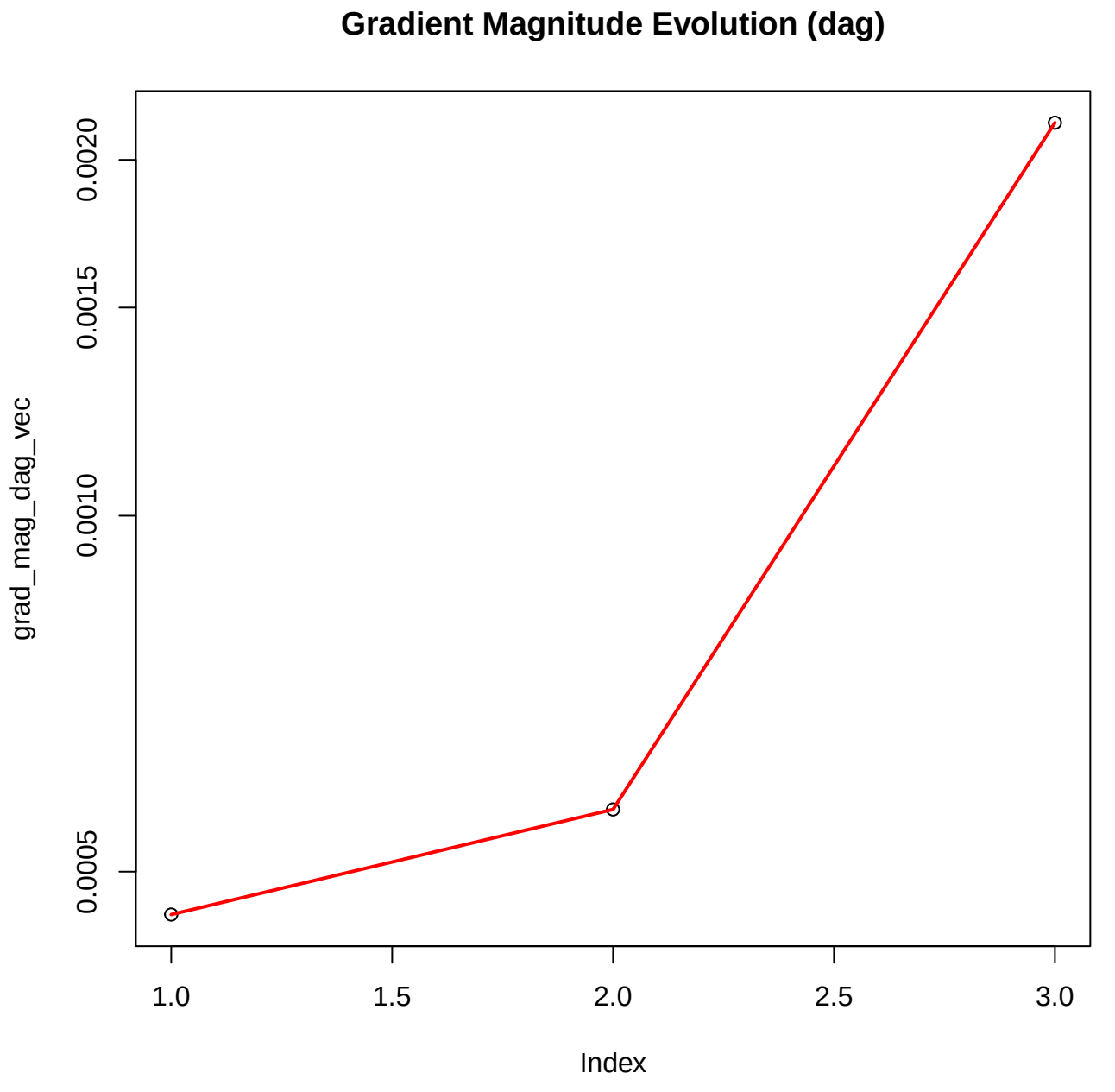
Gradient Magnitude Evolution (dag)
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2.5
3.0
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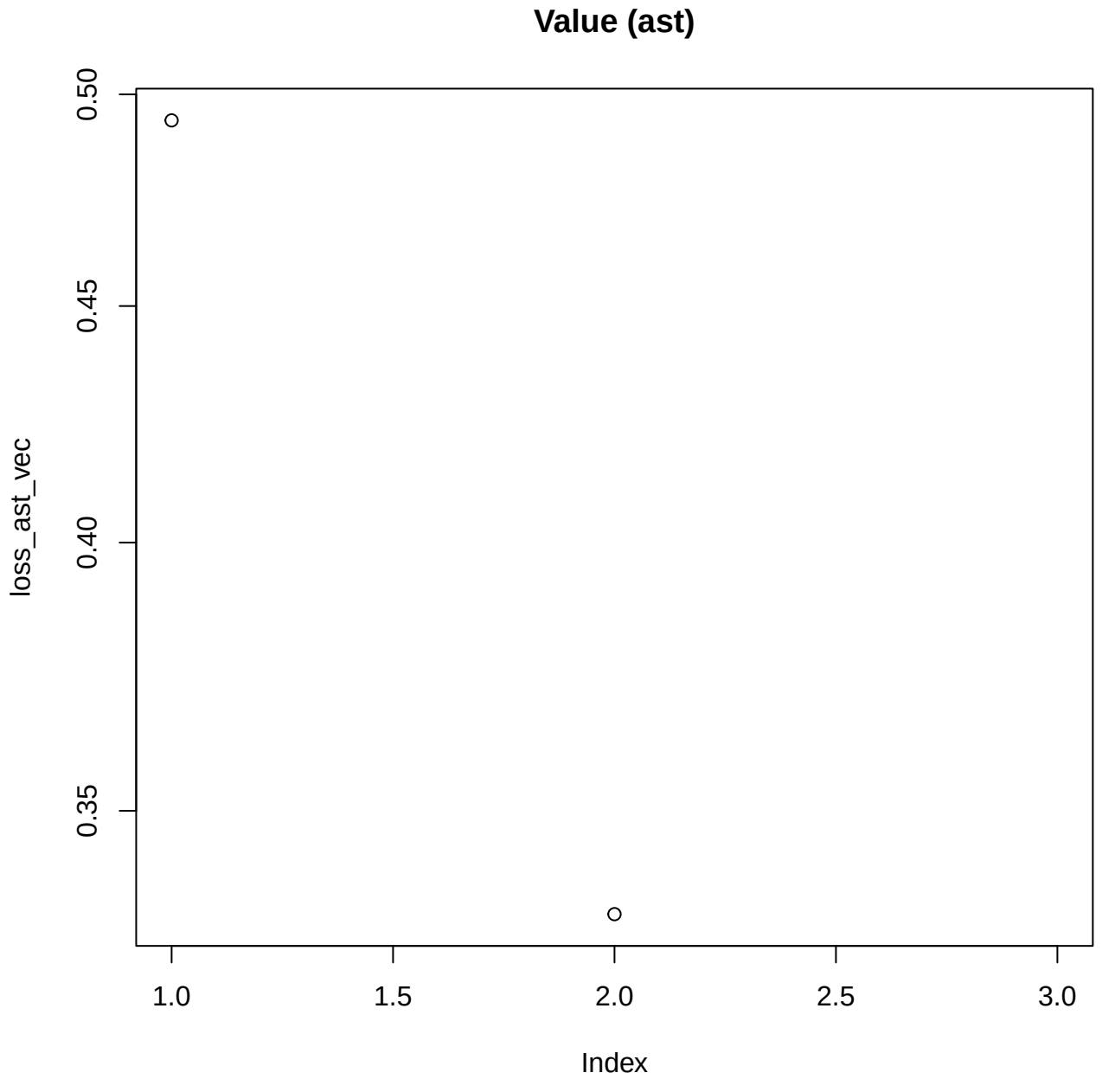
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2.5
3.0
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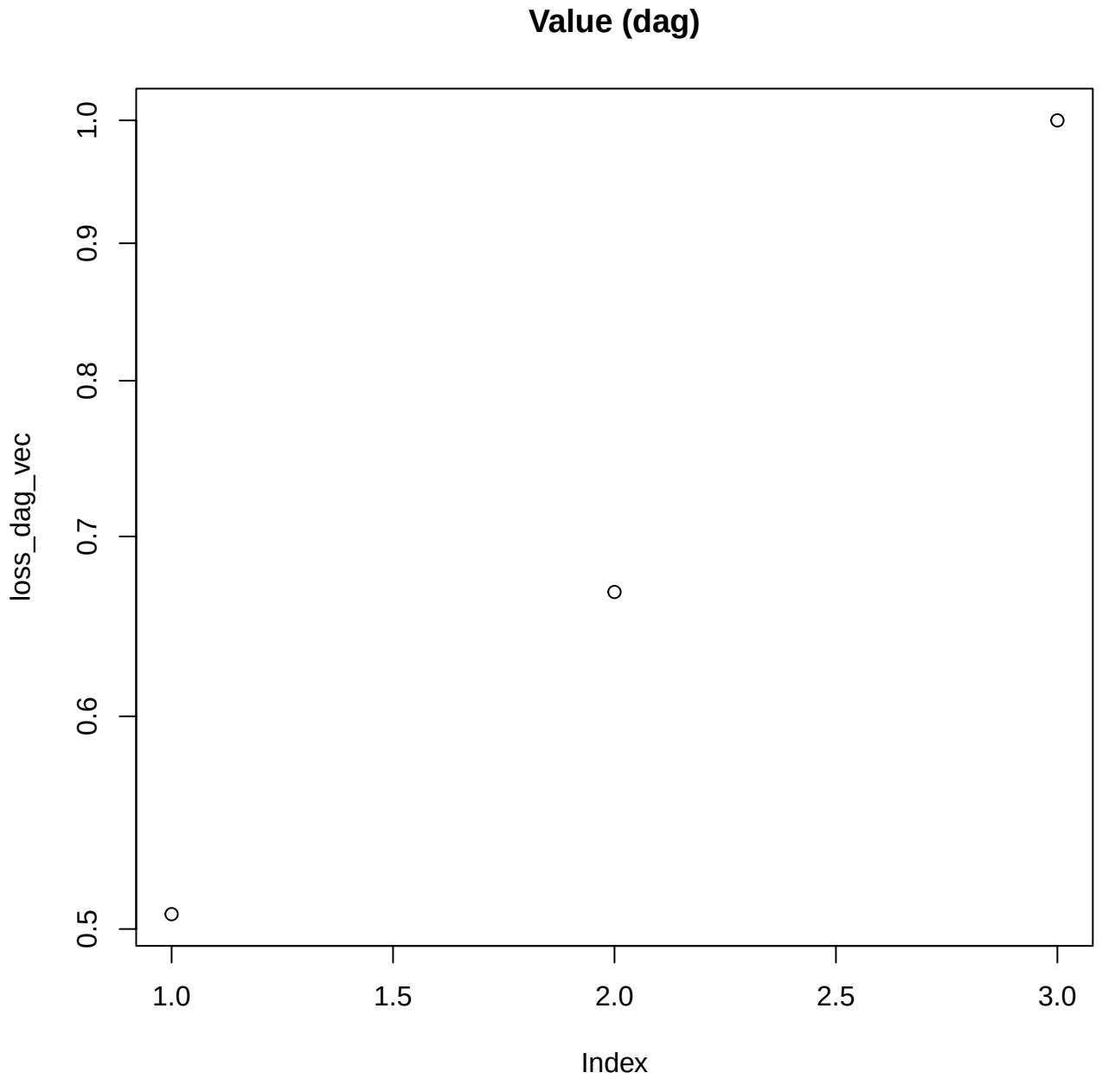
Value (dag)
loss_dag_vec
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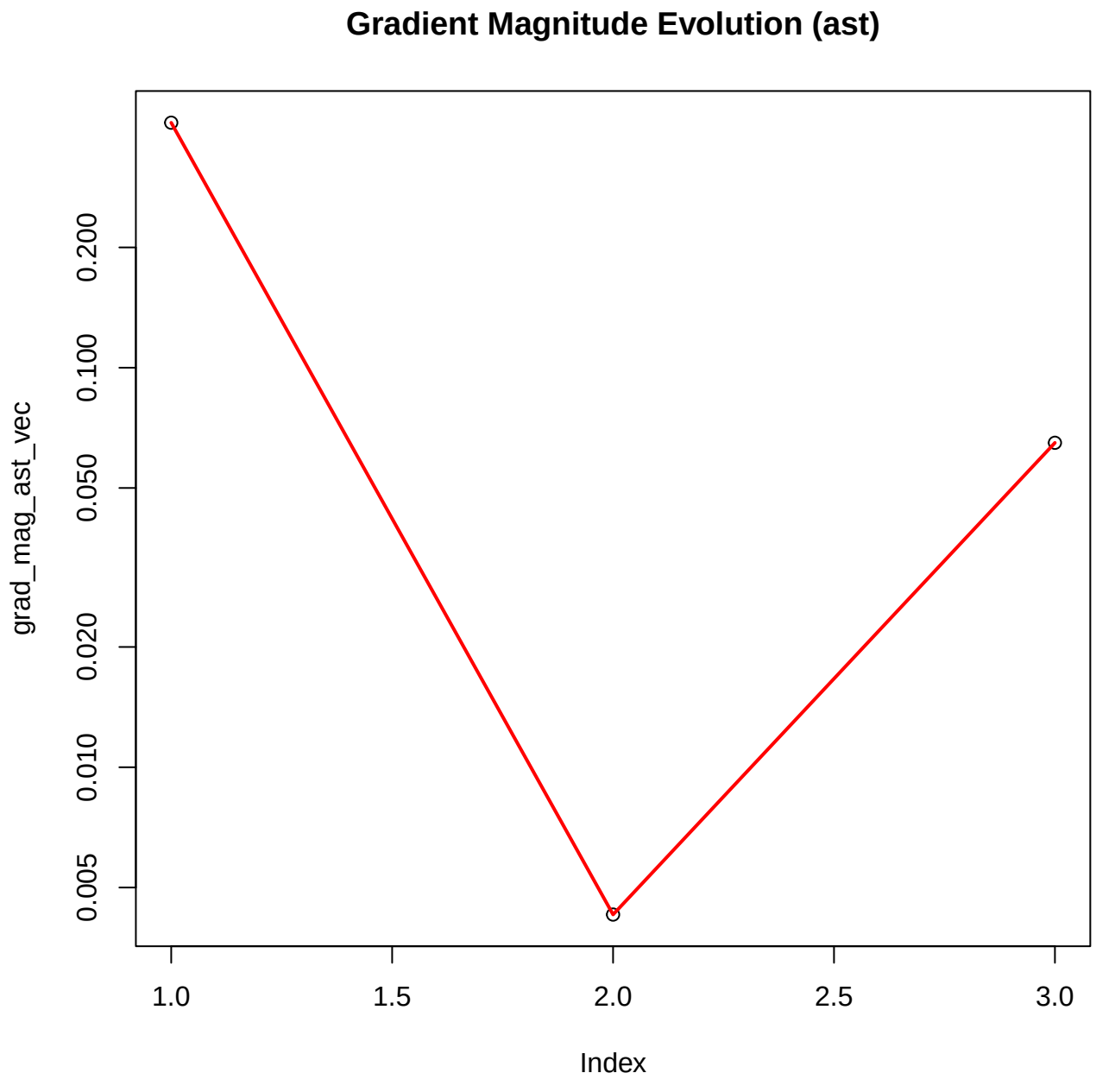
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.020
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1.5
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2.5
3.0
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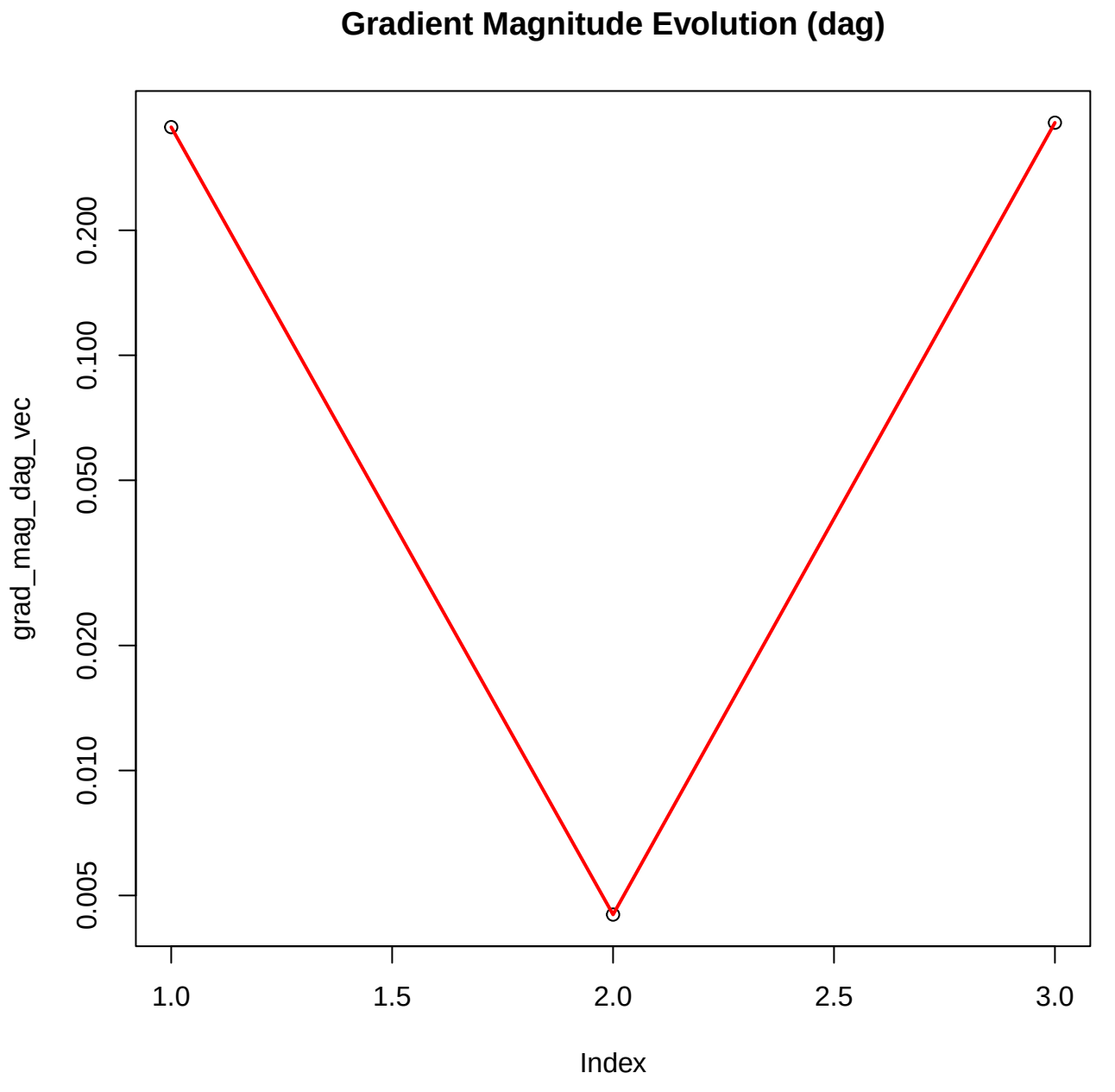
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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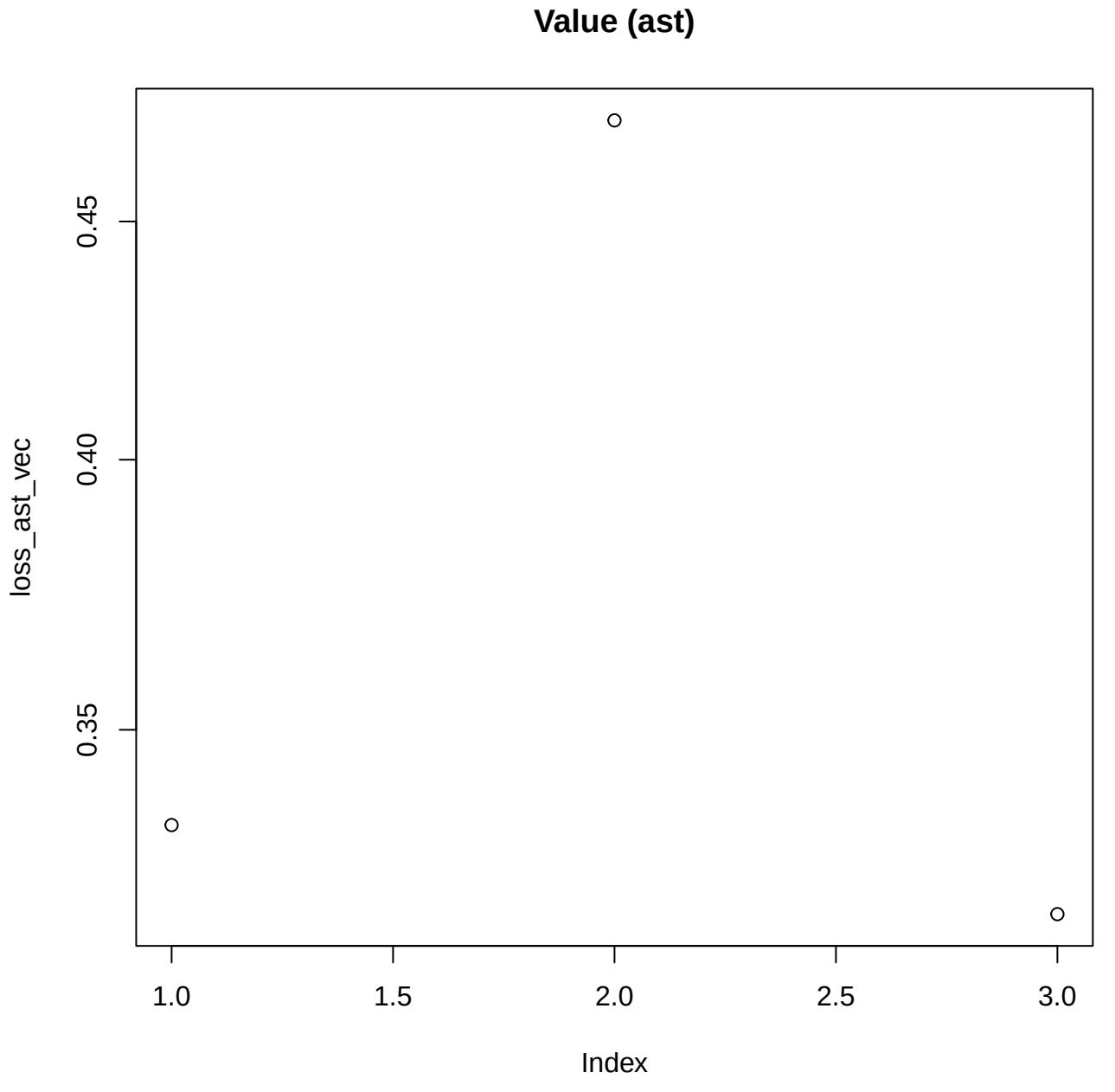
3.0
Index


Value (ast)
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3.0
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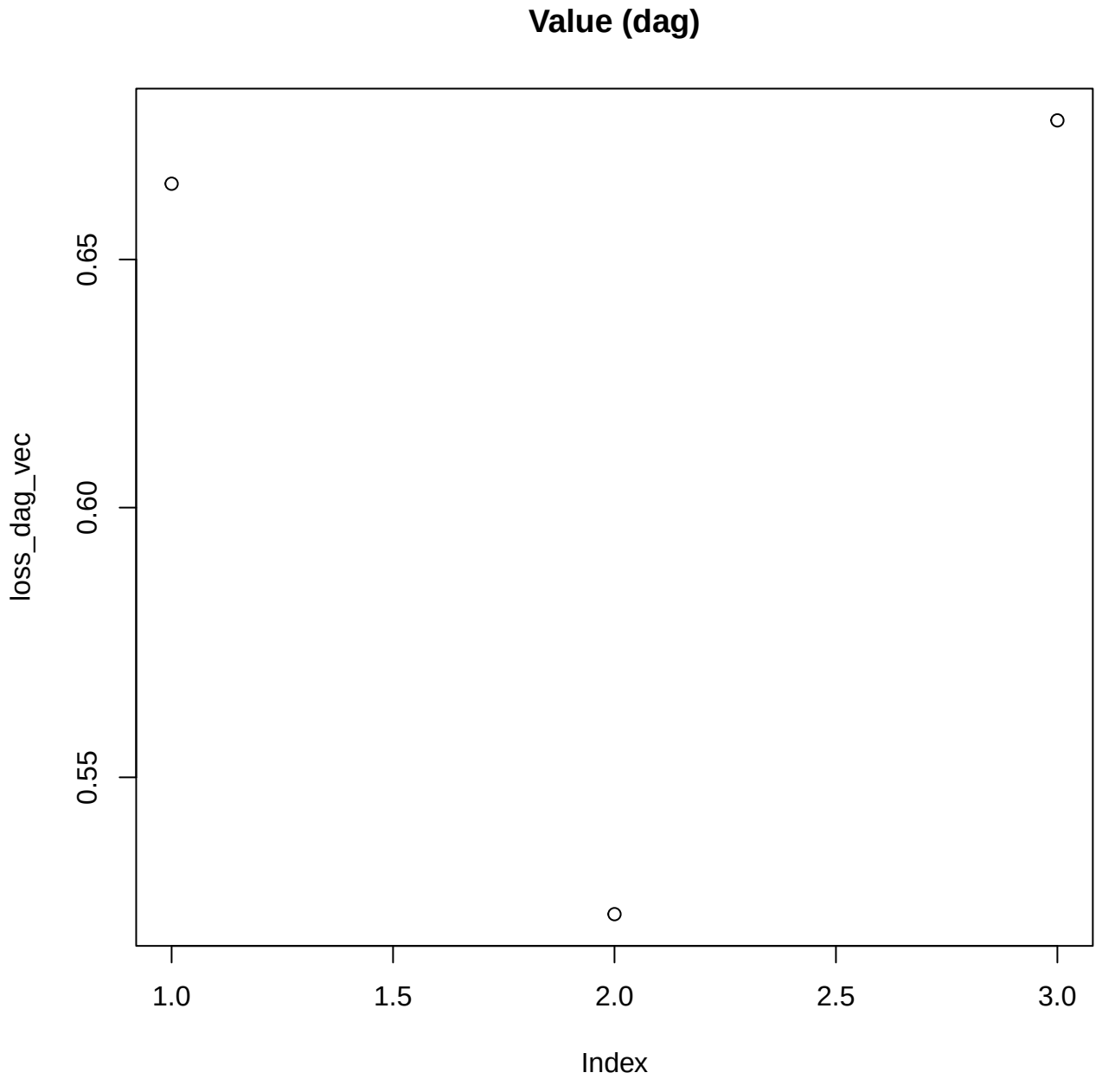
Value (dag)
loss_dag_vec
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2.5
3.0
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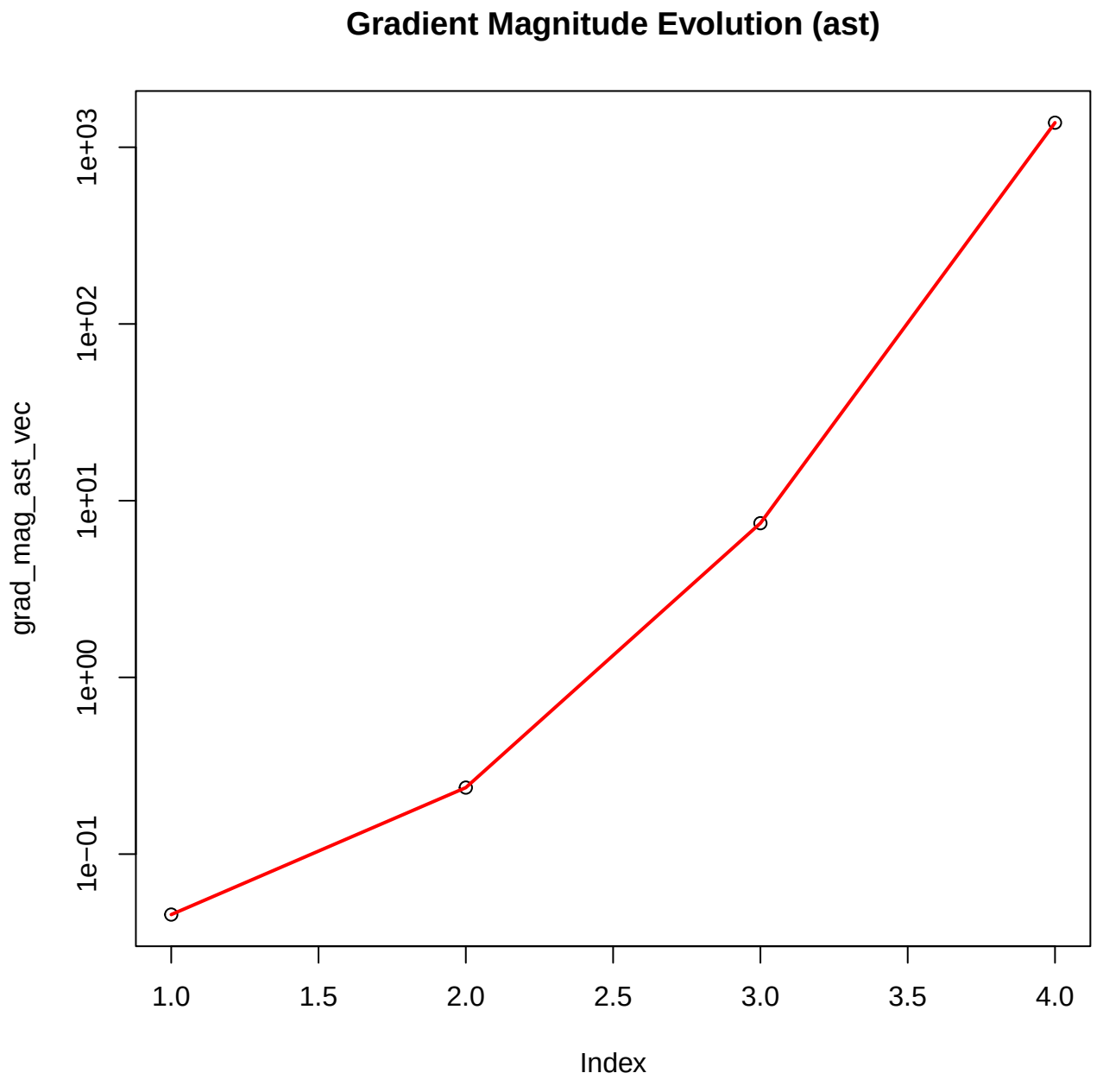
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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1e+01
1e+02
1e+03
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2.5
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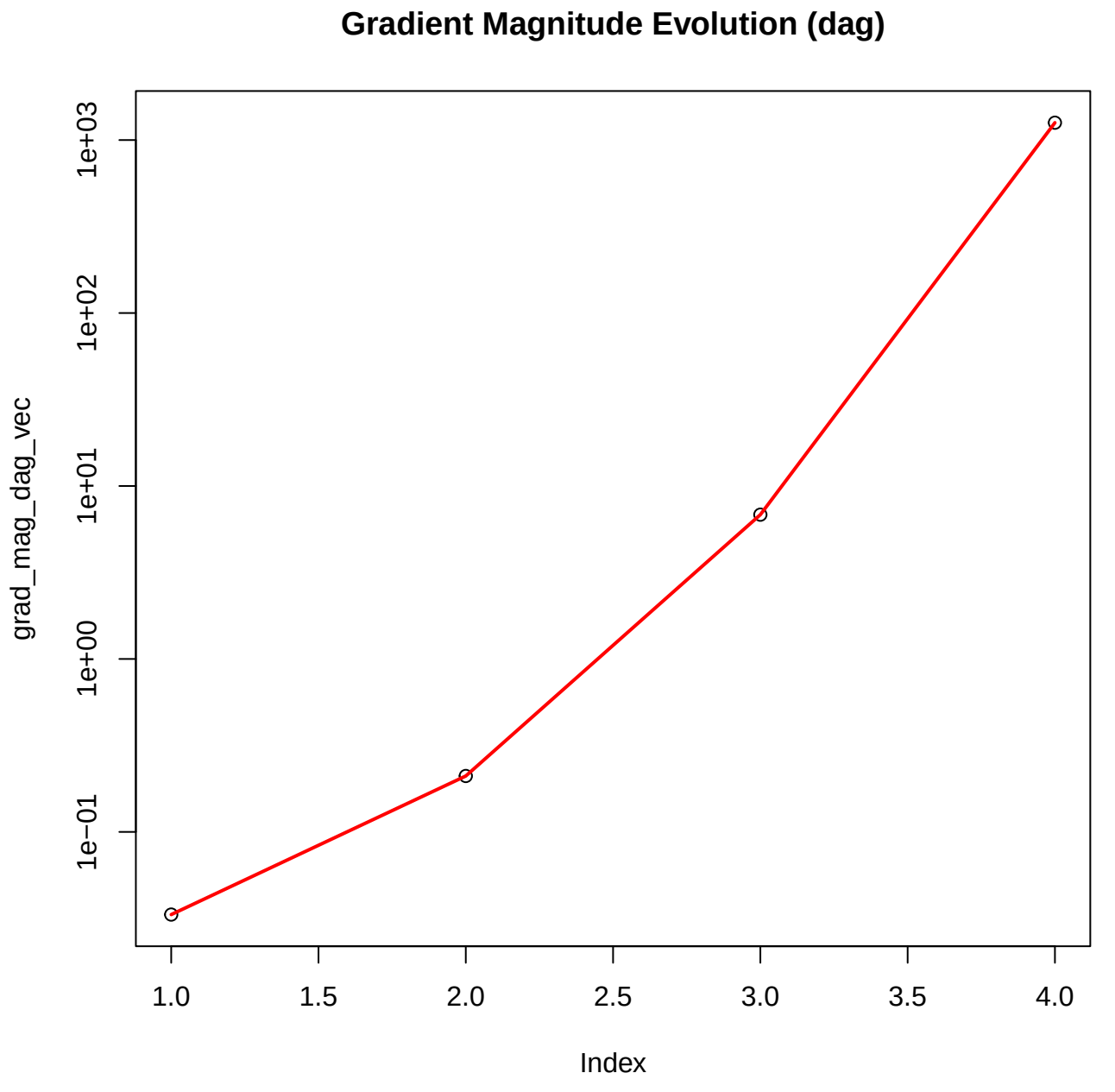
3.5
4.0
Index


Gradient Magnitude Evolution (dag)
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1e+01
1e+02
1e+03
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1.5
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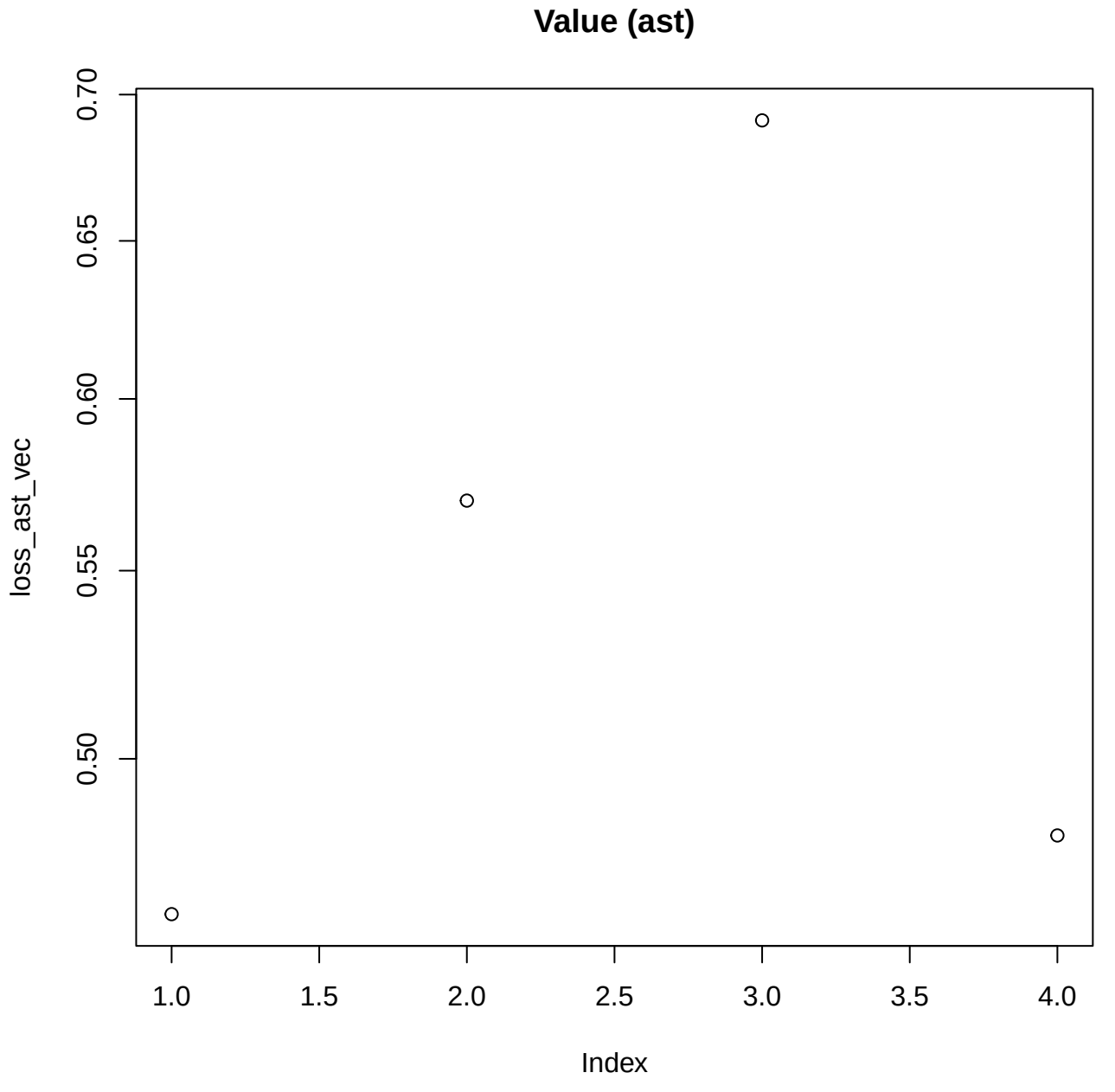
2.5
3.0
3.5
4.0
Index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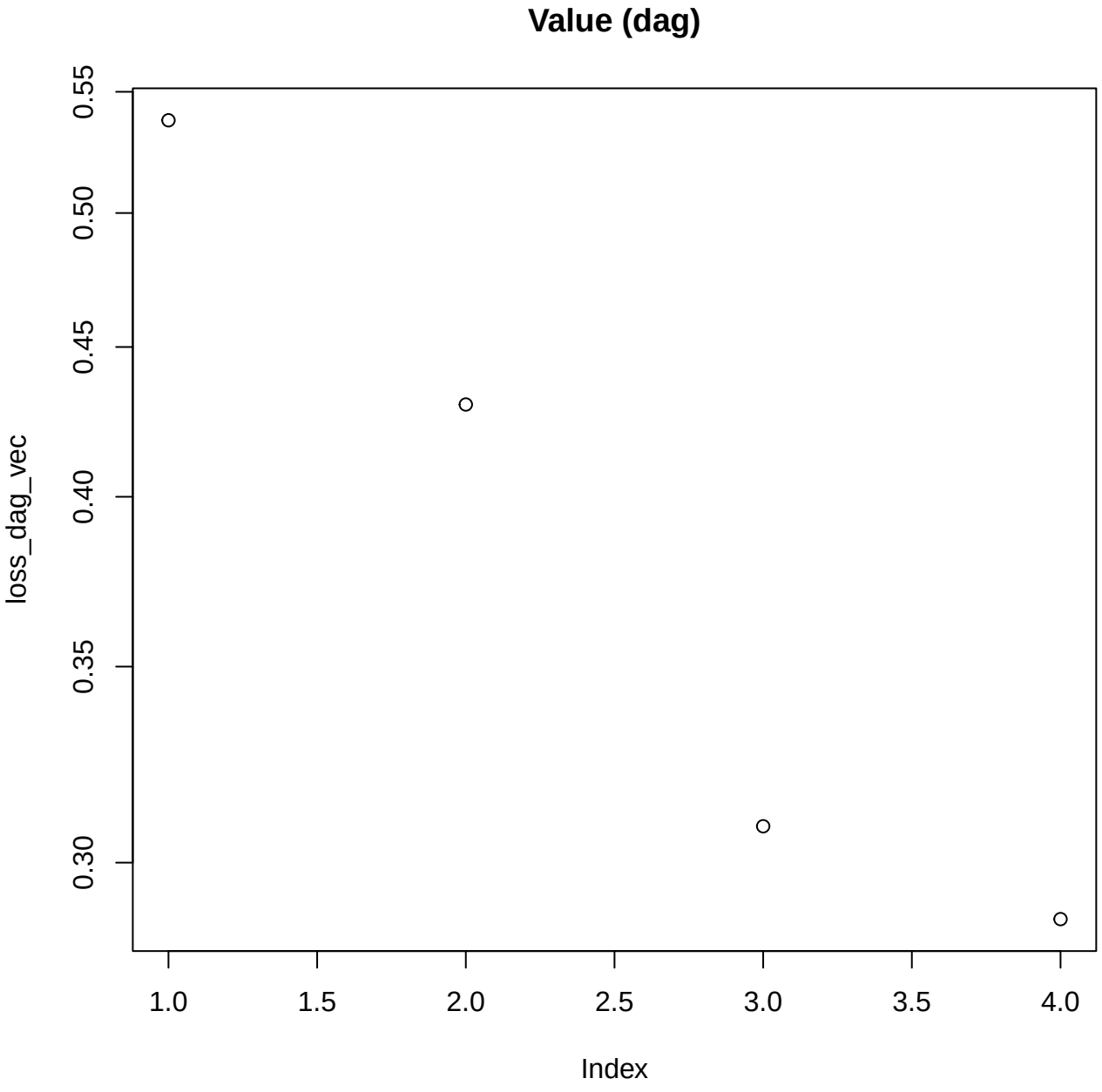
4.0
Index


Value (dag)
loss_dag_vec
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2.5
3.0
3.5
4.0
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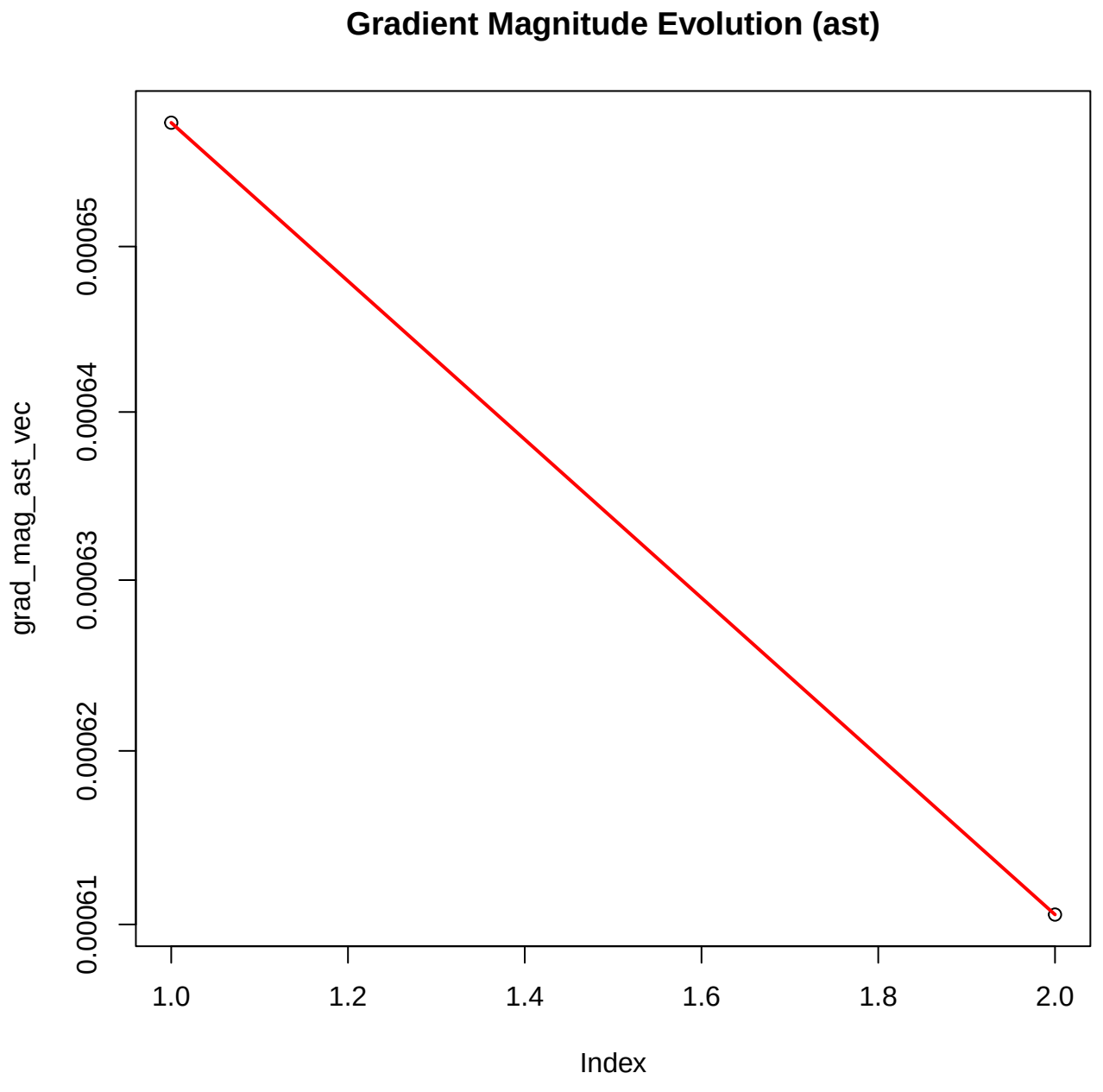
Gradient Magnitude Evolution (ast)
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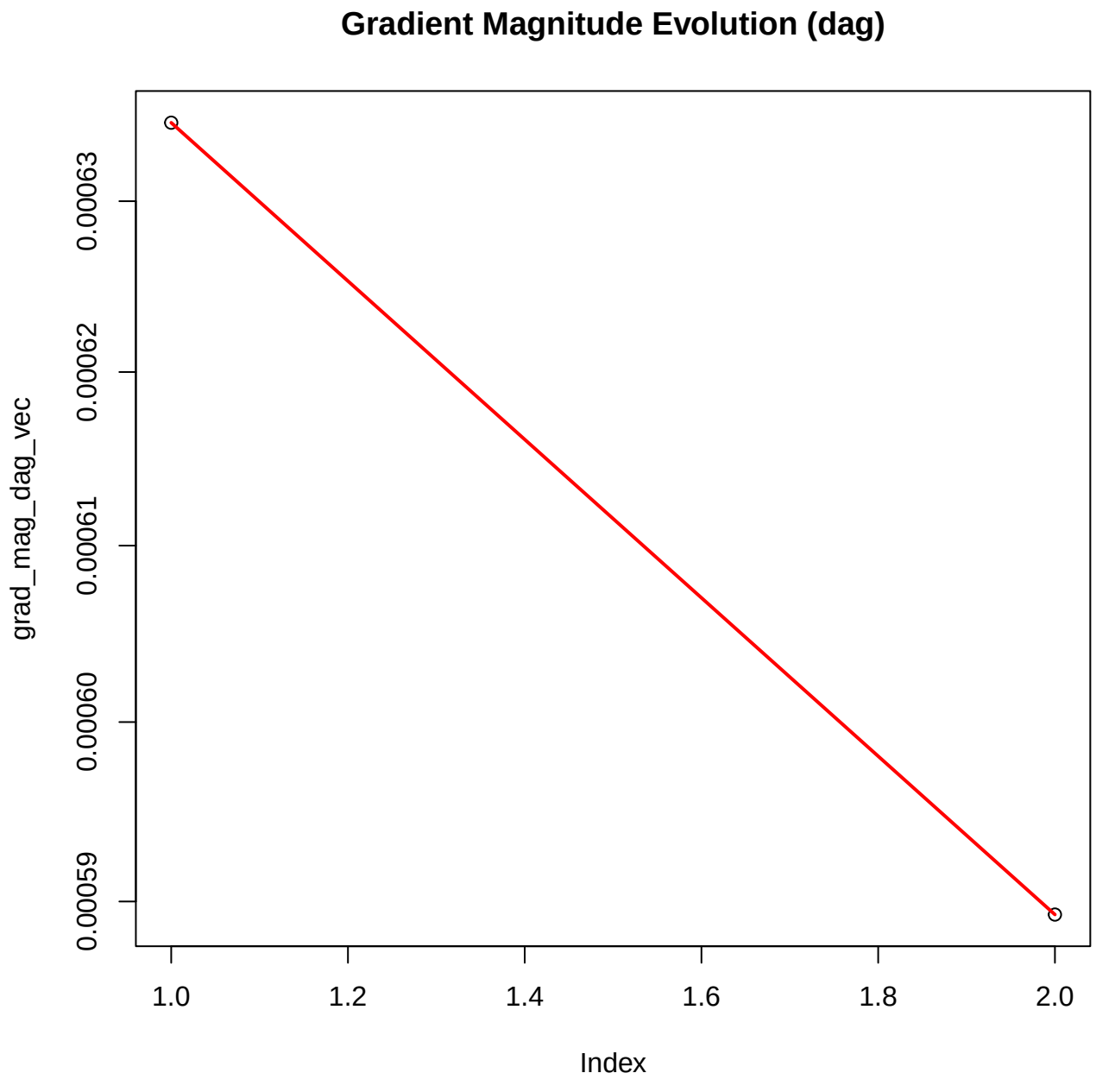
Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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2.0
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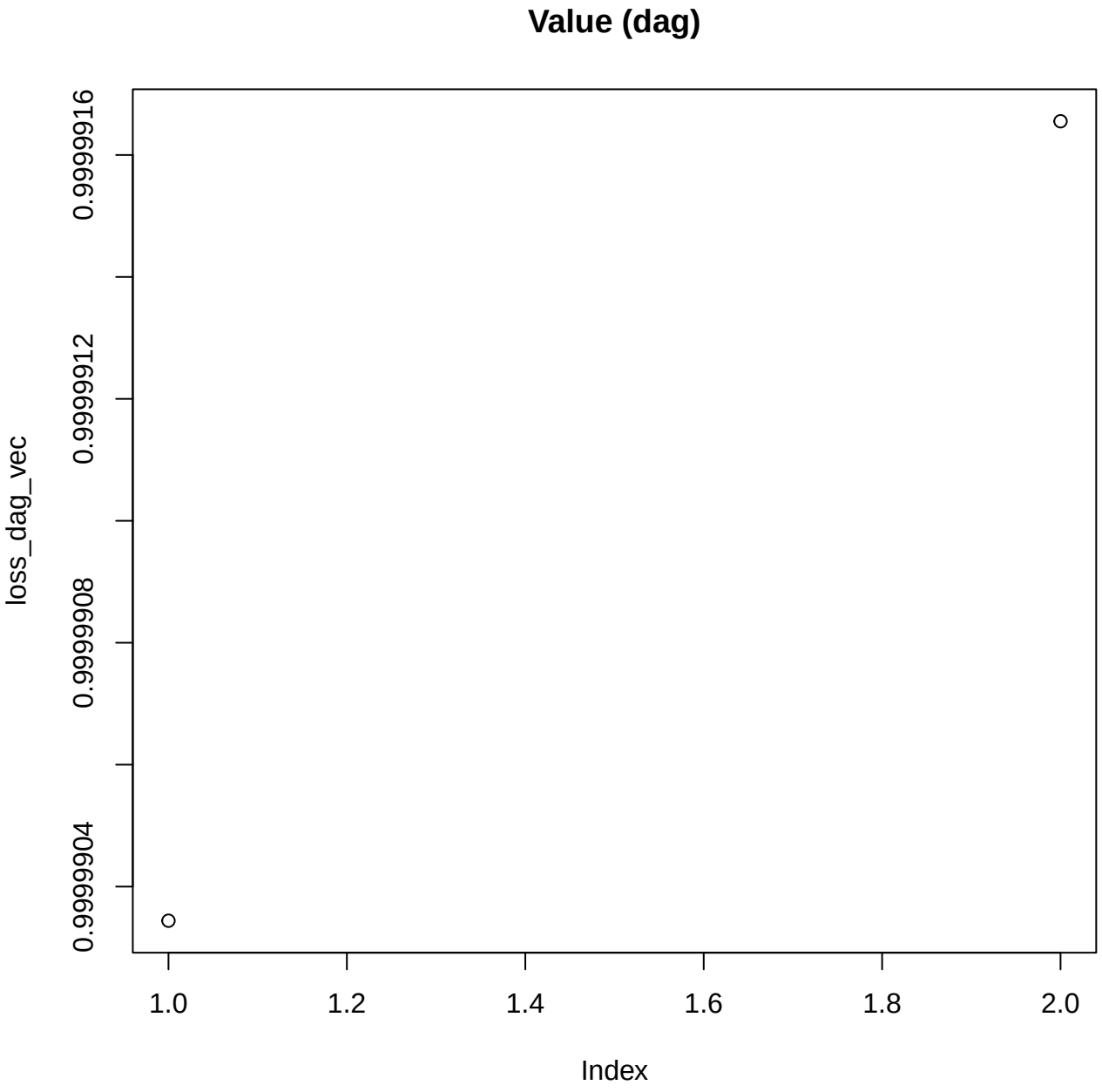
Value (dag)
loss_dag_vec
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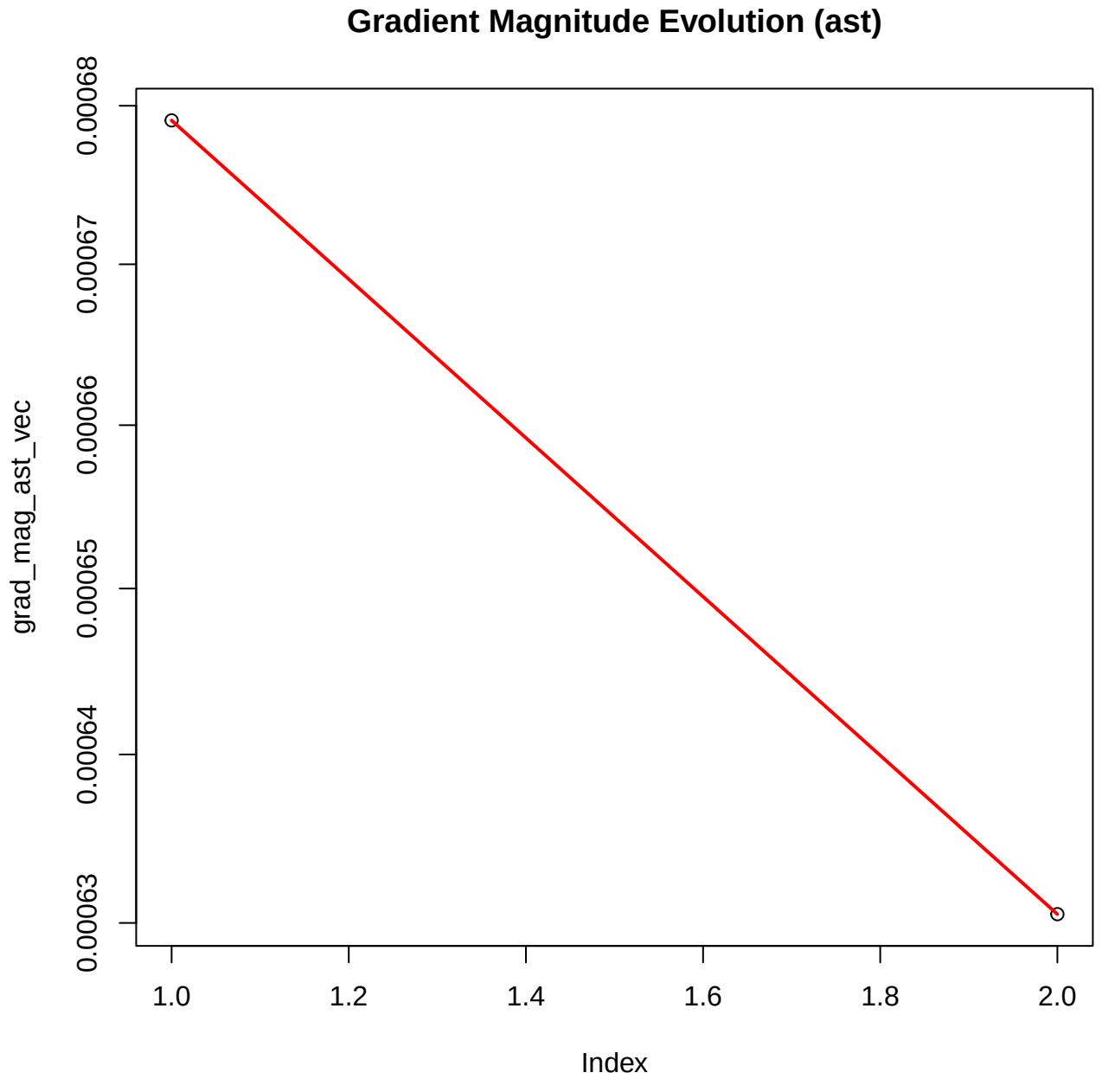
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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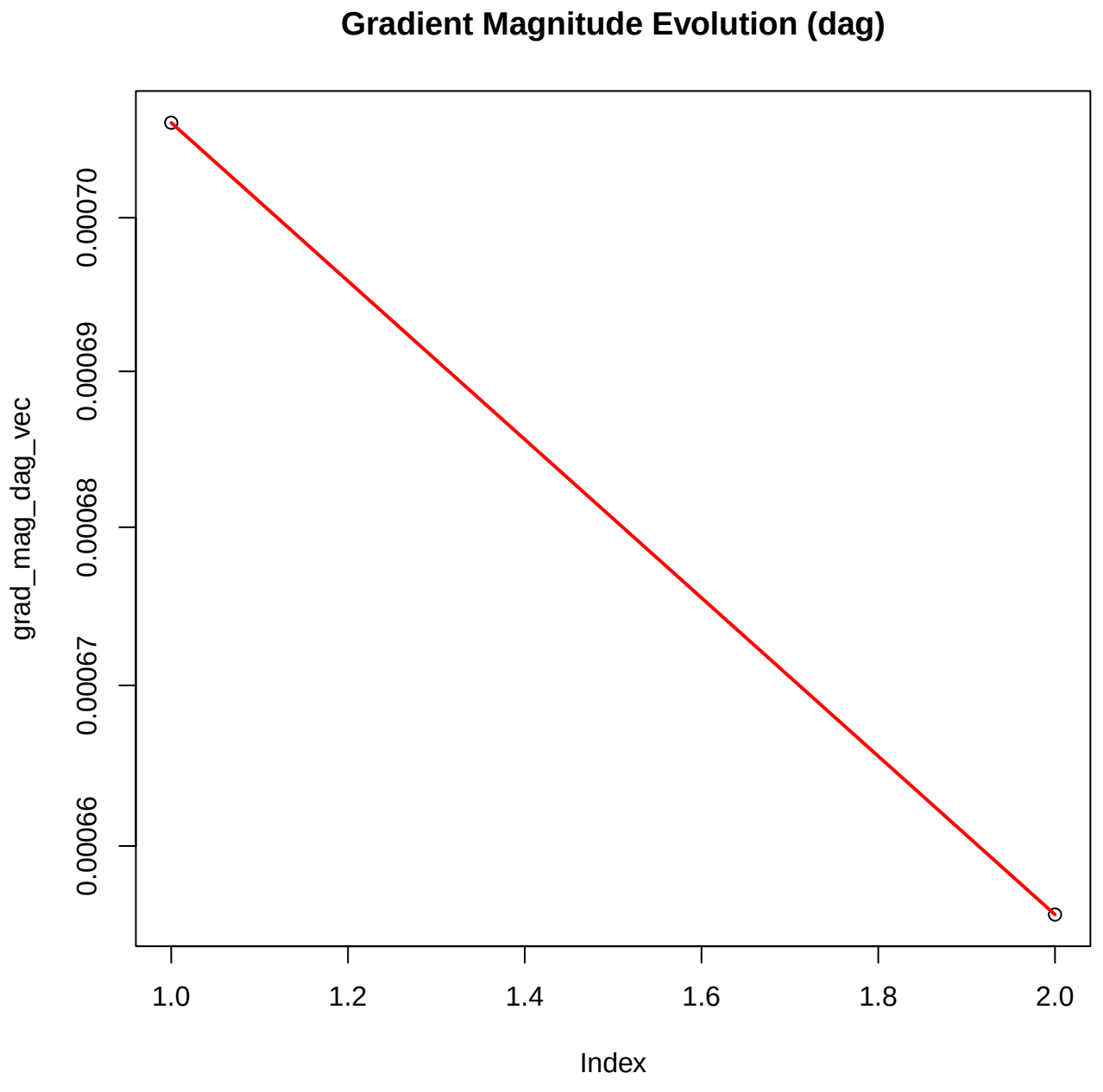
2.0
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
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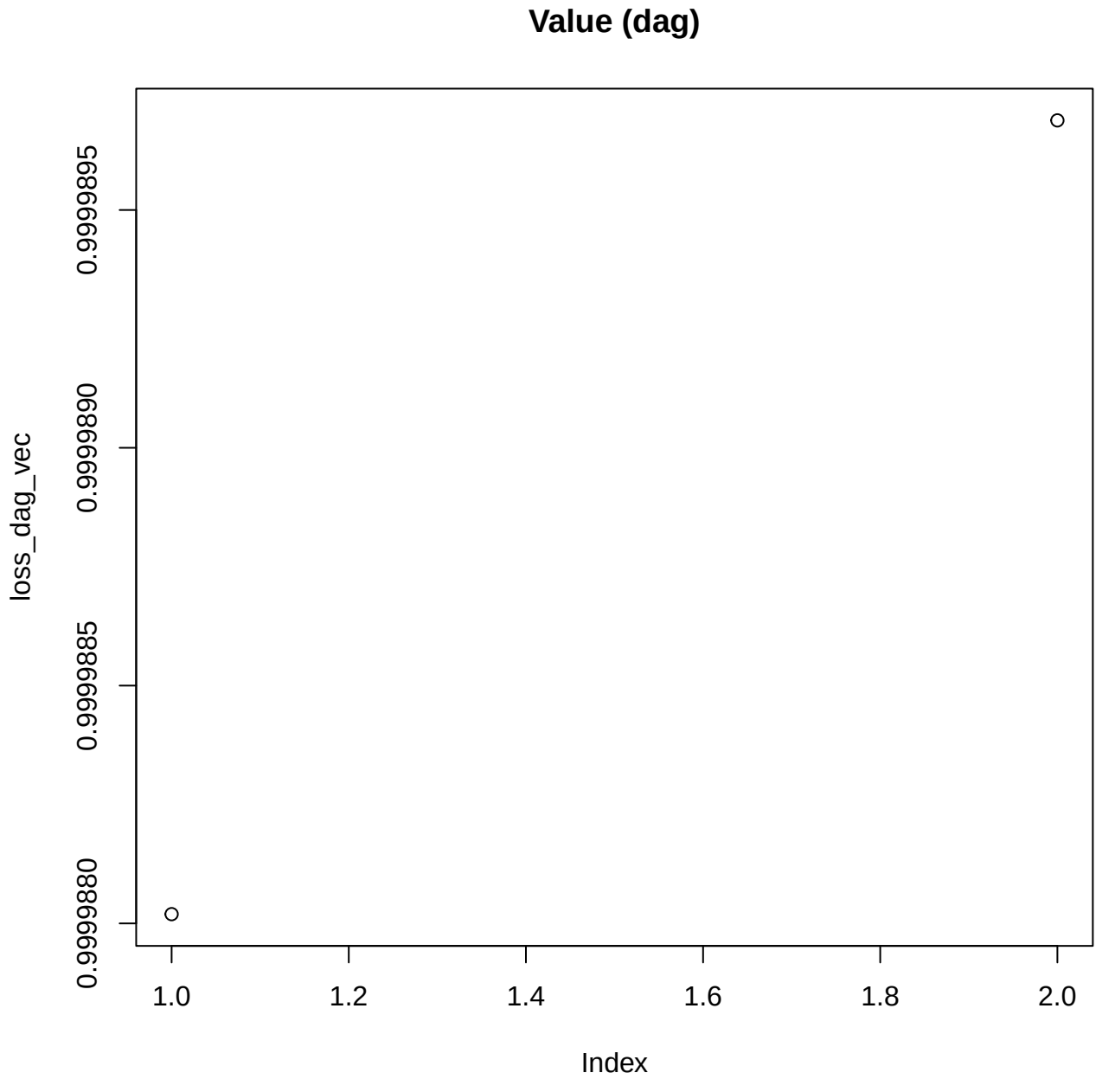
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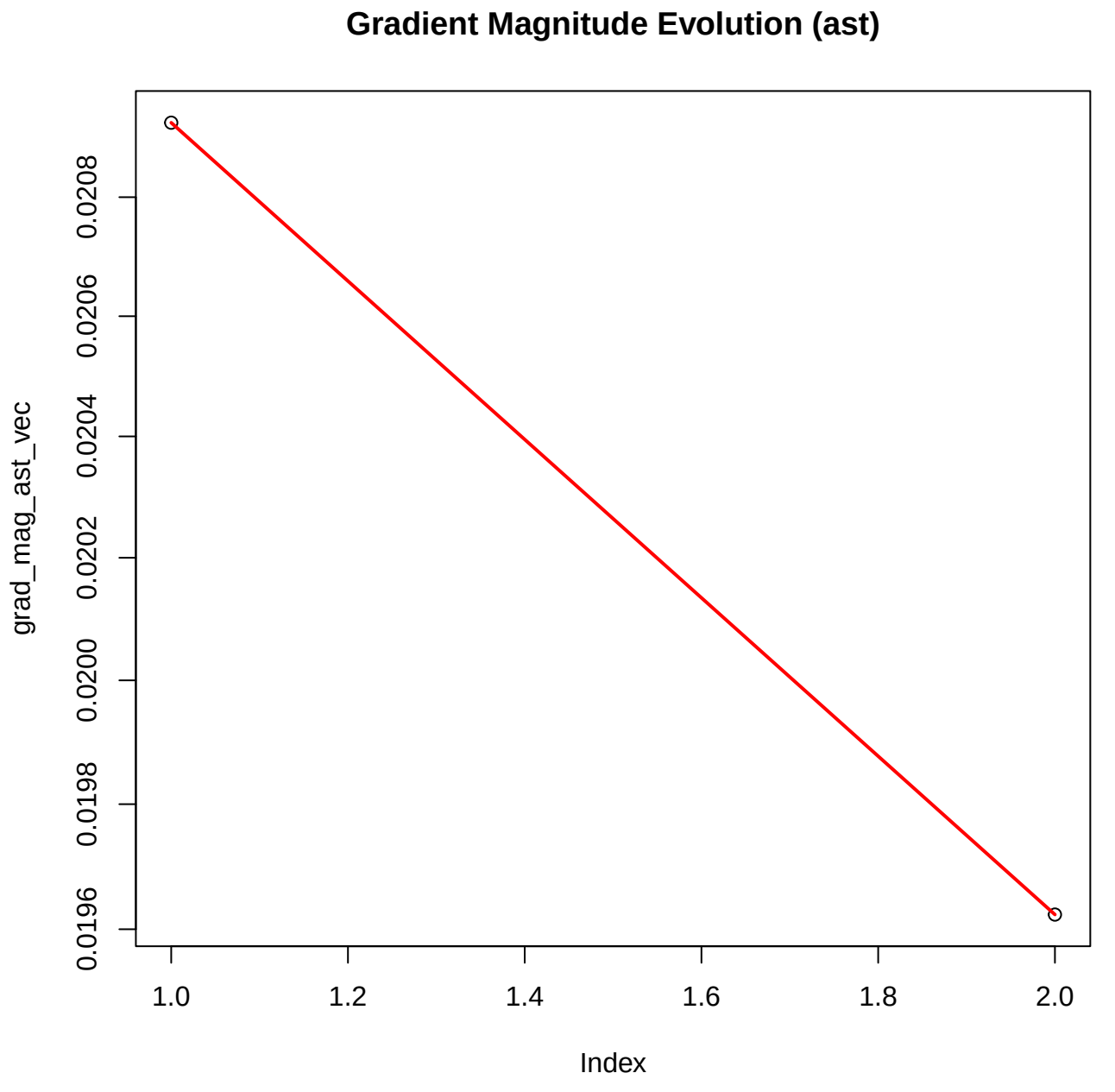
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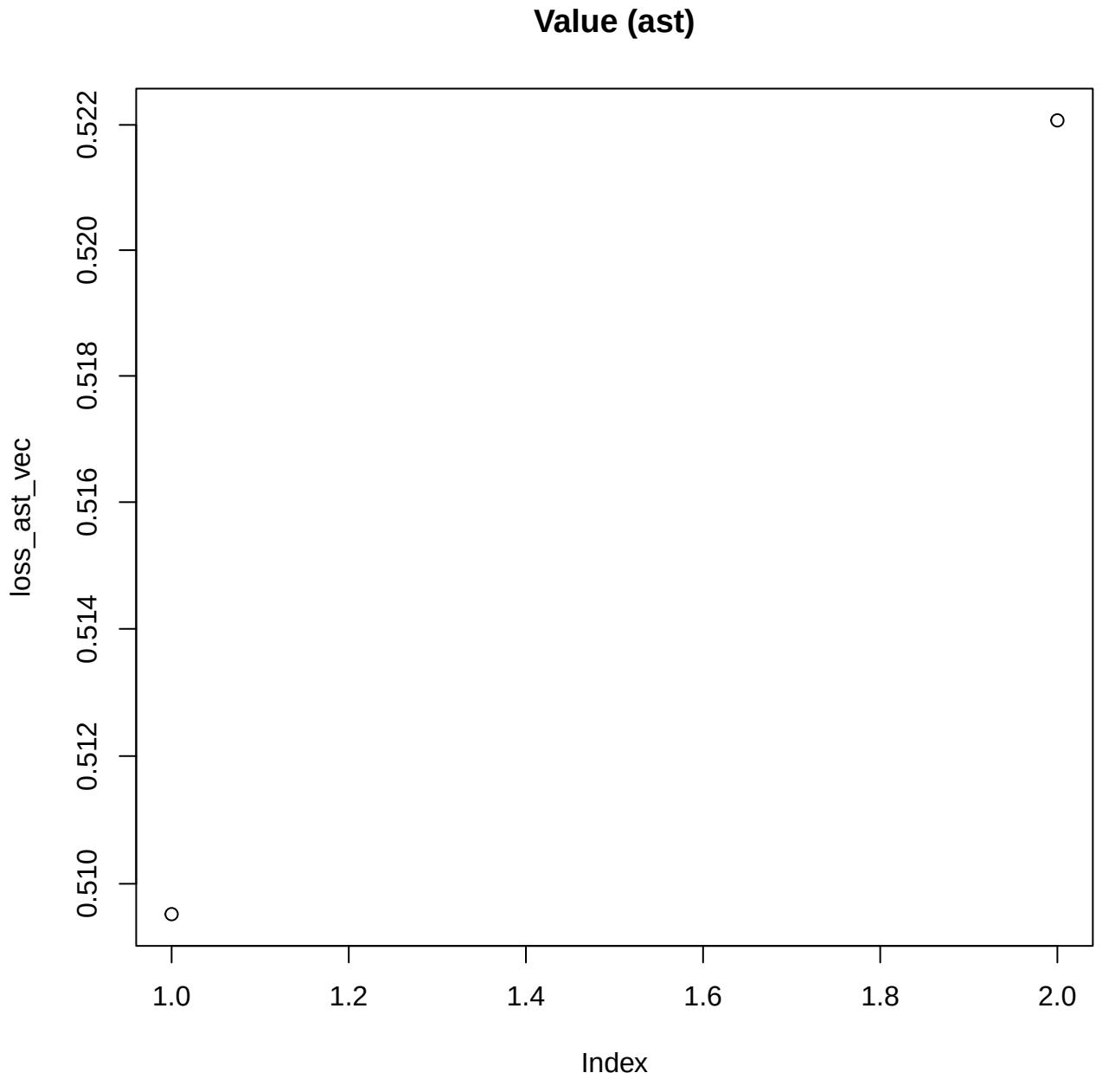
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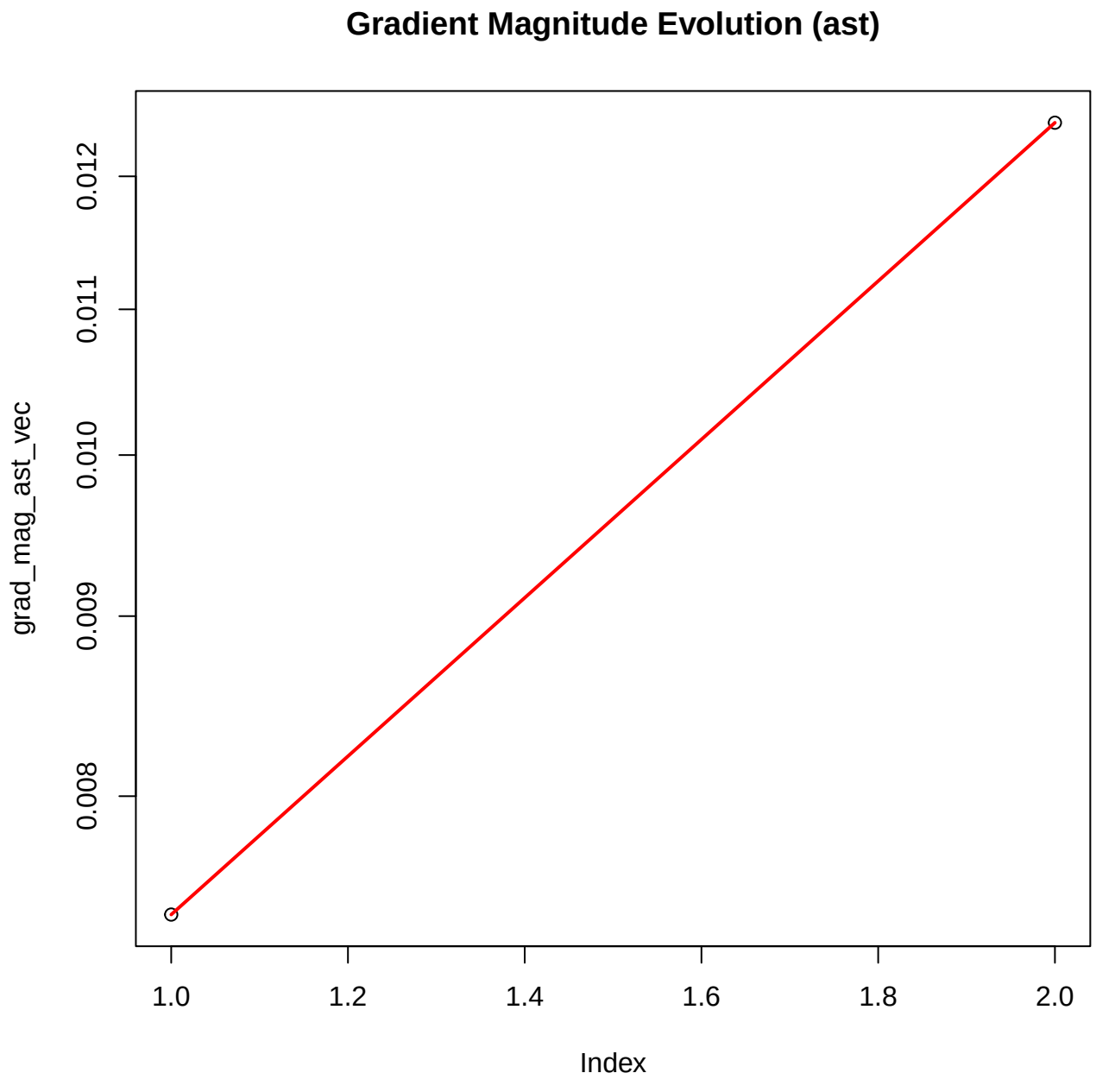
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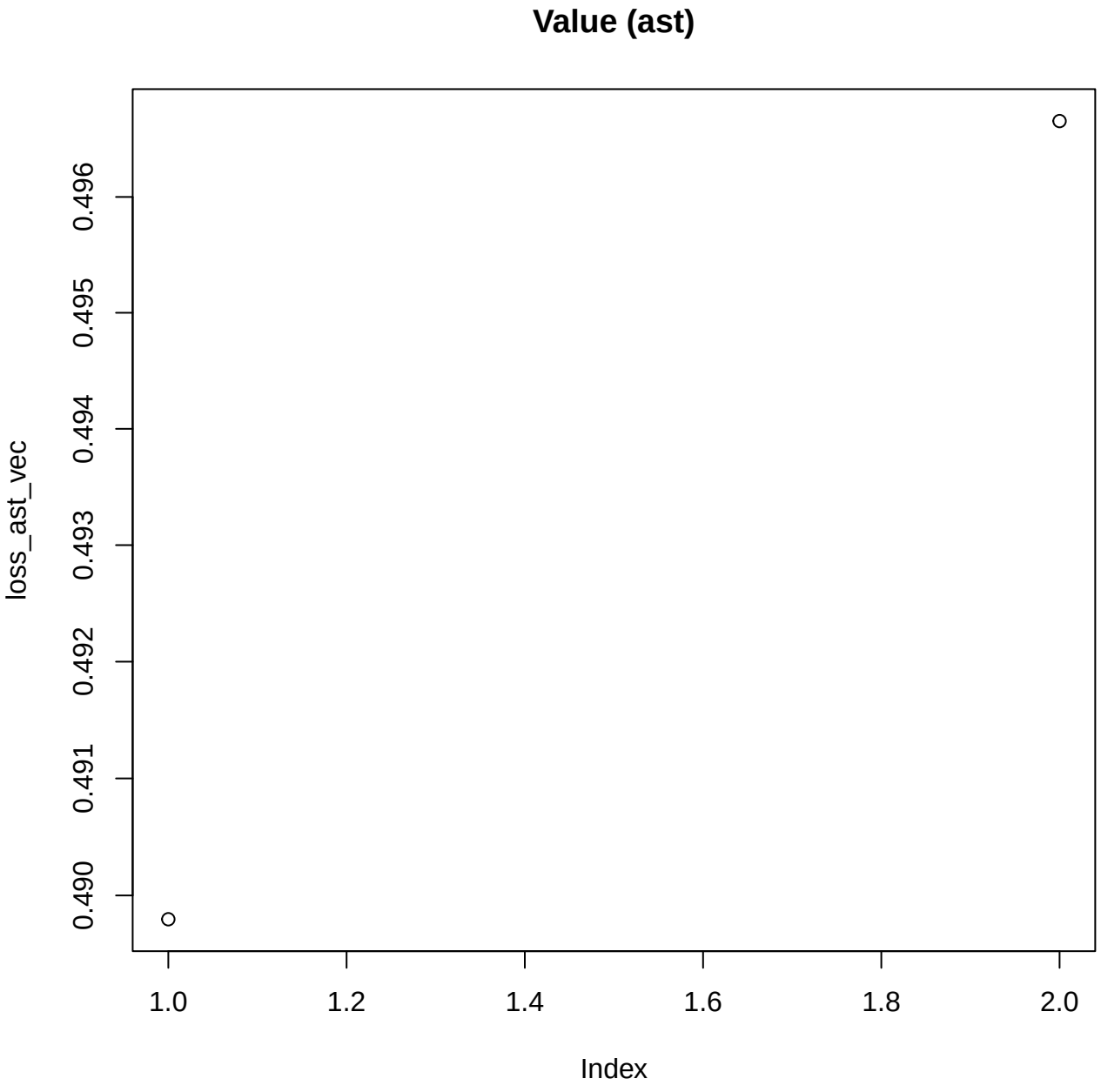
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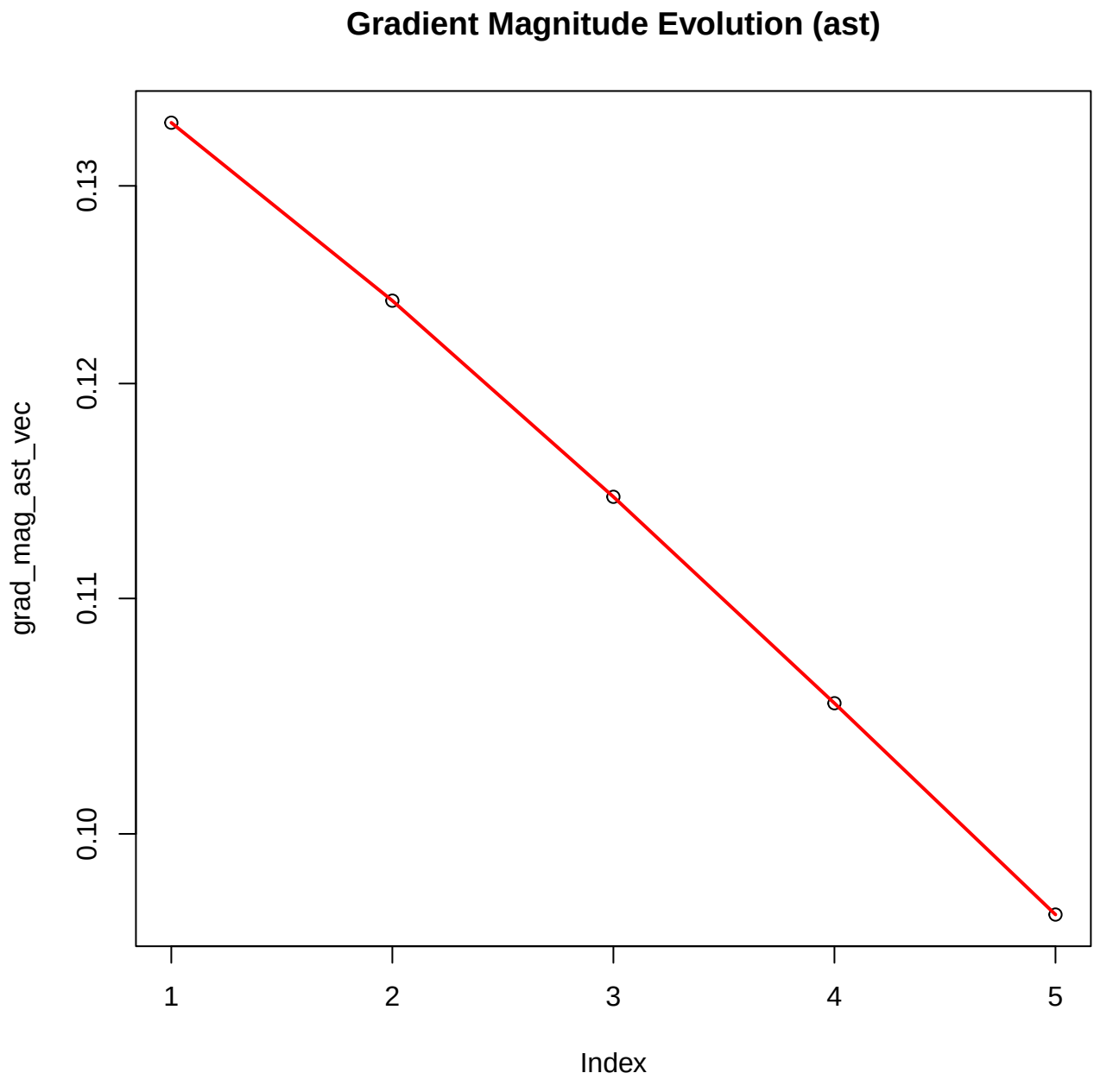
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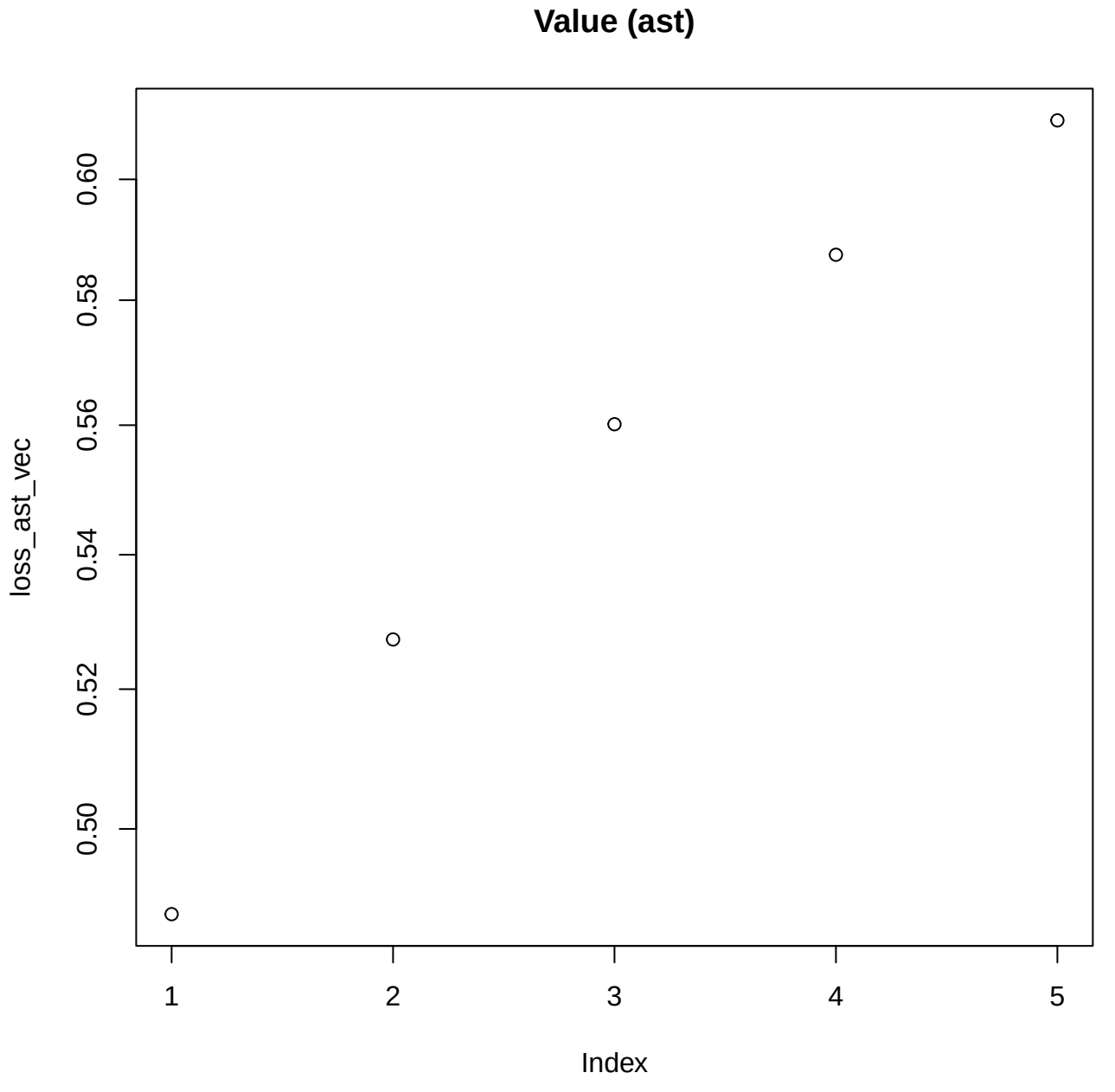
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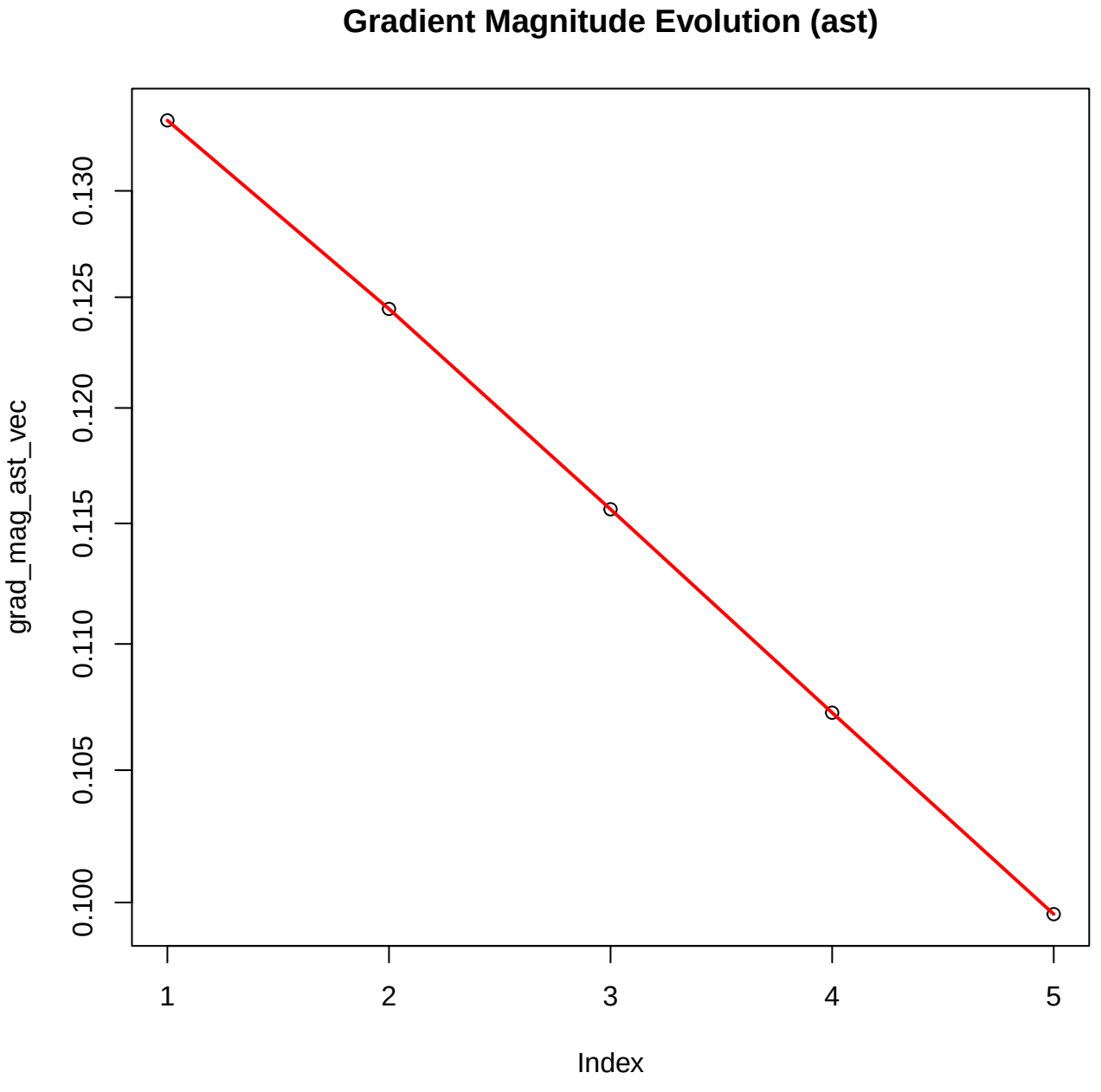
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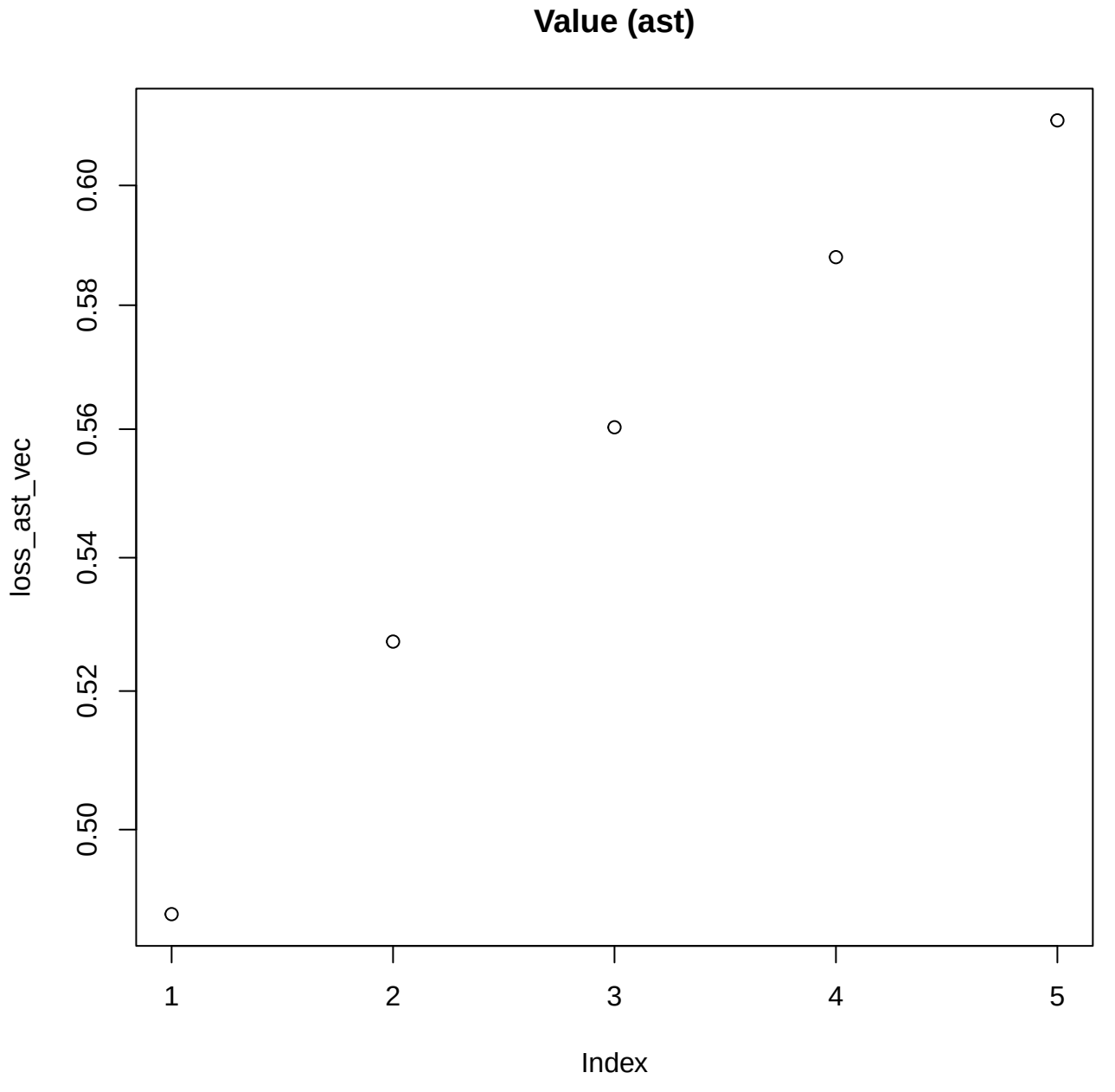
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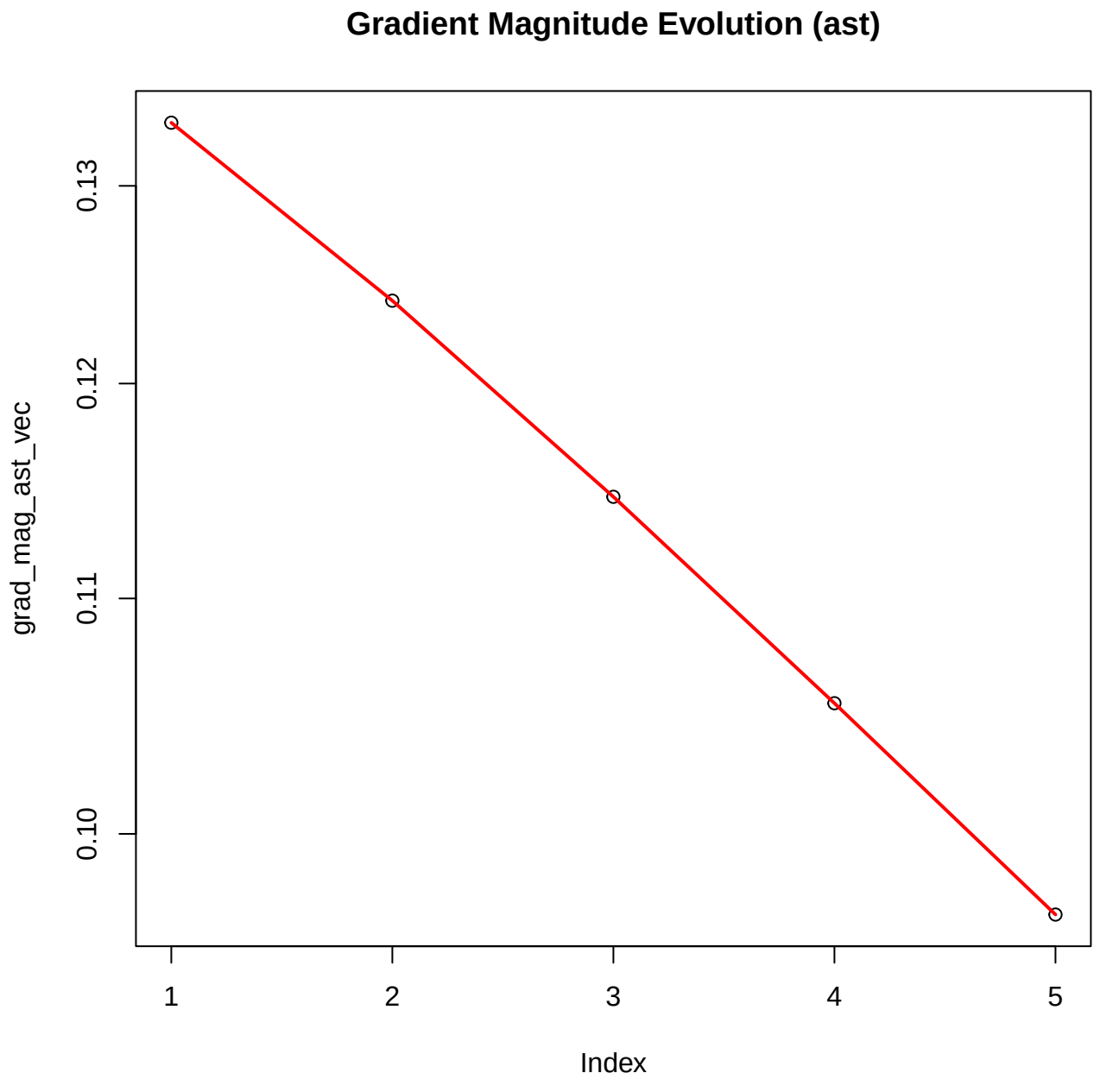
Gradient Magnitude Evolution (ast)
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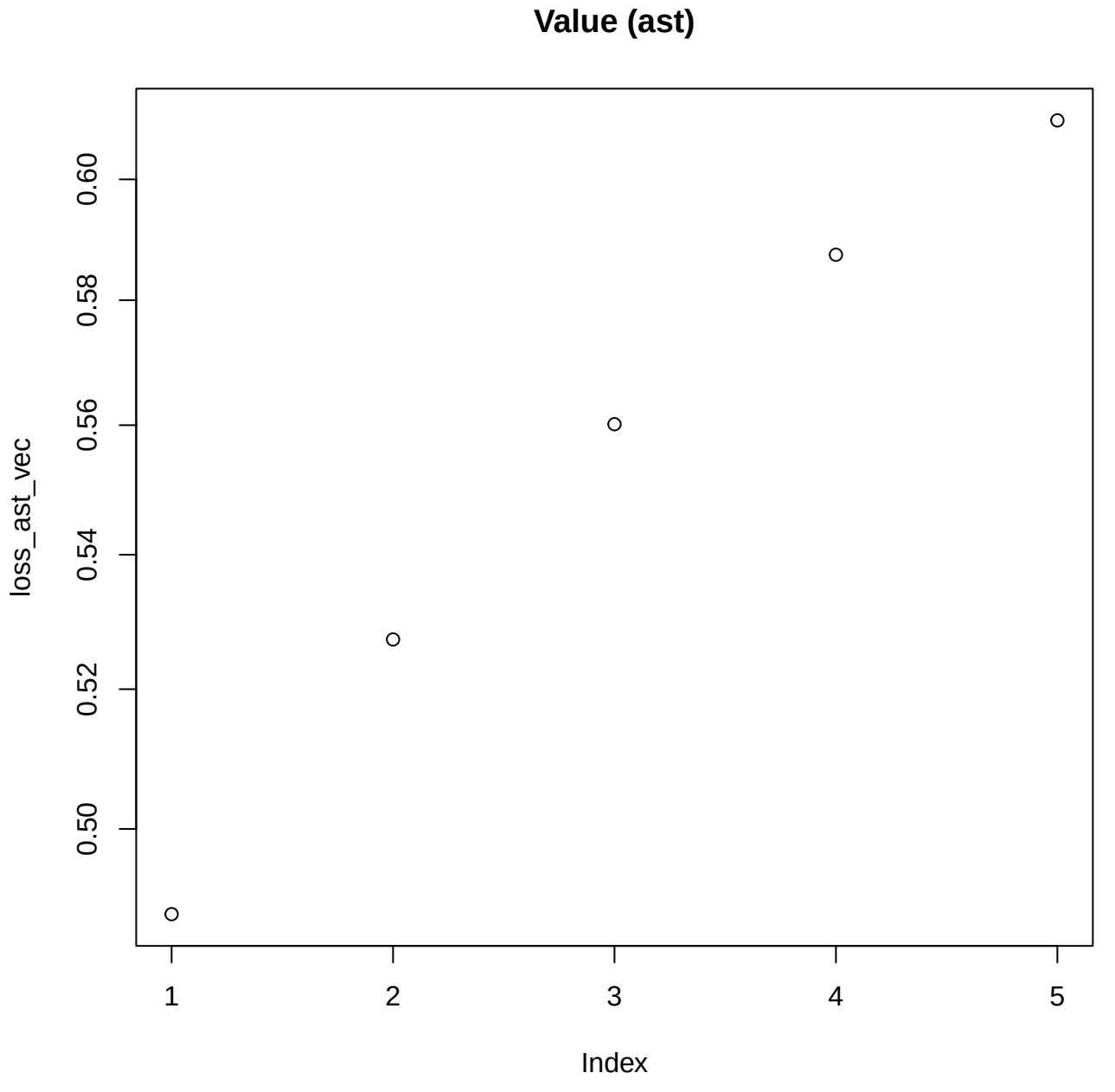
Value (ast)
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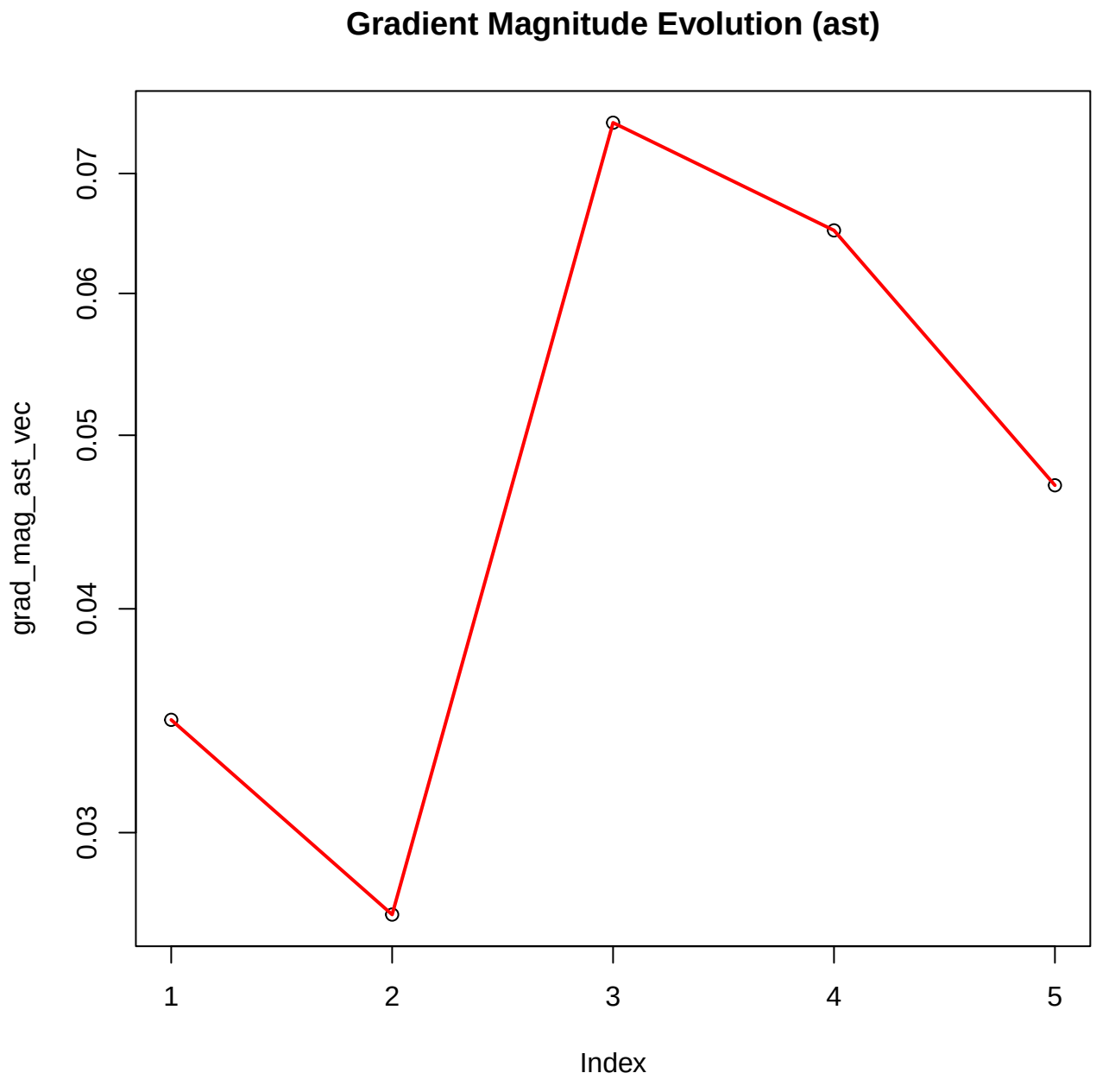
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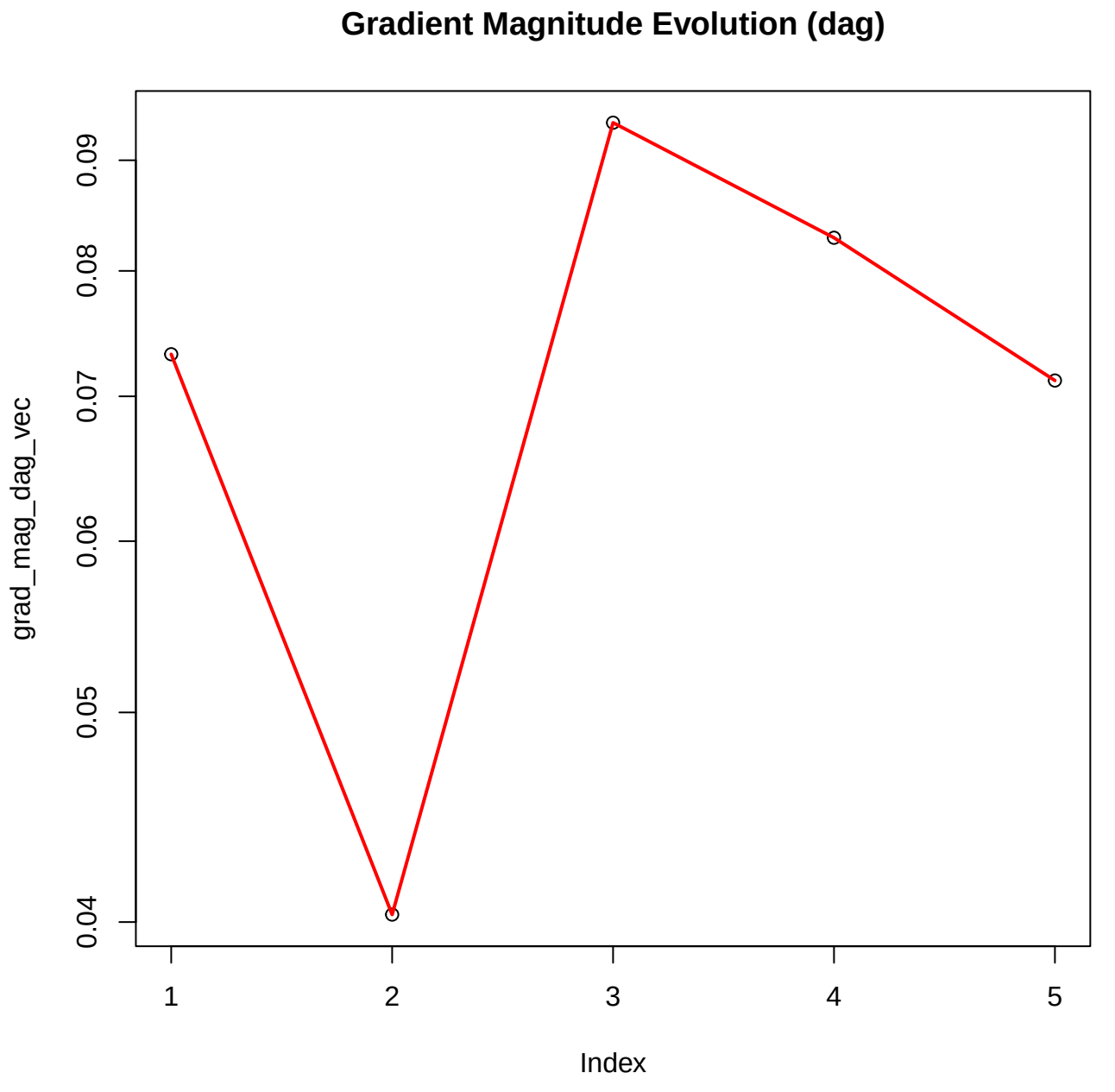
Gradient Magnitude Evolution (dag)
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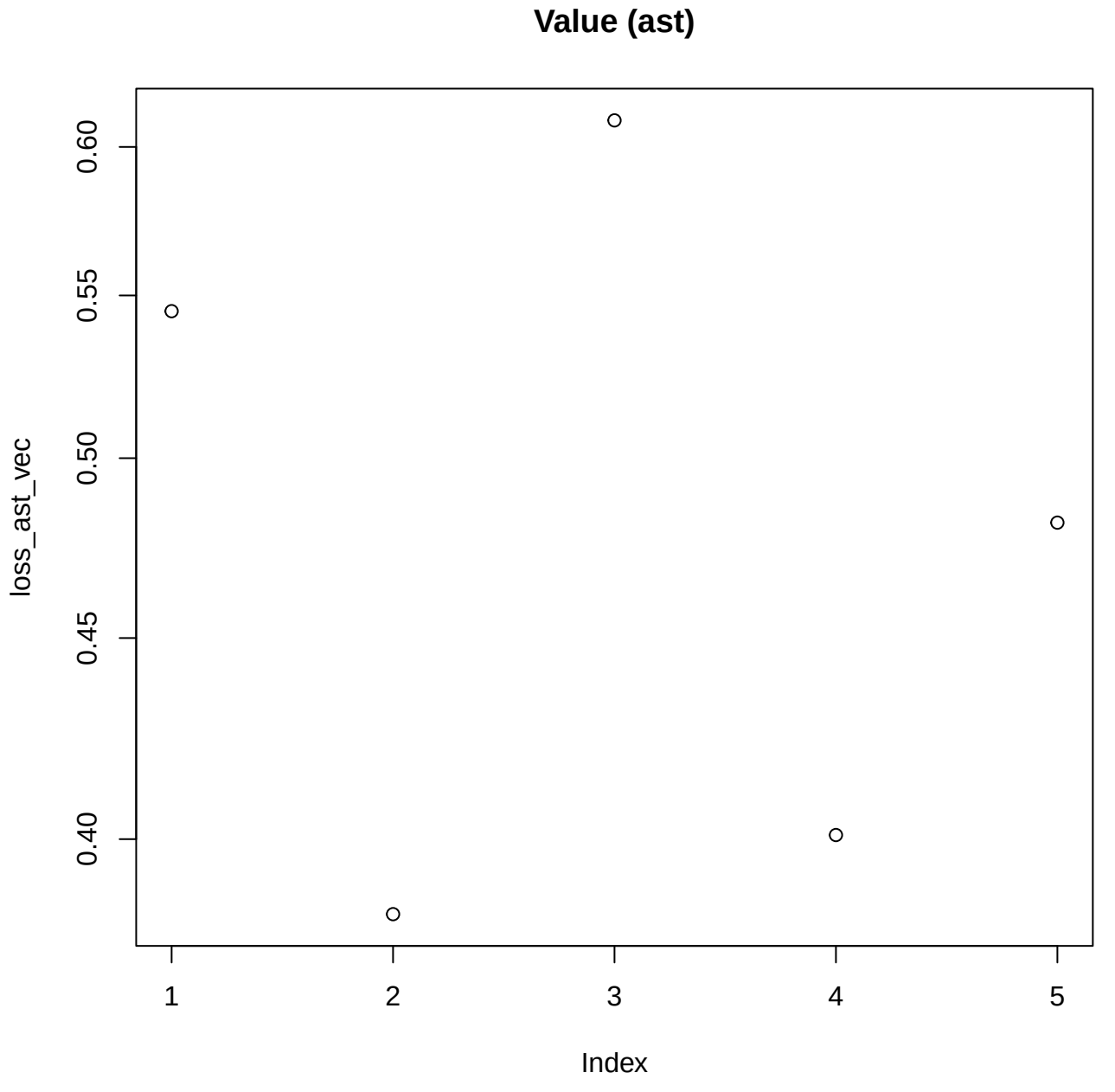
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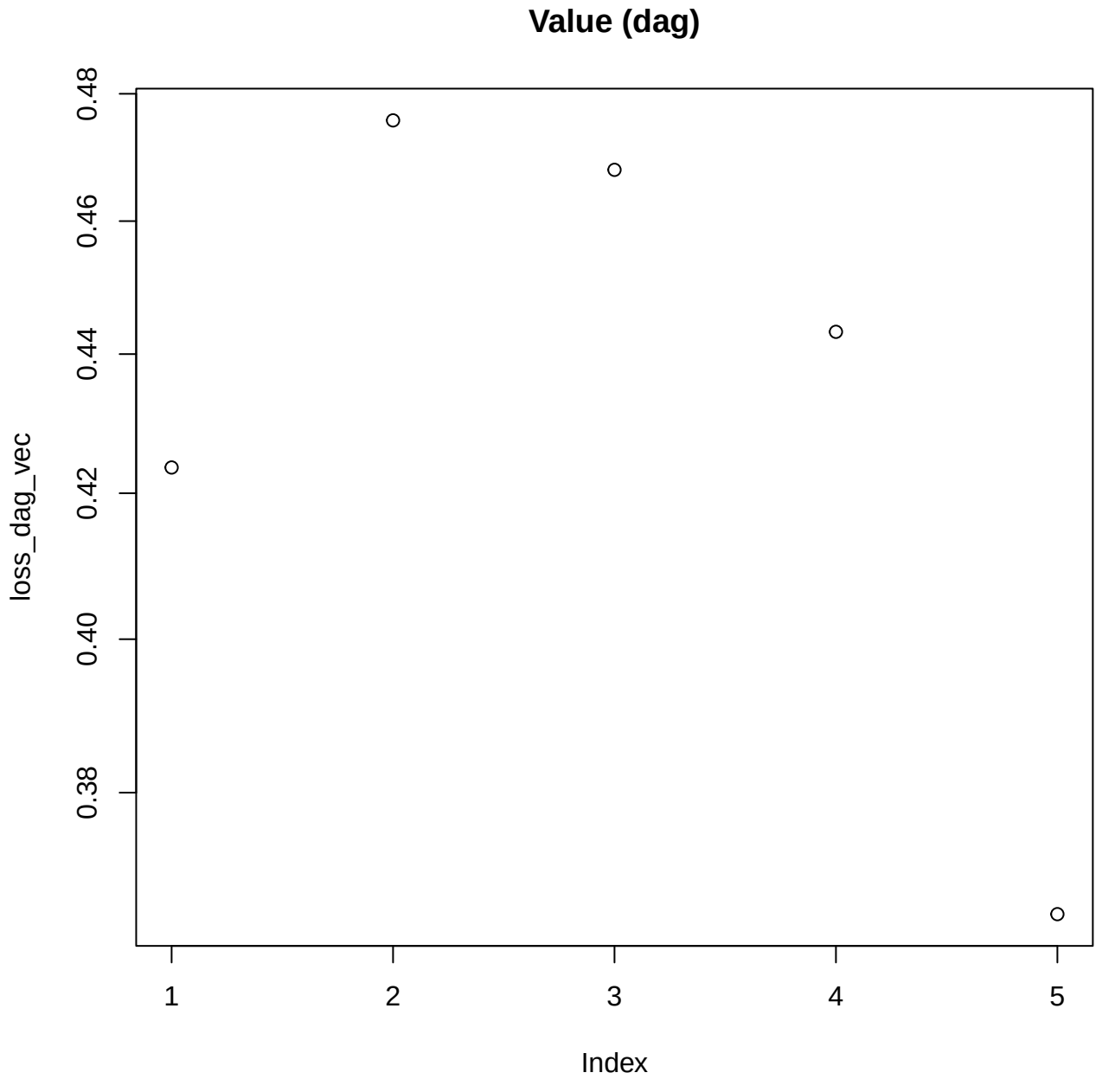
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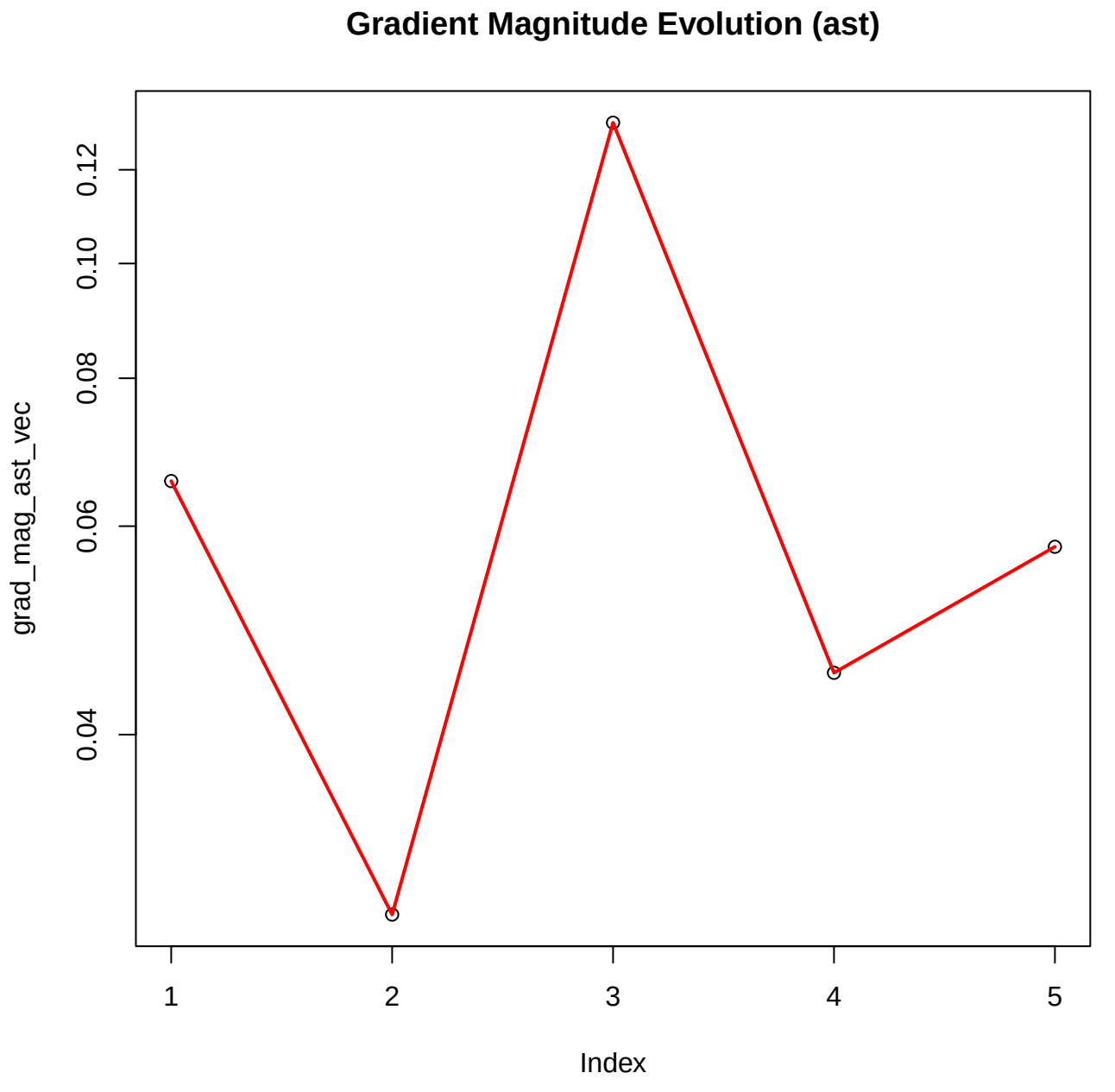
Gradient Magnitude Evolution (ast)
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0.08
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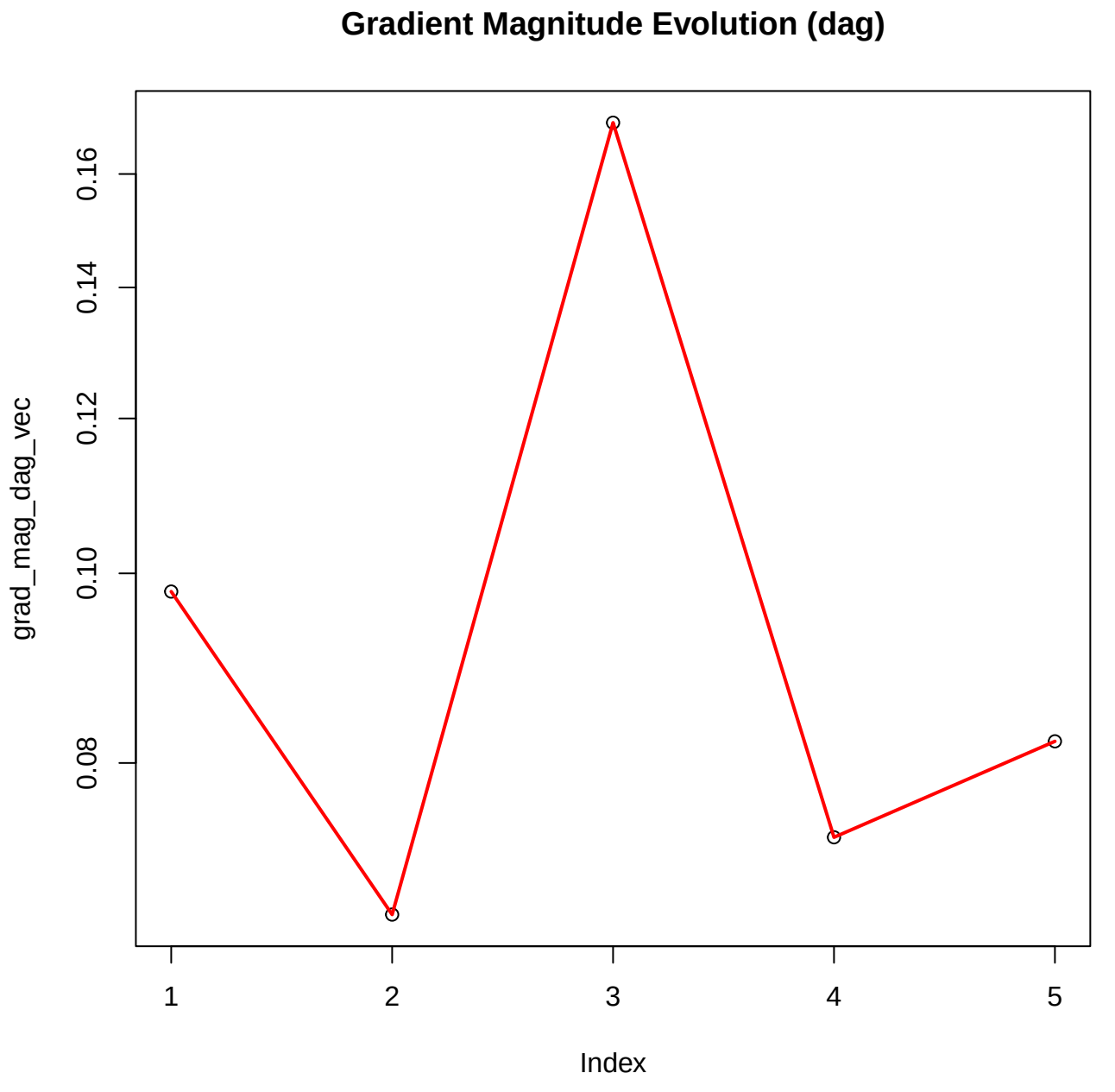
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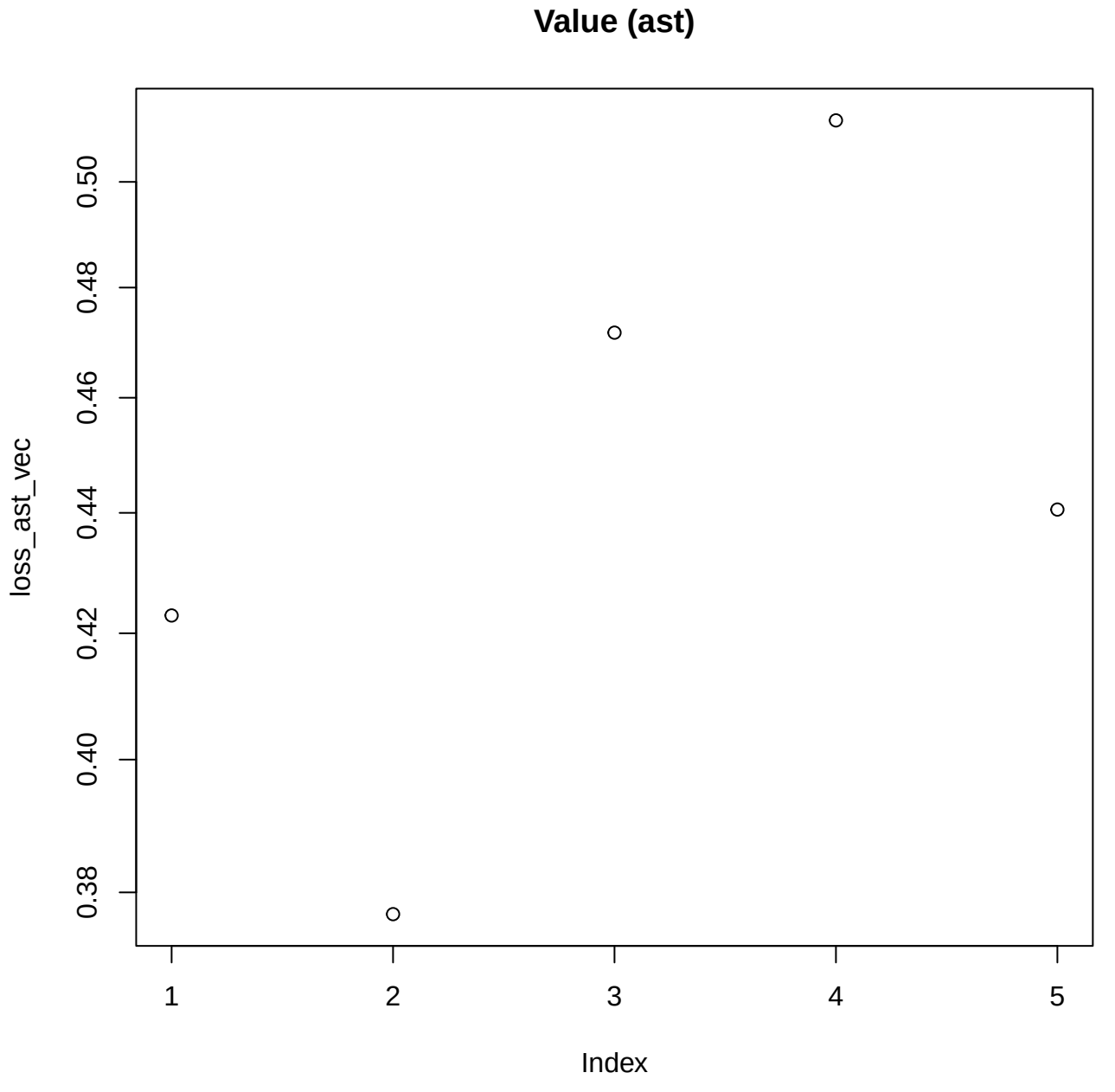
Value (ast)
loss_ast_vec
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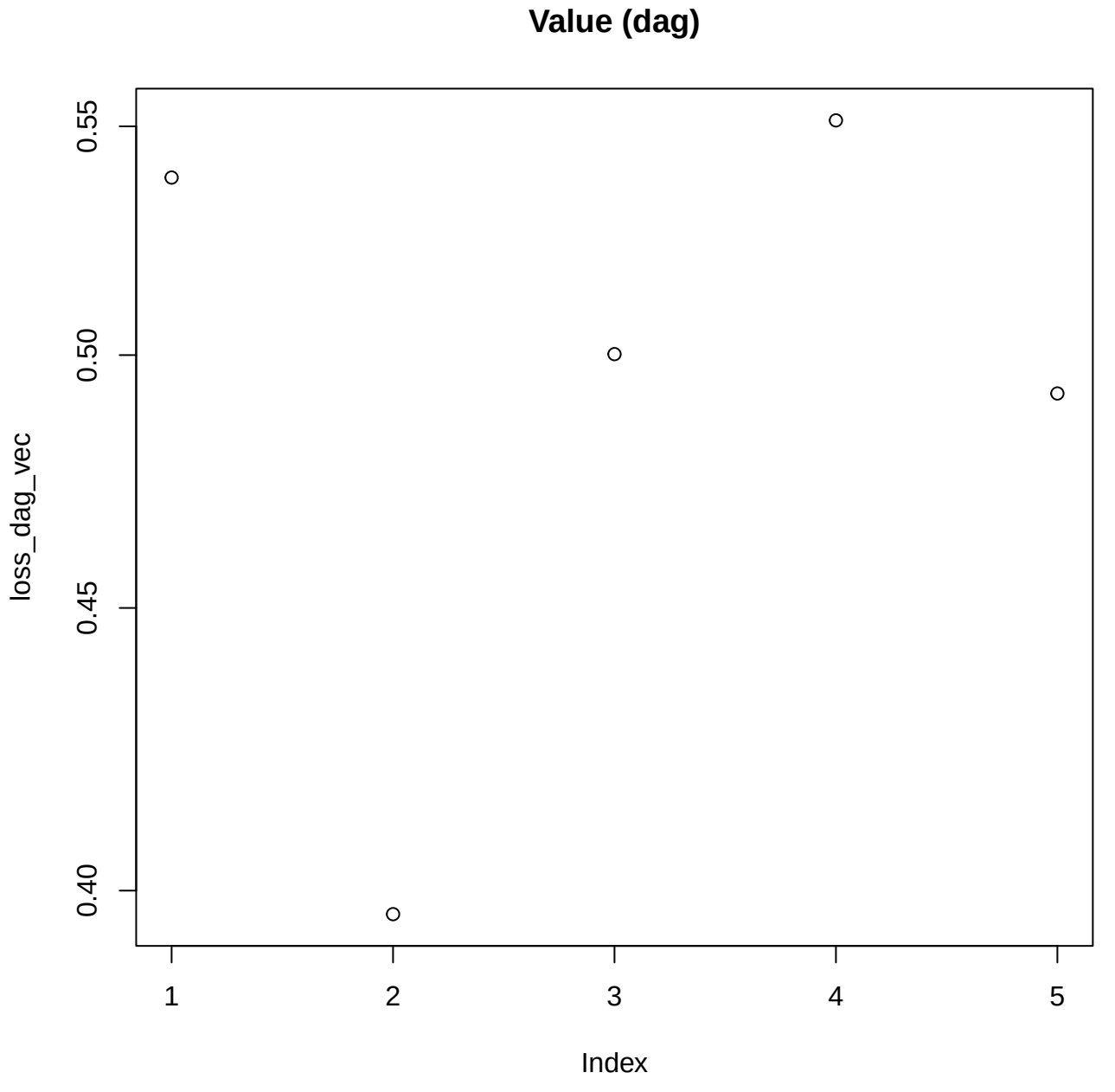
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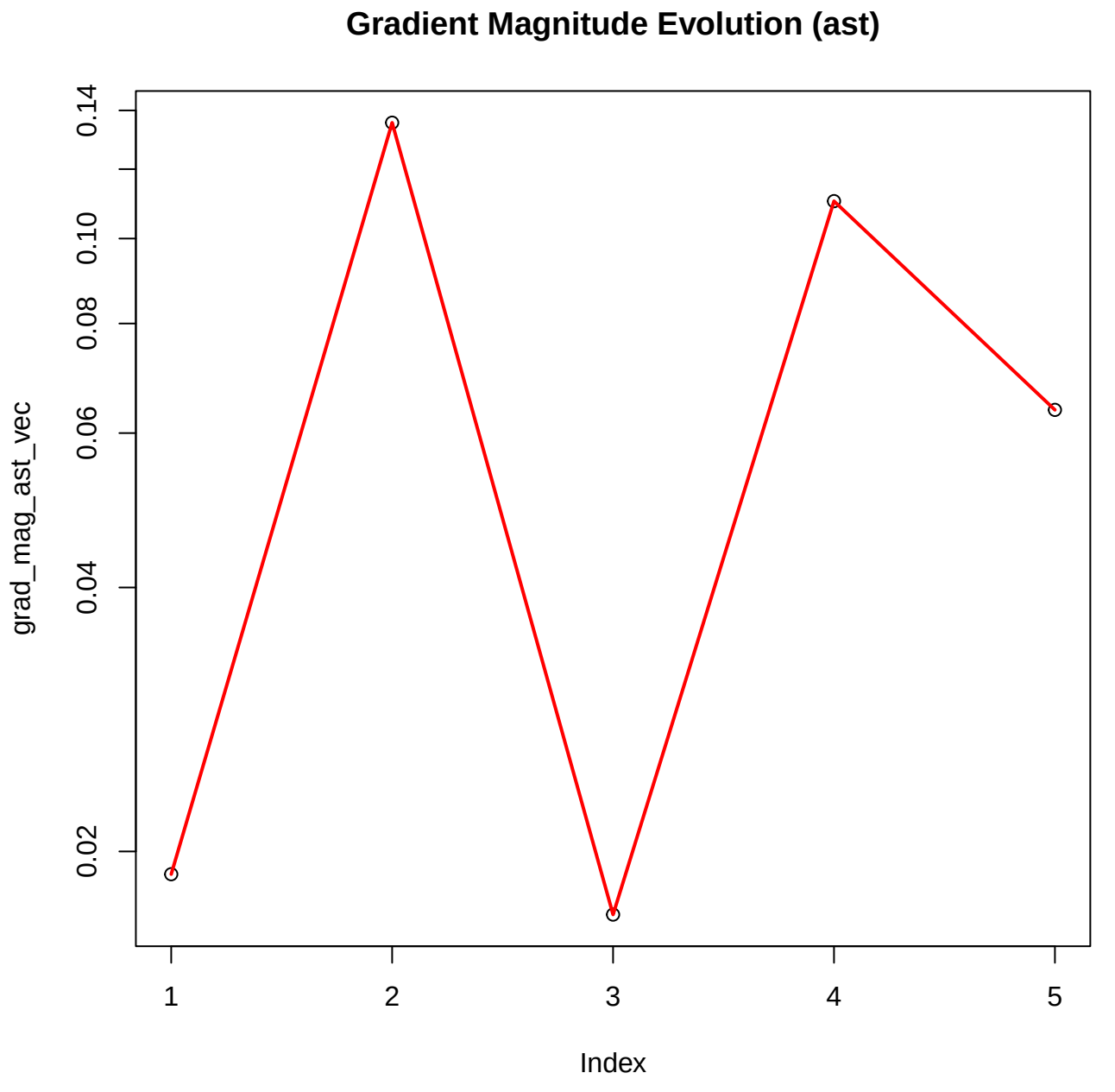
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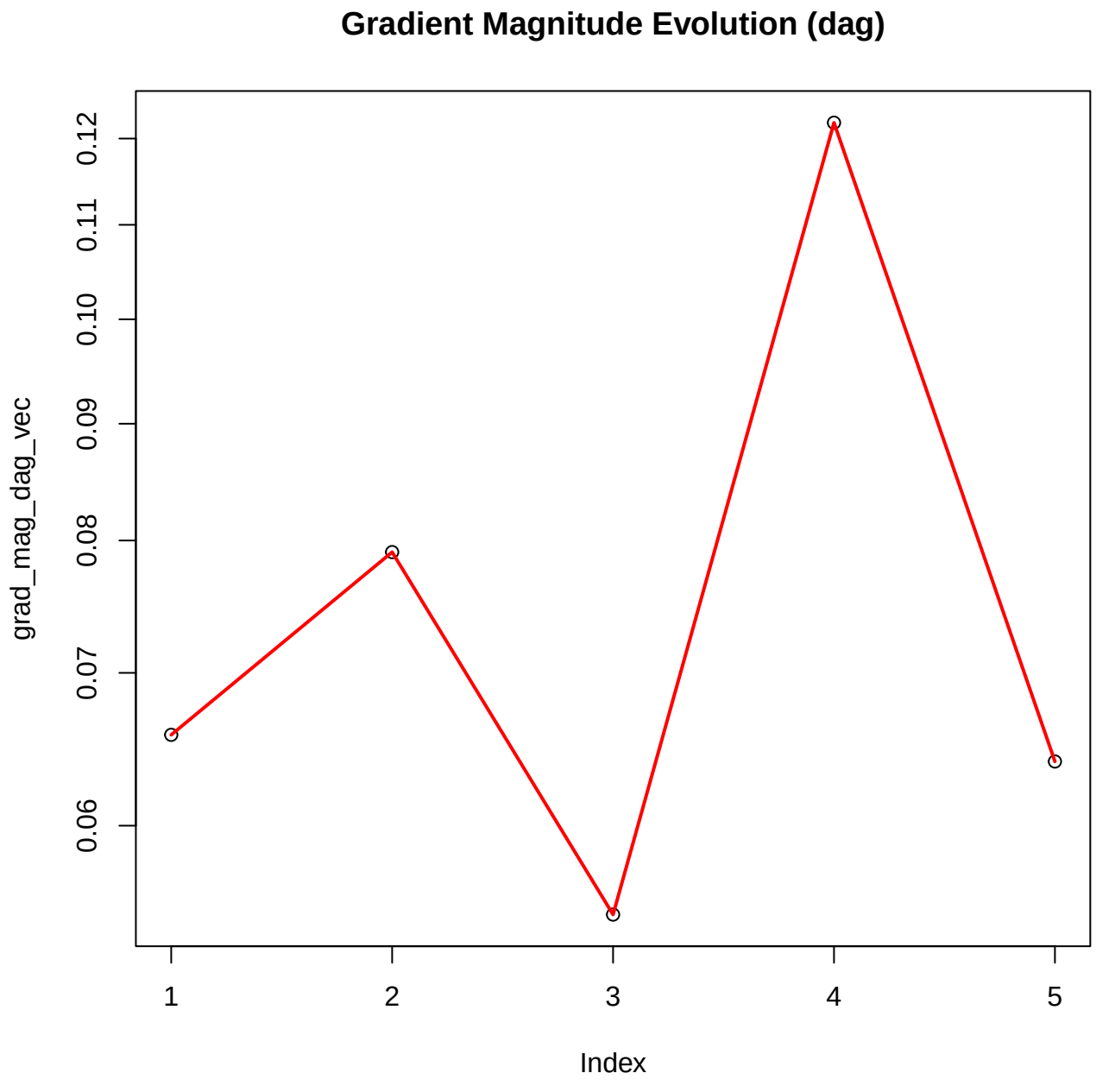
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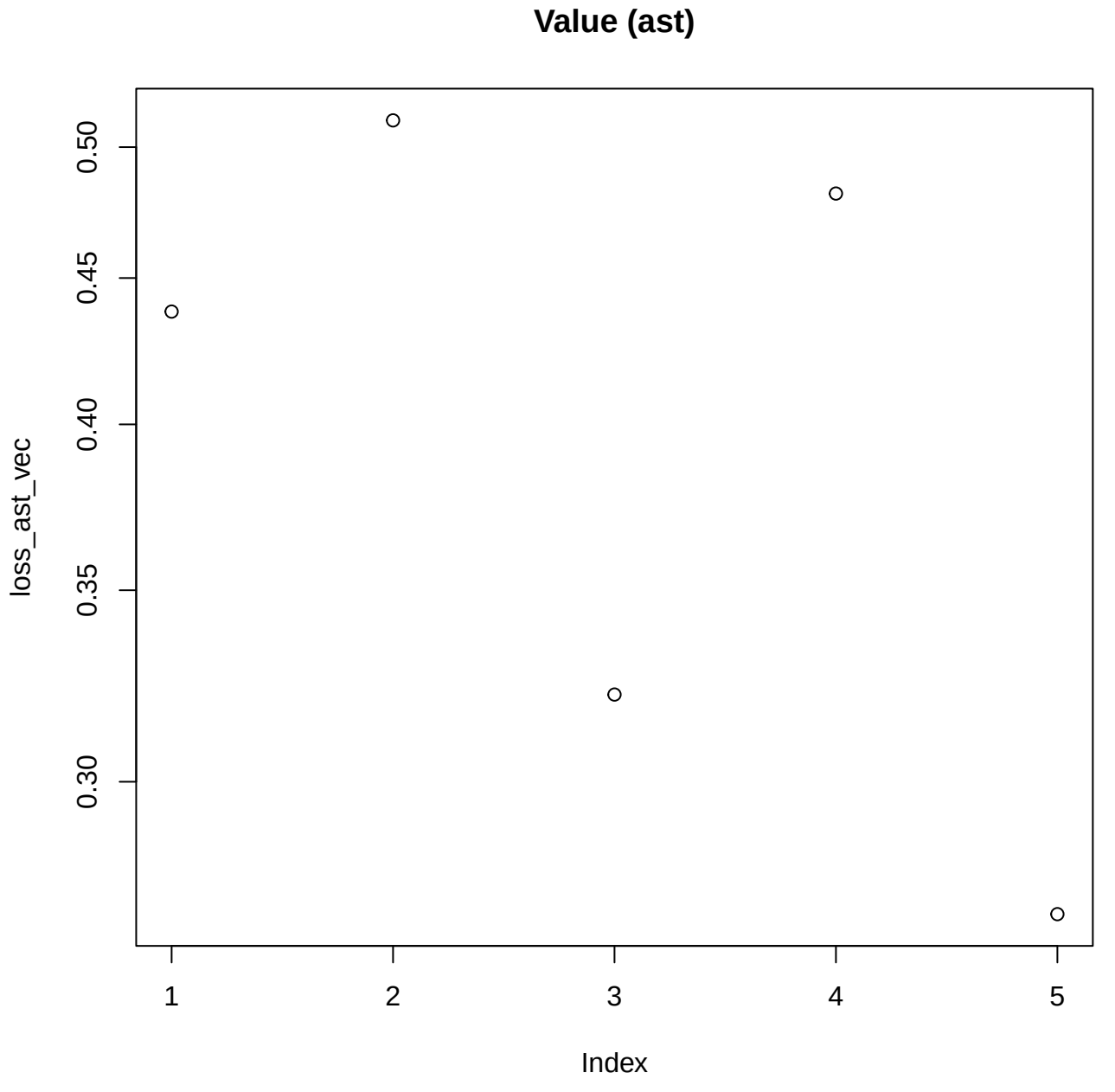
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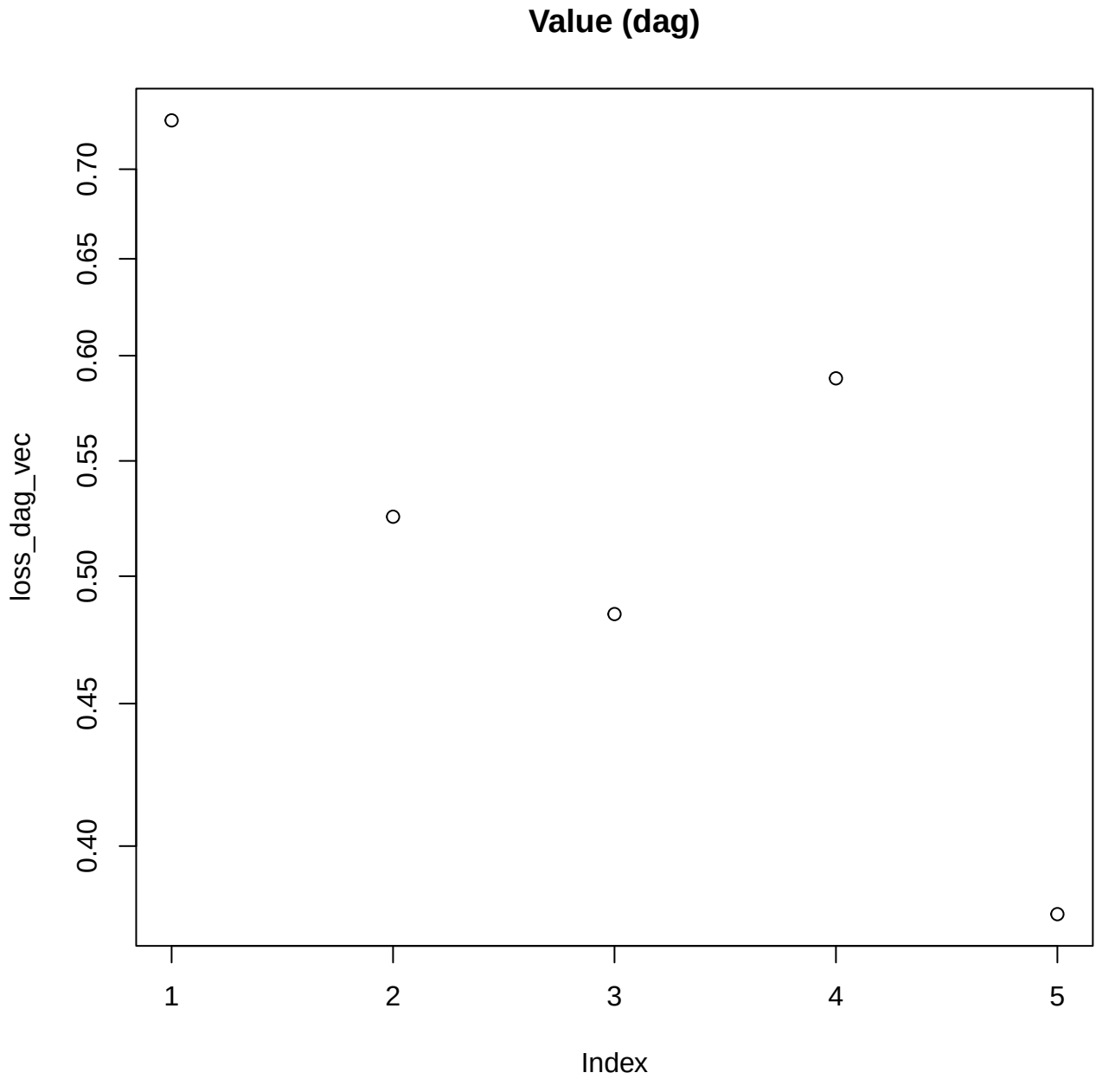
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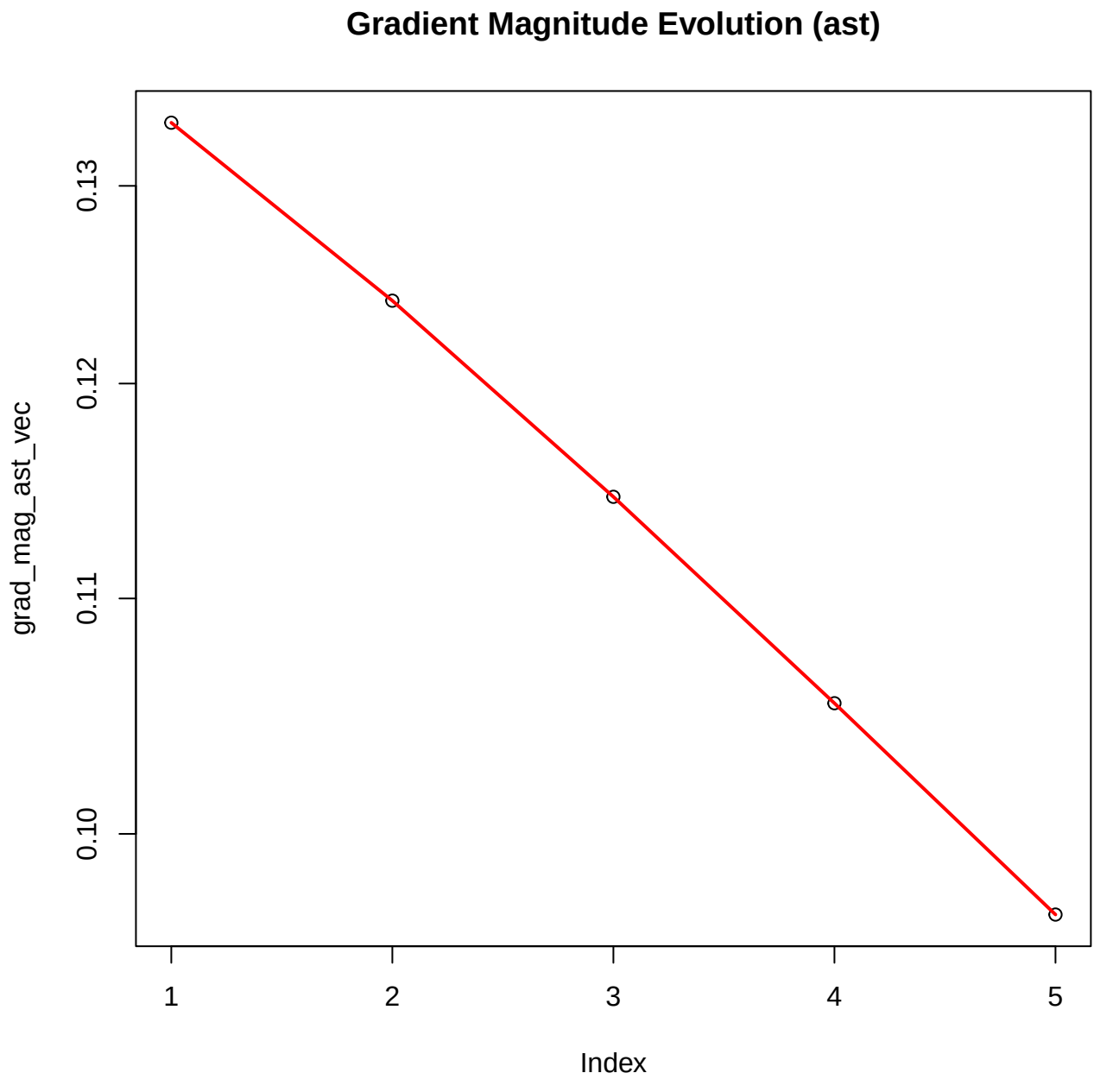
Gradient Magnitude Evolution (ast)
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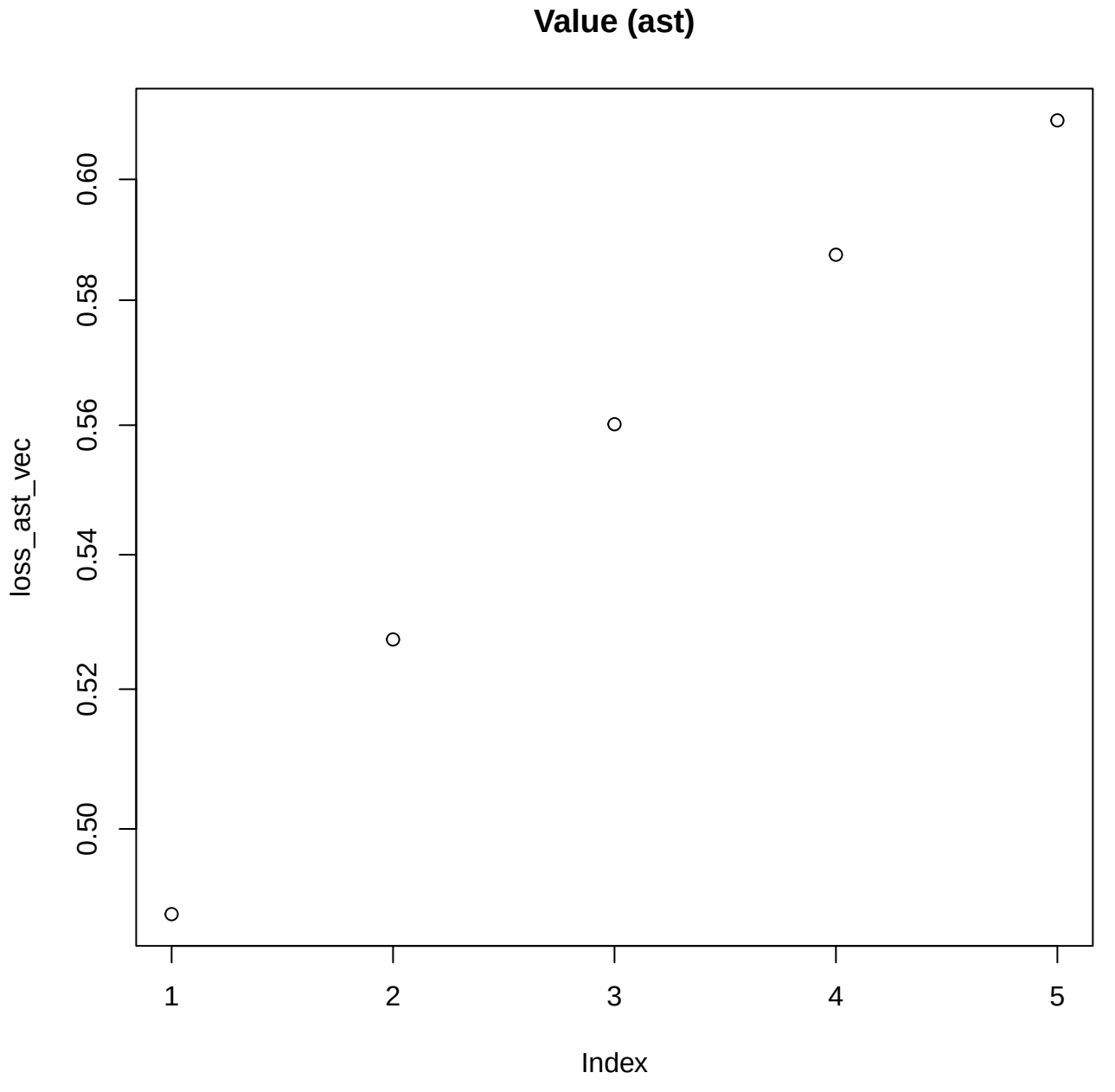
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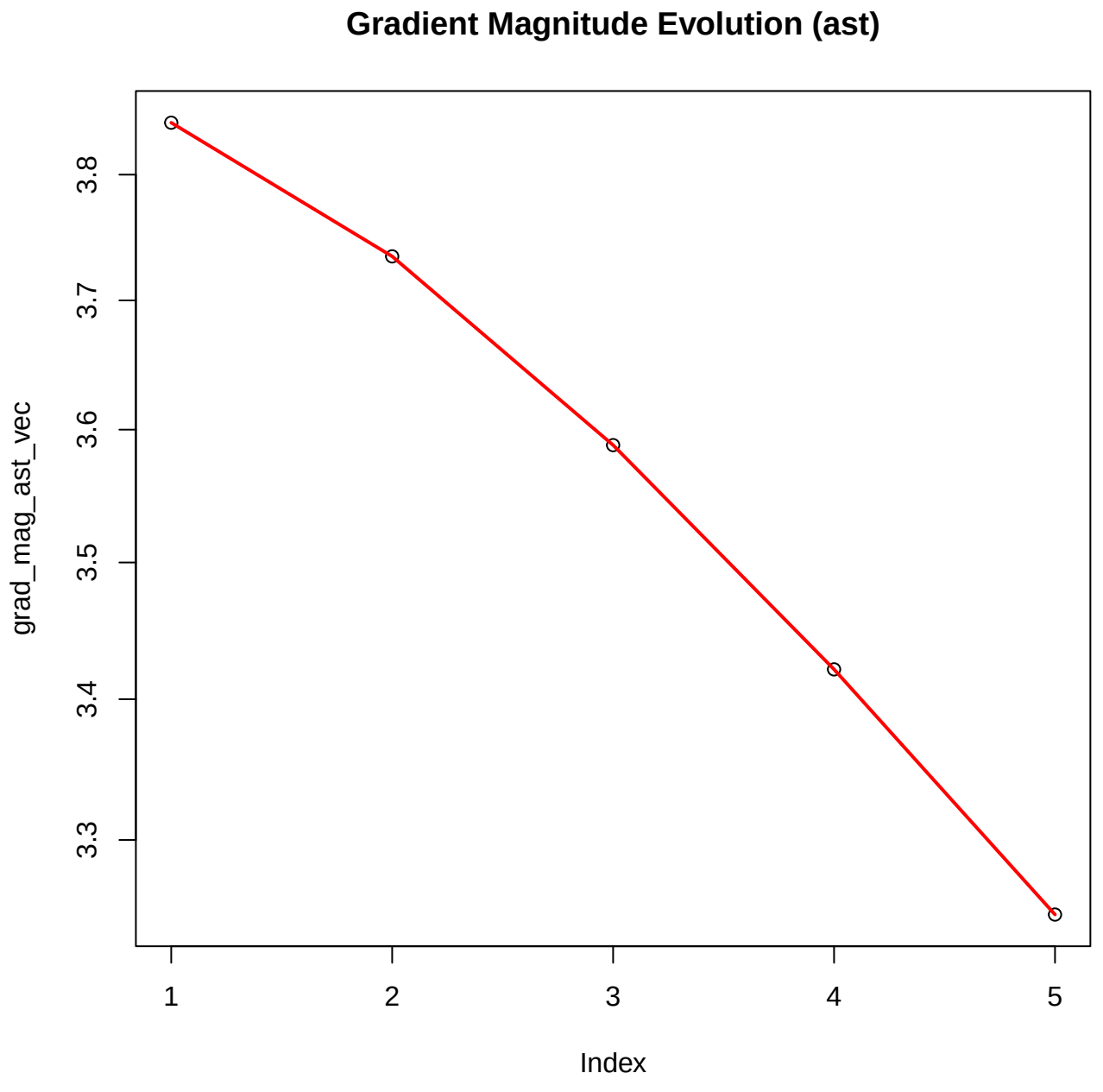
Gradient Magnitude Evolution (ast)
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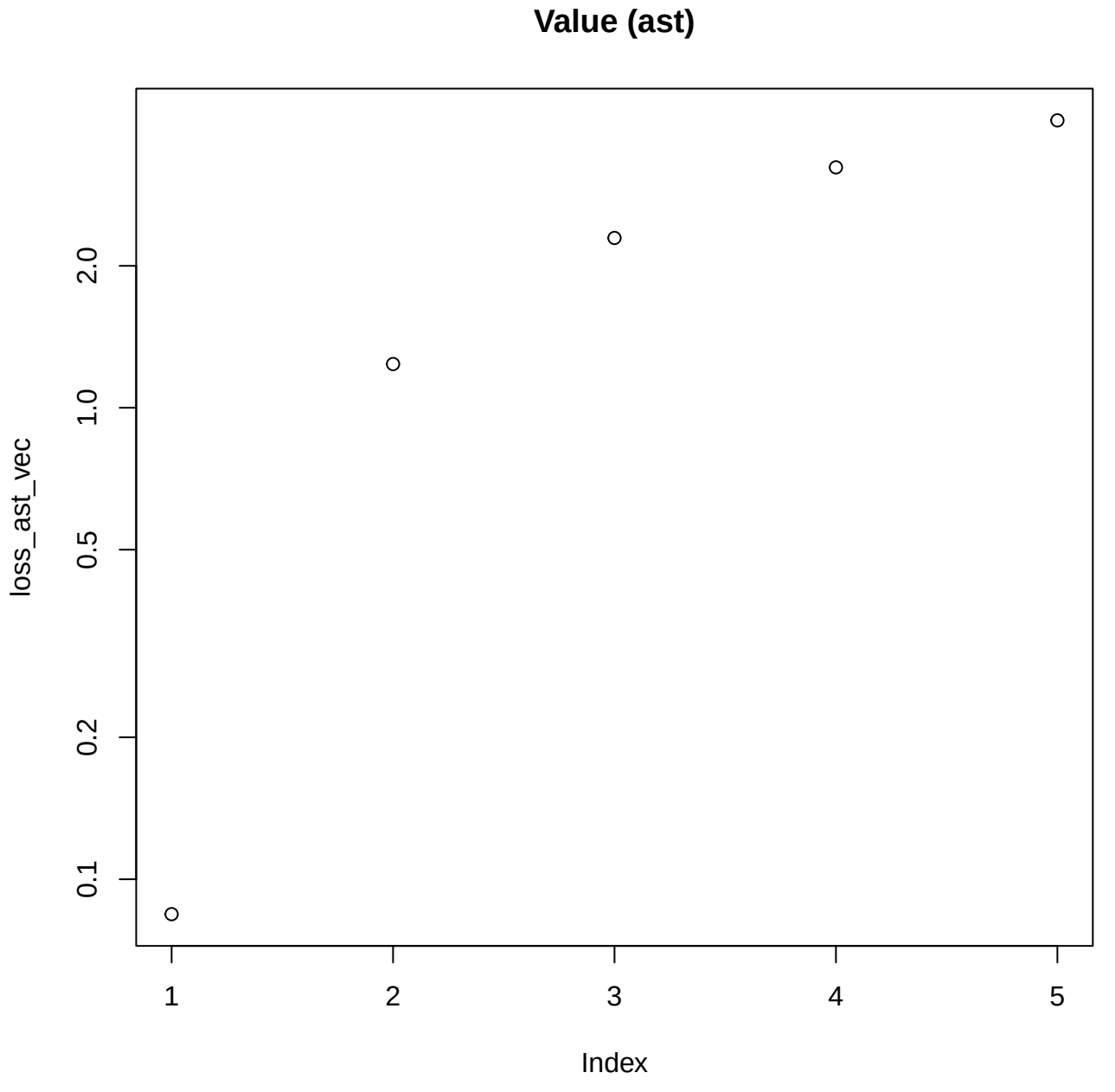
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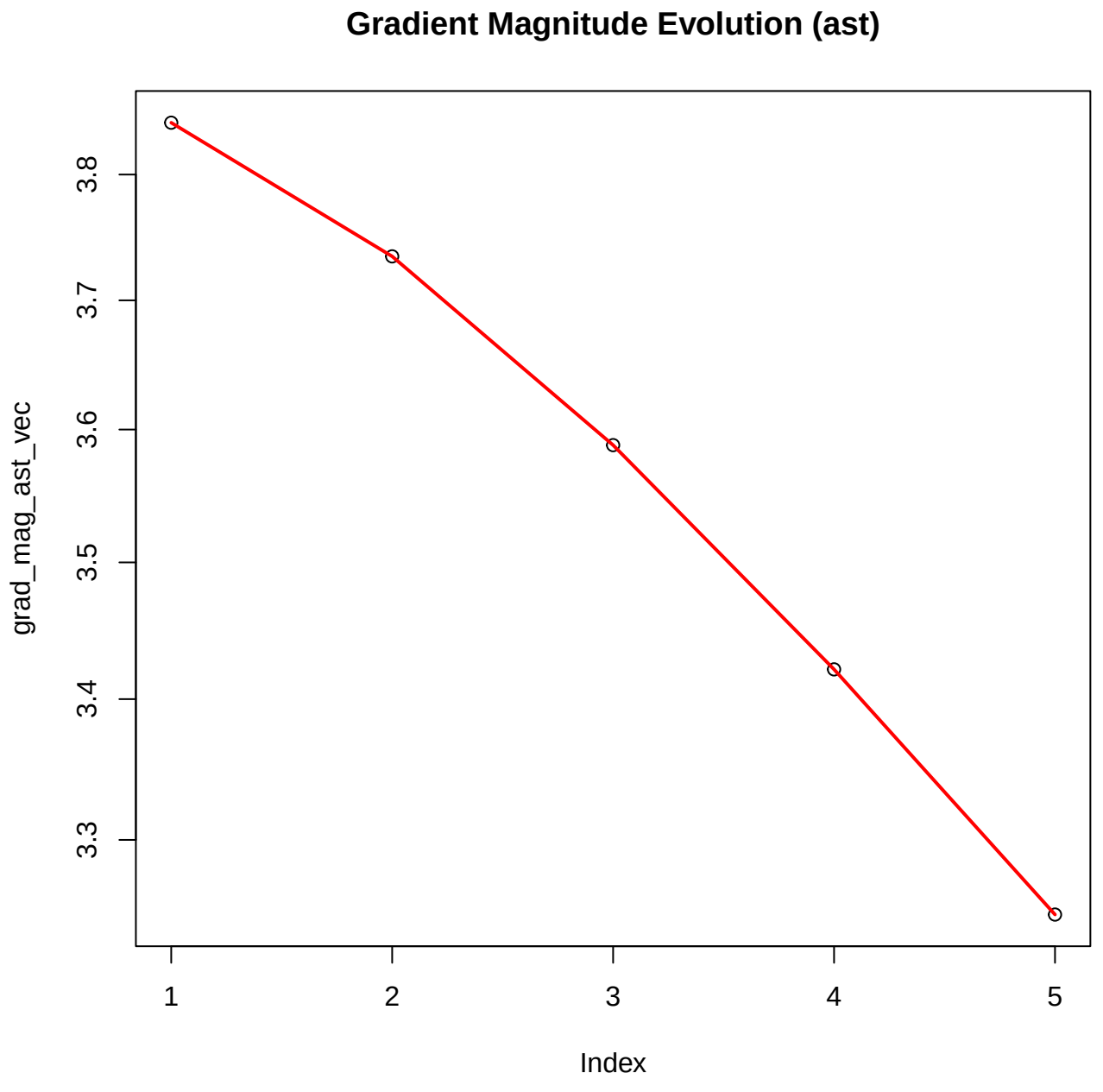
Gradient Magnitude Evolution (ast)
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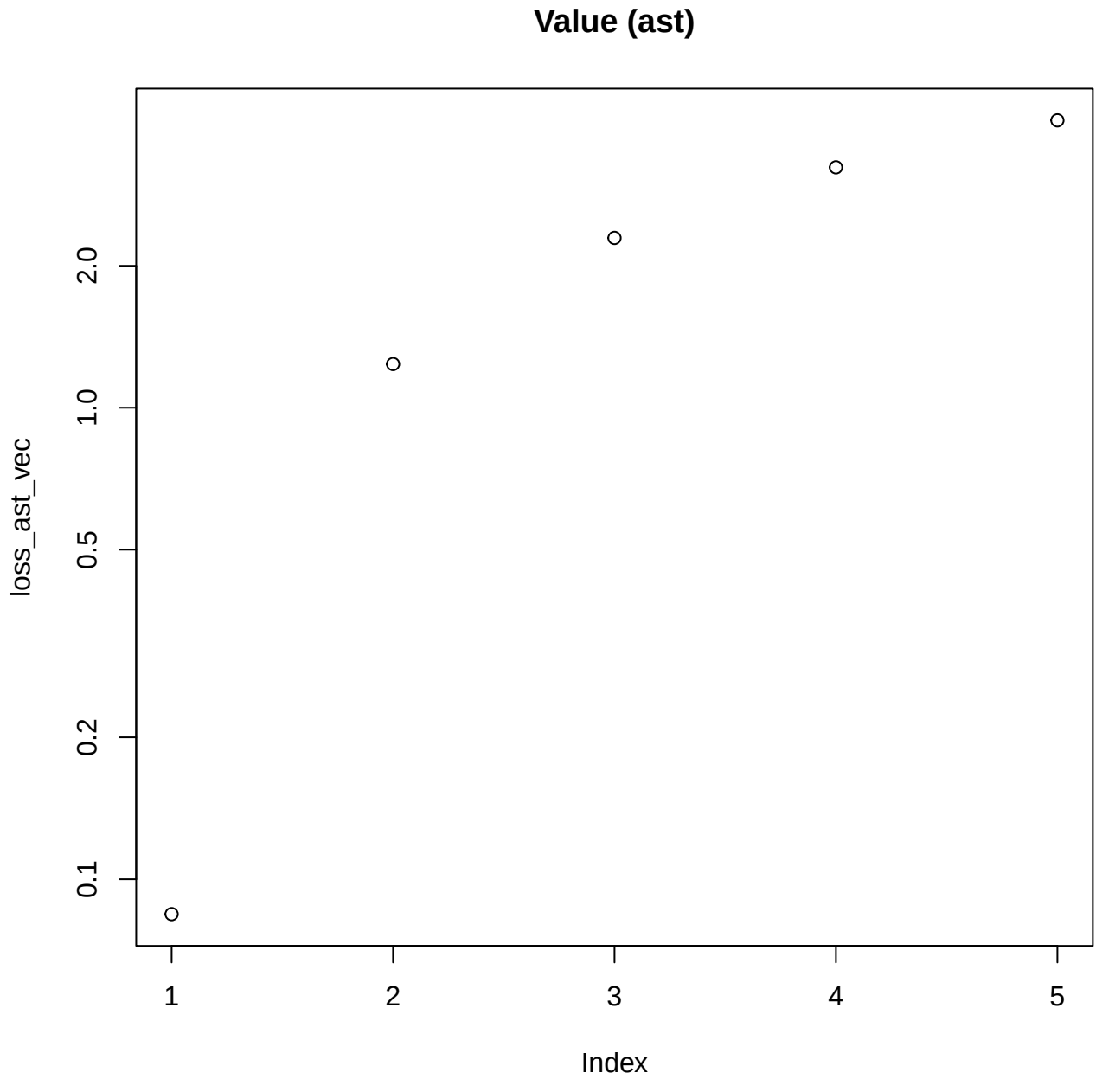
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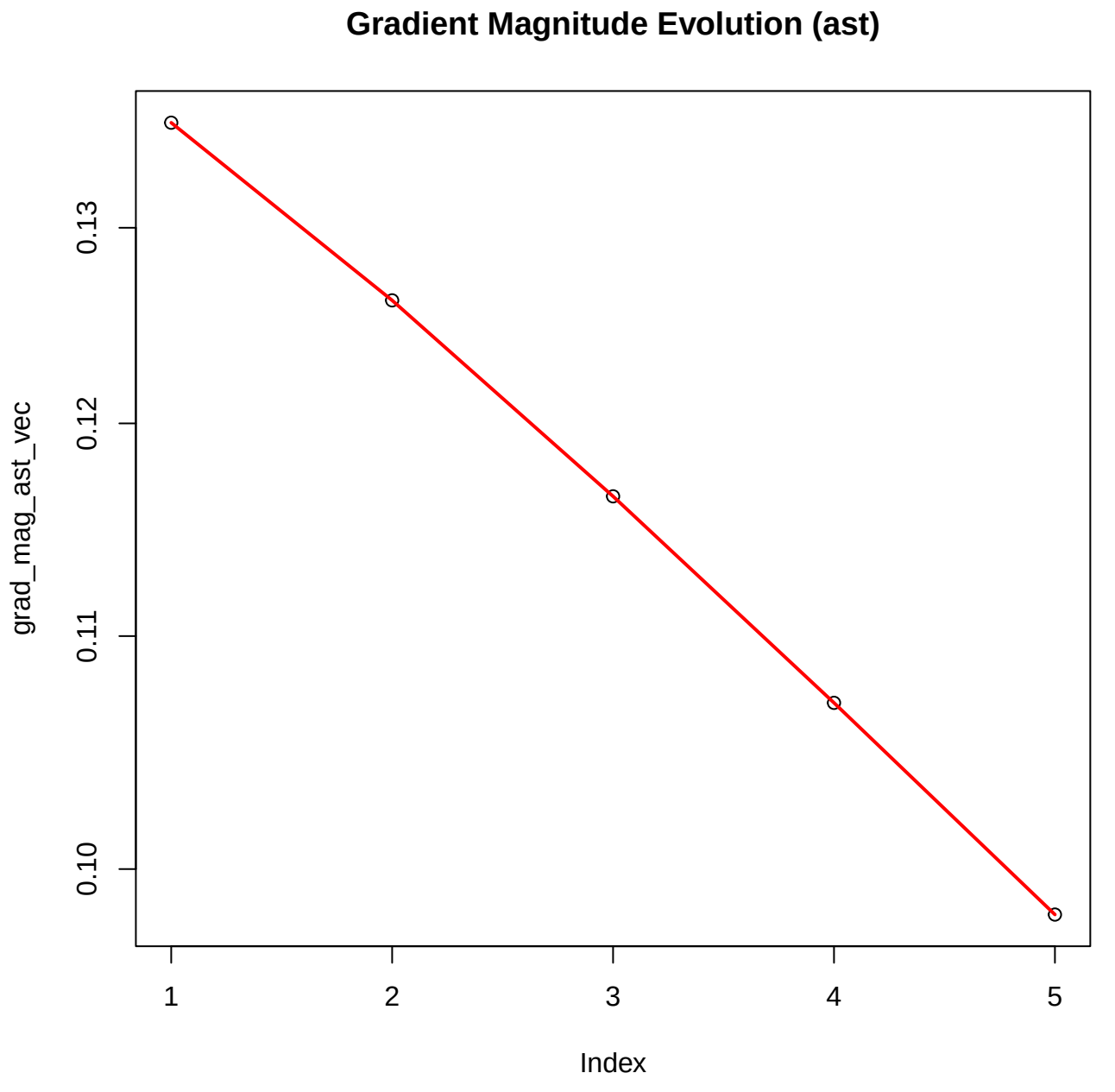
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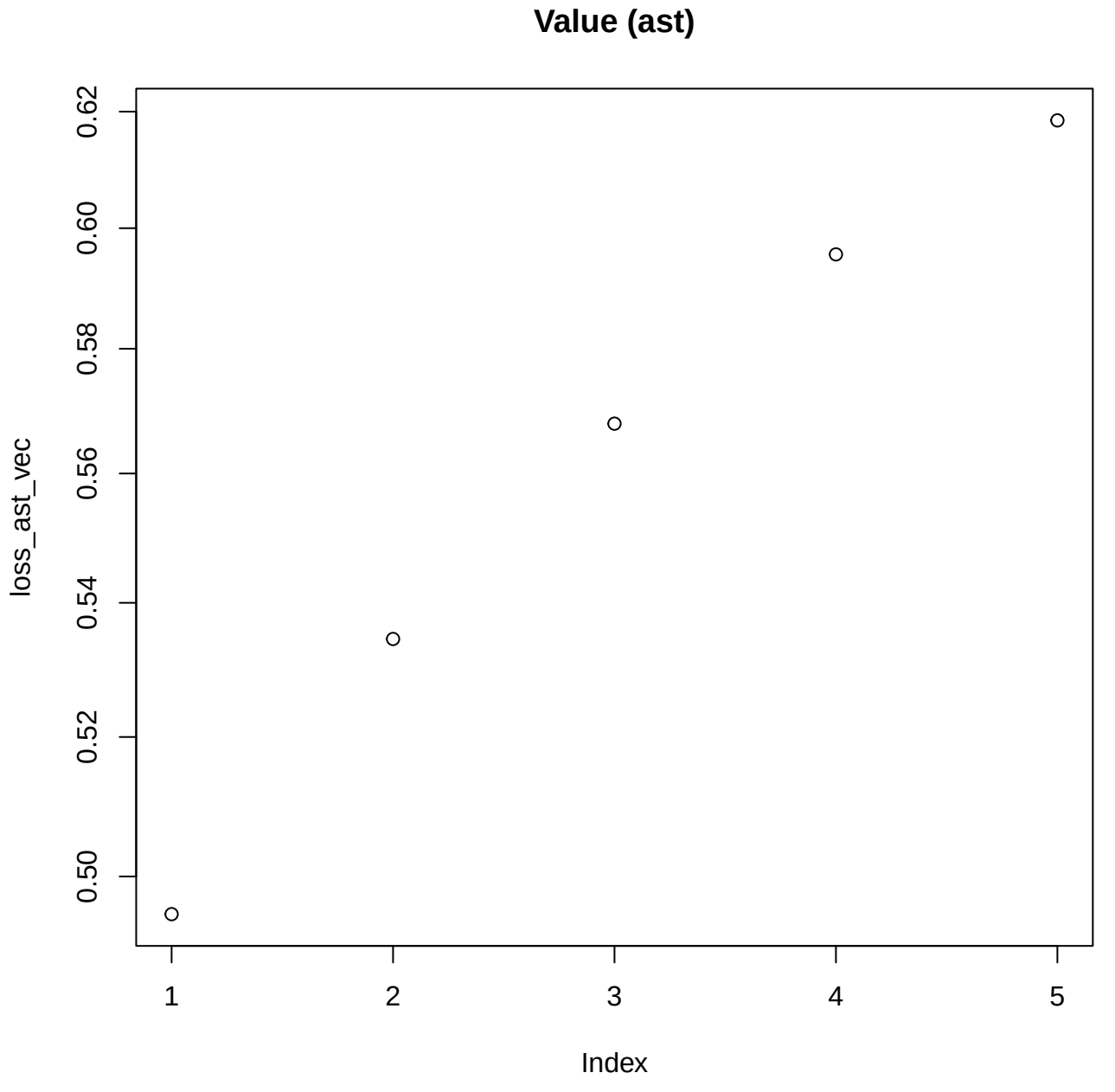
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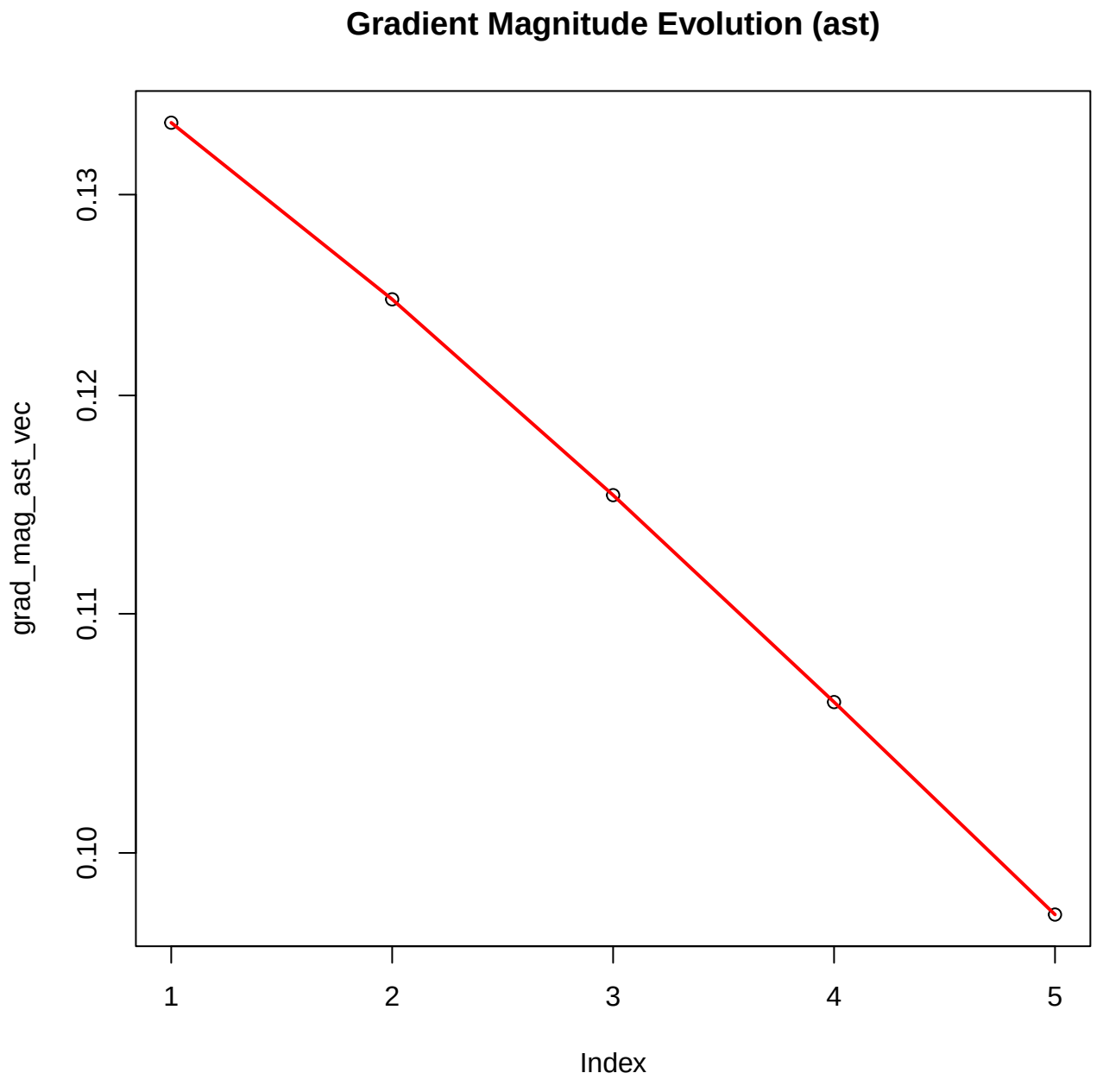
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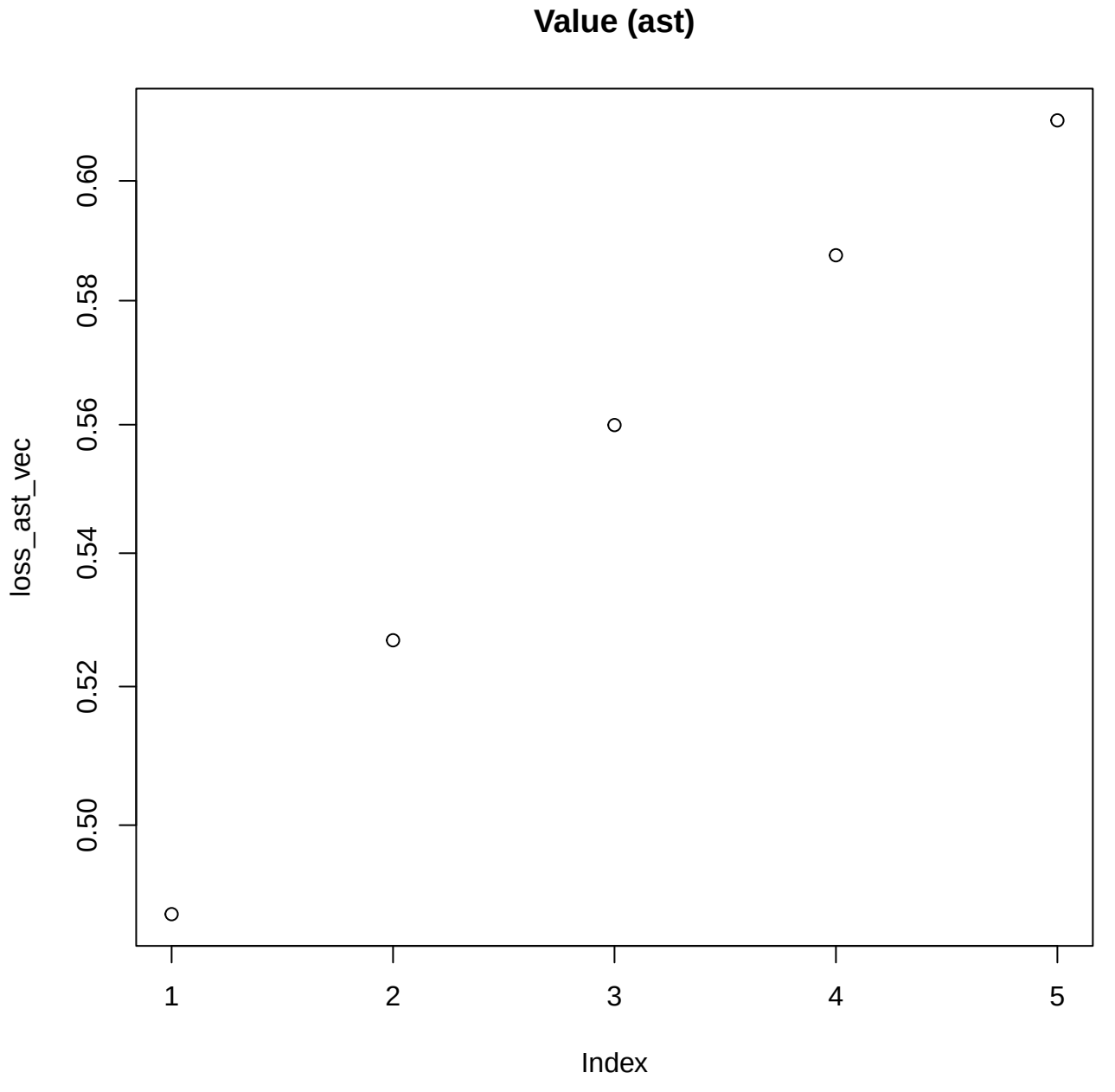
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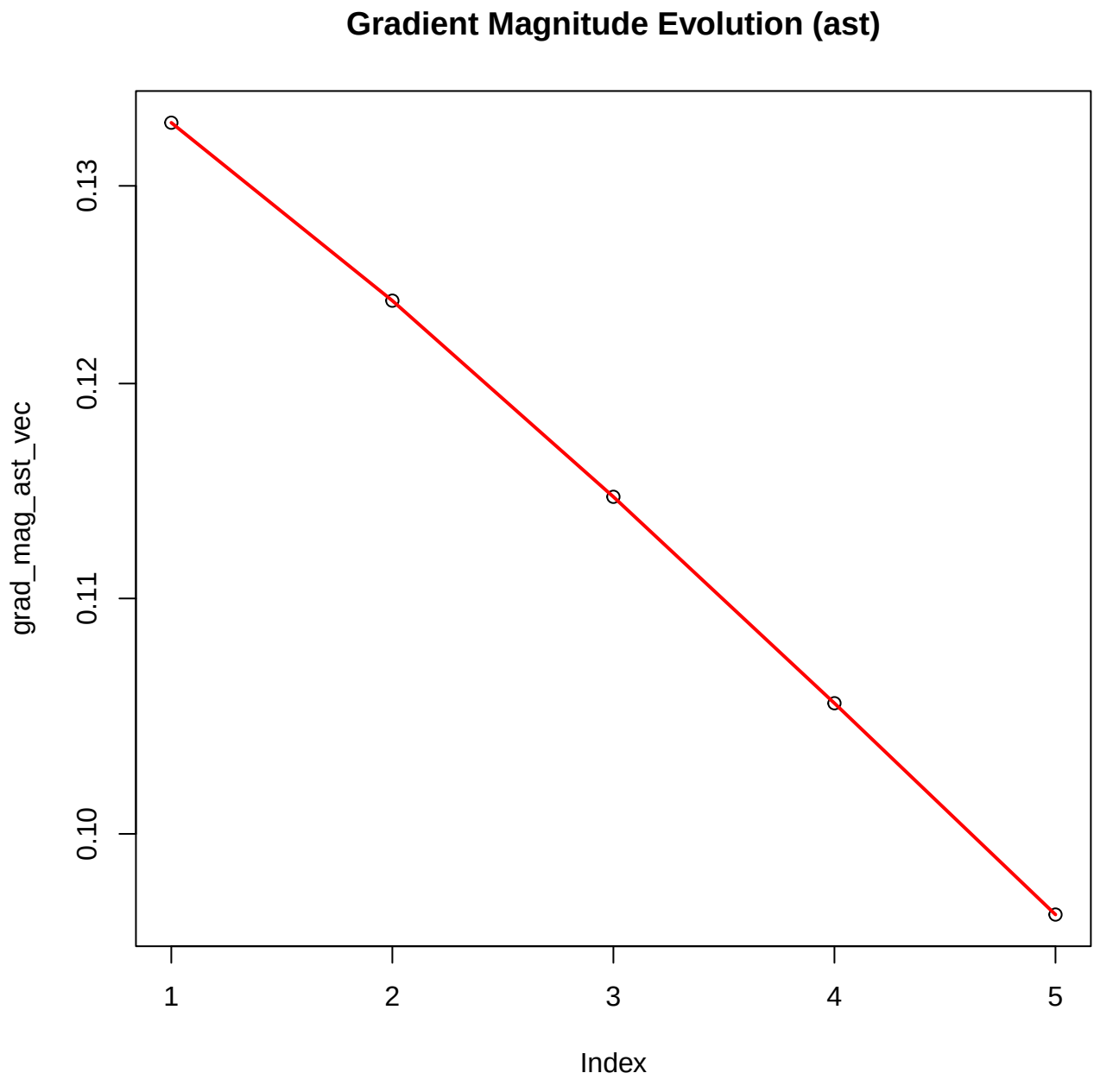
Gradient Magnitude Evolution (ast)
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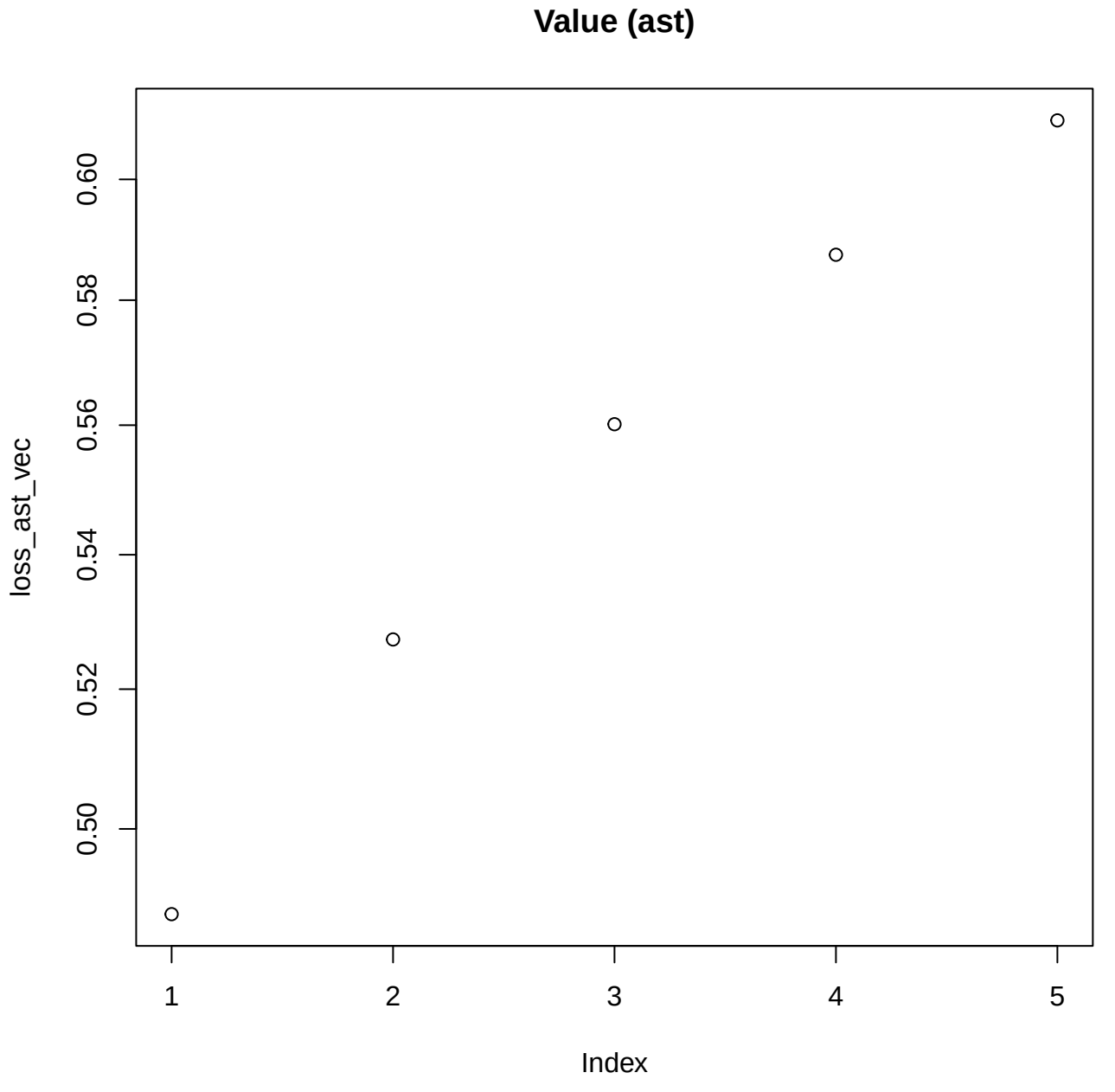
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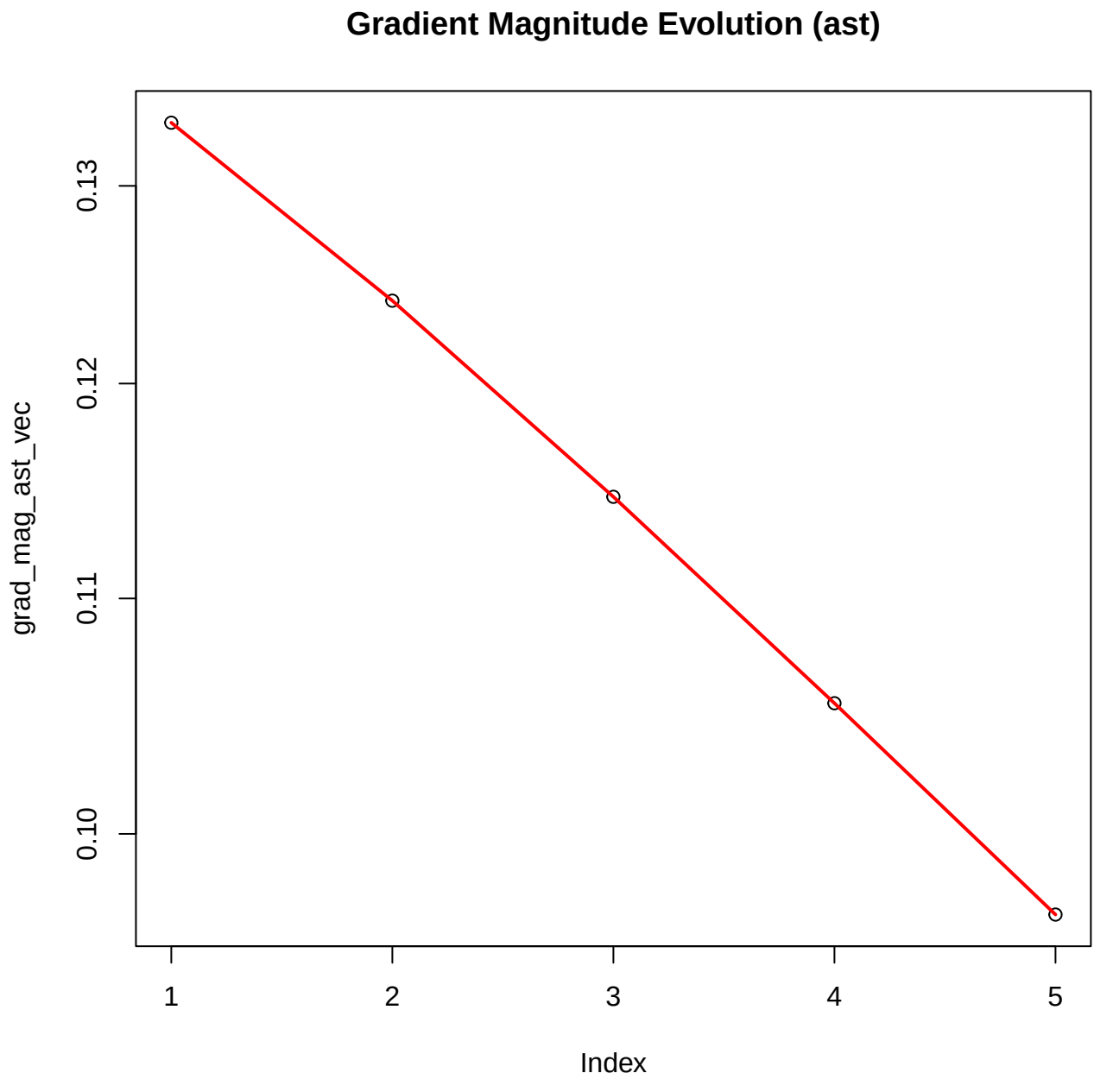
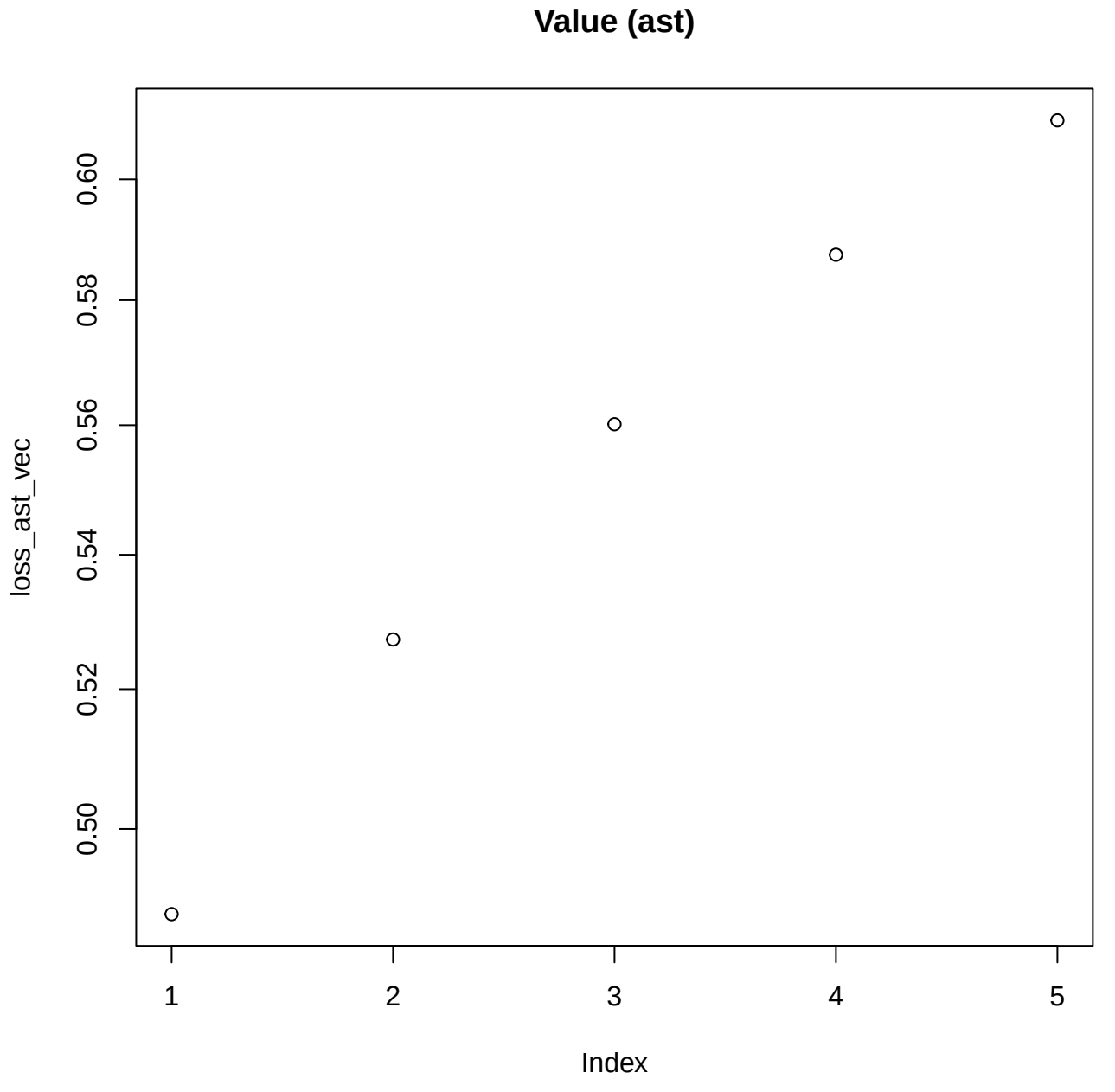
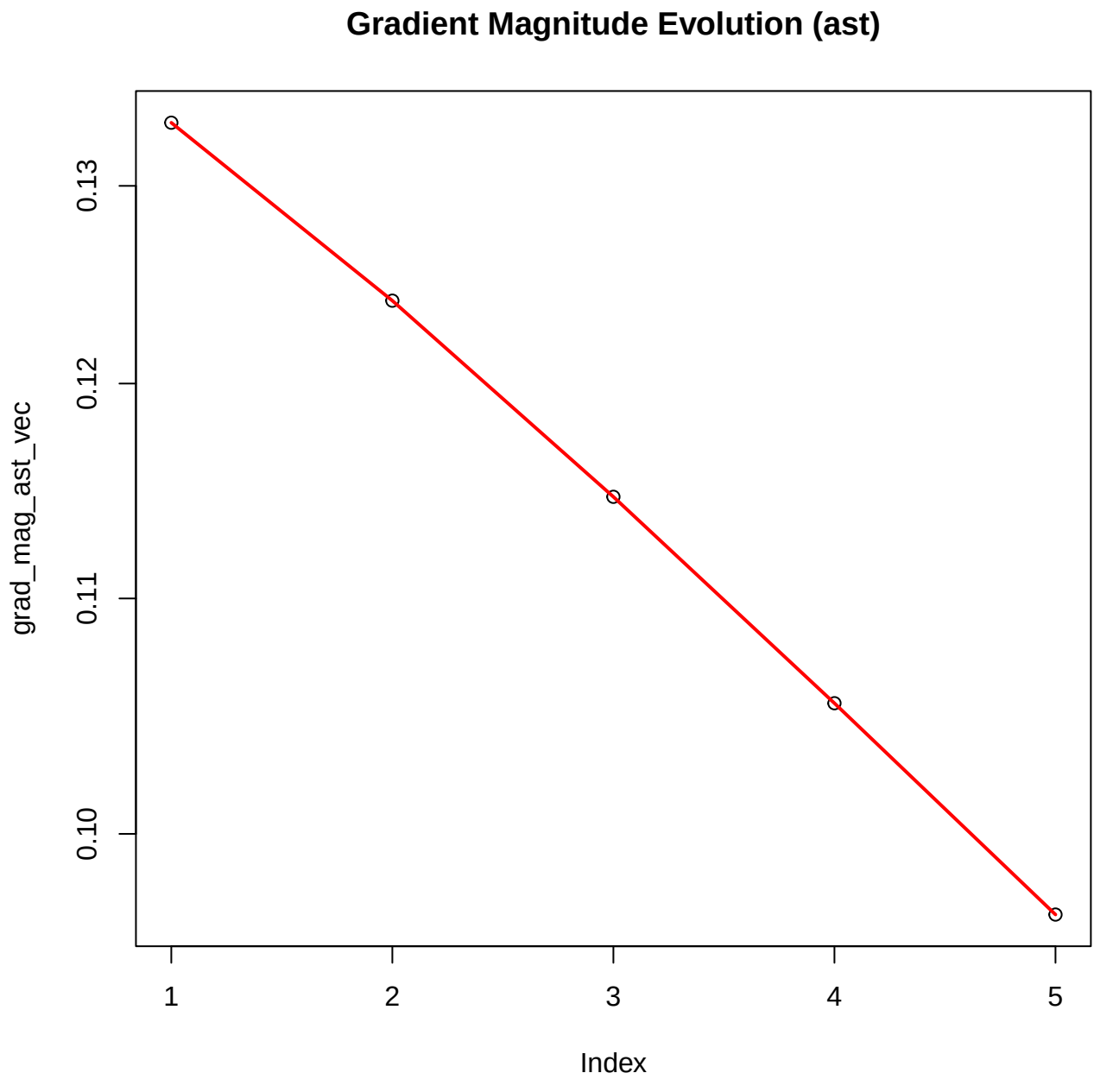
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
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1
2
3
4
5
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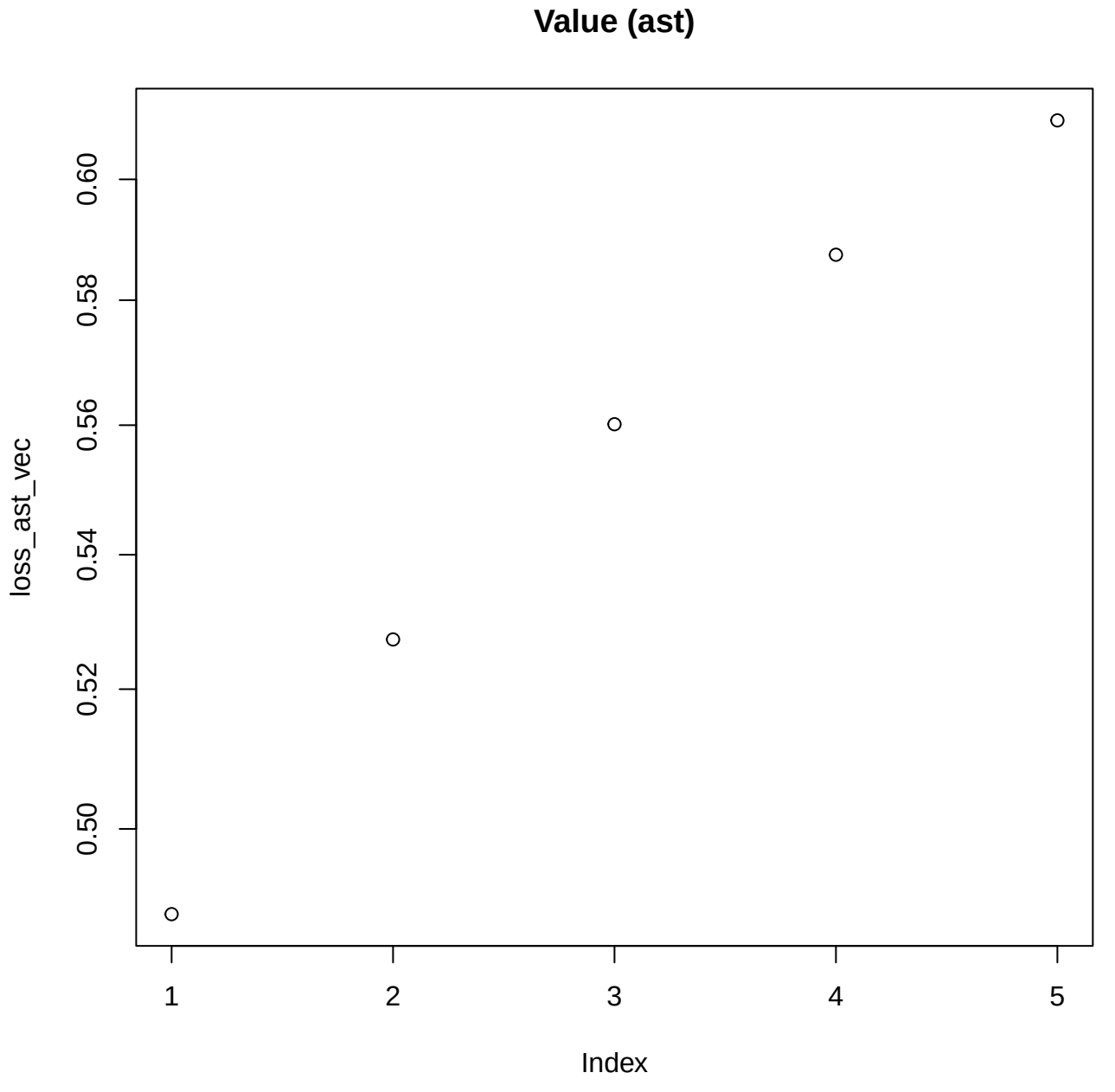
Value (ast)
loss_ast_vec
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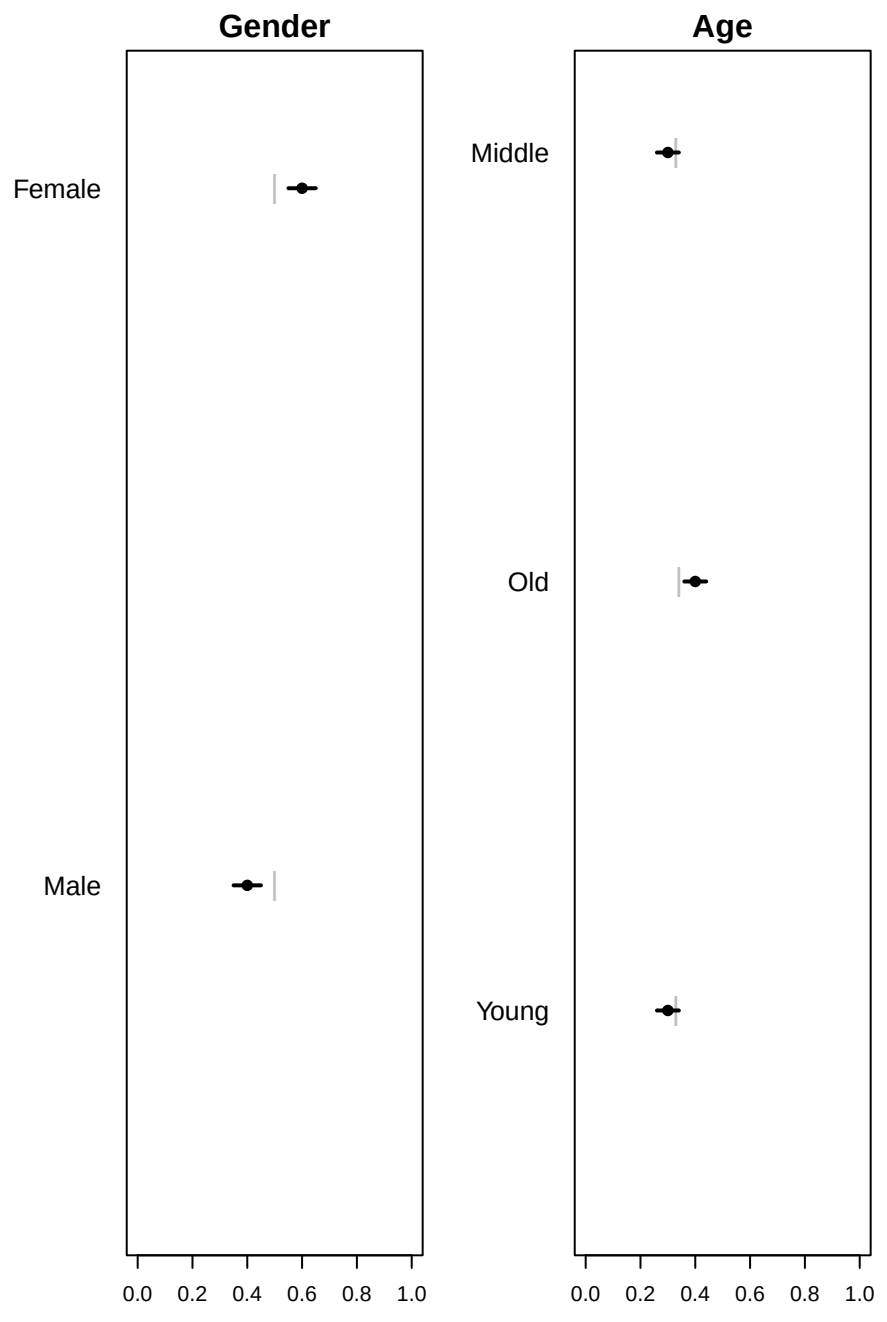
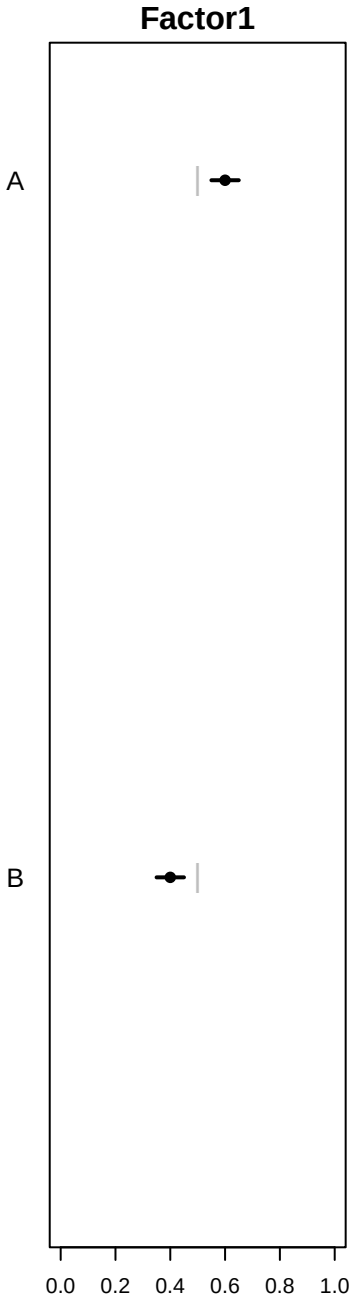
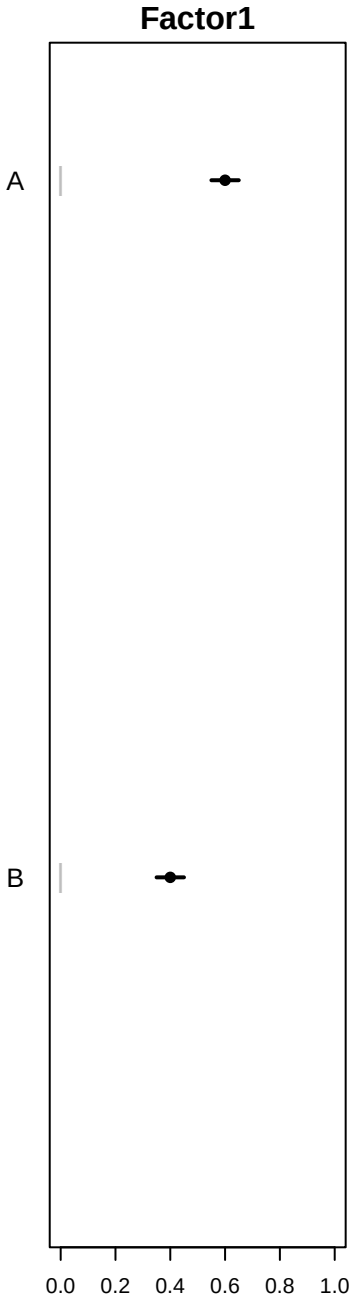
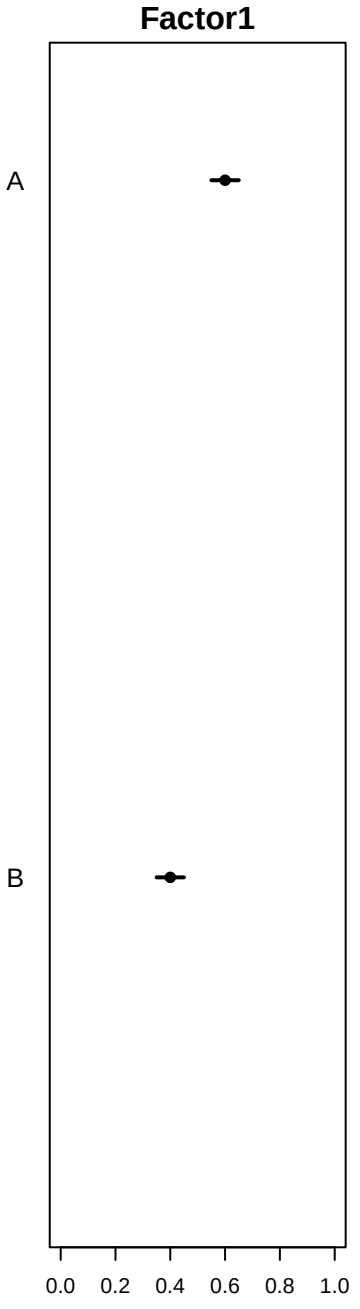
0.52
0.54
0.56
0.58
0.60
1
2
3
4
5
Index


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0
Age
Middle
Old
Young
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
0.0
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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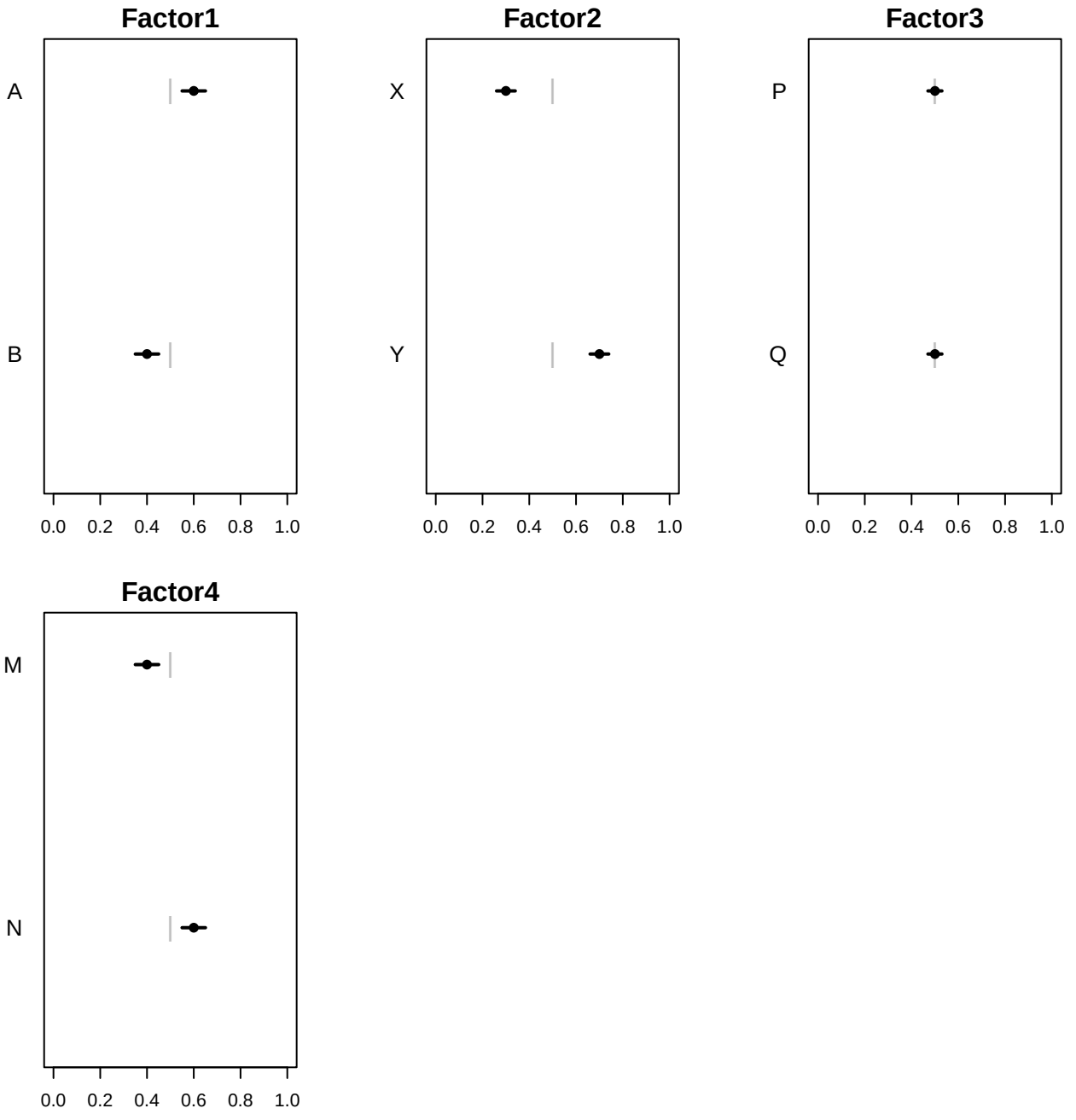
0.2
0.4
0.6
0.8
1.0


Factor1
A
B
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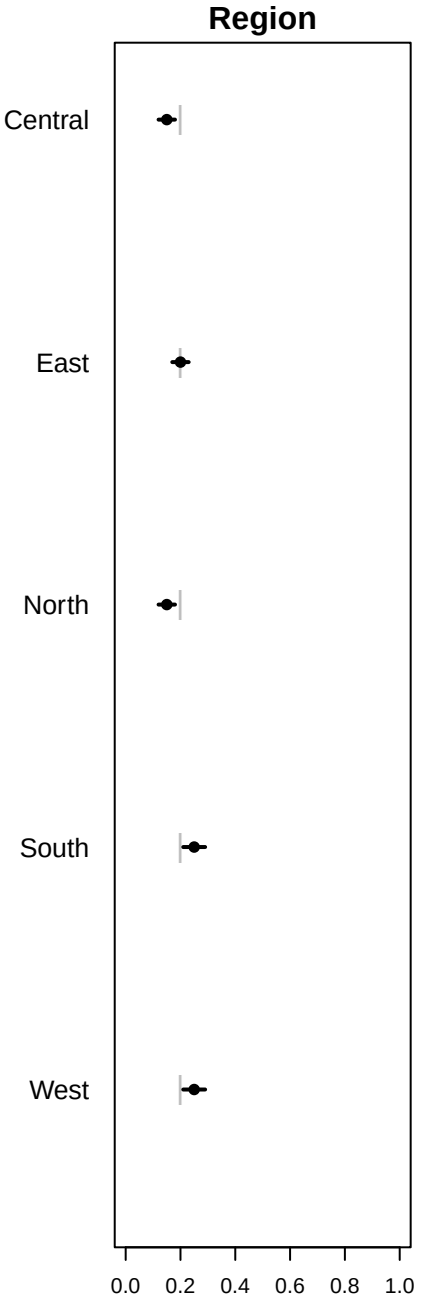
0.2
0.4
0.6
0.8
1.0
Factor2
X
Y
0.0
0.2
0.4
0.6
0.8
1.0
Factor3
P
Q
0.0
0.2
0.4
0.6
0.8
1.0
Factor4
M
N
0.0
0.2
0.4
0.6
0.8
1.0


Region
Central
East
North
South
West
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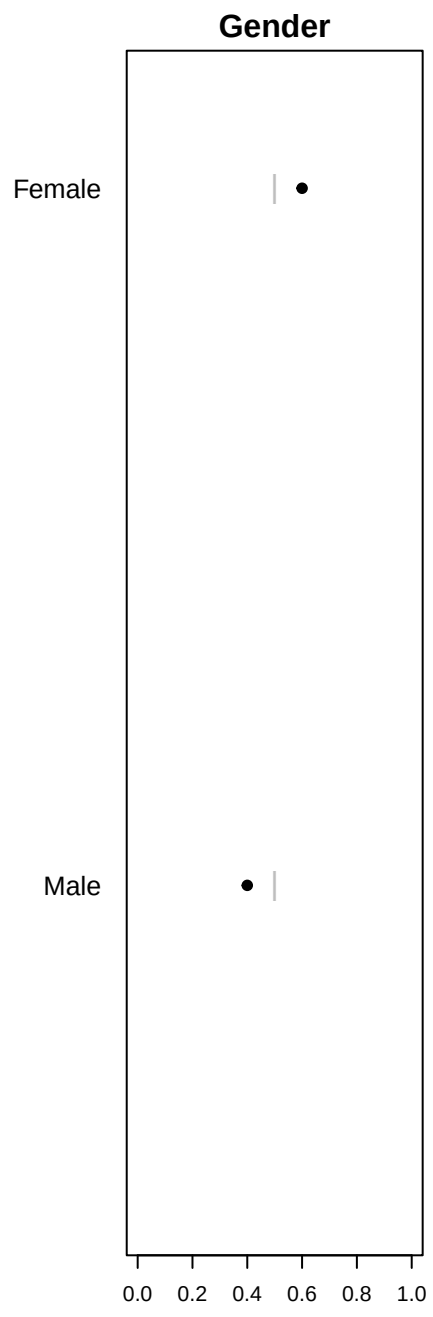
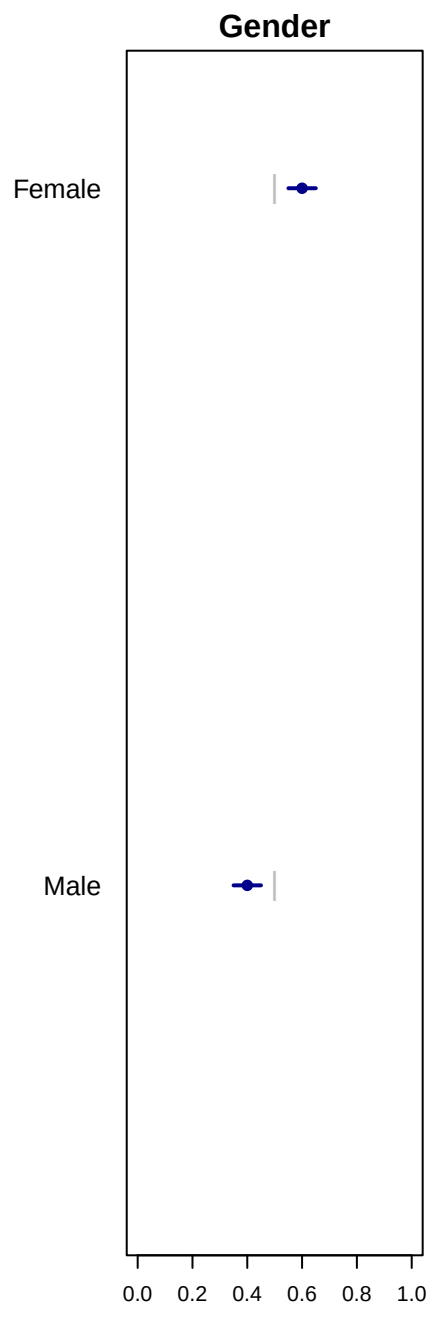
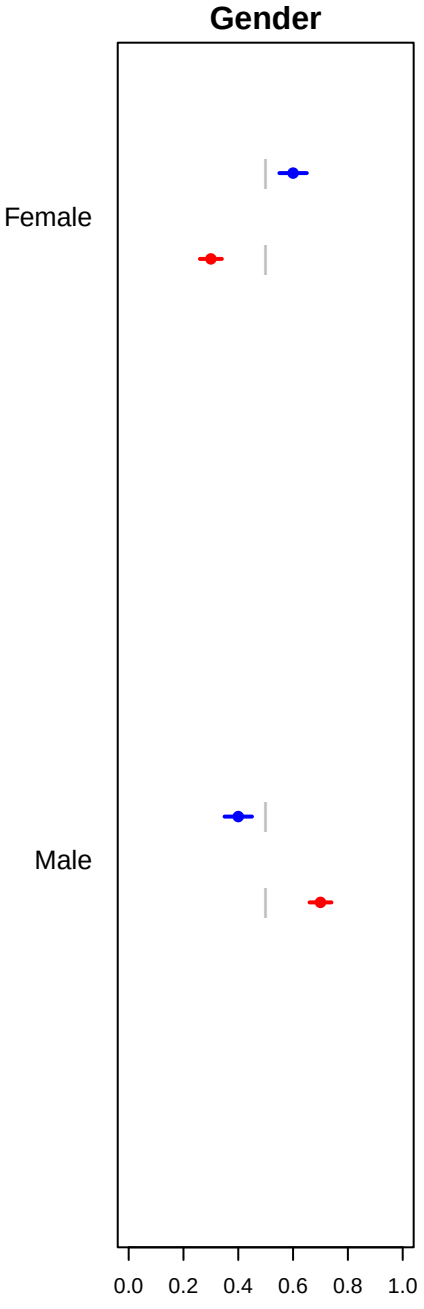
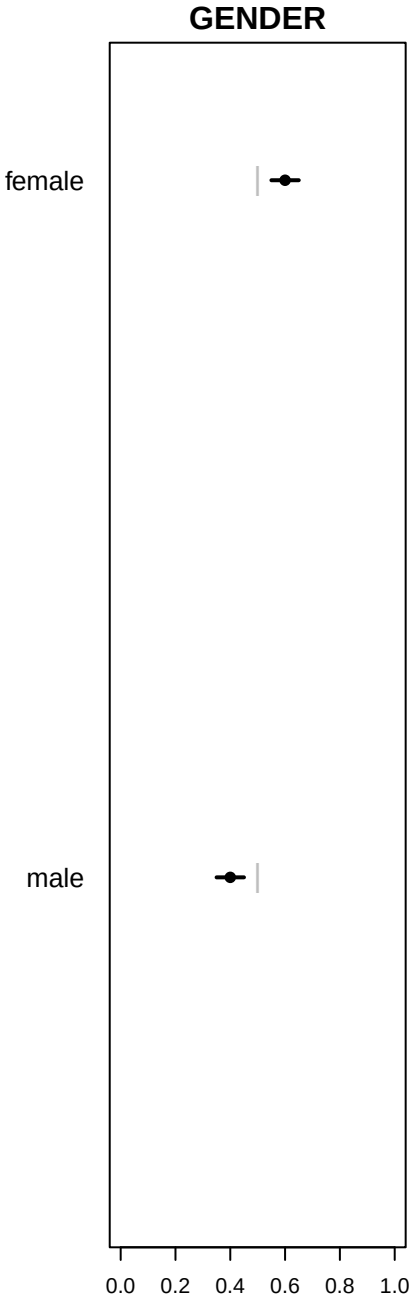
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0


GENDER
female
male
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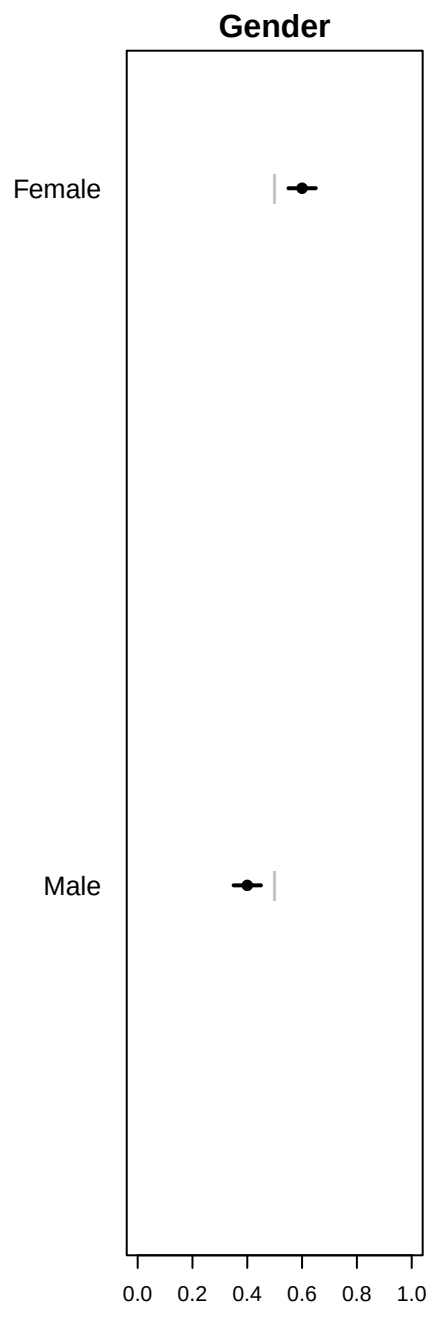
0.2
0.4
0.6
0.8
1.0


Gender
Female
Male
0.0
0.2
0.4
0.6
0.8
1.0